SM-X818U_REV0.5_SVC

2023.05.02

sheet01 : RF (1/3)
sheet02 : RF (2/3)
sheet03 : RF (3/3)
sheet04 : CONNECTIVITY - BT/WIFI(WCN6856), GPS
sheet05 : AP_SM8550(Secondary PCB), JTAG, UFS
sheet06 : PMIC(Secondary PCB), IF PMIC, D/C, OVP
sheet07 : AUDIO
sheet08 : CAMERA, DISPLAY
sheet09 : SENSOR, SD/SIM, KEY, I/F, eSIM
sheet10 : RF PMIC, 5G IFIC
sheet11 : WACOM PCB (Ext.) , ANT SUB PCB (Ext.)
sheet12 : USB PCB (Ext.)
sheet13 : WIFI TR PCB (Ext.) / TL PCB (Ext.), CLIP, SMR, SUS, SHIELDCAN
sheet14 : Sencondary PCB PAD (MAIN PCB)

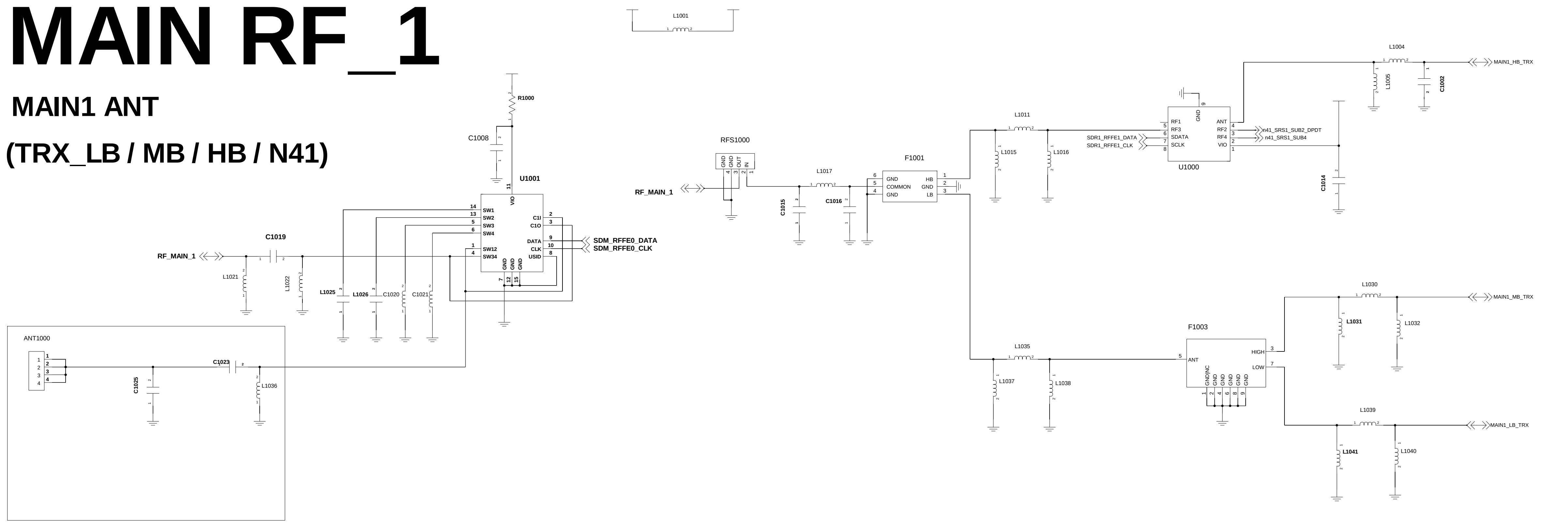
mmW,NA : X818U
Global,EUR : X816
WIFI : X810

TITLE TabS7P
SIZE A1
DWG NO PDM NUMBER
REV Rev
DRAWN BY User Name
SHEET 0 of 15
Date Drawing Date

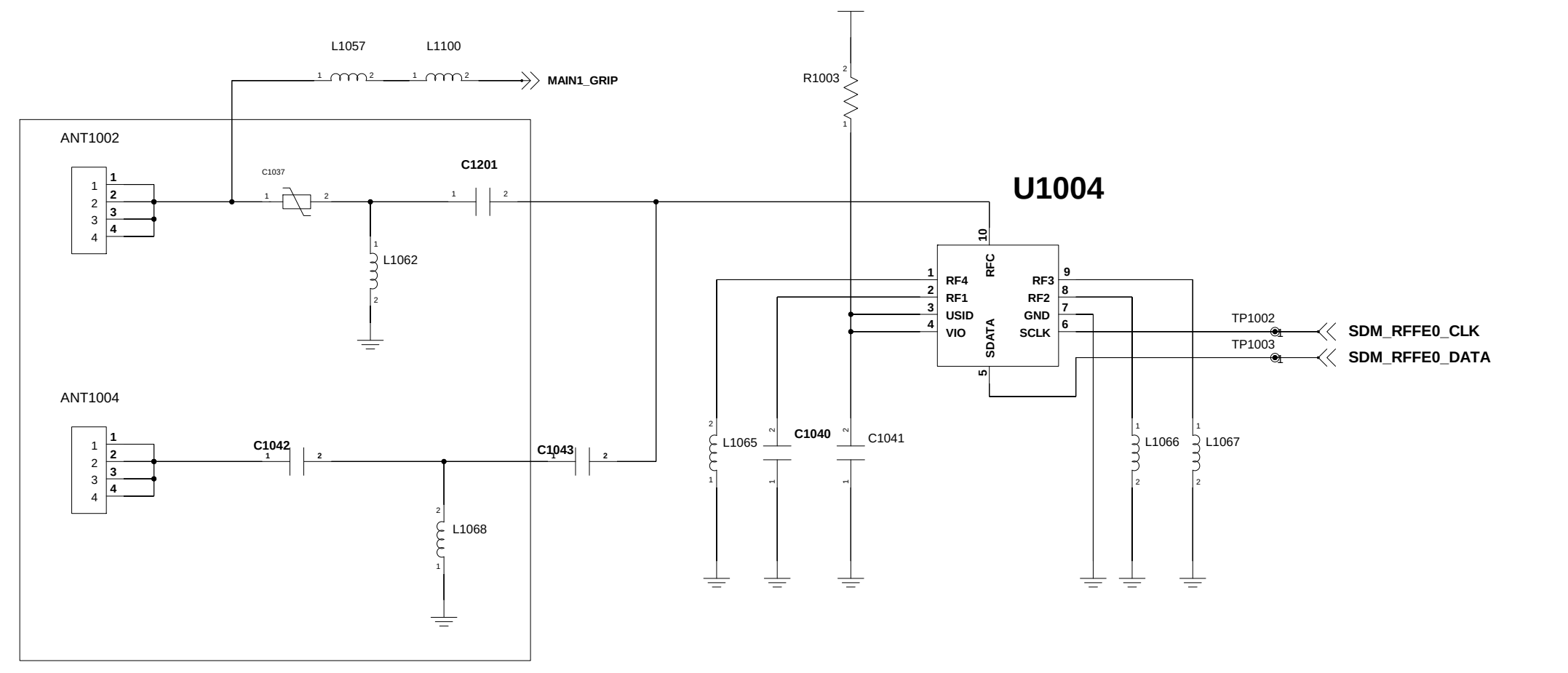
MAIN RF_1

MAIN1 ANT

(TRX_LB / MB / HB / N41)

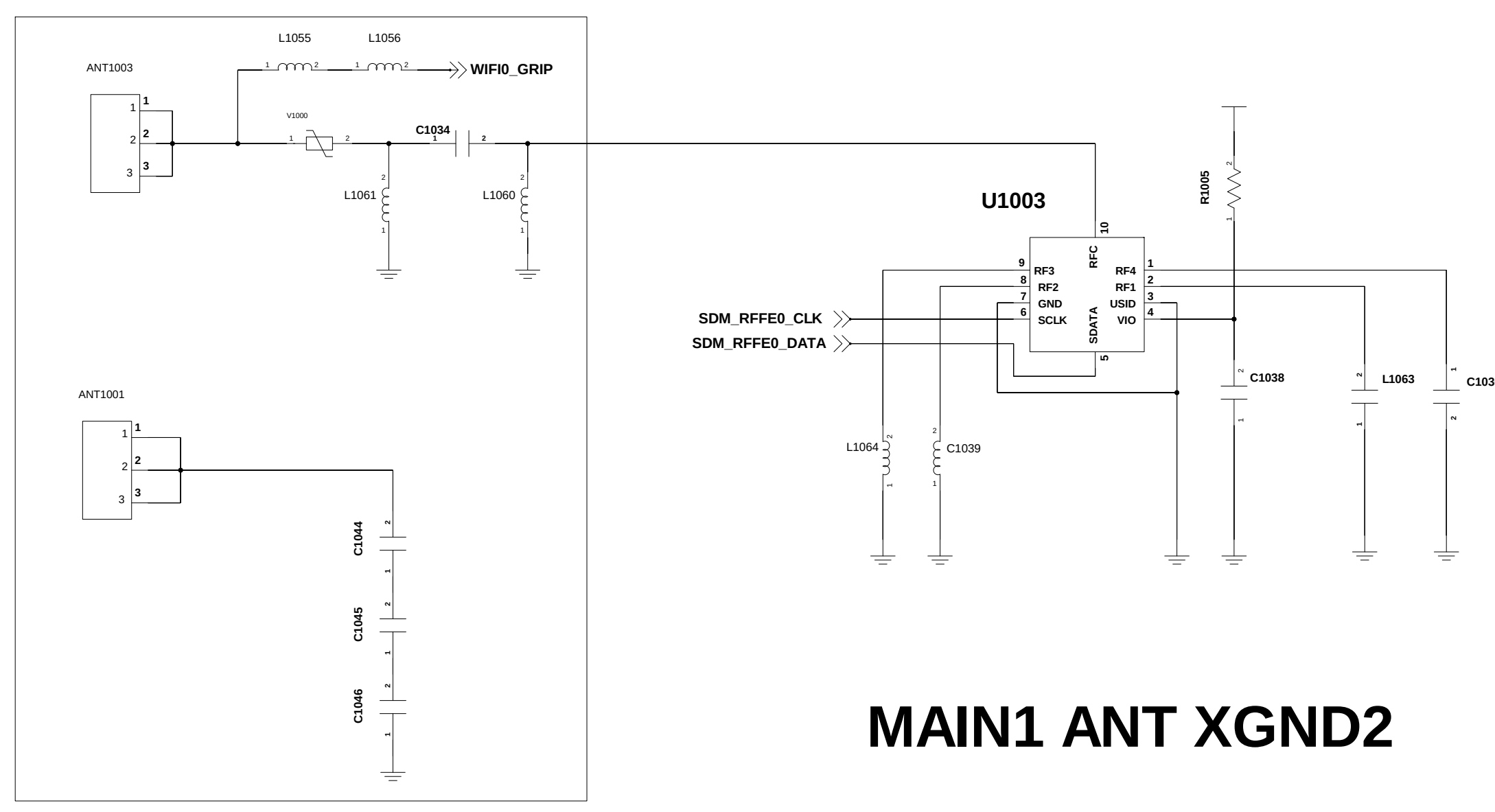

L1001
R1000
C1008
U1001
VIO
SW1
SW2
SW3
SW4
SW12
SW34
C1I
C1O
DATA
CLK
USID
GND
SDM_RFFE0_DATA
SDM_RFFE0_CLK
RF_MAIN_1
C1019
L1021
L1022
L1025
L1026
C1020
C1021
ANT1000
C1023
C1025
L1036
RFS1000
GND
OUT
IN
L1017
C1015
C1016
F1001
COMMON
HB
LB
L1011
L1015
L1016
L1035
L1037
L1038
SDR1_RFFE1_DATA
SDR1_RFFE1_CLK
U1000
RF1
RF3
SDATA
SCLK
ANT
RF2
RF4
n41_SRS1_SUB2_DPDT
n41_SRS1_SUB4
L1004
L1005
C1002
C1014
MAIN1_HB_TRX
F1003
HIGH
LOW
GND/NC
L1030
L1031
L1032
MAIN1_MB_TRX
L1039
L1041
L1040
MAIN1_LB_TRX

need for WIFI BOM


L1057
L1100
MAIN1_GRIP
ANT1002
C1037
C1201
L1062
ANT1004
C1042
C1043
L1068
R1003
U1004
RFC
RF4
RF1
USID
VIO
RF3
RF2
GND
SCLK
SDATA
TP1002
TP1003
SDM_RFFE0_CLK
SDM_RFFE0_DATA
L1065
C1040
C1041
L1066
L1067

need for WIFI BOM

MAIN1 ANT XGND1

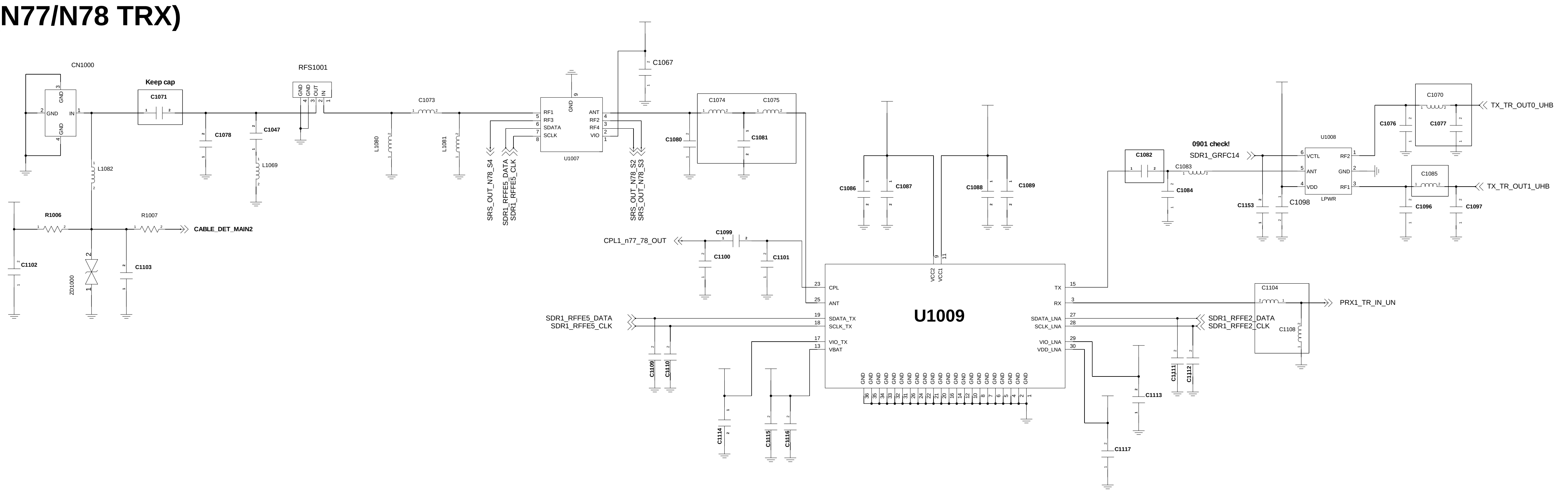

ANT1003
L1055
L1056
WIFI0_GRIP
V1000
C1034
L1061
L1060
ANT1001
C1044
C1045
C1046
U1003
RFC
RF3
RF2
GND
SCLK
SDATA
RF4
RF1
USID
VIO
R1005
SDM_RFFE0_CLK
SDM_RFFE0_DATA
C1038
L1063
C1036
L1064
C1039

need for WIFI BOM

MAIN1 ANT XGND2

MAIN2 ANT

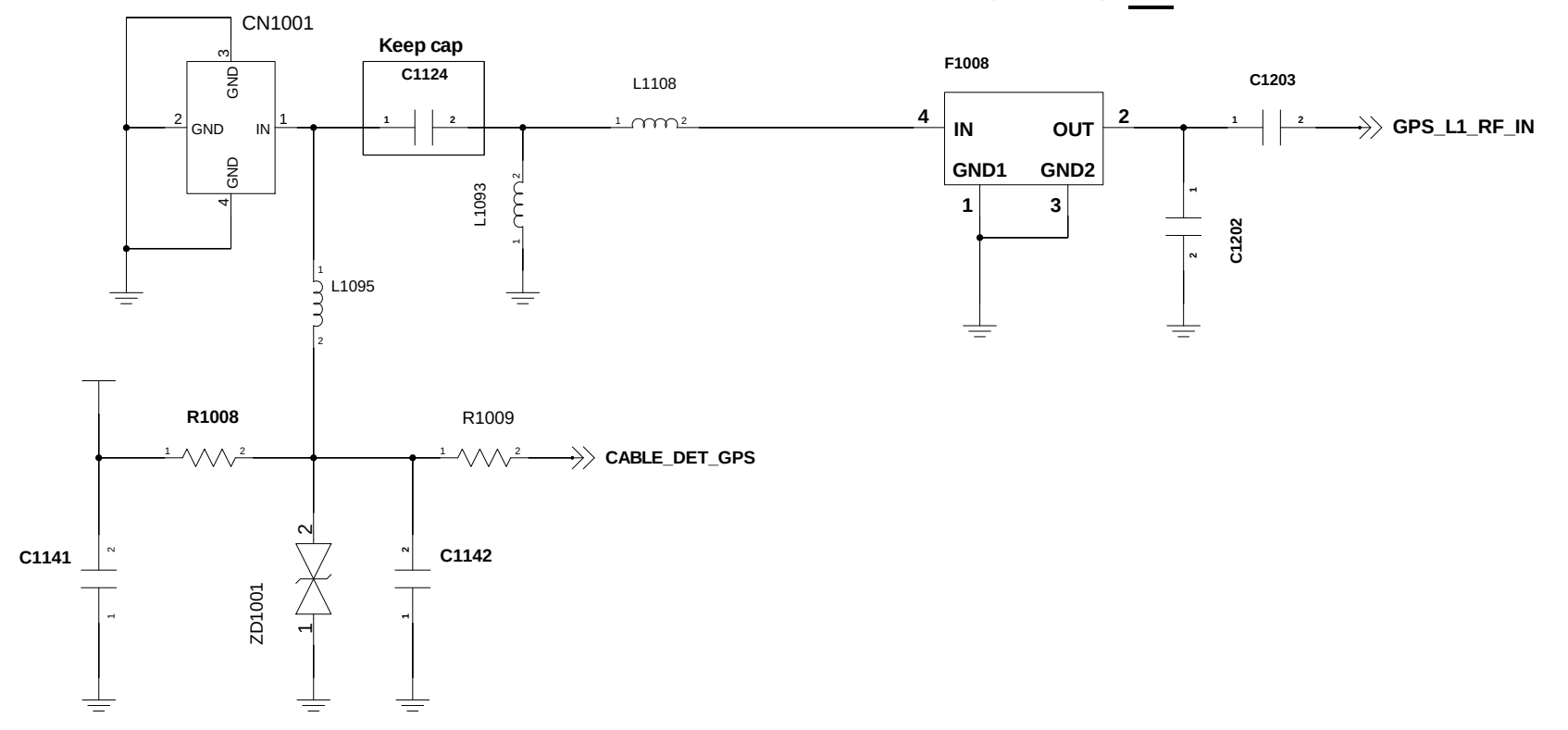
(N77/N78 TRX)


CN1000
GND
IN
Keep cap
C1071
C1078
C1047
L1069
RFS1001
L1082
R1006
R1007
CABLE_DET_MAIN2
C1102
ZD1000
C1103
C1073
L1080
L1081
U1007
RF1
RF3
SDATA
SCLK
ANT
RF2
RF4
VIO
SRS_OUT_N78_S4
SDR1_RFFE5_DATA
SDR1_RFFE5_CLK
SRS_OUT_N78_S2
SRS_OUT_N78_S3
C1067
C1074
C1075
C1080
C1081
C1086
C1087
C1088
C1089
CPL1_n77_78_OUT
C1099
C1100
C1101
U1009
CPL
ANT
SDATA_TX
SCLK_TX
VIO_TX
VBAT
VCC2
VCC1
TX
RX
SDATA_LNA
SCLK_LNA
VIO_LNA
VDD_LNA
SDR1_RFFE5_DATA
SDR1_RFFE5_CLK
C1109
C1110
C1114
C1115
C1116
C1117
C1113
C1111
C1112
C1082
0901 check!
SDR1_GRFC14
C1083
C1084
U1008
VCTL
ANT
VDD
RF2
GND
RF1
LPWR
C1153
C1098
C1070
C1076
C1077
TX_TR_OUT0_UHB
C1085
C1096
C1097
TX_TR_OUT1_UHB
C1104
C1108
PRX1_TR_IN_UN
SDR1_RFFE2_DATA
SDR1_RFFE2_CLK

SUB3 ANT

(GPS_L1)

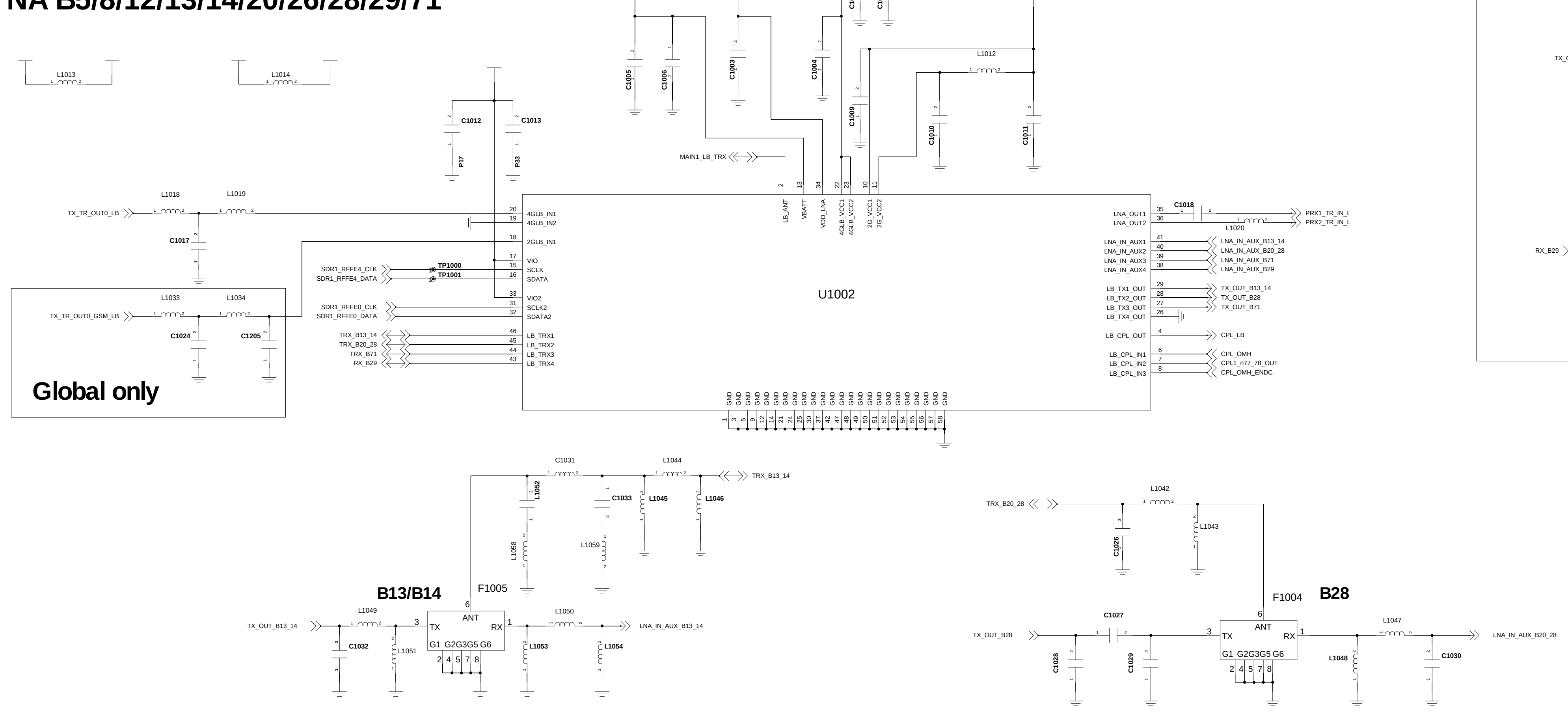
GPS_L1

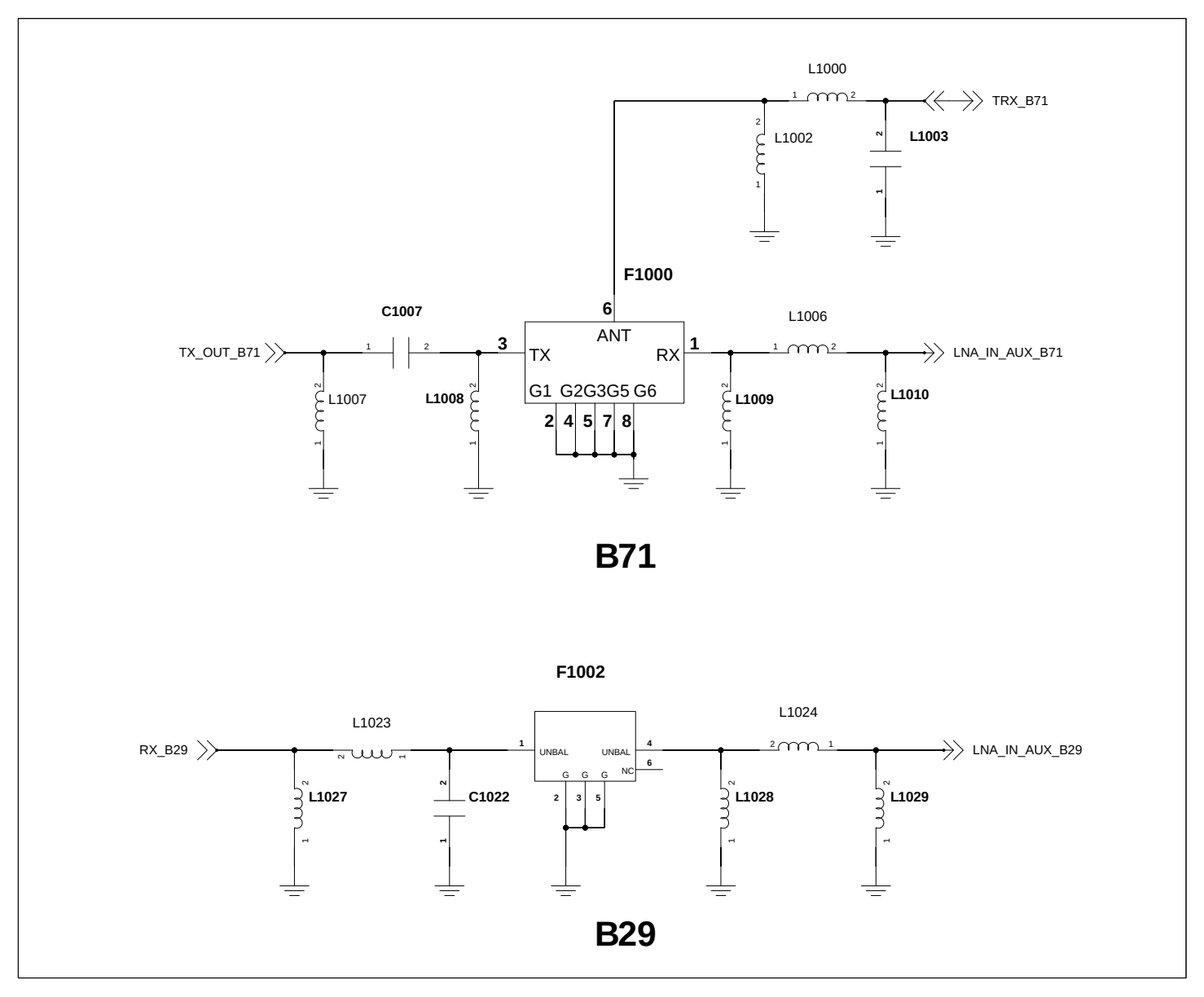

CN1001
GND
IN
Keep cap
C1124
L1093
L1095
L1108
R1008
R1009
CABLE_DET_GPS
C1141
ZD1001
C1142
F1008
IN
OUT
GND1
GND2
C1203
C1202
GPS_L1_RF_IN

LB LPAMID

Global B5/8/12/13/17/20/26/28, G850/900

NA B5/8/12/13/14/20/26/28/29/71


L1013
L1014
C1012
C1013
P17
P33
TX_TR_OUT0_LB
L1018
L1019
C1017
TX_TR_OUT0_GSM_LB
L1033
L1034
C1024
C1205
Global only
SDR1_RFFE4_CLK
SDR1_RFFE4_DATA
TP1000
TP1001
SDR1_RFFE0_CLK
SDR1_RFFE0_DATA
TRX_B13_14
TRX_B20_28
TRX_B71
RX_B29
4GLB_IN1
4GLB_IN2
2GLB_IN1
VIO
SCLK
SDATA
VIO2
SCLK2
SDATA2
LB_TRX1
LB_TRX2
LB_TRX3
LB_TRX4
U1002
C1005
C1006
C1003
C1004
C1000
C1001
C1009
C1010
C1011
L1012
MAIN1_LB_TRX
LB_ANT
VBATT
VDD_LNA
4GLB_VCC1
4GLB_VCC2
2G_VCC1
2G_VCC2
LNA_OUT1
LNA_OUT2
C1018
L1020
PRX1_TR_IN_L
PRX2_TR_IN_L
LNA_IN_AUX1
LNA_IN_AUX2
LNA_IN_AUX3
LNA_IN_AUX4
LNA_IN_AUX_B13_14
LNA_IN_AUX_B20_28
LNA_IN_AUX_B71
LNA_IN_AUX_B29
LB_TX1_OUT
LB_TX2_OUT
LB_TX3_OUT
LB_TX4_OUT
TX_OUT_B13_14
TX_OUT_B28
TX_OUT_B71
LB_CPL_OUT
CPL_LB
LB_CPL_IN1
LB_CPL_IN2
LB_CPL_IN3
CPL_OMH
CPL1_n77_78_OUT
CPL_OMH_ENDC
GND
C1031
L1044
TRX_B13_14
L1052
C1033
L1045
L1046
L1058
L1059
B13/B14
F1005
TX_OUT_B13_14
L1049
C1032
L1051
ANT
TX
RX
G1 G2G3G5 G6
L1050
L1053
L1054
LNA_IN_AUX_B13_14
TRX_B20_28
L1042
C1026
L1043
F1004
B28
TX_OUT_B28
C1027
C1028
C1029
L1047
L1048
C1030
LNA_IN_AUX_B20_28

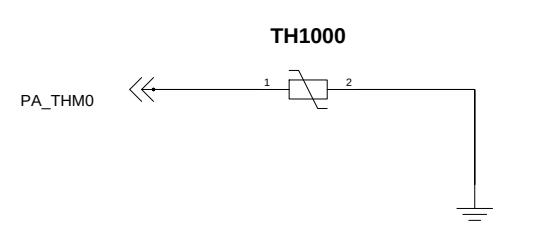

L1000
TRX_B71
L1002
L1003
F1000
C1007
TX_OUT_B71
L1007
L1008
ANT
TX
RX
G1 G2G3G5 G6
L1006
L1009
L1010
LNA_IN_AUX_B71
B71
F1002
RX_B29
L1023
L1027
C1022
UNBAL
L1024
L1028
L1029
LNA_IN_AUX_B29
B29

NA only

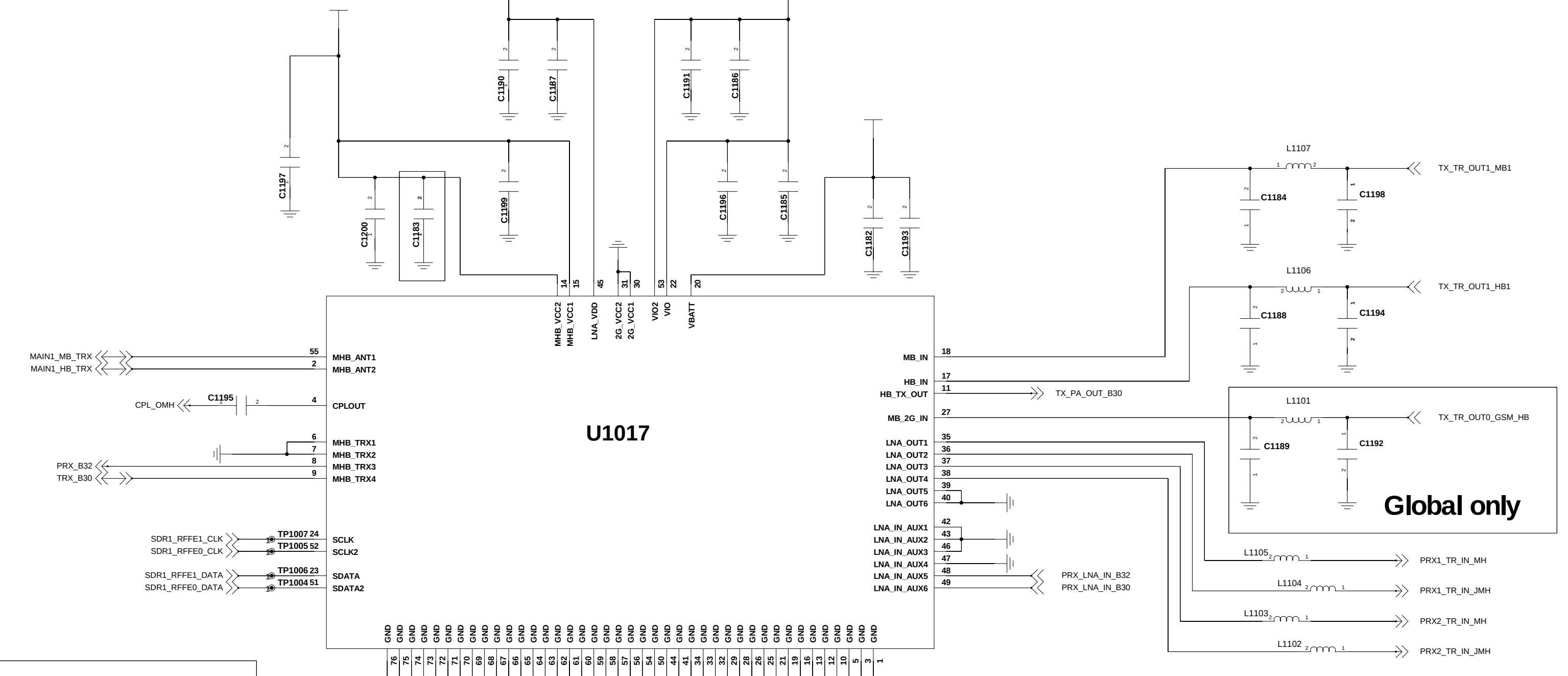
OMH LPAMID

Global 1/2/3/4/7/25/32/38/40/41/66, G1800/1900, N41

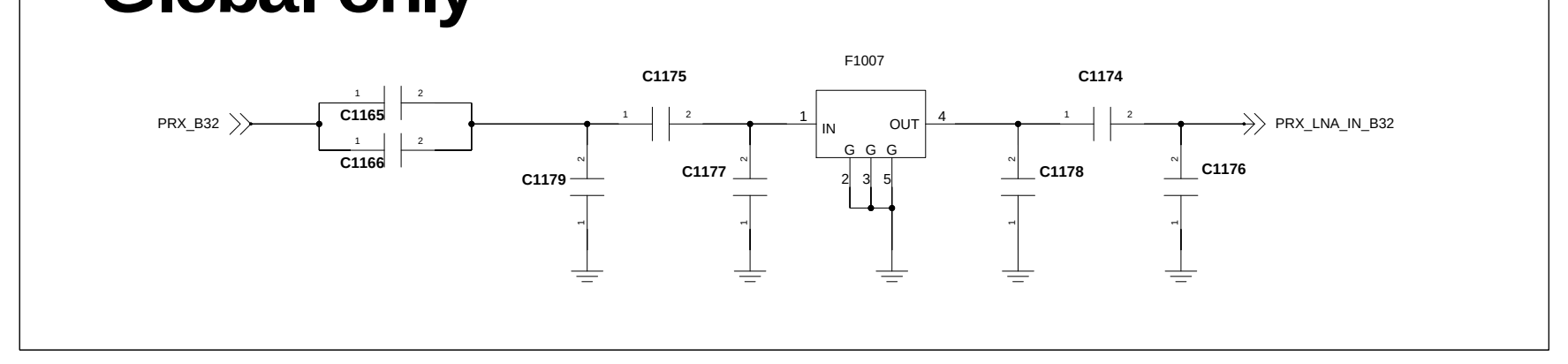
NA 1/2/3/4/7/25/30/40/41/66/70, N41


TH1000
PA_THM0

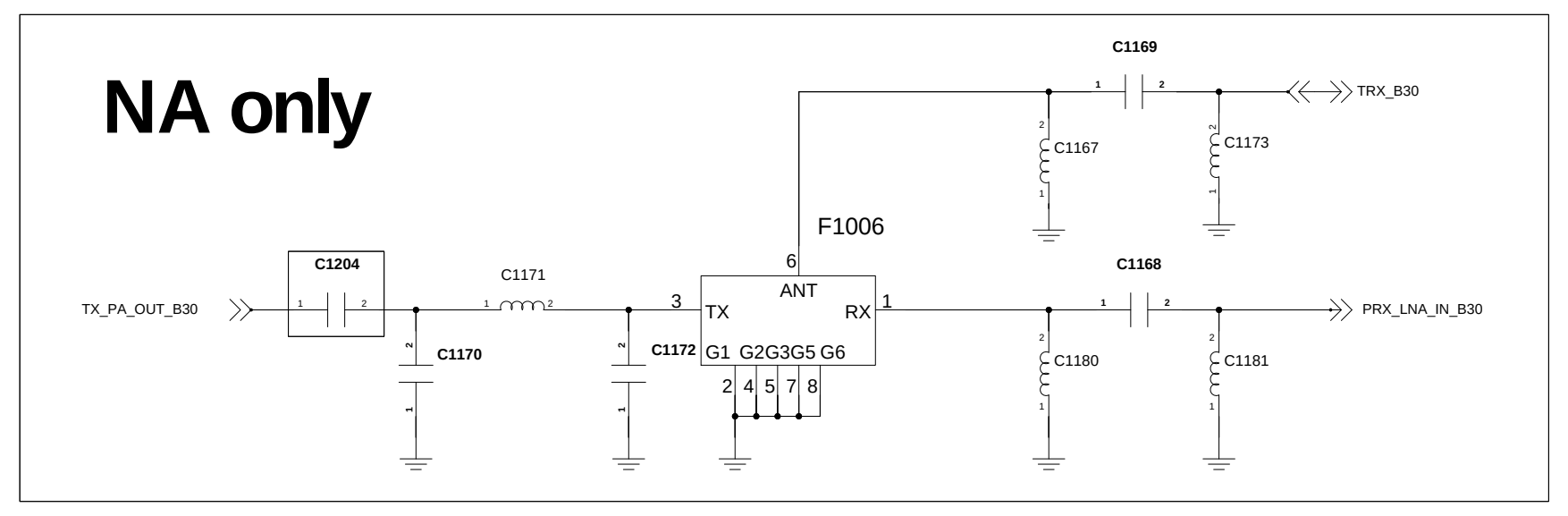
THERMISTERS

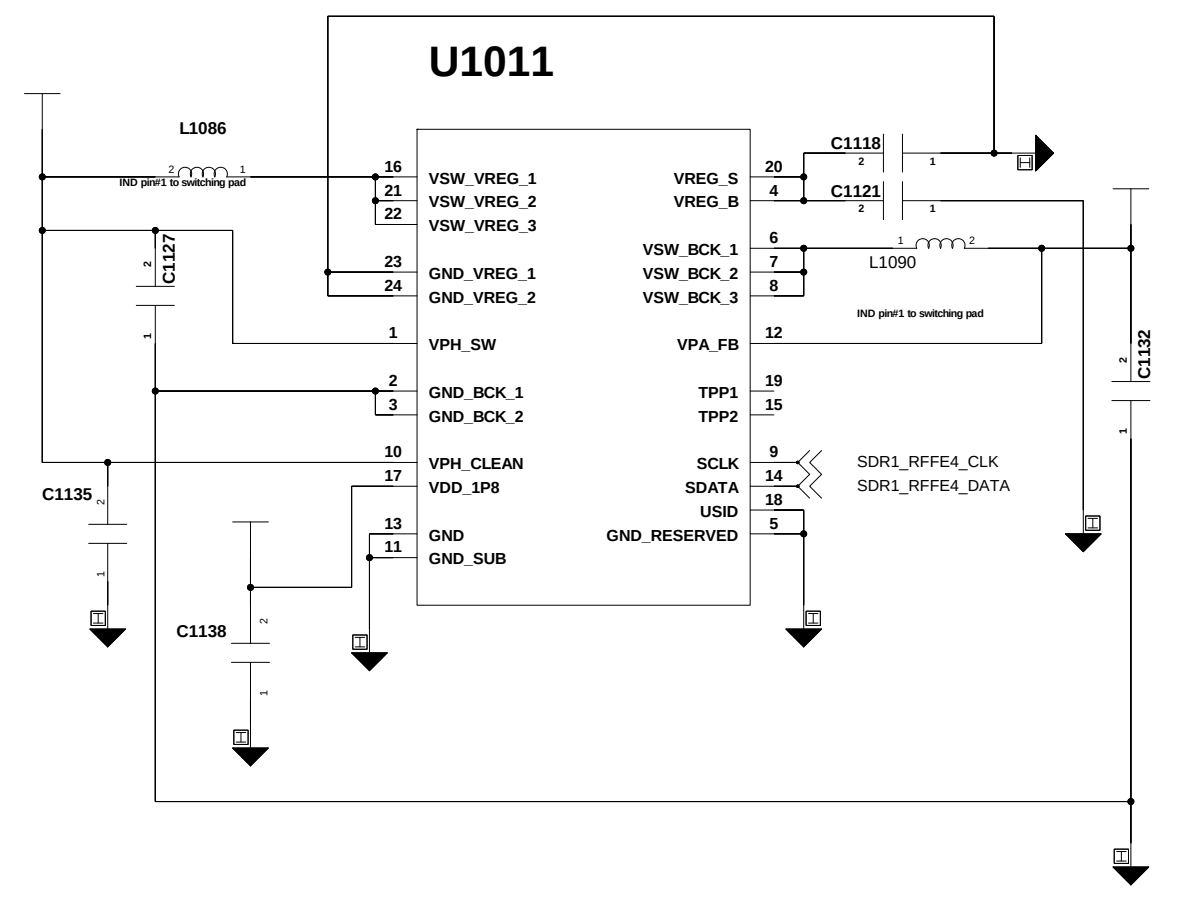

C1197
C1200
C1183
C1190
C1187
C1199
C1191
C1186
C1196
C1185
C1182
C1193
MHB_VCC2
MHB_VCC1
LNA_VDD
2G_VCC2
2G_VCC1
VIO2
VIO
VBATT
U1017
MAIN1_MB_TRX
MAIN1_HB_TRX
MHB_ANT1
MHB_ANT2
CPL_OMH
C1195
CPLOUT
MHB_TRX1
MHB_TRX2
MHB_TRX3
MHB_TRX4
PRX_B32
TRX_B30
SDR1_RFFE1_CLK
SDR1_RFFE0_CLK
SDR1_RFFE1_DATA
SDR1_RFFE0_DATA
TP1007
TP1005
TP1006
TP1004
SCLK
SCLK2
SDATA
SDATA2
MB_IN
HB_IN
HB_TX_OUT
TX_PA_OUT_B30
MB_2G_IN
LNA_OUT1
LNA_OUT2
LNA_OUT3
LNA_OUT4
LNA_OUT5
LNA_OUT6
LNA_IN_AUX1
LNA_IN_AUX2
LNA_IN_AUX3
LNA_IN_AUX4
LNA_IN_AUX5
LNA_IN_AUX6
PRX_LNA_IN_B32
PRX_LNA_IN_B30
GND
L1107
C1184
C1198
TX_TR_OUT1_MB1
L1106
C1188
C1194
TX_TR_OUT1_HB1
L1101
C1189
C1192
TX_TR_OUT0_GSM_HB
Global only
L1105
PRX1_TR_IN_MH
L1104
PRX1_TR_IN_JMH
L1103
PRX2_TR_IN_MH
L1102
PRX2_TR_IN_JMH

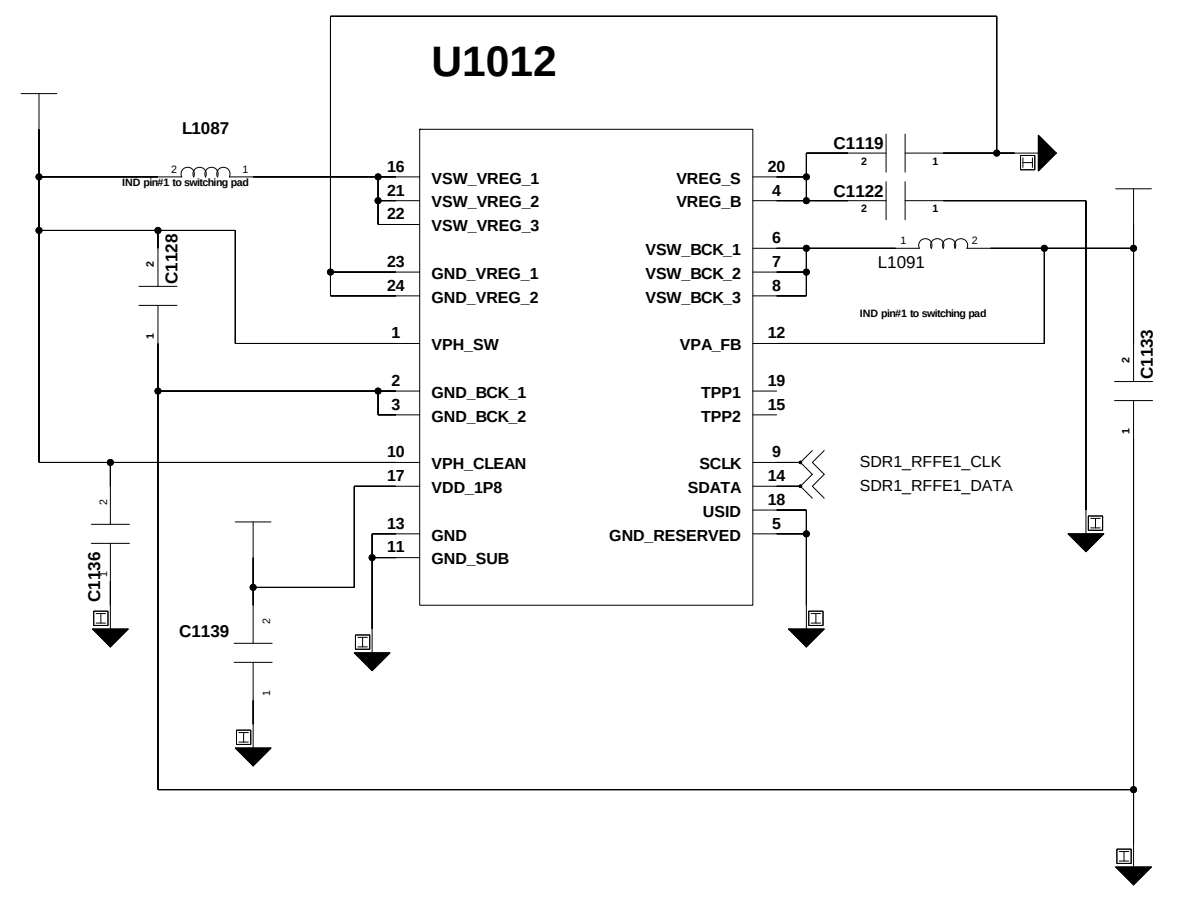
Global only


PRX_B32
C1165
C1166
C1175
C1179
C1177
F1007
IN
OUT
G G G
C1174
C1178
C1176
PRX_LNA_IN_B32

NA only


C1169
TRX_B30
C1167
C1173
F1006
TX_PA_OUT_B30
C1204
C1171
C1170
C1172
ANT
TX
RX
G1 G2G3G5 G6
C1168
C1180
C1181
PRX_LNA_IN_B30


U1011
L1086
IND pin#1 to switching pad
VSW_VREG_1
VSW_VREG_2
VSW_VREG_3
GND_VREG_1
GND_VREG_2
VPH_SW
GND_BCK_1
GND_BCK_2
VPH_CLEAN
VDD_1P8
GND
GND_SUB
VREG_S
VREG_B
VSW_BCK_1
VSW_BCK_2
VSW_BCK_3
VPA_FB
TPP1
TPP2
SCLK
SDATA
USID
GND_RESERVED
C1118
C1121
L1090
C1127
C1135
C1138
C1132
SDR1_RFFE4_CLK
SDR1_RFFE4_DATA


U1012
L1087
IND pin#1 to switching pad
VSW_VREG_1
VSW_VREG_2
VSW_VREG_3
GND_VREG_1
GND_VREG_2
VPH_SW
GND_BCK_1
GND_BCK_2
VPH_CLEAN
VDD_1P8
GND
GND_SUB
VREG_S
VREG_B
VSW_BCK_1
VSW_BCK_2
VSW_BCK_3
VPA_FB
TPP1
TPP2
SCLK
SDATA
USID
GND_RESERVED
C1119
C1122
L1091
C1128
C1136
C1139
C1133
SDR1_RFFE1_CLK
SDR1_RFFE1_DATA

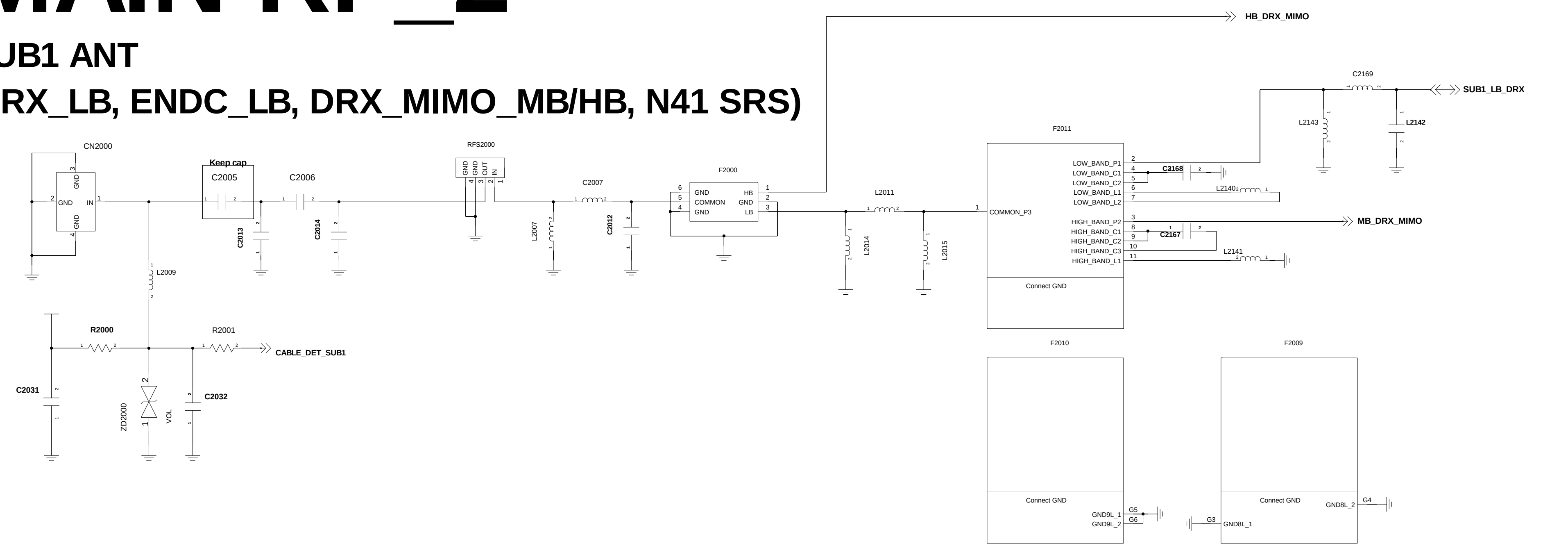
LB/UHB APT Modulator(QET6105#0)

MHB/UHB APT Modulator(QET6105#1)

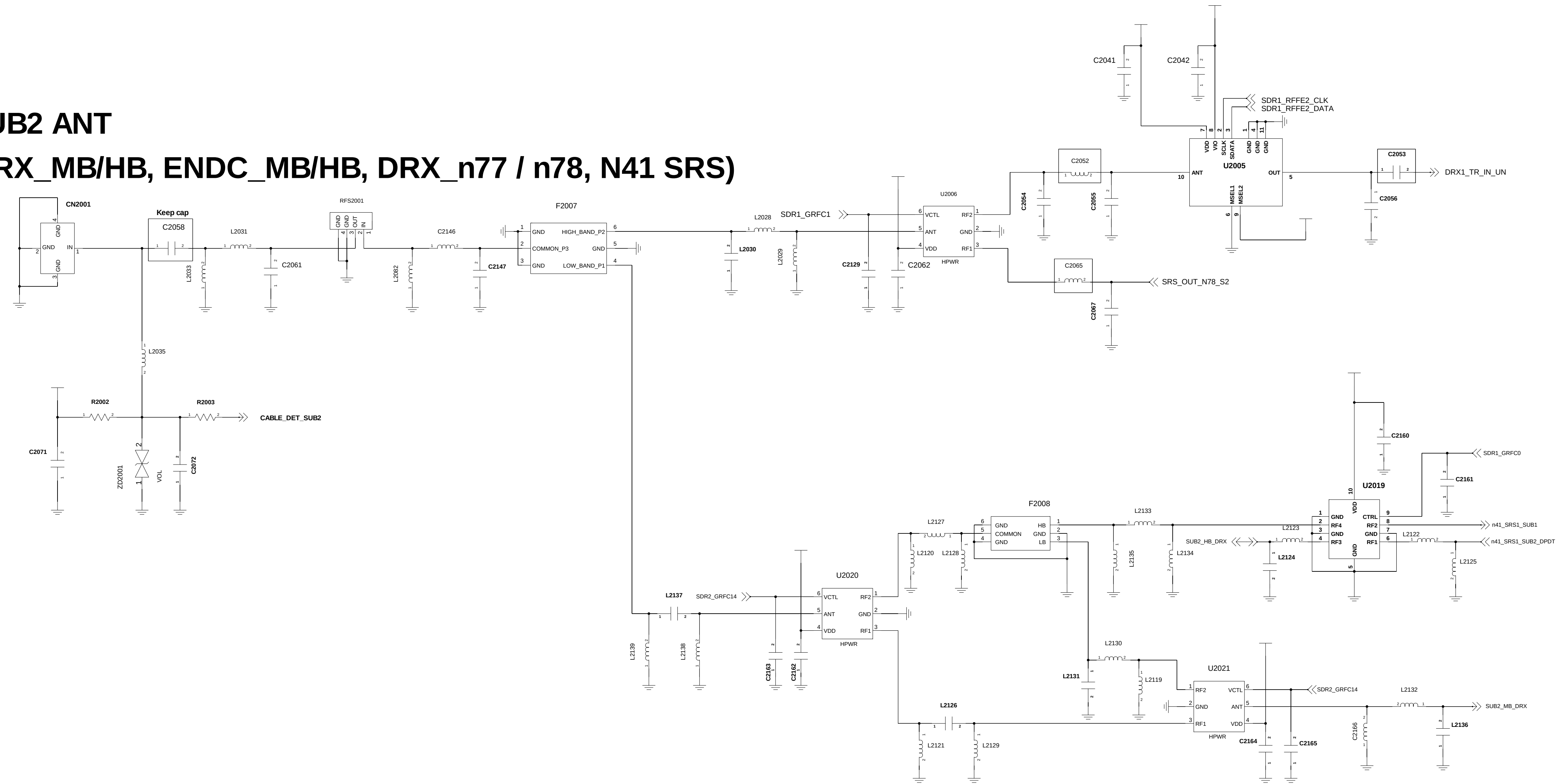
TITLE TabS7P
SIZE A1 DWG NO REV
DRAWN BY User Name
SHEET 1 of 15 Date

MAIN RF_2

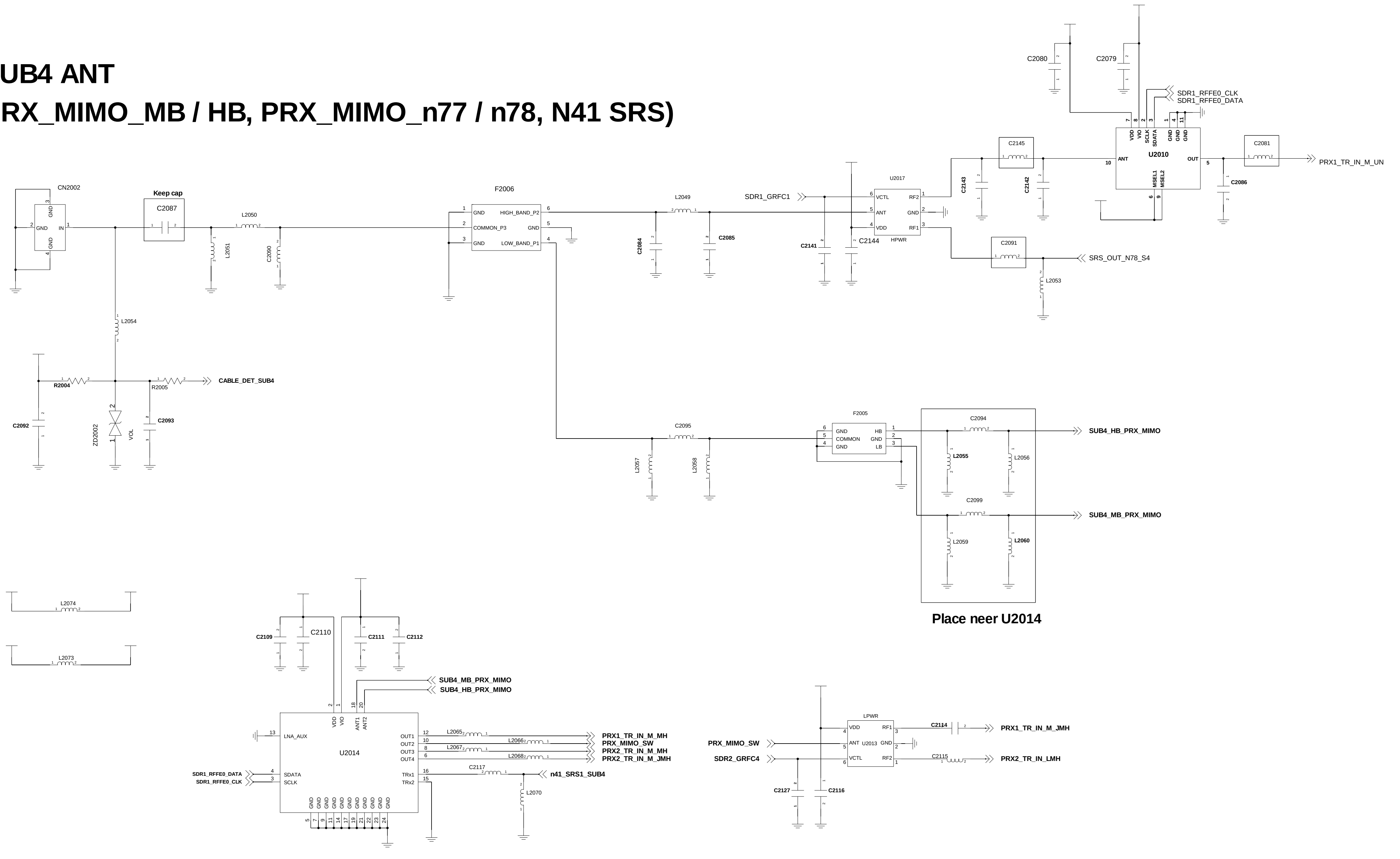
SUB1 ANT
(DRX_LB, ENDC_LB, DRX_MIMO_MB/HB, N41 SRS)


CN2000
GND
IN
Keep cap
C2005
C2006
C2013
C2014
RFS2000
GND GND OUT IN
C2007
L2007
C2012
F2000
GND HB
COMMON GND
GND LB
HB_DRX_MIMO
L2011
L2014
L2015
F2011
COMMON_P3
LOW_BAND_P1
LOW_BAND_C1
LOW_BAND_C2
LOW_BAND_L1
LOW_BAND_L2
HIGH_BAND_P2
HIGH_BAND_C1
HIGH_BAND_C2
HIGH_BAND_C3
HIGH_BAND_L1
C2168
C2167
L2140
L2141
Connect GND
C2169
L2143
L2142
SUB1_LB_DRX
MB_DRX_MIMO
F2010
F2009
Connect GND
GND9L_1
GND9L_2
G5
G6
G3
GND8L_1
GND8L_2
G4
L2009
R2000
R2001
CABLE_DET_SUB1
C2031
ZD2000
VOL
C2032

SUB2 ANT
(DRX_MB/HB, ENDC_MB/HB, DRX_n77 / n78, N41 SRS)

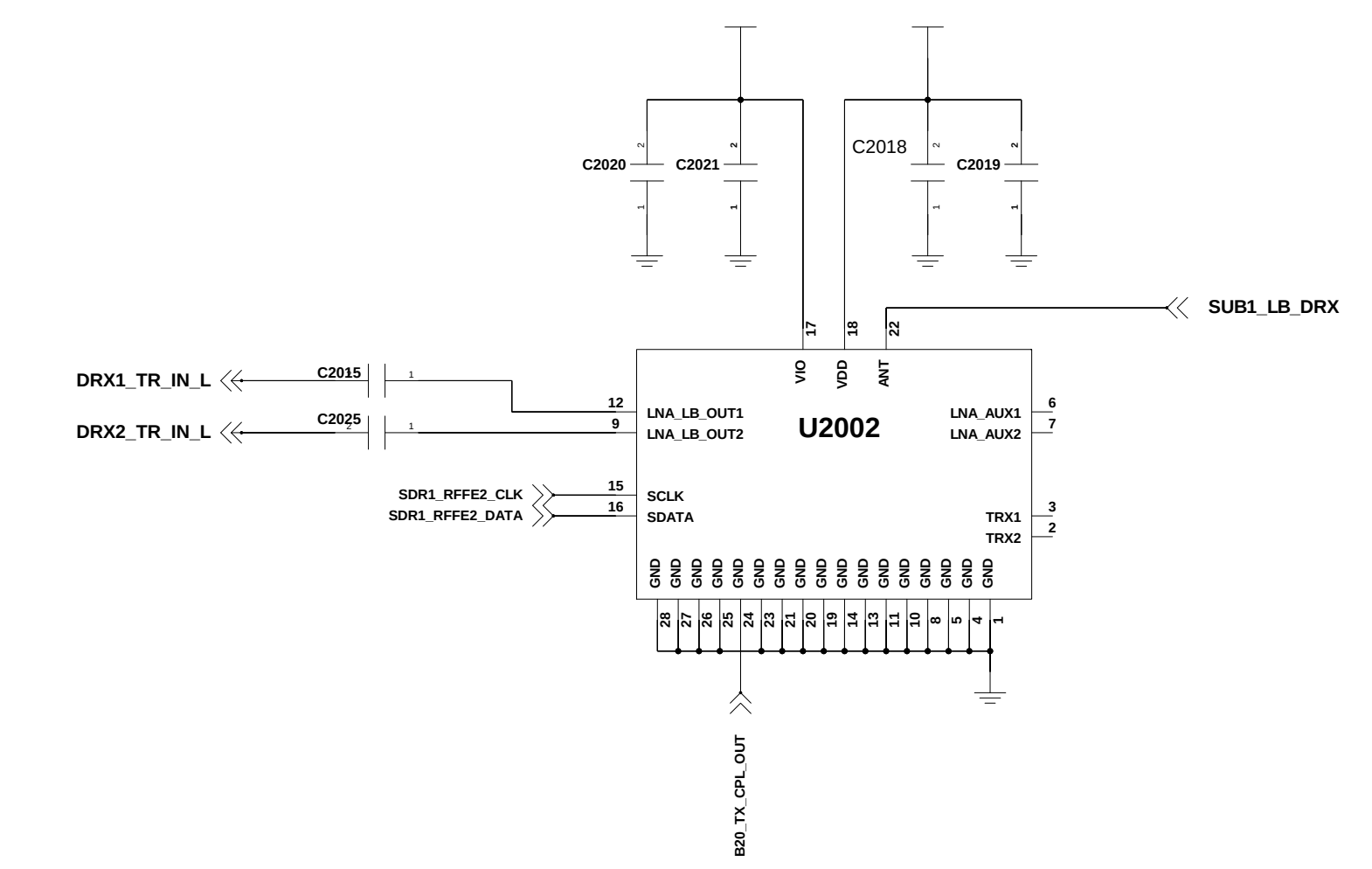

CN2001
Keep cap
C2058
L2031
L2033
C2061
RFS2001
C2146
L2082
C2147
F2007
GND HIGH_BAND_P2
COMMON_P3 GND
GND LOW_BAND_P1
L2028
SDR1_GRFC1
L2030
L2029
C2129
C2062
U2006
VCTL RF2
ANT GND
VDD RF1
HPWR
C2052
C2054
C2055
C2065
SRS_OUT_N78_S2
C2067
C2041
C2042
SDR1_RFFE2_CLK
SDR1_RFFE2_DATA
VDD VIO SCLK SDATA GND GND GND
U2005
ANT
OUT
MSEL1 MSEL2
C2053
DRX1_TR_IN_UN
C2056
L2035
R2002
R2003
CABLE_DET_SUB2
C2071
ZD2001
VOL
C2072
F2008
GND HB
COMMON GND
GND LB
L2127
L2120
L2128
L2133
L2135
L2134
SUB2_HB_DRX
L2123
L2124
U2019
GND CTRL
RF4 RF2
GND GND
RF3 RF1
VDD
C2160
SDR1_GRFC0
C2161
n41_SRS1_SUB1
L2122
n41_SRS1_SUB2_DPDT
L2125
L2137
SDR2_GRFC14
U2020
VCTL RF2
ANT GND
VDD RF1
HPWR
L2139
L2138
C2163
C2162
L2130
L2131
L2119
L2126
L2121
L2129
U2021
RF2 VCTL
GND ANT
RF1 VDD
HPWR
C2164
C2165
SDR2_GRFC14
L2132
C2166
L2136
SUB2_MB_DRX

SUB4 ANT
(PRX_MIMO_MB / HB, PRX_MIMO_n77 / n78, N41 SRS)

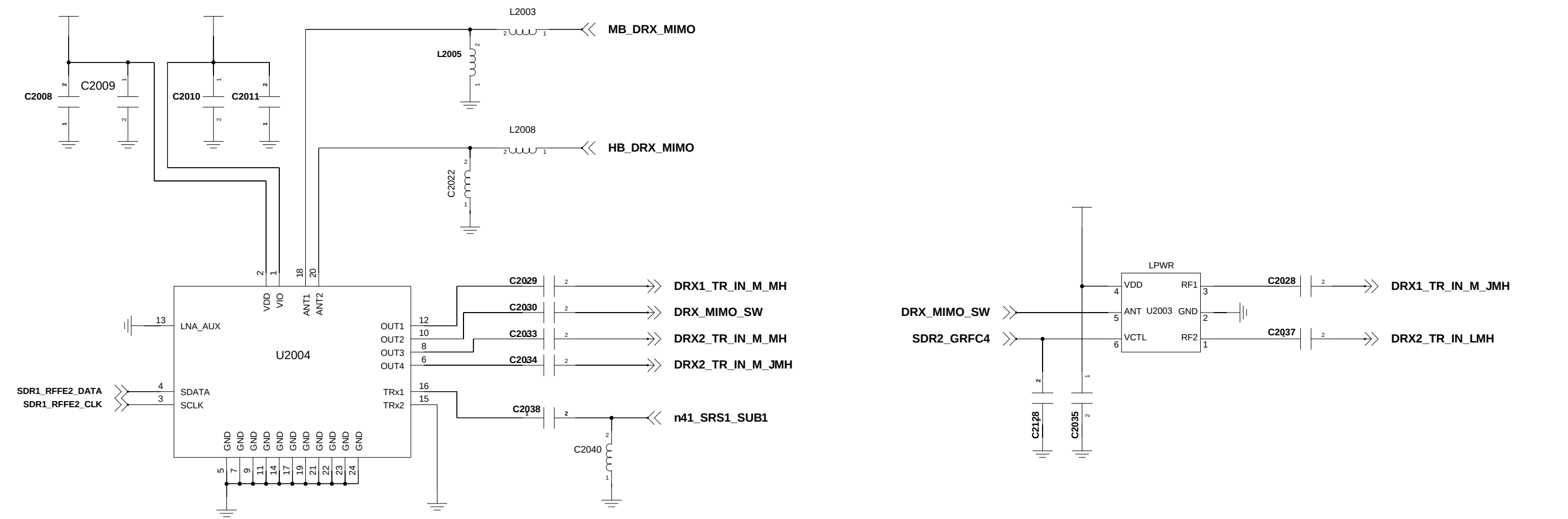

CN2002
Keep cap
C2087
L2050
L2051
C2090
F2006
GND HIGH_BAND_P2
COMMON_P3 GND
GND LOW_BAND_P1
L2049
SDR1_GRFC1
C2084
C2085
C2141
C2144
U2017
VCTL RF2
ANT GND
VDD RF1
HPWR
C2145
C2143
C2142
C2091
SRS_OUT_N78_S4
L2053
C2080
C2079
SDR1_RFFE0_CLK
SDR1_RFFE0_DATA
U2010
ANT
OUT
MSEL1 MSEL2
C2081
PRX1_TR_IN_M_UN
C2086
L2054
R2004
R2005
CABLE_DET_SUB4
C2092
ZD2002
VOL
C2093
C2095
L2057
L2058
F2005
GND HB
COMMON GND
GND LB
C2094
L2055
L2056
C2099
L2059
L2060
SUB4_HB_PRX_MIMO
SUB4_MB_PRX_MIMO
Place neer U2014
L2074
L2073
C2109
C2110
C2111
C2112
SUB4_MB_PRX_MIMO
SUB4_HB_PRX_MIMO
U2014
LNA_AUX
VDD VIO ANT1 ANT2
OUT1 OUT2 OUT3 OUT4
SDATA
SCLK
TRx1
TRx2
SDR1_RFFE0_DATA
SDR1_RFFE0_CLK
L2065
L2066
L2067
L2068
PRX1_TR_IN_M_MH
PRX_MIMO_SW
PRX2_TR_IN_M_MH
PRX2_TR_IN_M_JMH
C2117
n41_SRS1_SUB4
L2070
LPWR
PRX_MIMO_SW
SDR2_GRFC4
VDD RF1
ANT U2013 GND
VCTL RF2
C2114
C2115
PRX1_TR_IN_M_JMH
PRX2_TR_IN_LMH
C2127
C2116

MHB MIMO_LNA OUT

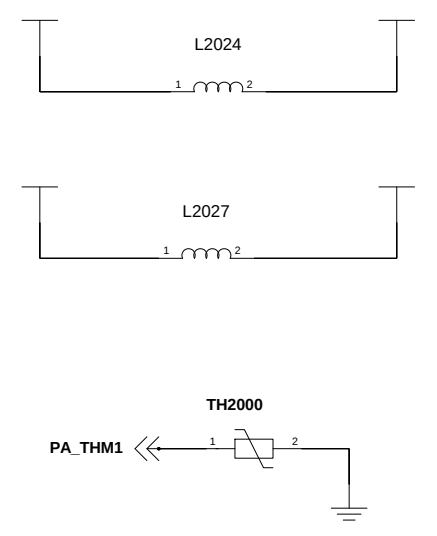
DRX_LB
(B5/8/12/13/14/17/18/19/20/26/28/29/71_G850/900)


C2020
C2021
C2018
C2019
SUB1_LB_DRX
VIO VDD ANT
DRX1_TR_IN_L
C2015
DRX2_TR_IN_L
C2025
LNA_LB_OUT1
LNA_LB_OUT2
U2002
LNA_AUX1
LNA_AUX2
SDR1_RFFE2_CLK
SDR1_RFFE2_DATA
SCLK
SDATA
TRX1
TRX2
B20_TX_CPL_OUT

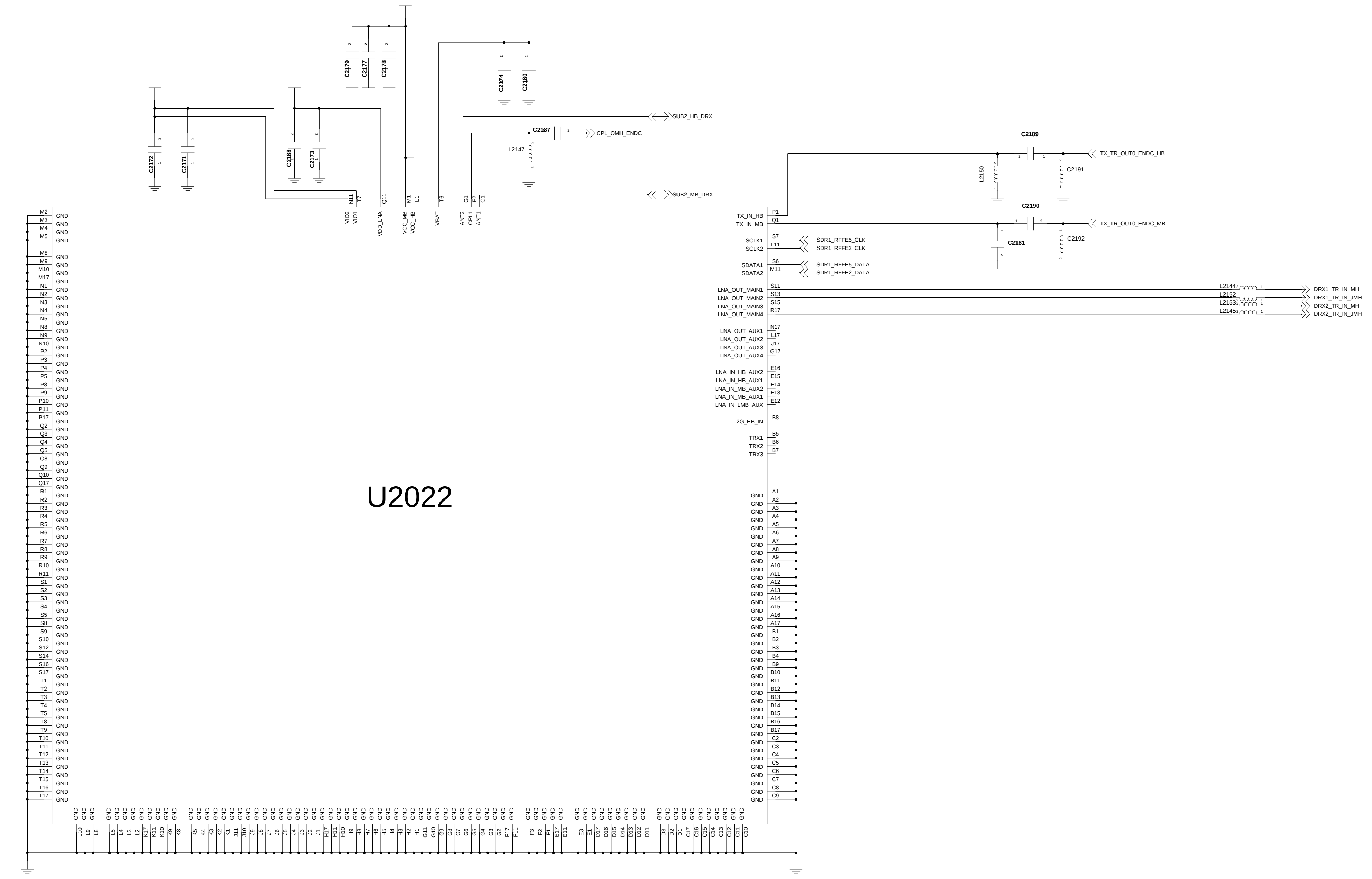
DRX_MHB_2
(B2/4/7/25/30/41/66)

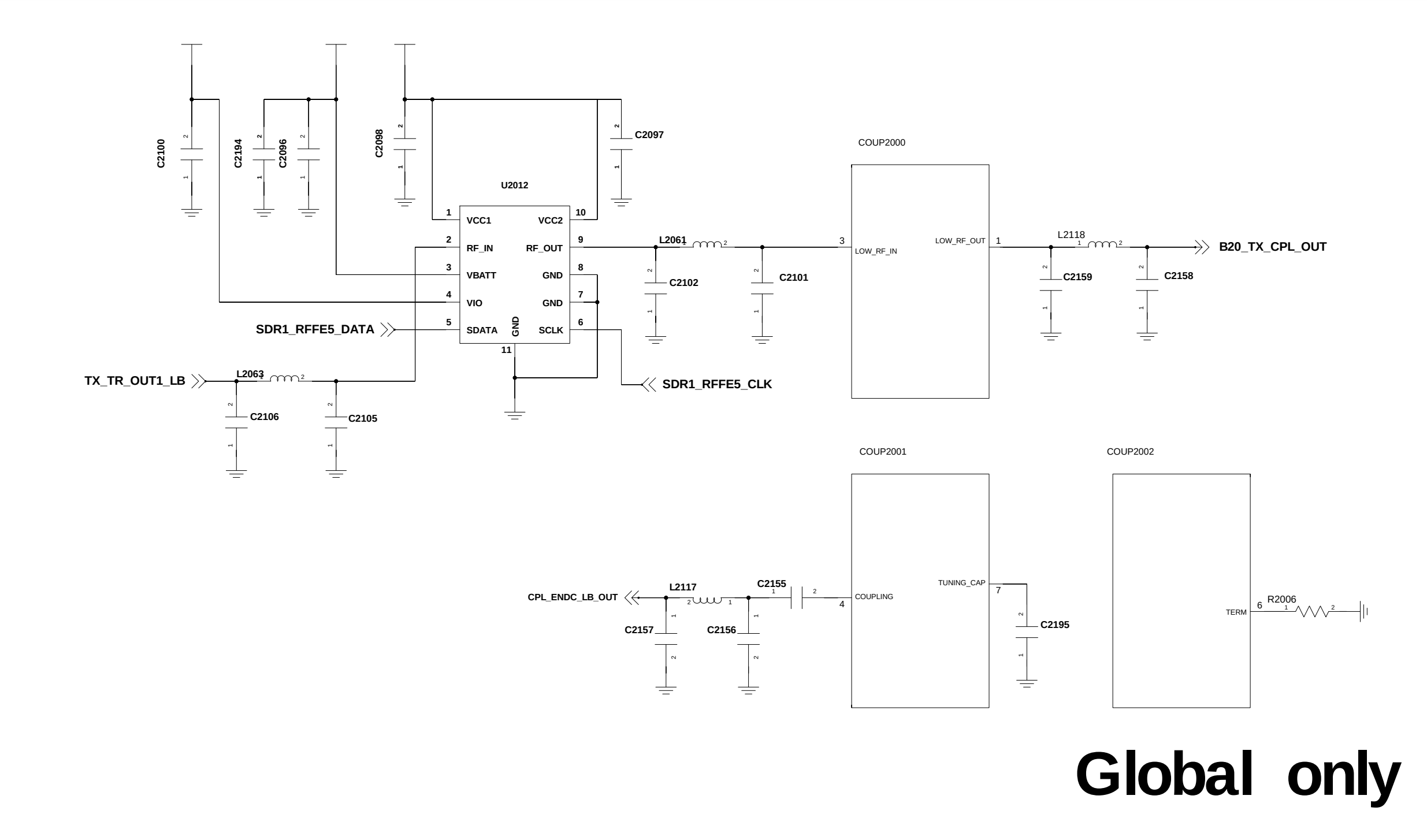

C2008
C2009
C2010
C2011
L2003
MB_DRX_MIMO
L2005
L2008
HB_DRX_MIMO
C2022
VDD VIO ANT1 ANT2
LNA_AUX
U2004
OUT1 OUT2 OUT3 OUT4
SDATA
SCLK
SDR1_RFFE2_DATA
SDR1_RFFE2_CLK
TRx1
TRx2
C2029
DRX1_TR_IN_M_MH
C2030
DRX_MIMO_SW
C2033
DRX2_TR_IN_M_MH
C2034
DRX2_TR_IN_M_JMH
C2038
n41_SRS1_SUB1
C2040
LPWR
DRX_MIMO_SW
SDR2_GRFC4
VDD RF1
ANT U2003 GND
VCTL RF2
C2028
DRX1_TR_IN_M_JMH
C2037
DRX2_TR_IN_LMH
C2128
C2035

ENDC module
(B1/2/3/4/7/25/30/39/41/66/70_DCS/PCS)

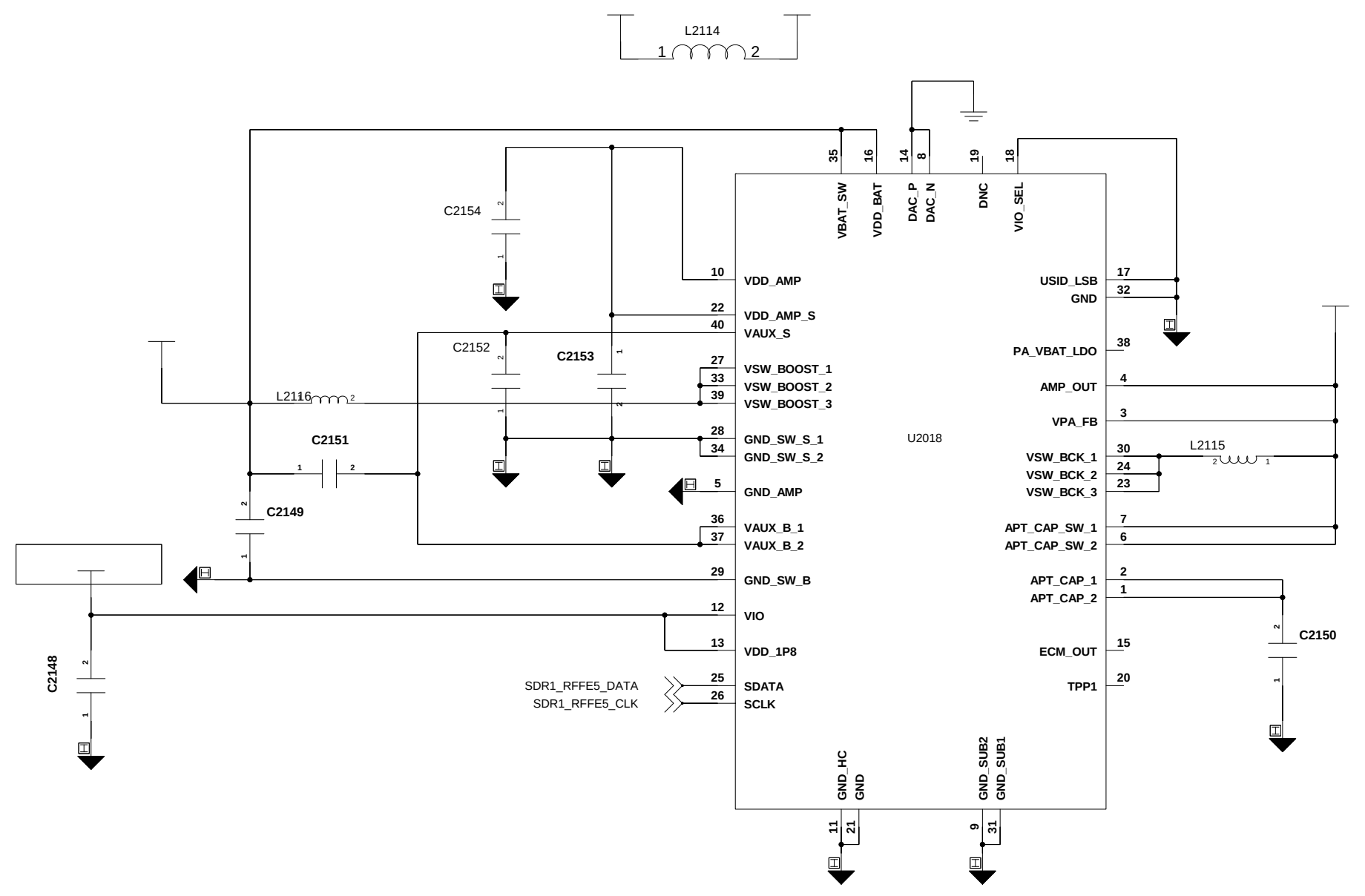

L2024
L2027
TH2000
PA_THM1

THERMISTERS


U2022
C2179
C2177
C2178
C2172
C2171
C2188
C2173
C2174
C2180
VIO2 VIO1
VDD_LNA
VCC_MB VCC_HB
VBAT
ANT2 CPL1 ANT1
SUB2_HB_DRX
C2187
CPL_OMH_ENDC
L2147
SUB2_MB_DRX
C2189
TX_TR_OUT0_ENDC_HB
L2150
C2191
C2190
TX_TR_OUT0_ENDC_MB
C2181
C2192
TX_IN_HB
TX_IN_MB
SCLK1
SCLK2
SDATA1
SDATA2
SDR1_RFFE5_CLK
SDR1_RFFE2_CLK
SDR1_RFFE5_DATA
SDR1_RFFE2_DATA
LNA_OUT_MAIN1
LNA_OUT_MAIN2
LNA_OUT_MAIN3
LNA_OUT_MAIN4
L2144
L2152
L2153
L2145
DRX1_TR_IN_MH
DRX1_TR_IN_JMH
DRX2_TR_IN_MH
DRX2_TR_IN_JMH
LNA_OUT_AUX1
LNA_OUT_AUX2
LNA_OUT_AUX3
LNA_OUT_AUX4
LNA_IN_HB_AUX2
LNA_IN_HB_AUX1
LNA_IN_MB_AUX2
LNA_IN_MB_AUX1
LNA_IN_LMB_AUX
2G_HB_IN
TRX1
TRX2
TRX3
GND


C2100
C2194
C2096
C2098
C2097
U2012
VCC1 VCC2
RF_IN RF_OUT
VBATT GND
VIO GND
SDATA SCLK
GND
SDR1_RFFE5_DATA
SDR1_RFFE5_CLK
TX_TR_OUT1_LB
L2063
C2106
C2105
L2061
C2102
C2101
COUP2000
LOW_RF_IN
LOW_RF_OUT
L2118
C2159
C2158
B20_TX_CPL_OUT
COUP2001
COUPLING
TUNING_CAP
C2195
COUP2002
TERM
R2006
CPL_ENDC_LB_OUT
L2117
C2155
C2157
C2156

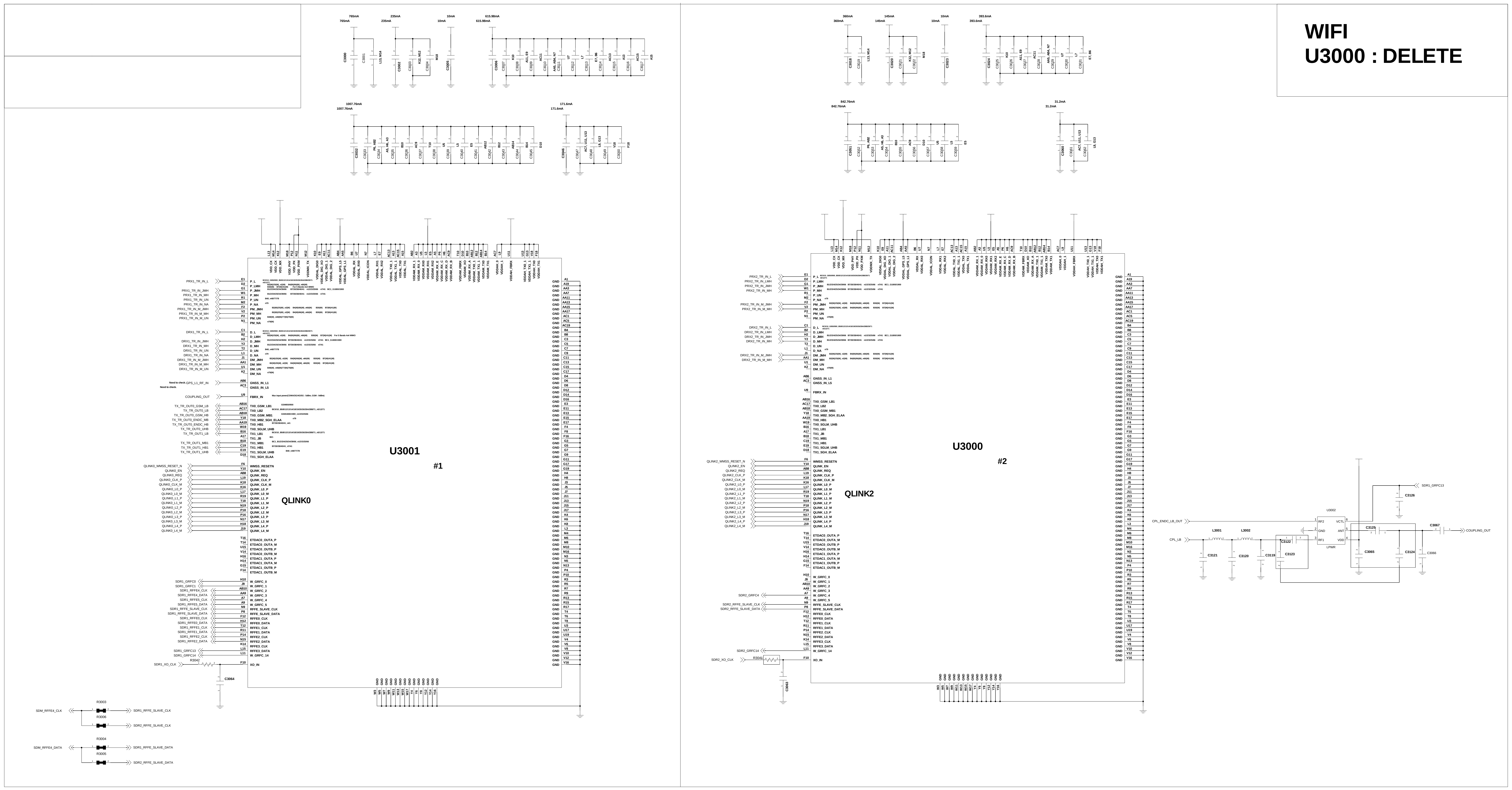
Global only


L2114
C2154
C2152
C2153
L2116
C2151
C2149
C2148
U2018
VBAT_SW VDD_BAT DAC_P DAC_N DNC VIO_SEL
VDD_AMP
VDD_AMP_S
VAUX_S
VSW_BOOST_1
VSW_BOOST_2
VSW_BOOST_3
GND_SW_S_1
GND_SW_S_2
GND_AMP
VAUX_B_1
VAUX_B_2
GND_SW_B
VIO
VDD_1P8
SDATA
SCLK
SDR1_RFFE5_DATA
SDR1_RFFE5_CLK
USID_LSB
GND
PA_VBAT_LDO
AMP_OUT
VPA_FB
VSW_BCK_1
VSW_BCK_2
VSW_BCK_3
L2115
APT_CAP_SW_1
APT_CAP_SW_2
APT_CAP_1
APT_CAP_2
C2150
ECM_OUT
TPP1
GND_HC GND
GND_SUB2 GND_SUB1

ENDC APT Modulator(QET5100)

TITLE	TabS7P		
SIZE A1	DWG NO		REV
DRAWN BY User Name	SHEET 2 of 15	Date Drawing Date	

MAIN RF_3


765mA
235mA
10mA
615.98mA
C3000
C3001
L13, M14
C3002
C3003
K12, M12
C3004
M18
C3005
C3006
C3007
K10
C3008
A11, E9
C3009
AC11
C3010
AA5, AB4, N7
C3011
U7
C3012
L7
C3013
E7, B6
C3014
AC13
C3015
A13
C3016
AC15
C3017
A15
1007.76mA
C3032
C3033
P6, AB2
C3034
A5, H6, A3
C3035
B10
C3036
AC9
C3037
T10
C3038
U5
C3039
L5
C3040
E5
C3041
AB12
C3042
B12
C3043
AB14
C3044
B14
C3045
D10
171.6mA
C3046
C3047
AC7, U11, U13
C3048
L9, G13
C3049
V18
C3050
F18
360mA
145mA
10mA
393.6mA
C3018
C3019
C3020
C3021
C3022
C3023
C3024
C3025
C3026
C3027
C3028
C3029
C3030
C3031
842.76mA
C3051
C3052
C3053
C3054
C3055
C3056
C3057
C3058
C3059
31.2mA
C3060
C3061
C3062
VDD_CX
VDD_MX
VDD_PHY
VDD_PX
VDD_PXM
VDDMX_TX
VDDAL_DIG0
VDDAL_DIG_XO
VDDAL_DIG_1
VDDAL_DIG_2
VDDAL_GPS_L5
VDDAL_GPS_L1
VDDAL_RX
VDDAL_RX0
VDDAL_ICON
VDDAL_RX1
VDDAL_RX2
VDDAL_TX0_1
VDDAL_TX1_1
VDDAL_TX0
VDDAL_TX1
VDDAM_RX_1
VDDAM_RX_0
VDDAM_RX0
VDDAM_RX1
VDDAM_RX2
VDDAM_RX_E
VDDAM_RX_C
VDDAM_RX_D
VDDAM_RX_B
VDDAM_FBRX
VDDAM_XO
VDDAM_RX_A
VDDAM_TX0_1
VDDAM_TX1_1
VDDAM_TX0
VDDAM_TX1
VDDAH_0
VDDAH_1
VDDAH_FBRX
VDDAH_TX0_1
VDDAH_TX1_1
VDDAH_TX0
VDDAH_TX1
PRX1_TR_IN_L
PRX1_TR_IN_JMH
PRX1_TR_IN_MH
PRX1_TR_IN_UN
PRX1_TR_IN_NA
PRX1_TR_IN_M_JMH
PRX1_TR_IN_M_MH
PRX1_TR_IN_M_UN
DRX1_TR_IN_L
DRX1_TR_IN_JMH
DRX1_TR_IN_MH
DRX1_TR_IN_UN
DRX1_TR_IN_NA
DRX1_TR_IN_M_JMH
DRX1_TR_IN_M_MH
DRX1_TR_IN_M_UN
Need to check.
GPS_L1_RF_IN
Need to check.
COUPLING_OUT
TX_TR_OUT0_GSM_LB
TX_TR_OUT0_LB
TX_TR_OUT0_GSM_HB
TX_TR_OUT0_ENDC_MB
TX_TR_OUT0_ENDC_HB
TX_TR_OUT0_UHB
TX_TR_OUT1_LB
TX_TR_OUT1_MB1
TX_TR_OUT1_HB1
TX_TR_OUT1_UHB
P_L
P_LMH
P_JMH
P_MH
P_UN
P_NA
PM_JMH
PM_MH
PM_UN
PM_NA
D_L
D_LMH
D_JMH
D_MH
D_UN
D_NA
DM_JMH
DM_MH
DM_UN
DM_NA
GNSS_IN_L1
GNSS_IN_L5
FBRX_IN
Max input power(CDMA/3G/4G/5G : 5dBm, GSM : 9dBm)
TX0_GSM_LB1
TX0_LB2
TX0_GSM_MB1
TX0_MB2_5GH_ELAA
TX0_HB1
TX0_5GLM_UHB
TX1_LB1
TX1_JB
TX1_MB1
TX1_HB1
TX1_5GLM_UHB
TX1_5GH_ELAA
U3001
#1
QLINK0_WMSS_RESET_N
QLINK0_EN
QLINK0_REQ
QLINK0_CLK_P
QLINK0_CLK_M
QLINK0_L0_P
QLINK0_L0_M
QLINK0_L1_P
QLINK0_L1_M
QLINK0_L2_P
QLINK0_L2_M
QLINK0_L3_P
QLINK0_L3_M
QLINK0_L4_P
QLINK0_L4_M
WMSS_RESETN
QLINK_EN
QLINK_REQ
QLINK_CLK_P
QLINK_CLK_M
QLINK_L0_P
QLINK_L0_M
QLINK_L1_P
QLINK_L1_M
QLINK_L2_P
QLINK_L2_M
QLINK_L3_P
QLINK_L3_M
QLINK_L4_P
QLINK_L4_M
QLINK0
ETDAC0_OUTA_P
ETDAC0_OUTA_M
ETDAC0_OUTB_P
ETDAC0_OUTB_M
ETDAC1_OUTA_P
ETDAC1_OUTA_M
ETDAC1_OUTB_P
ETDAC1_OUTB_M
SDR1_GRFC0
SDR1_GRFC1
SDR1_RFFE4_CLK
SDR1_RFFE4_DATA
SDR1_RFFE5_CLK
SDR1_RFFE5_DATA
SDR1_RFFE_SLAVE_CLK
SDR1_RFFE_SLAVE_DATA
SDR1_RFFE0_CLK
SDR1_RFFE0_DATA
SDR1_RFFE1_CLK
SDR1_RFFE1_DATA
SDR1_RFFE2_CLK
SDR1_RFFE2_DATA
SDR1_GRFC13
SDR1_GRFC14
W_GRFC_0
W_GRFC_1
W_GRFC_2
W_GRFC_3
W_GRFC_4
W_GRFC_5
RFFE_SLAVE_CLK
RFFE_SLAVE_DATA
RFFE0_CLK
RFFE0_DATA
RFFE1_CLK
RFFE1_DATA
RFFE2_CLK
RFFE2_DATA
RFFE3_CLK
RFFE3_DATA
W_GRFC_14
R3042
SDR1_XO_CLK
XO_IN
C3064
R3003
SDM_RFFE4_CLK
SDR1_RFFE_SLAVE_CLK
R3006
SDR2_RFFE_SLAVE_CLK
R3004
SDM_RFFE4_DATA
SDR1_RFFE_SLAVE_DATA
R3005
SDR2_RFFE_SLAVE_DATA
GND
PRX2_TR_IN_L
PRX2_TR_IN_LMH
PRX2_TR_IN_JMH
PRX2_TR_IN_MH
PRX2_TR_IN_M_JMH
PRX2_TR_IN_M_MH
DRX2_TR_IN_L
DRX2_TR_IN_LMH
DRX2_TR_IN_JMH
DRX2_TR_IN_MH
DRX2_TR_IN_M_JMH
DRX2_TR_IN_M_MH
U3000
#2
QLINK2_WMSS_RESET_N
QLINK2_EN
QLINK2_REQ
QLINK2_CLK_P
QLINK2_CLK_M
QLINK2_L0_P
QLINK2_L0_M
QLINK2_L1_P
QLINK2_L1_M
QLINK2_L2_P
QLINK2_L2_M
QLINK2_L3_P
QLINK2_L3_M
QLINK2_L4_P
QLINK2_L4_M
QLINK2
SDR2_GRFC4
SDR2_RFFE_SLAVE_CLK
SDR2_RFFE_SLAVE_DATA
SDR2_GRFC14
R3041
SDR2_XO_CLK
C3063
SDR1_GRFC13
C3126
U3002
CPL_ENDC_LB_OUT
RF2
VCTL
GND
ANT
RF1
VDD
LPWR
C3125
C3067
COUPLING_OUT
CPL_LB
L3001
L3002
C3121
C3120
C3119
C3122
C3123
C3065
C3124
C3066

WIFI
U3000 : DELETE

TRANSCEIVER(Dual SDR735)

TITLE TabS7P
SIZE A1
DWG NO
REV
DRAWN BY User Name
SHEET 3 of 15
Date

Connectivity

WIFI0 (WiFi2.4G / 5G / 6G)
CN4000
U4000
F4000
F4002
CABLE_DET_WIFI0
WL_TRX_5G_CH0
WL_TRX_2G_CH0
WIFI1 (WiFi2.4G / 5G / 6G/N77,N78)
CN4001
F4008
F4004
F4007
F4006
CABLE_DET_WIFI1
WL_TRX_5G_CH1
WL_TRX_2G_CH1
SDR1_GRFC1
U4014
U4015
SDR1_RFFE0_CLK
SDR1_RFFE0_DATA
DRX1_TR_IN_M_UN
SRS_OUT_N78_S3
U4001
WL_RFI_2G_CH1
WL_RFO_2G_CH1
BT_RFO_TX1
WL_RFI_2G_CH0
WL_RFO_2G_CH0
BT_RFO_TX0
BT_EN_CH0
BT_EN_CH1
WL_LNA_EN_2G_CH0
WL_LNA_EN_2G_CH1
WL_RF_PDET_2G_IN_CH1
WL_TRX_2G_CH1
WL_RF_PDET_2G_IN_CH0
WL_TRX_2G_CH0
WL_PA_EN_2G_CH0
WL_PA_EN_2G_CH1
U4002
WL_RFO_5G_CH1
WL_RFI_5G_CH1
5G_LTE_U_IN1
WL_RFO_5G_CH0
WL_RFI_5G_CH0
5G_LTE_U_IN0
WL_BIAS_SEL_5G_CH0
WL_BIAS_SEL_5G_CH1
WL_5G_RFFE_SEL4_CH1
WL_5G_RFFE_SEL4_CH0
WL_RF_PDET_5G_IN_CH1
WL_TRX_5G_CH1
WL_RF_PDET_5G_IN_CH0
WL_TRX_5G_CH0
WL_PA_EN_5G_CH0
WL_PA_EN_5G_CH1
WL_LNA_EN_5G_CH0
WL_LNA_EN_5G_CH1
U4011
U4010
U4013
U4012
MAIN_GRIP_SENSOR
U4004
GRIP1_SCL_1P8
GRIP1_SDA_1P8
GRIP_INT_1
WIFI0_GRIP
MAIN1_GRIP
U4008
GRIP1_LDO_EN
GRIP1 LDO
U4005
U4006
PRX1_TR_IN_NA
DRX1_TR_IN_NA
SDR1_RFFE2_DATA
SDR1_RFFE2_CLK
SDR1_RFFE0_DATA
SDR1_RFFE0_CLK
LAA LNA
LAA Option
mmW : O
SUB6, WIFI : X
U4007
U4009
OSC4000
OPTION
BT_RFO_TX0
BT_RFO_TX1
BT_SW_CTRL
BT_EN
WLAN_EN
LAA_AS_EN
RF_CLK1_WLAN
WIFI_THM
BT_UART_TXD
BT_UART_RXD
BT_UART_RTS
BT_UART_CTS
WLAN_BT_SLIMBUS_DATA
WLAN_BT_SLIMBUS_CLK
WLAN_COEX_SDXUART_TX
WLAN_COEX_SDXUART_RX
WLAN_SLEEP_CLK
LAA_EN_0
WCN_SW_CTRL
WCN_HOST_SOL_OUT
PCIE0_RST_N
PCIE0_WAKE_N
PCIE0_CLK_REQ_N
WCN6856 Decap
VDD08_RFA_CMN1
K4
VDD08_TOP_AON
P18 W13
VDD08_WL_CX
VDD08_WL_MX
W19 Y10 Y16 AA19
VDD08_CX_BT
K20
VDD12_WLBBPLL
L11
VDD12_WLADDA5G_CH0
L7
VDD12_BTSYN
F20
VDD12_BTTXBB_CH0
VDD12_WLBB2G_CH1
VDD12_WLSYN2G_PLL
VDD12_WLSYN2G_VCO
E23 H12 J13 G15
VDD12_WLBB2G_CH0
H16
VDD12_WLBB5G_CH1
VDD12_WLSYN5G_TXLO_CH0/1
VDD12_WLSYN5G_PLL
H2 G9 J5
VDD12_WLBB5G_CH0
H8
VDD12_BTPA_CH0
D22
VDDA_PCIE_1P8
AB16
VDDA_PCIE_CORE
AA17
VDD08_BTADDA_CH0
F22
VDD08_BTRX_CH0
VDD08_BTRXBB_CH0
VDD08_WLRX2G_CH1
VDD08_WLRX5G_CH1
VDD08_WLSYN2G_LOGEN
D20 E19 F12 F4 G13
VDD08_WLRX2G_CH0
D16
VDD08_WLRX5G_CH0
E7
VDD08_WLSYN5G_LOGEN
H4
VDD08_WLLPSYN5G
F6
VDD17_WLTX2G_CH0
D18
VDD17_WLTX5G_CH0
F8
VDD17_WLSYN5G_VCO
H6
VDD17_BTPA_CH0
C23
VDD17_XTAL
H20

WIFI		
	L4046	NC
	R4012, R4010	NC
	D4001	NC

TITLE TabS7P
SIZE A1
DWG NO
REV
DRAWN BY User Name
SHEET 4 of 15
Date

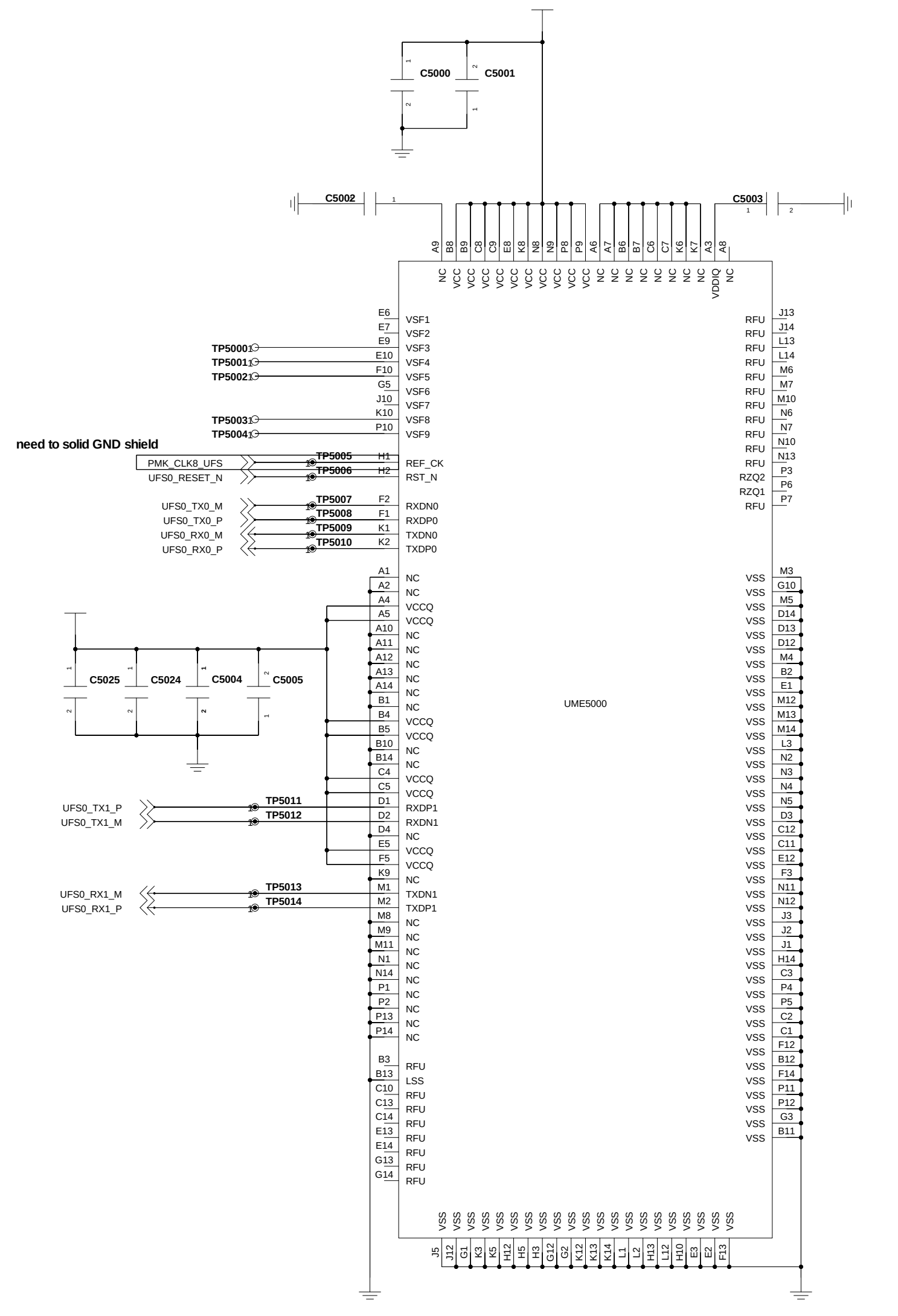
SM8550

(INTERFACE/GPIO)

on SECONDARY PCB

System

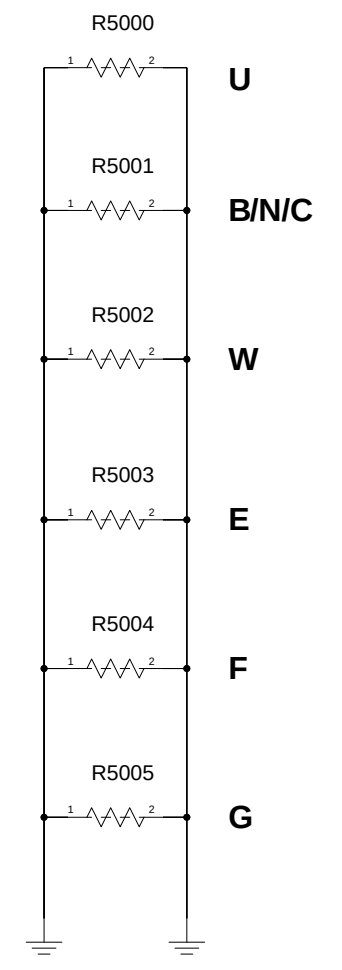
System


C5000
C5001
C5002
C5003
TP5000
TP5001
TP5002
TP5003
TP5004
need to solid GND shield
PMK_CLK8_UFS
UFS0_RESET_N
TP5005
TP5006
UFS0_TX0_M
UFS0_TX0_P
UFS0_RX0_M
UFS0_RX0_P
TP5007
TP5008
TP5009
TP5010
C5025
C5024
C5004
C5005
UFS0_TX1_P
UFS0_TX1_M
TP5011
TP5012
UFS0_RX1_M
UFS0_RX1_P
TP5013
TP5014
UME5000
VSF1
VSF2
VSF3
VSF4
VSF5
VSF6
VSF7
VSF8
VSF9
REF_CK
RST_N
RXDN0
RXDP0
TXDN0
TXDP0
VCCQ
RXDP1
RXDN1
TXDN1
TXDP1
LSS
RFU
RZQ2
RZQ1
VDDIQ
VCC
VSS
NC

UFS 4.0 256G

REGION

ANT PCB	SHORT
X818U (U)	R5000
X816B/N/C (B/N/C)	R5001
X810 (W)	R5002
ETC (E)	R5003
ETC (F)	R5004
ETC (G)	R5005

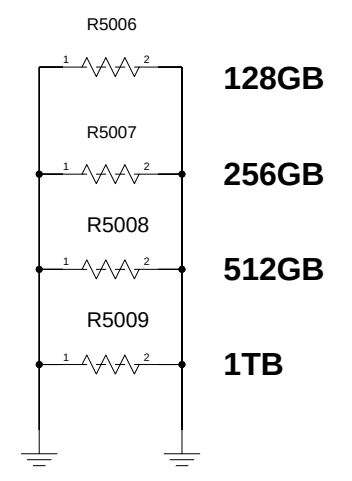

R5000
U
R5001
B/N/C
R5002
W
R5003
E
R5004
F
R5005
G

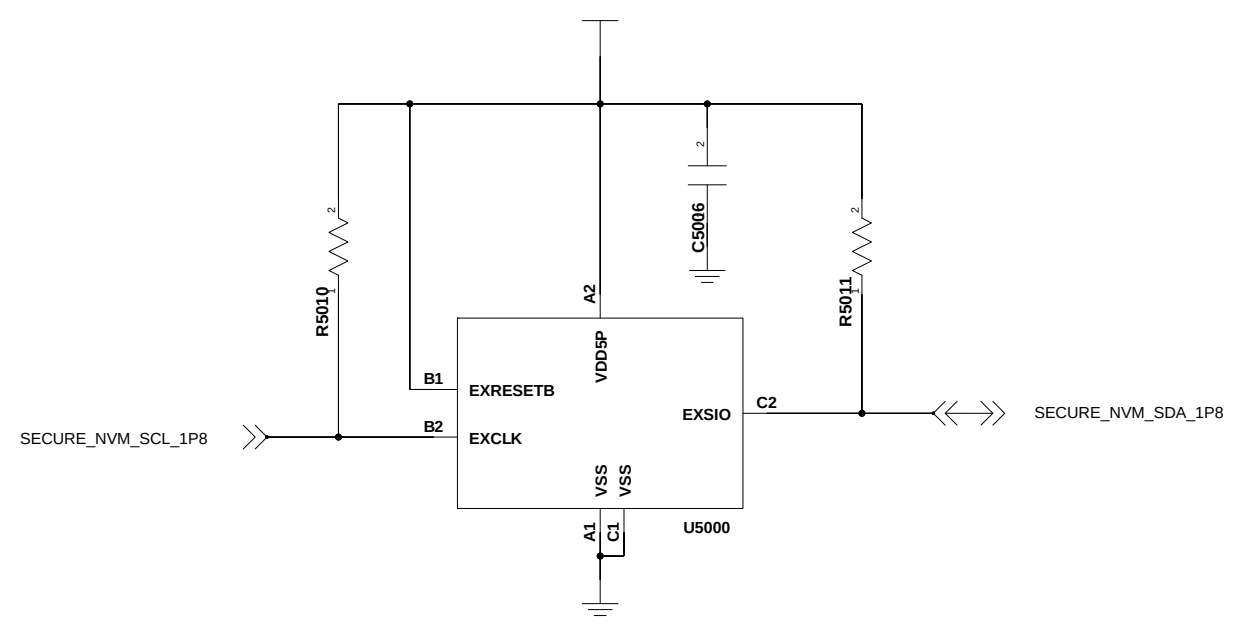
Memory_POP (CORE)

12GB	S9P

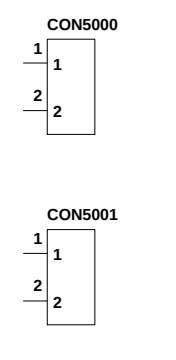
UFS Storage (MAIN)

256GB	S9P
512GB	S9P


R5006
128GB
R5007
256GB
R5008
512GB
R5009
1TB

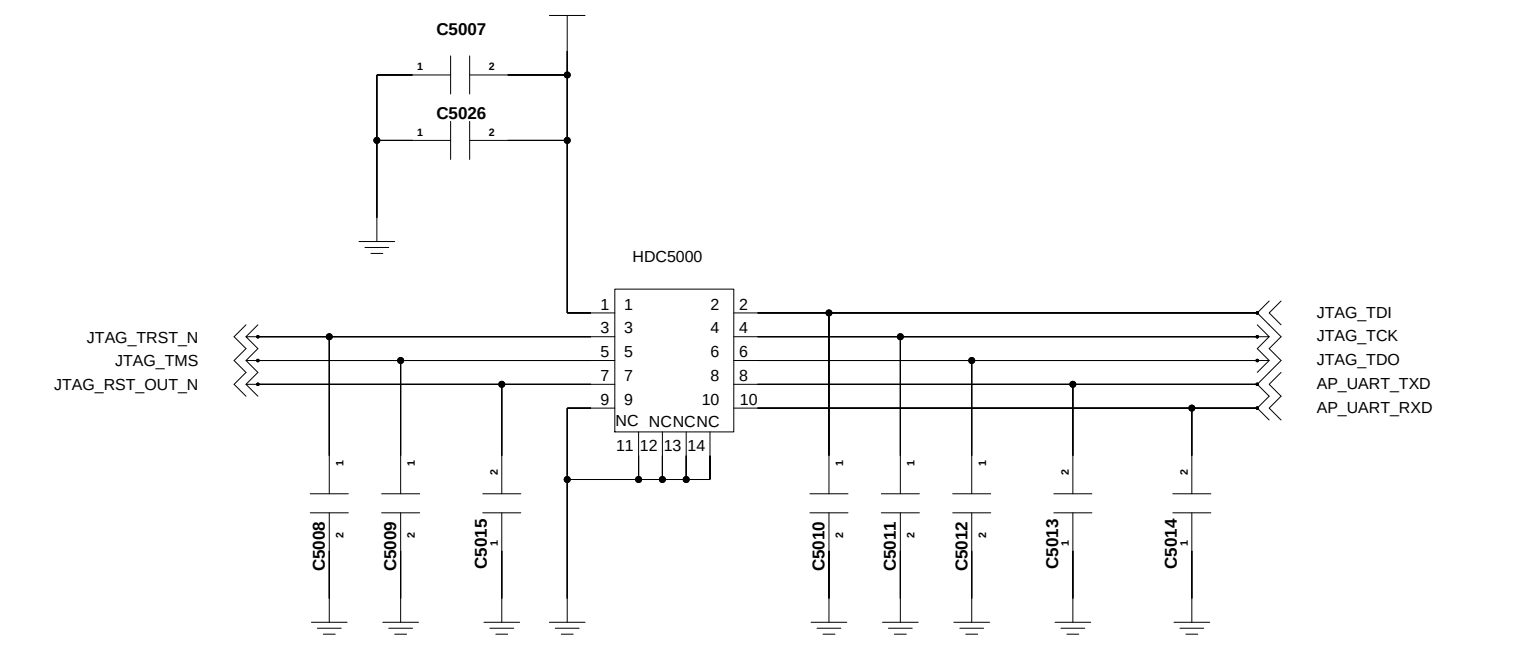

R5010
R5011
C5006
A2
VDDSP
B1
EXRESETB
B2
EXCLK
EXSIO
C2
VSS
VSS
A1
C1
U5000
SECURE_NVM_SCL_1P8
SECURE_NVM_SDA_1P8

SECURE NVM


CON5000
CON5001
1
2

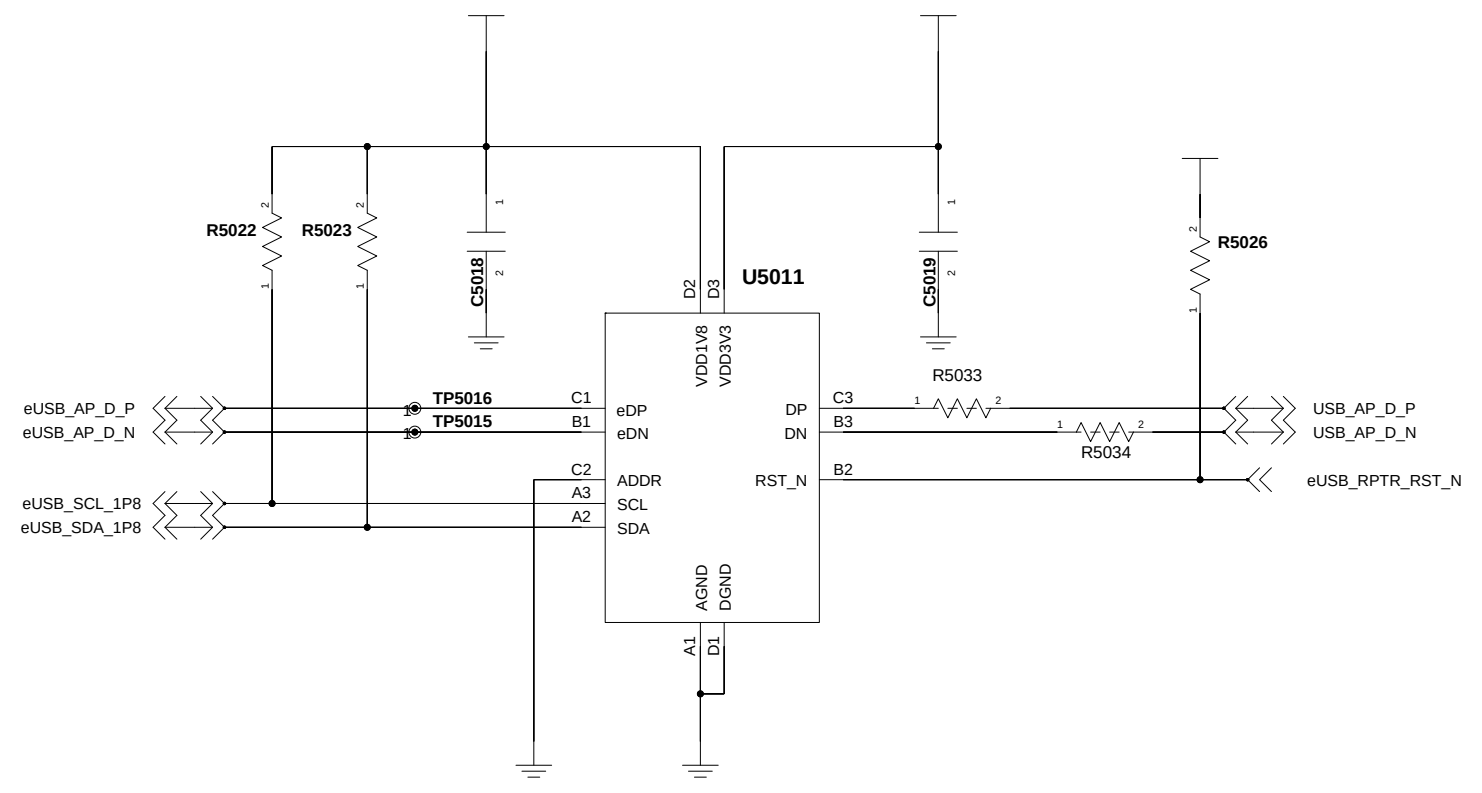
2D BARCODE

SMD NC

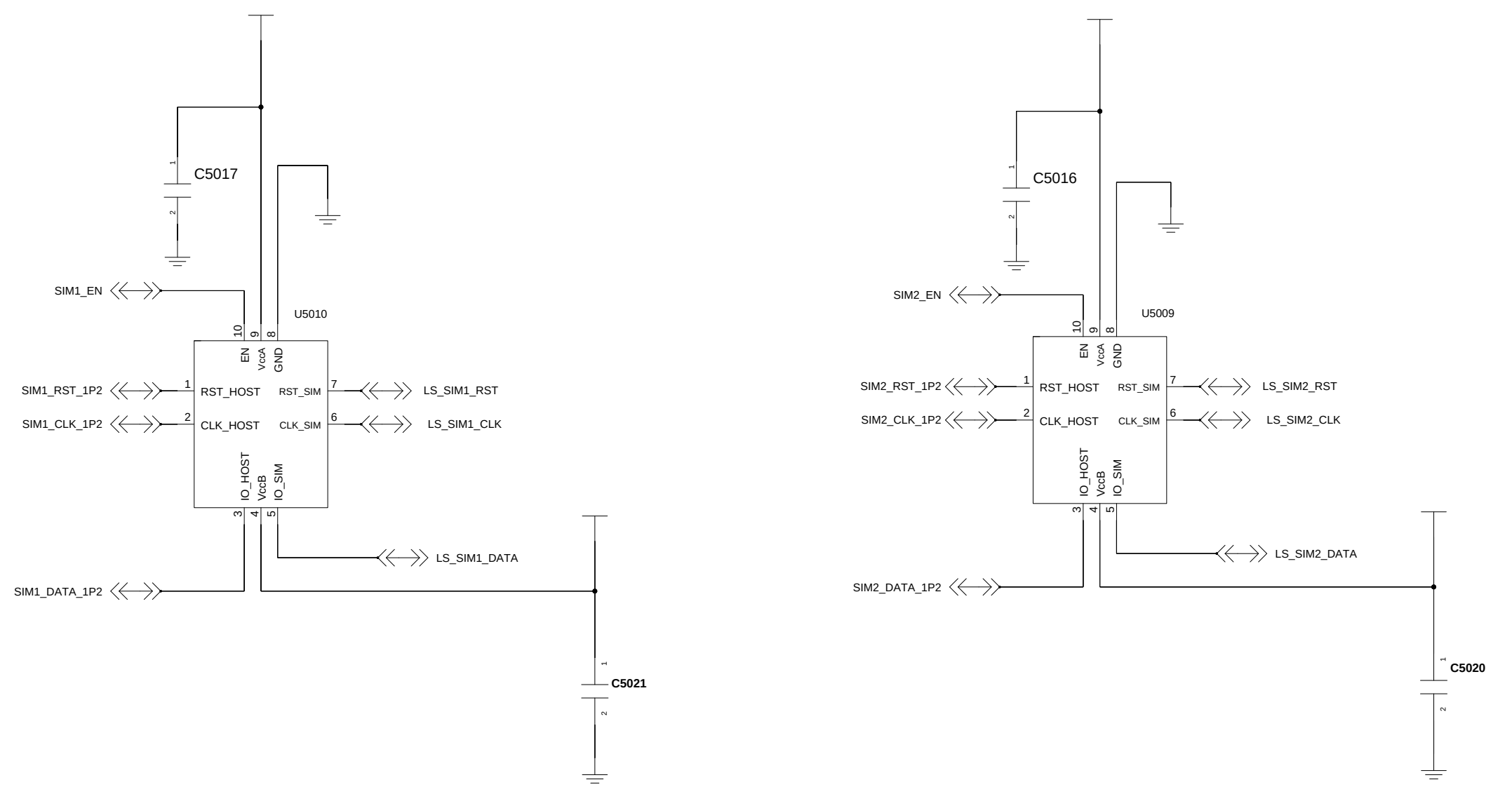

C5007
C5026
HDC5000
JTAG_TRST_N
JTAG_TMS
JTAG_RST_OUT_N
JTAG_TDI
JTAG_TCK
JTAG_TDO
AP_UART_TXD
AP_UART_RXD
NC
C5008
C5009
C5015
C5010
C5011
C5012
C5013
C5014

BOM NC

AP JTAG CONN

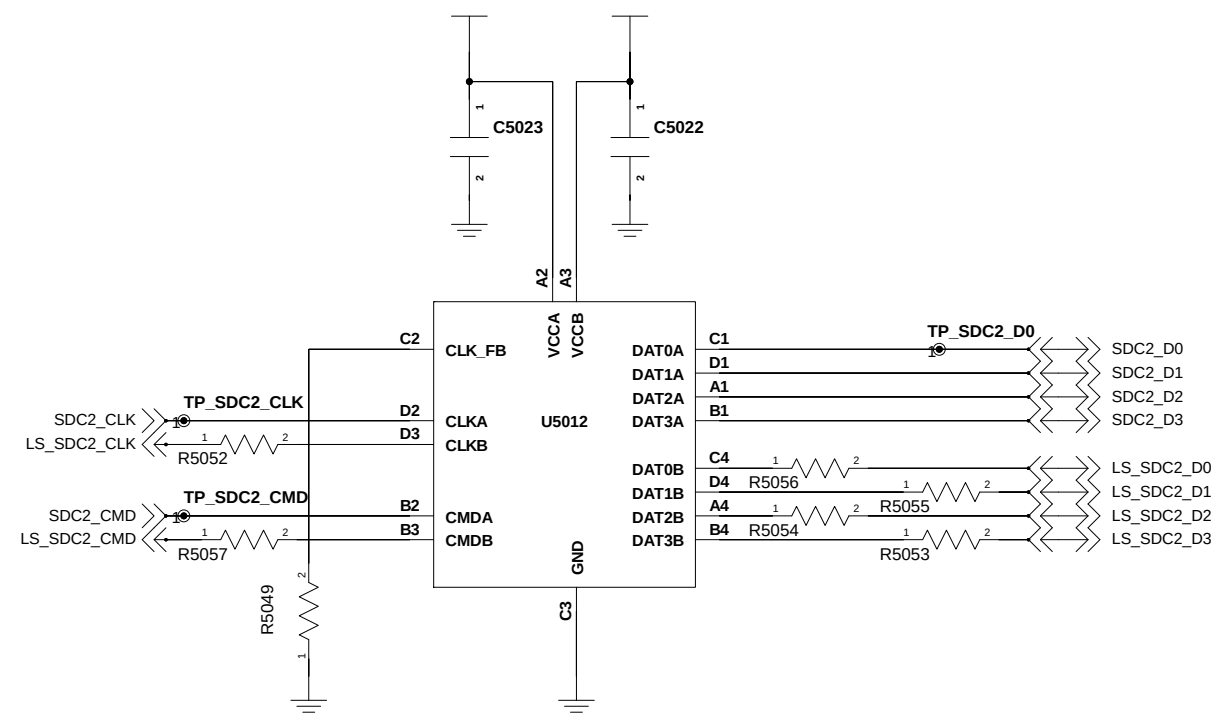

R5022
R5023
C5018
C5019
R5026
U5011
TP5016
TP5015
eUSB_AP_D_P
eUSB_AP_D_N
eUSB_SCL_1P8
eUSB_SDA_1P8
eDP
eDN
ADDR
SCL
SDA
VDD1V8
VDD3V3
DP
DN
RST_N
AGND
DGND
R5033
R5034
USB_AP_D_P
USB_AP_D_N
eUSB_RPTR_RST_N

eUSB REPEATER


C5017
U5010
SIM1_EN
SIM1_RST_1P2
SIM1_CLK_1P2
SIM1_DATA_1P2
LS_SIM1_RST
LS_SIM1_CLK
LS_SIM1_DATA
C5021
C5016
U5009
SIM2_EN
SIM2_RST_1P2
SIM2_CLK_1P2
SIM2_DATA_1P2
LS_SIM2_RST
LS_SIM2_CLK
LS_SIM2_DATA
C5020
EN
VccA
GND
RST_HOST
CLK_HOST
RST_SIM
CLK_SIM
IO_HOST
VccB
IO_SIM

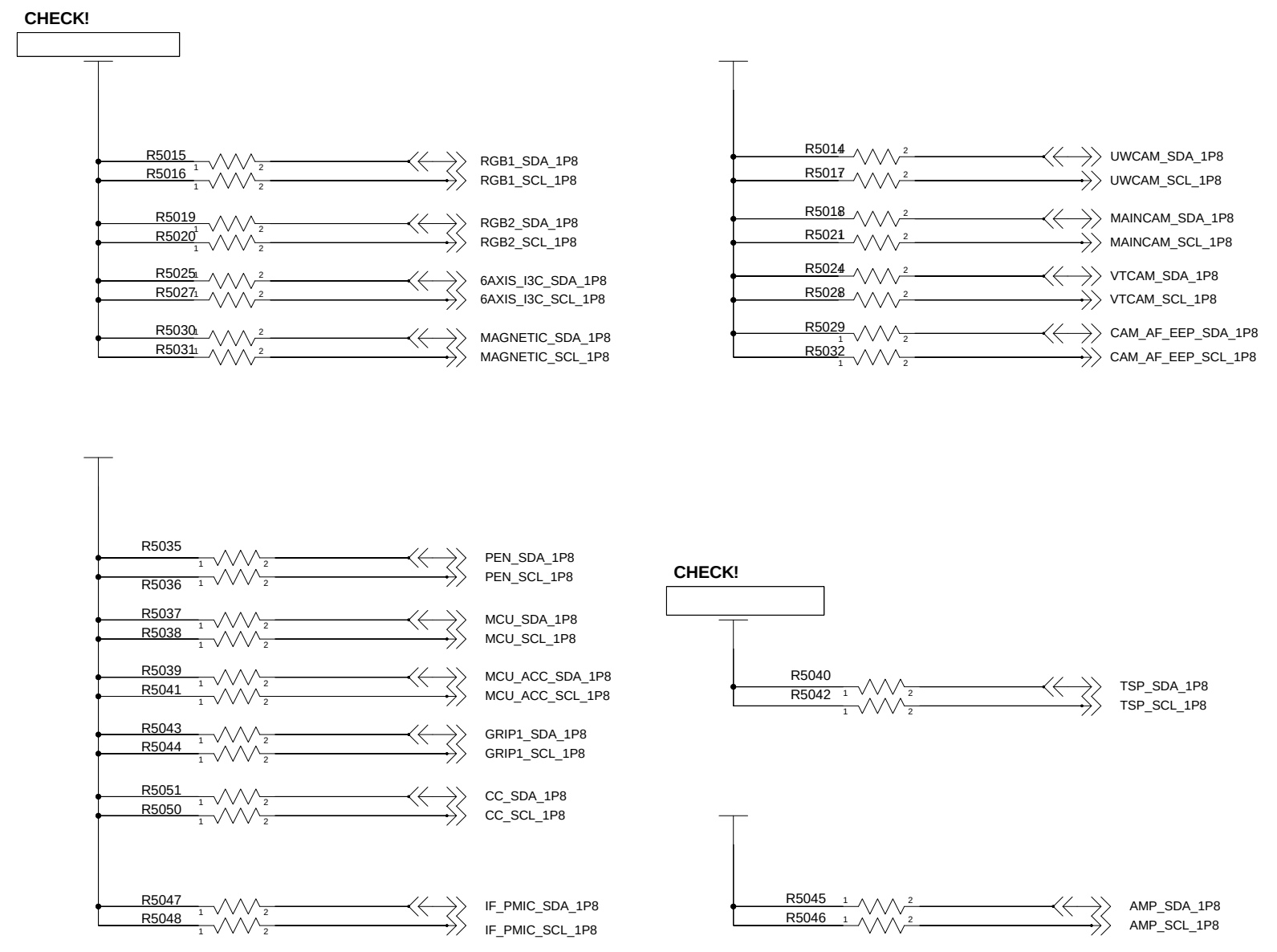
SIM Level Shifter

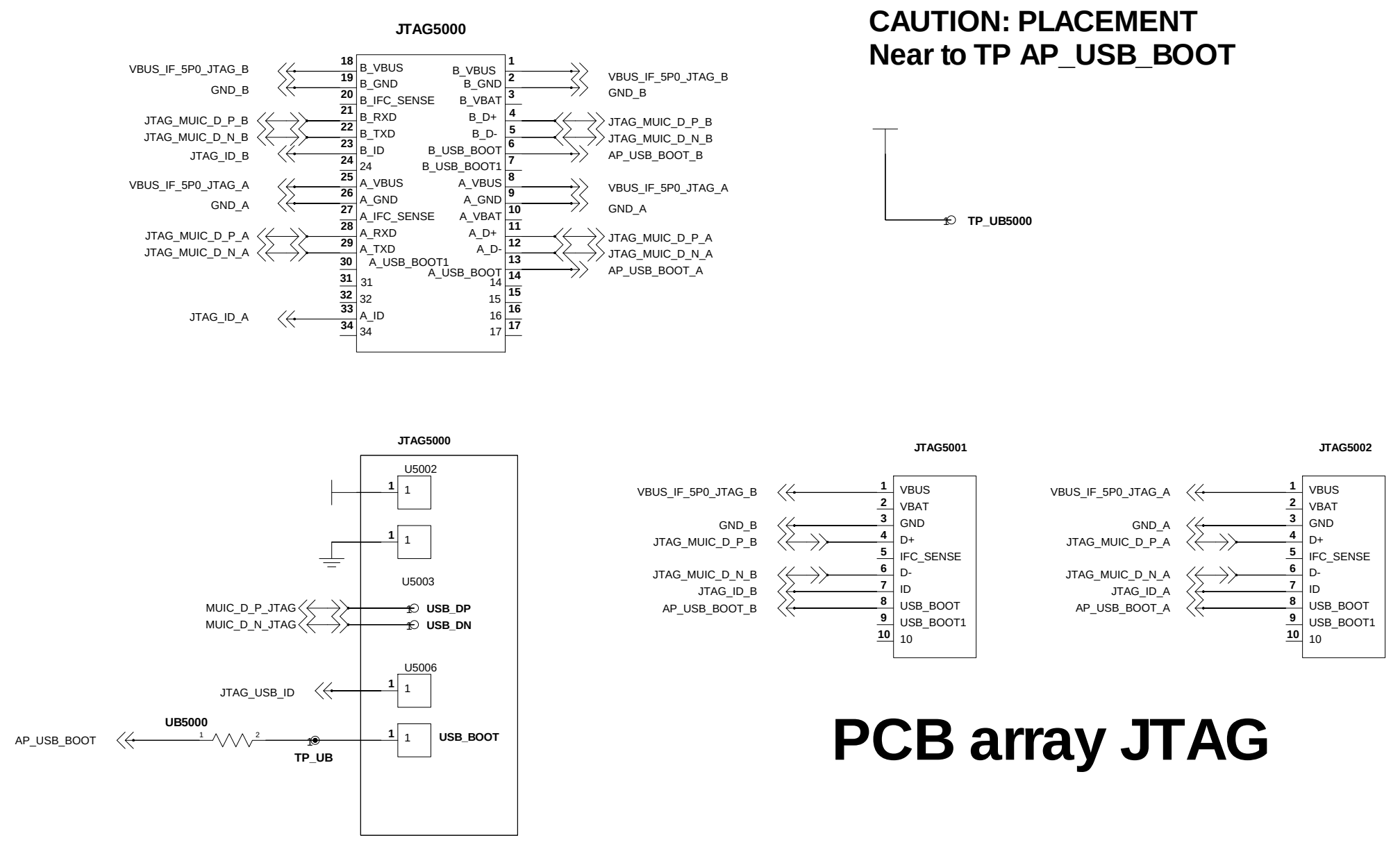
WIFI delete


C5023
C5022
U5012
TP_SDC2_CLK
TP_SDC2_CMD
TP_SDC2_D0
SDC2_CLK
LS_SDC2_CLK
SDC2_CMD
LS_SDC2_CMD
R5052
R5057
R5049
R5056
R5055
R5054
R5053
CLK_FB
CLKA
CLKB
CMDA
CMDB
VCCA
VCCB
GND
DAT0A
DAT1A
DAT2A
DAT3A
DAT0B
DAT1B
DAT2B
DAT3B
SDC2_D0
SDC2_D1
SDC2_D2
SDC2_D3
LS_SDC2_D0
LS_SDC2_D1
LS_SDC2_D2
LS_SDC2_D3

SD Level Shifter

Place close to connector


CHECK!
R5015
R5016
RGB1_SDA_1P8
RGB1_SCL_1P8
R5019
R5020
RGB2_SDA_1P8
RGB2_SCL_1P8
R5025
R5027
6AXIS_I3C_SDA_1P8
6AXIS_I3C_SCL_1P8
R5030
R5031
MAGNETIC_SDA_1P8
MAGNETIC_SCL_1P8
R5014
R5017
UWCAM_SDA_1P8
UWCAM_SCL_1P8
R5018
R5021
MAINCAM_SDA_1P8
MAINCAM_SCL_1P8
R5024
R5028
VTCAM_SDA_1P8
VTCAM_SCL_1P8
R5029
R5032
CAM_AF_EEP_SDA_1P8
CAM_AF_EEP_SCL_1P8
R5035
R5036
PEN_SDA_1P8
PEN_SCL_1P8
R5037
R5038
MCU_SDA_1P8
MCU_SCL_1P8
R5039
R5041
MCU_ACC_SDA_1P8
MCU_ACC_SCL_1P8
R5043
R5044
GRIP1_SDA_1P8
GRIP1_SCL_1P8
R5051
R5050
CC_SDA_1P8
CC_SCL_1P8
R5047
R5048
IF_PMIC_SDA_1P8
IF_PMIC_SCL_1P8
CHECK!
R5040
R5042
TSP_SDA_1P8
TSP_SCL_1P8
R5045
R5046
AMP_SDA_1P8
AMP_SCL_1P8


JTAG5000
VBUS_IF_5P0_JTAG_B
GND_B
JTAG_MUIC_D_P_B
JTAG_MUIC_D_N_B
JTAG_ID_B
AP_USB_BOOT_B
VBUS_IF_5P0_JTAG_A
GND_A
JTAG_MUIC_D_P_A
JTAG_MUIC_D_N_A
AP_USB_BOOT_A
JTAG_ID_A
U5002
U5003
U5006
MUIC_D_P_JTAG
MUIC_D_N_JTAG
USB_DP
USB_DN
JTAG_USB_ID
UB5000
AP_USB_BOOT
TP_UB
USB_BOOT
TP_UB5000
JTAG5001
JTAG5002
VBUS
VBAT
GND
D+
IFC_SENSE
D-
ID
USB_BOOT1

CAUTION: PLACEMENT
Near to TP AP_USB_BOOT

PCB array JTAG

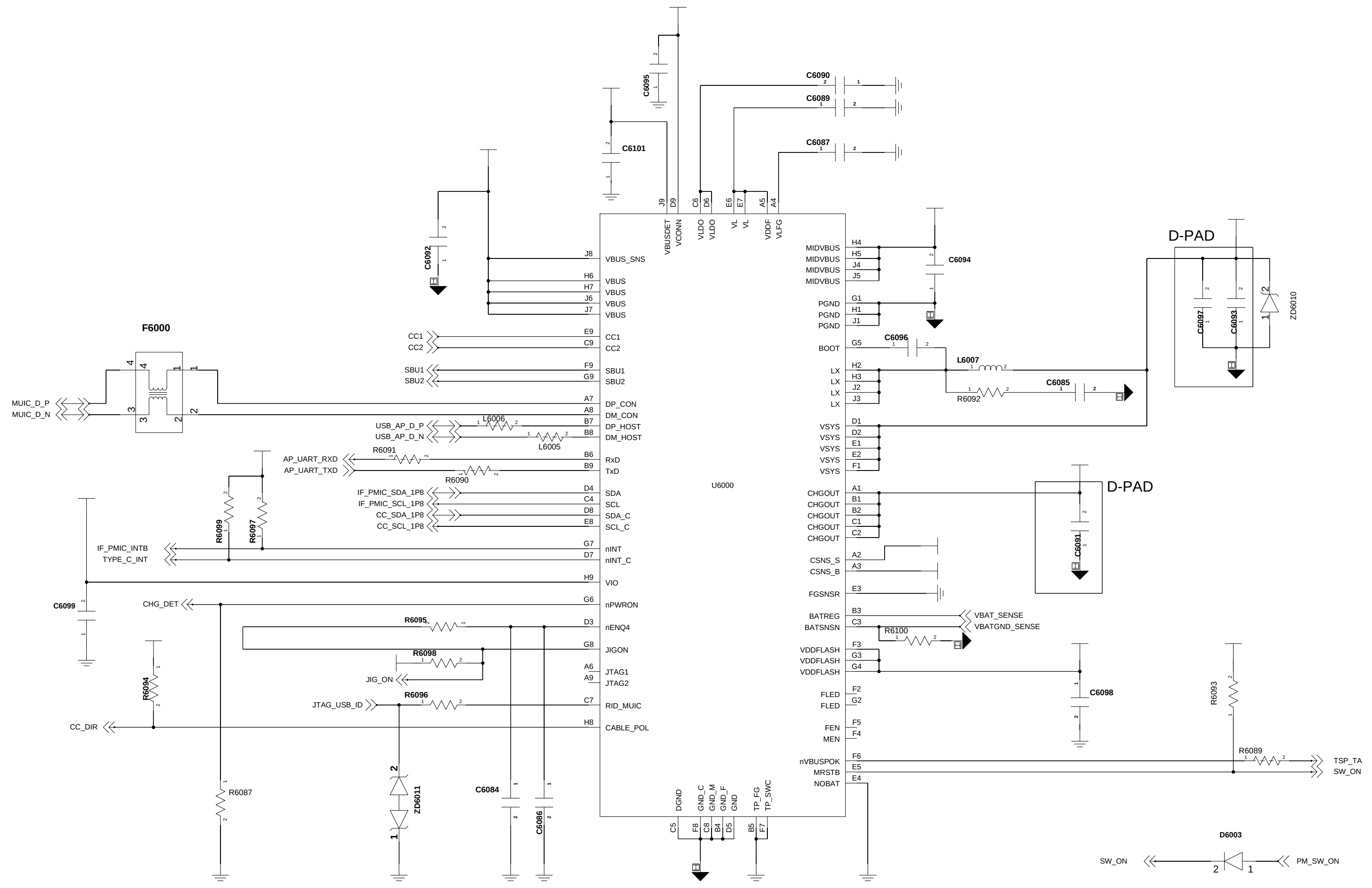
TITLE	TabS7P
SIZE	A1
DWG NO	
REV	
DRAWN BY	User Name
SHEET	5 of 15
Date	

AP PMIC set & CLK Generator

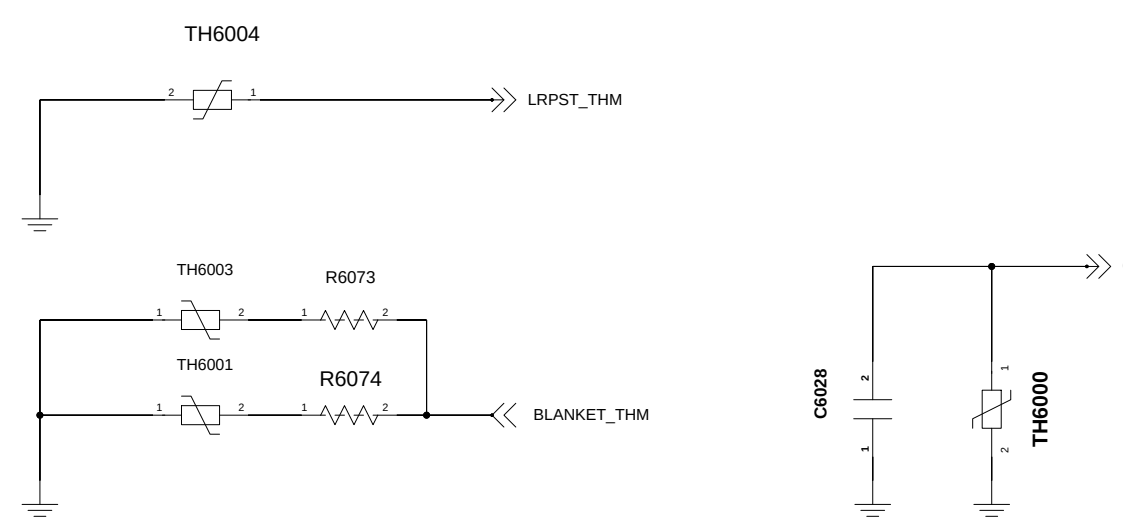
Power

PMIC (PM8550/8550VS/8550VE)

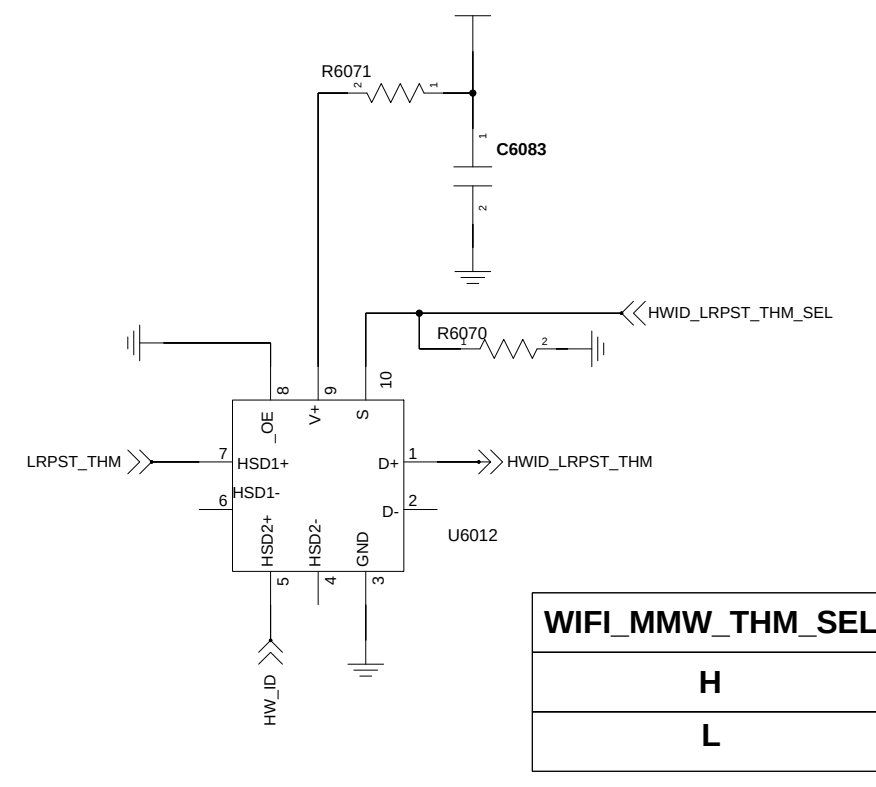
on SECONDARY PCB


C6095
C6101
C6090
C6089
C6087
C6092
C6094
D-PAD
C6097
C6093
ZD6010
F6000
MUIC_D_P
MUIC_D_N
CC1
CC2
SBU1
SBU2
USB_AP_D_P
USB_AP_D_N
L6006
L6005
AP_UART_RXD
AP_UART_TXD
R6091
R6090
IF_PMIC_SDA_1P8
IF_PMIC_SCL_1P8
CC_SDA_1P8
CC_SCL_1P8
R6099
R6097
IF_PMIC_INTB
TYPE_C_INT
C6099
CHG_DET
R6095
R6098
JIG_ON
R6096
JTAG_USB_ID
R6094
CC_DIR
R6087
ZD6011
C6084
C6086
VBUS_SNS
VBUS
DP_CON
DM_CON
DP_HOST
DM_HOST
RxD
TxD
SDA
SCL
SDA_C
SCL_C
nINT
nINT_C
VIO
nPWRON
nENQ4
JIGON
JTAG1
JTAG2
RID_MUIC
CABLE_POL
U6000
VBUSDET
VCONN
VLDO
VL
VDDF
VLFG
MIDVBUS
PGND
BOOT
LX
VSYS
CHGOUT
CSNS_S
CSNS_B
FGSNSR
BATREG
BATSNSN
VDDFLASH
FLED
FEN
MEN
nVBUSPOK
MRSTB
NOBAT
DGND
GND_C
GND_M
GND_F
GND
TP_FG
TP_SWC
C6096
L6007
R6092
C6085
D-PAD
C6091
VBAT_SENSE
VBATGND_SENSE
R6100
C6098
R6093
R6089
TSP_TA
SW_ON
D6003
SW_ON
PM_SW_ON

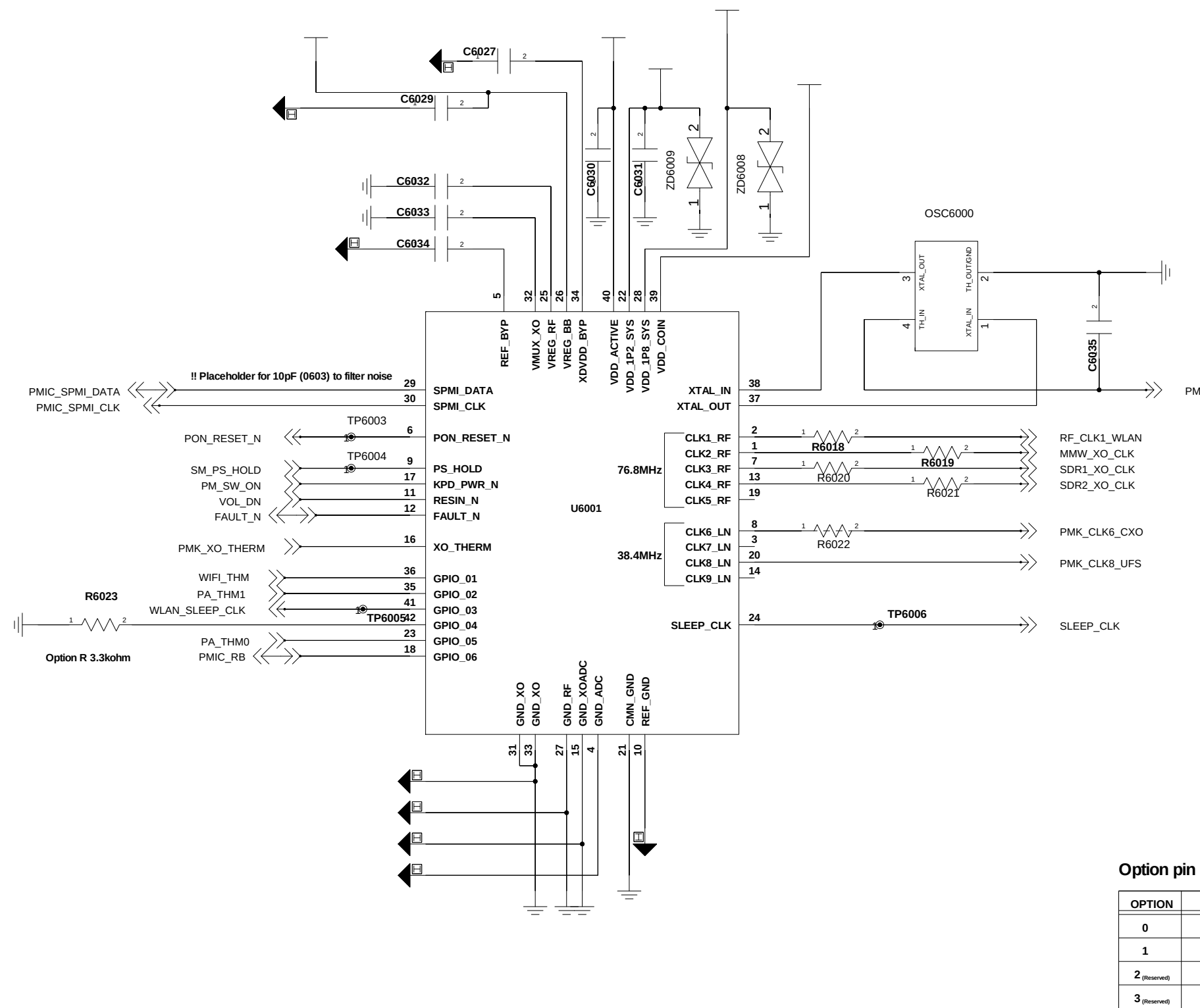
IF PMIC


TH6004
LRPST_THM
TH6003
R6073
TH6001
R6074
BLANKET_THM
CHG_THM
C6028
TH6000

THERMISTERS

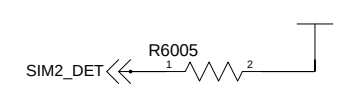

R6071
C6083
HWID_LRPST_THM_SEL
R6070
LRPST_THM
HSD1+
HSD1-
HSD2+
HSD2-
OE
V+
S
D+
D-
GND
HWID_LRPST_THM
U6012
HW_ID

WIFI_MMW_THM_SEL	RESULT
H	LRPST_THM
L	HW_ID

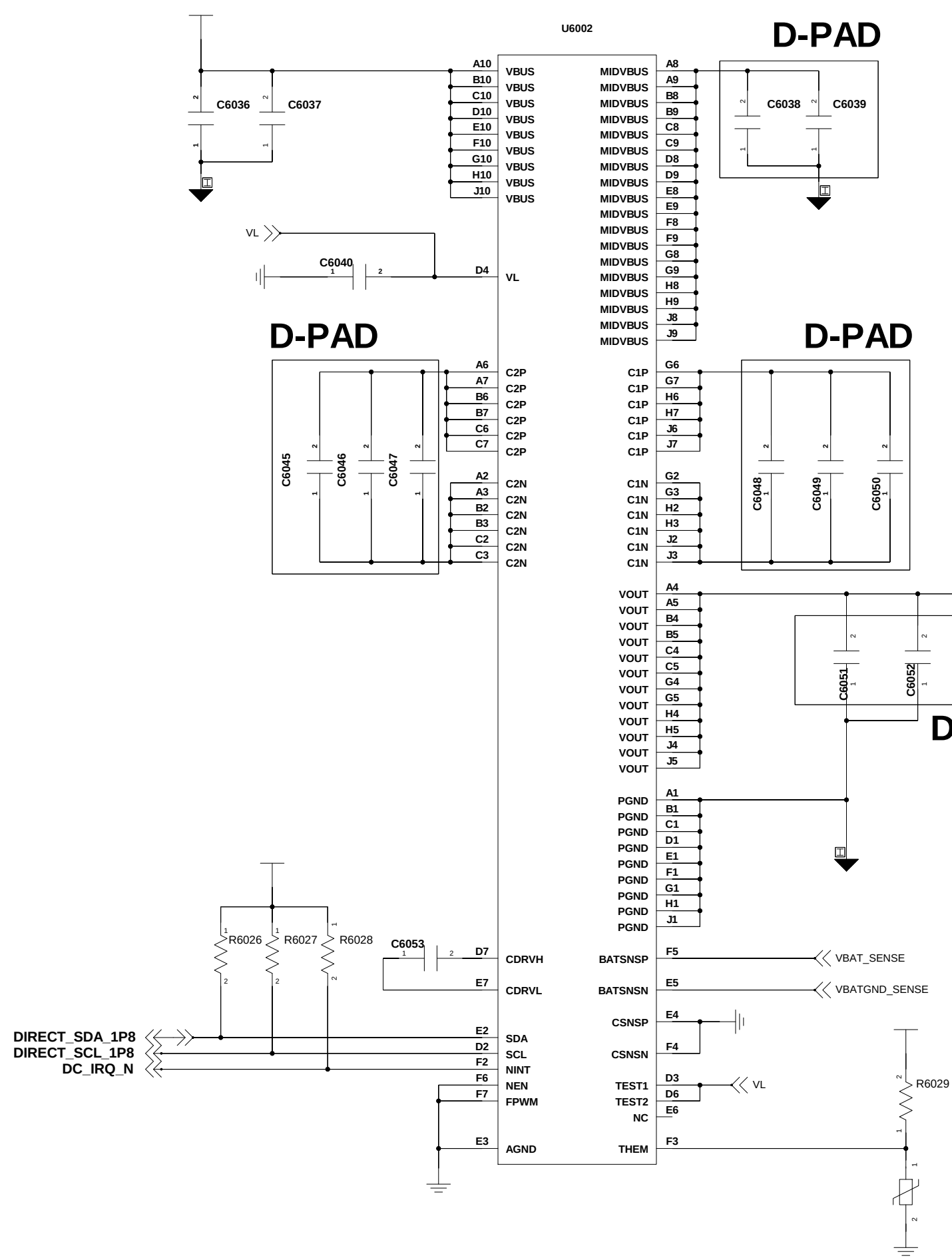

C6027
C6029
C6032
C6033
C6034
C6030
C6031
ZD6009
ZD6008
OSC6000
C6035
!! Placeholder for 10pF (0603) to filter noise
PMIC_SPMI_DATA
PMIC_SPMI_CLK
TP6003
PON_RESET_N
TP6004
SM_PS_HOLD
PM_SW_ON
VOL_DN
FAULT_N
PMK_XO_THERM
WIFI_THM
PA_THM1
WLAN_SLEEP_CLK
R6023
TP6005
Option R 3.3kohm
PA_THM0
PMIC_RB
REF_BYP
VMUX_XO
VREG_RF
VREG_BB
XDVDD_BYP
VDD_ACTIVE
VDD_1P2_SYS
VDD_1P8_SYS
VDD_COIN
SPMI_DATA
SPMI_CLK
PS_HOLD
KPD_PWR_N
RESIN_N
FAULT_N
XO_THERM
GPIO_01
GPIO_02
GPIO_03
GPIO_04
GPIO_05
GPIO_06
U6001
XTAL_IN
XTAL_OUT
76.8MHz
CLK1_RF
CLK2_RF
CLK3_RF
CLK4_RF
CLK5_RF
38.4MHz
CLK6_LN
CLK7_LN
CLK8_LN
CLK9_LN
SLEEP_CLK
GND_XO
GND_RF
GND_XOADC
GND_ADC
CMN_GND
REF_GND
R6018
R6019
R6020
R6021
R6022
TP6006
PMK_XO_THERM
RF_CLK1_WLAN
MMW_XO_CLK
SDR1_XO_CLK
SDR2_XO_CLK
PMK_CLK6_CXO
PMK_CLK8_UFS
SLEEP_CLK

Option pin state (PMK8550; 76.8MHz XO)

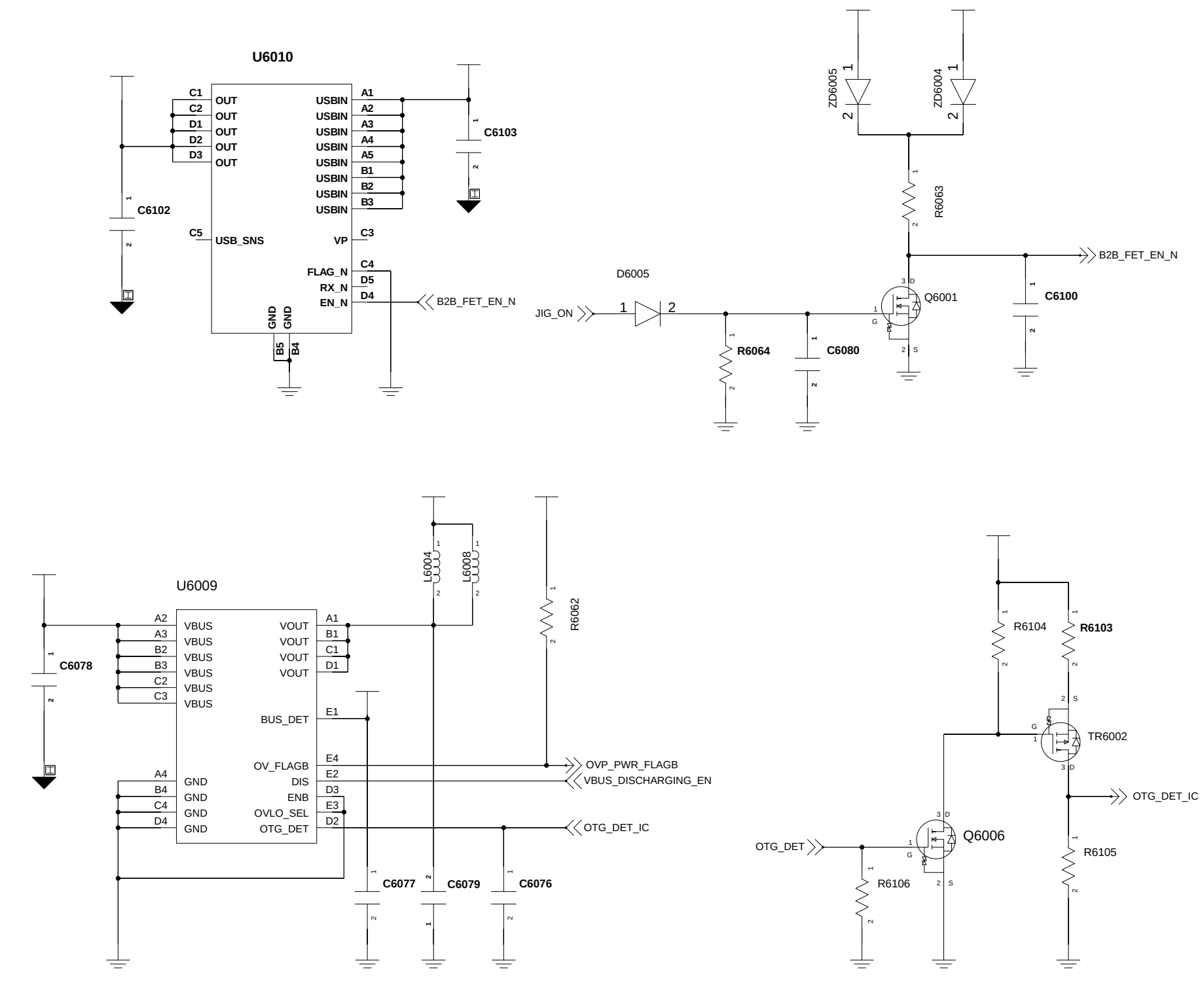
OPTION	LPDDR5	SD card LDO	GPIO05B Active	R value
0	O	enable	HIGH	
1	O	enable	LOW	
2 (Reserved)				
3 (Reserved)				
4	O	skip	HIGH	
5	O	skip	LOW	
6 (Reserved)				
7 (Reserved)				


SIM2_DET
R6005

PMK8550

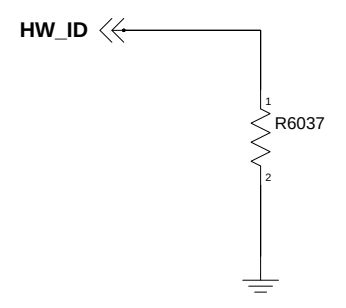

U6002
D-PAD
C6036
C6037
C6038
C6039
VL
C6040
D-PAD
C6045
C6046
C6047
C6048
C6049
C6050
C6051
C6052
D-PAD
VBUS
MIDVBUS
C2P
C2N
C1P
C1N
VOUT
PGND
R6026
R6027
R6028
C6053
CDRVH
CDRVL
DIRECT_SDA_1P8
DIRECT_SCL_1P8
DC_IRQ_N
SDA
SCL
NINT
NEN
FPWM
AGND
BATSNSP
BATSNSN
CSNSP
CSNSN
TEST1
TEST2
NC
THEM
VBAT_SENSE
VBATGND_SENSE
R6029
TH6002

Direct Charging


U6010
C6103
C6102
OUT
USBIN
USB_SNS
VP
FLAG_N
RX_N
EN_N
B2B_FET_EN_N
ZD6005
ZD6004
R6063
B2B_FET_EN_N
D6005
JIG_ON
R6064
C6080
Q6001
C6100
U6009
L6004
L6008
R6062
C6078
VBUS
VOUT
BUS_DET
OV_FLAGB
DIS
ENB
OVLO_SEL
OTG_DET
OVP_PWR_FLAGB
VBUS_DISCHARGING_EN
OTG_DET_IC
C6077
C6079
C6076
R6104
R6103
TR6002
OTG_DET_IC
OTG_DET
Q6006
R6106
R6105

OVP

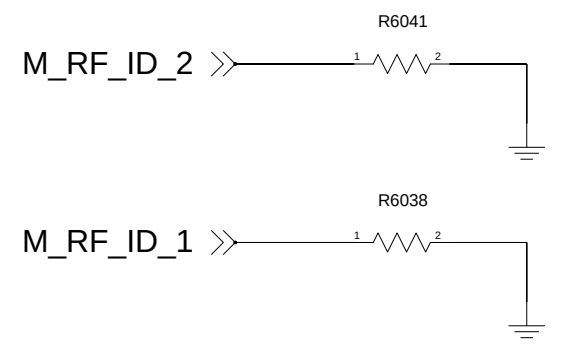
HW ID



R6037

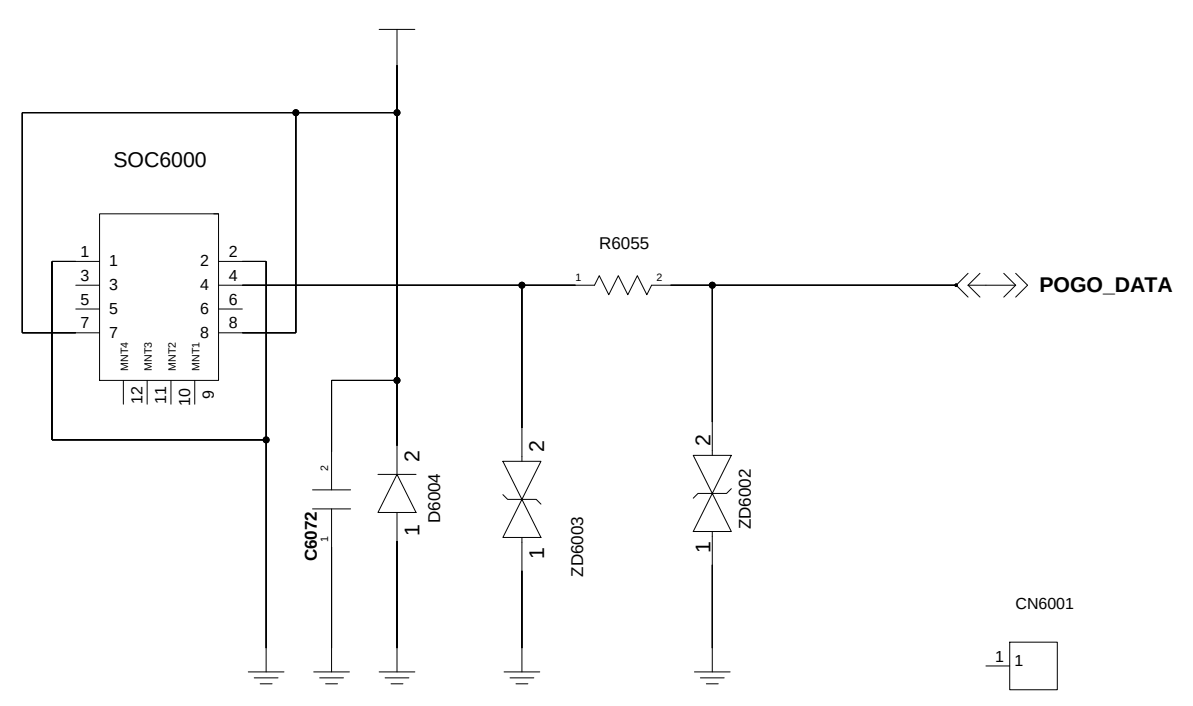
System

M_RF_ID

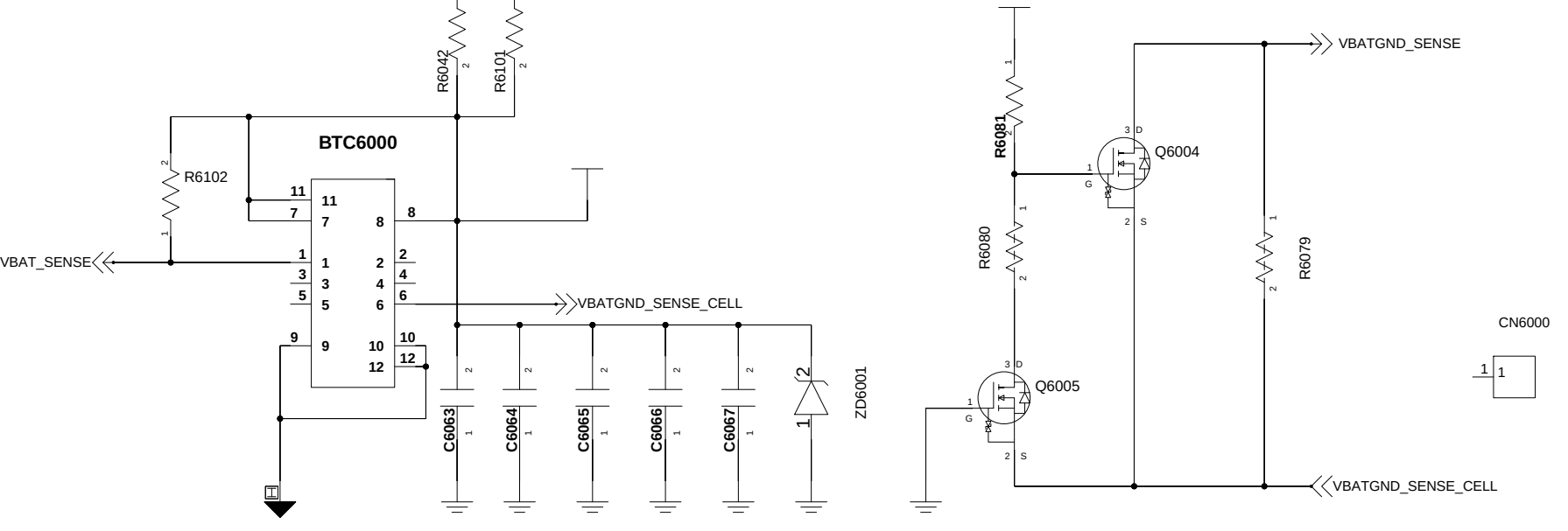

R6041
M_RF_ID_2
R6038
M_RF_ID_1

Check R11011 for M_RF_ID_0

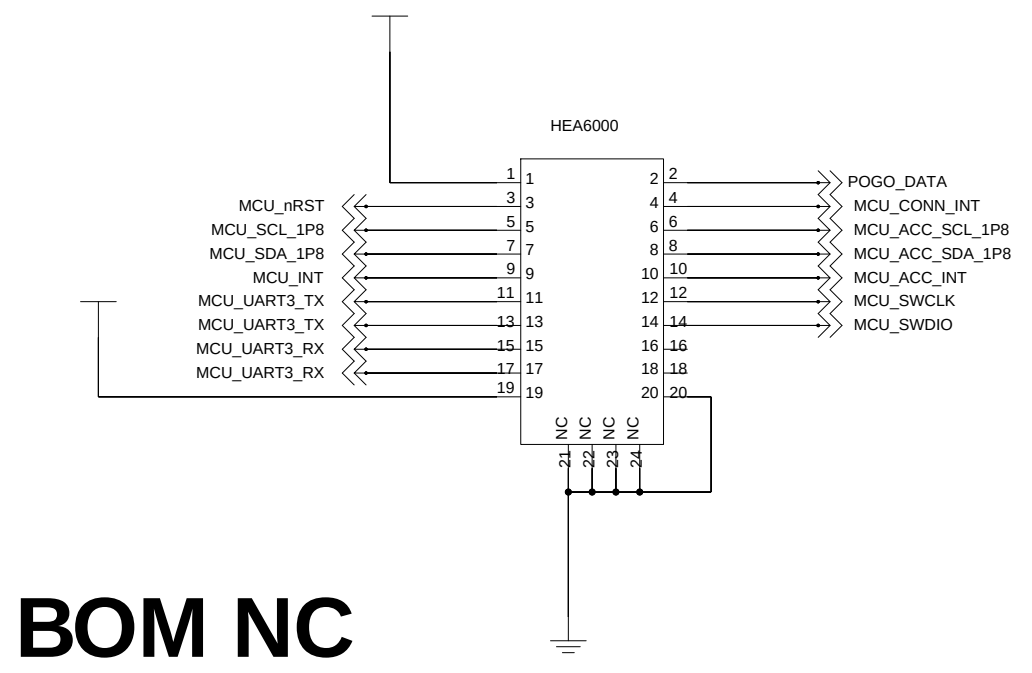
POGO


SOC6000
R6055
POGO_DATA
C6072
D6004
ZD6003
ZD6002
CN6001

POGO BtoB Con.

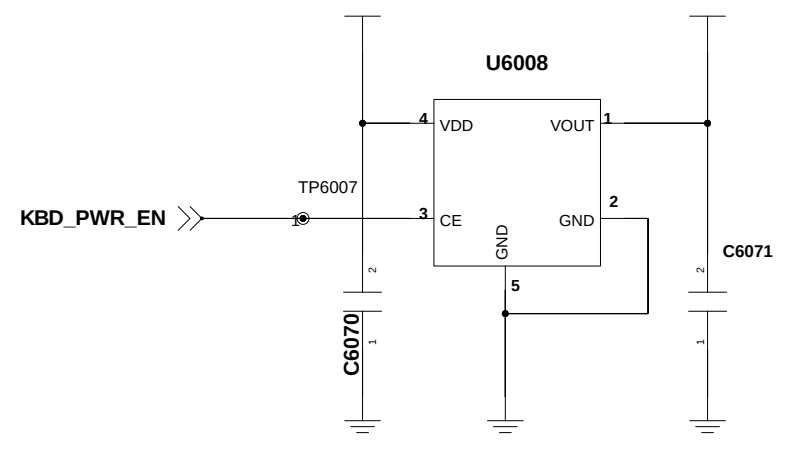

R6042
R6101
R6102
BTC6000
VBAT_SENSE
VBATGND_SENSE_CELL
C6063
C6064
C6065
C6066
C6067
ZD6001
R6081
Q6004
VBATGND_SENSE
R6080
R6079
Q6005
VBATGND_SENSE_CELL
CN6000

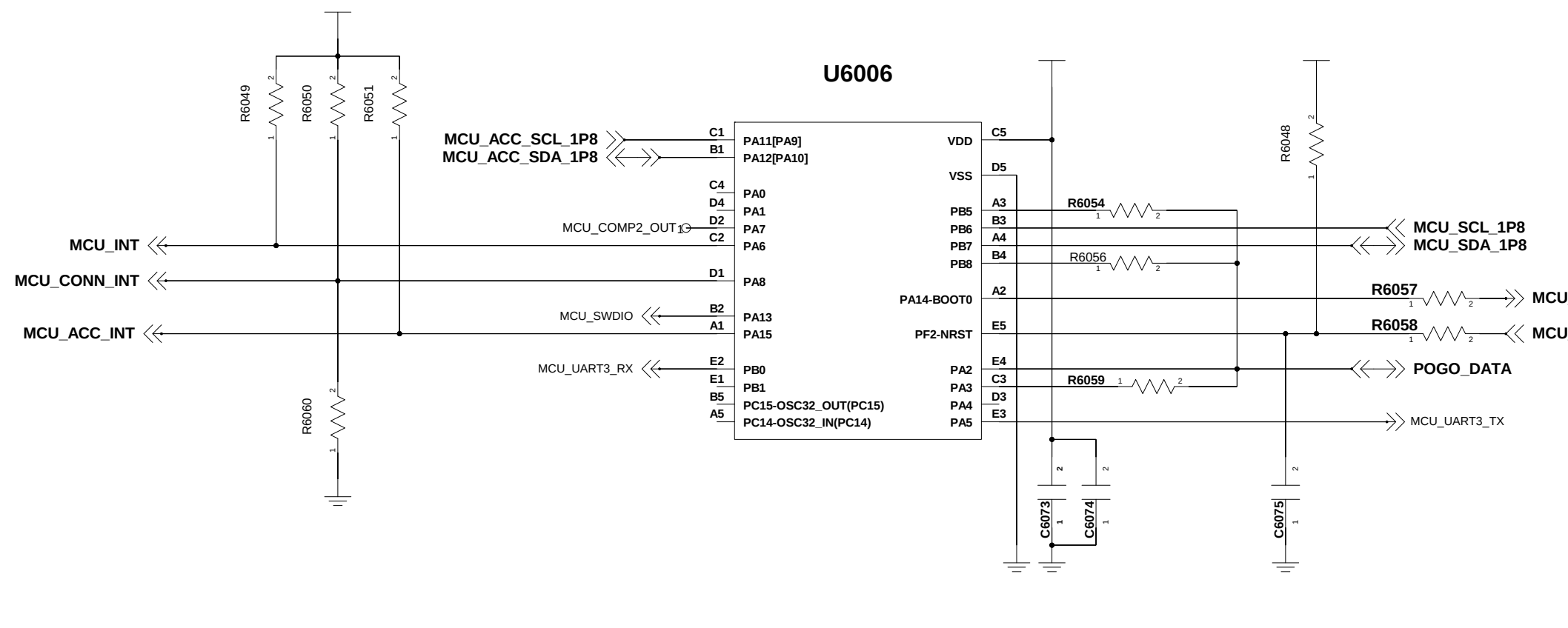
Battery Con


HEA6000
MCU_nRST
MCU_SCL_1P8
MCU_SDA_1P8
MCU_INT
MCU_UART3_TX
MCU_UART3_TX
MCU_UART3_RX
MCU_UART3_RX
POGO_DATA
MCU_CONN_INT
MCU_ACC_SCL_1P8
MCU_ACC_SDA_1P8
MCU_ACC_INT
MCU_SWCLK
MCU_SWDIO
NC

BOM NC

POGO Debug

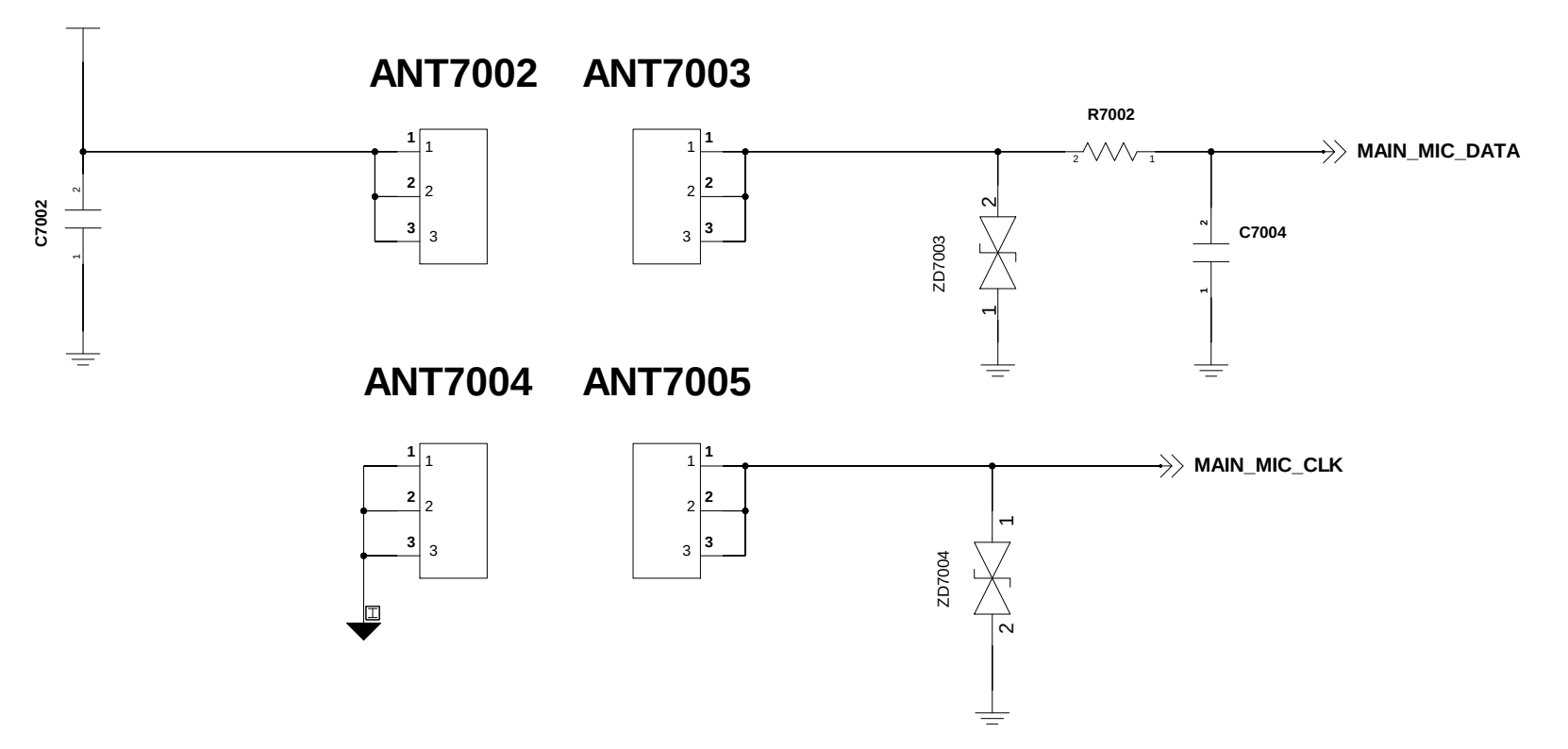

U6008
VDD
VOUT
TP6007
KBD_PWR_EN
CE
GND
C6070
C6071


U6006
R6049
R6050
R6051
R6060
MCU_ACC_SCL_1P8
MCU_ACC_SDA_1P8
MCU_COMP2_OUT
MCU_INT
MCU_CONN_INT
MCU_SWDIO
MCU_ACC_INT
MCU_UART3_RX
PA11[PA9]
PA12[PA10]
PA0
PA1
PA7
PA6
PA8
PA13
PA15
PB0
PB1
PC15-OSC32_OUT(PC15)
PC14-OSC32_IN(PC14)
VDD
VSS
PB5
PB6
PB7
PB8
PA14-BOOT0
PF2-NRST
PA2
PA3
PA4
PA5
R6054
R6056
R6057
R6058
R6059
R6048
MCU_SCL_1P8
MCU_SDA_1P8
MCU_SWCLK
MCU_nRST
POGO_DATA
MCU_UART3_TX
C6073
C6074
C6075

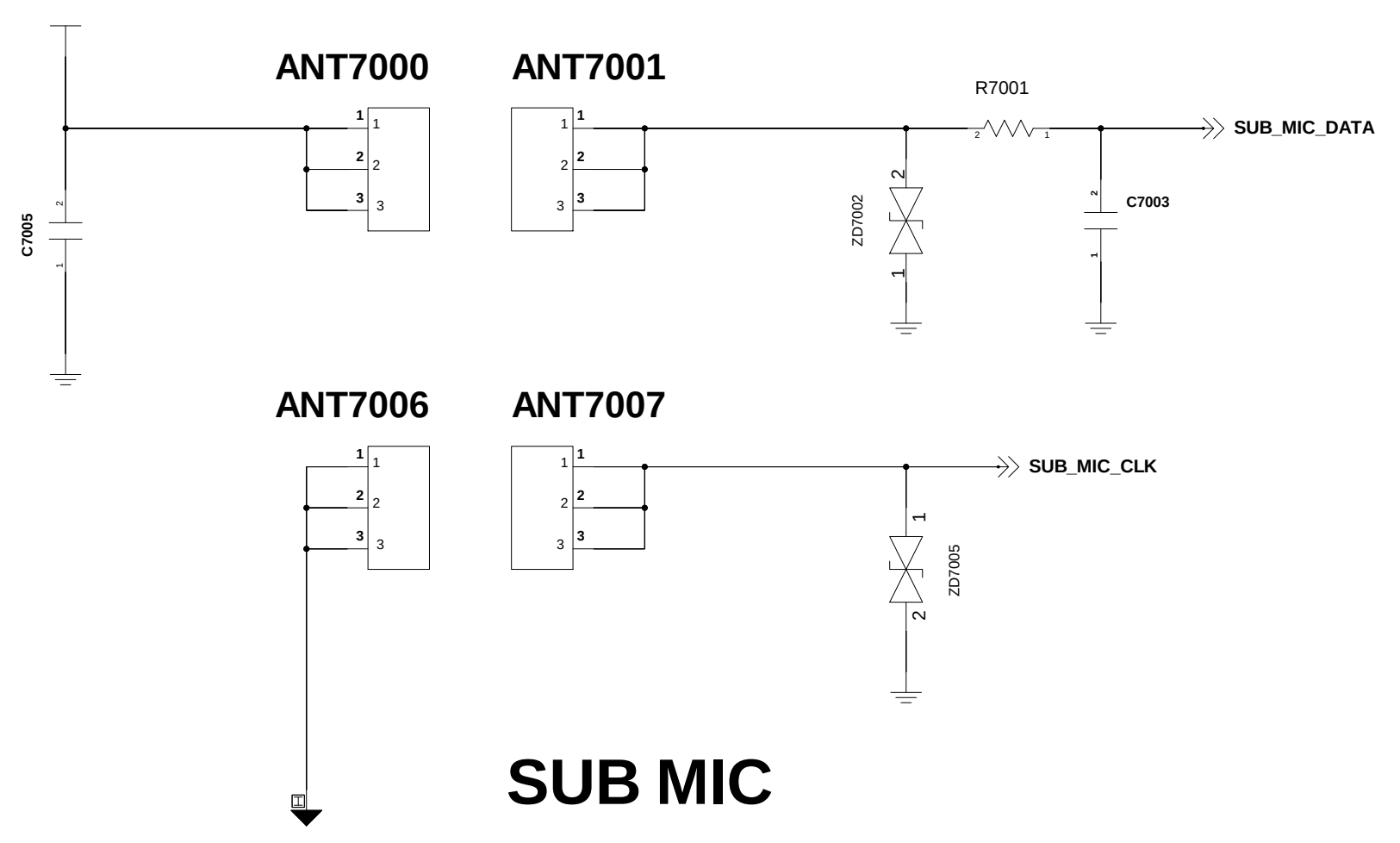
POGO MCU

TITLE TabS7P
SIZE A1
DWG NO
REV
DRAWN BY User Name
SHEET 6 of 15
Date

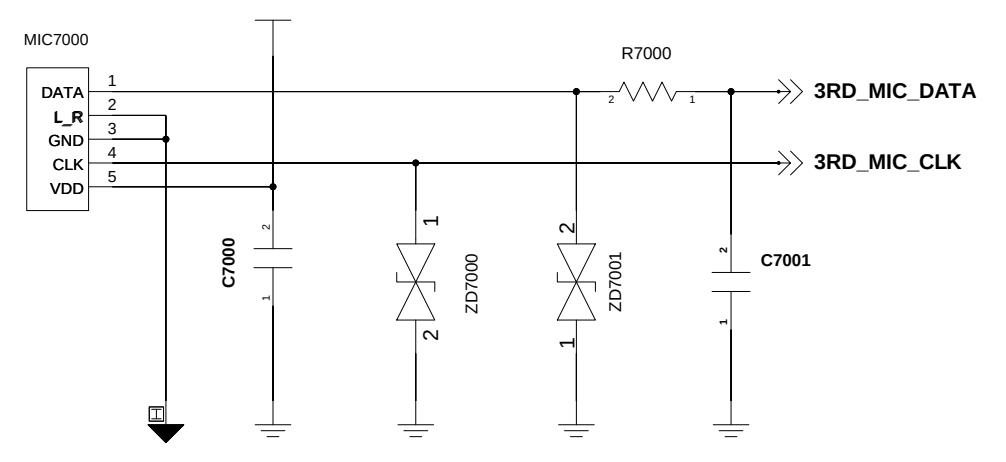
AUDIO


ANT7002
ANT7003
ANT7004
ANT7005
C7002
R7002
C7004
ZD7003
ZD7004
MAIN_MIC_DATA
MAIN_MIC_CLK

MAIN MIC

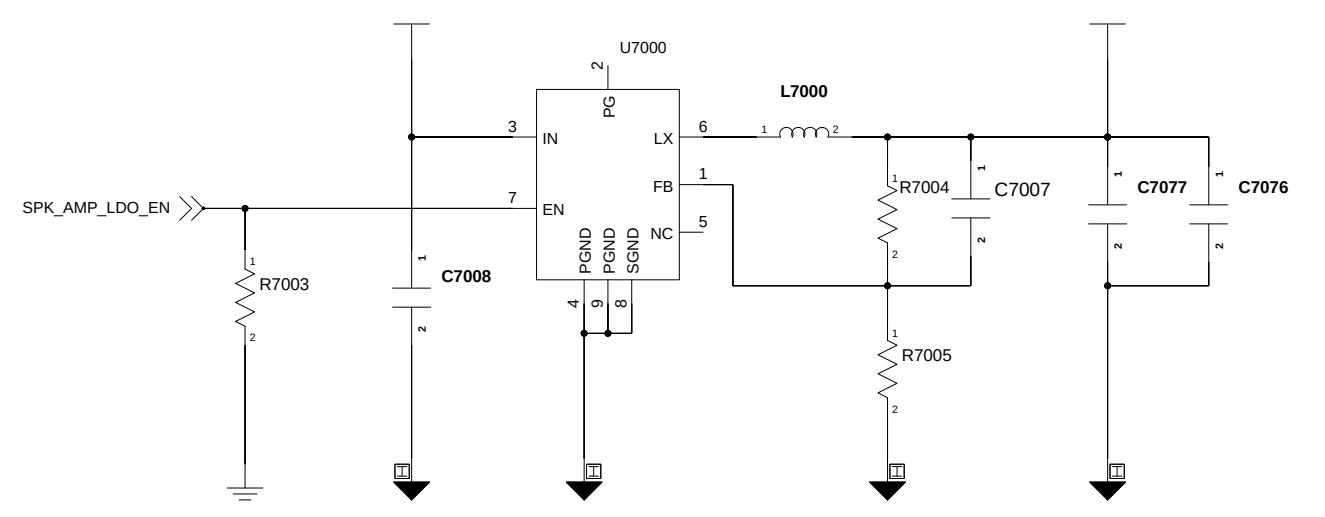

ANT7000
ANT7001
ANT7006
ANT7007
C7005
R7001
C7003
ZD7002
ZD7005
SUB_MIC_DATA
SUB_MIC_CLK

SUB MIC

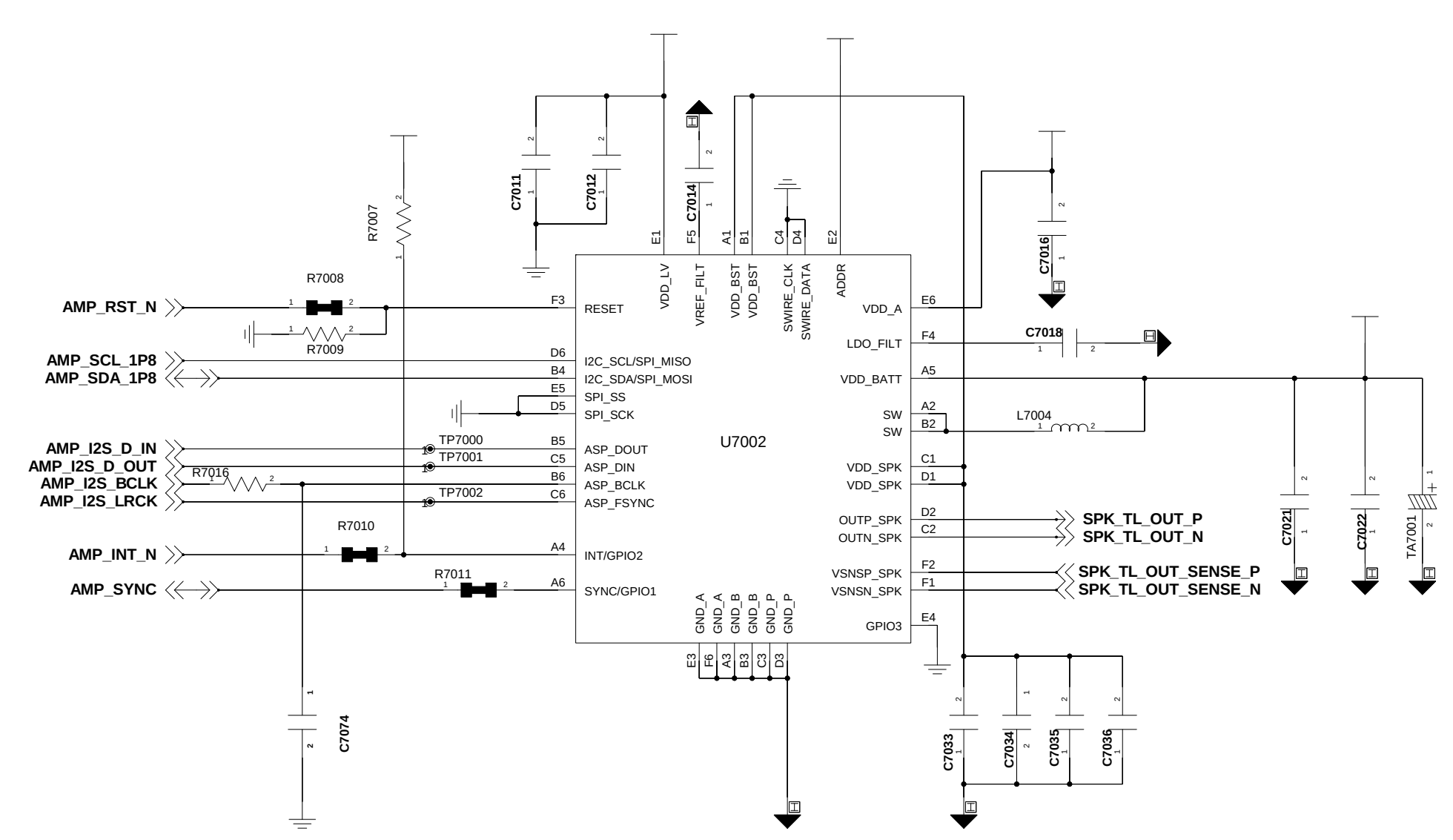

MIC7000
DATA
L_R
GND
CLK
VDD
R7000
C7000
ZD7000
ZD7001
C7001
3RD_MIC_DATA
3RD_MIC_CLK

3rd MIC

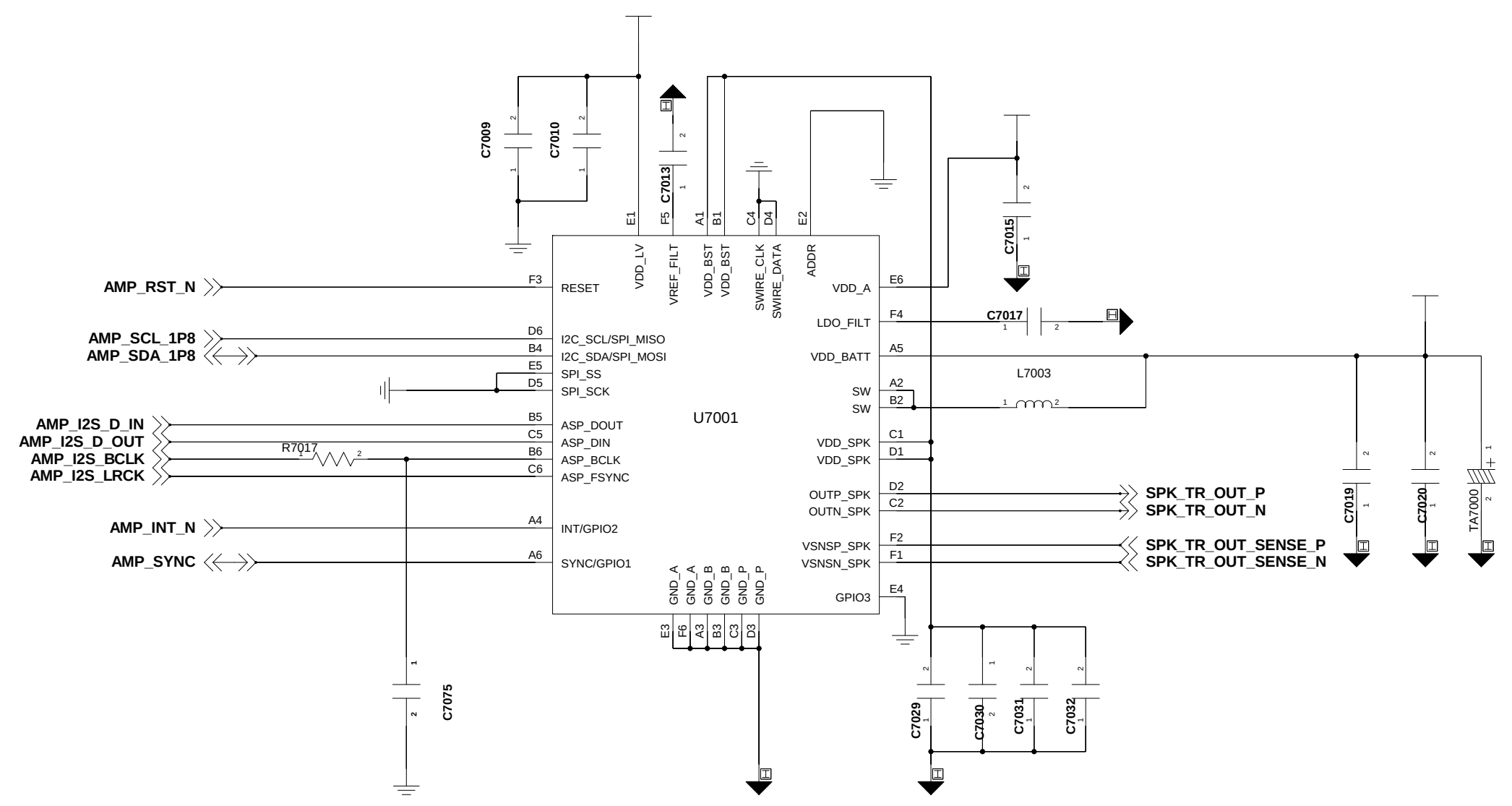
Place on array dummy
(SPK AMP SMD F/T)


U7000
IN
EN
PG
LX
FB
NC
PGND
PGND
SGND
L7000
R7004
C7007
C7077
C7076
R7005
R7003
C7008
SPK_AMP_LDO_EN

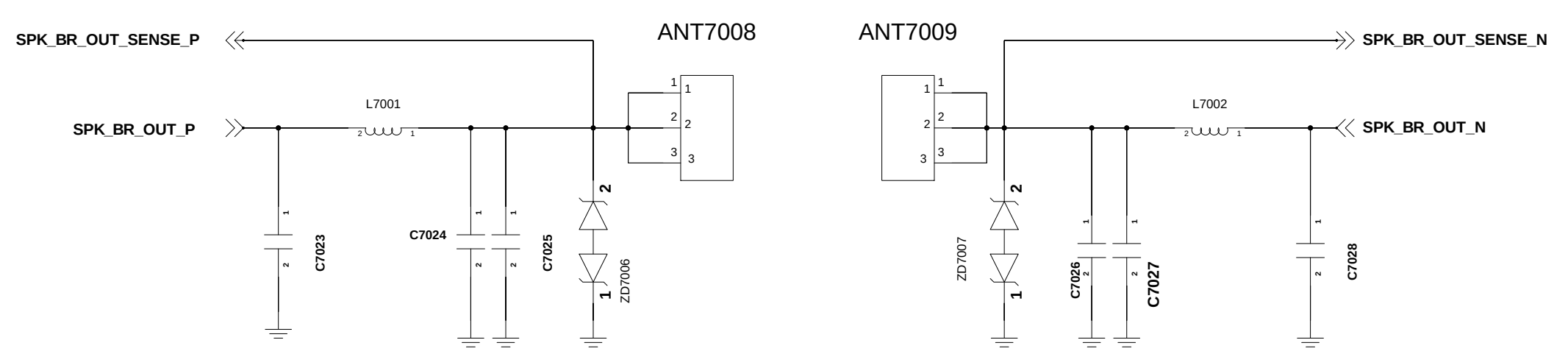
AUDIO DCDC


U7002
RESET
I2C_SCL/SPI_MISO
I2C_SDA/SPI_MOSI
SPI_SS
SPI_SCK
ASP_DOUT
ASP_DIN
ASP_BCLK
ASP_FSYNC
INT/GPIO2
SYNC/GPIO1
VDD_A
LDO_FILT
VDD_BATT
SW
VDD_SPK
OUTP_SPK
OUTN_SPK
VSNSP_SPK
VSNSN_SPK
GPIO3
AMP_RST_N
AMP_SCL_1P8
AMP_SDA_1P8
AMP_I2S_D_IN
AMP_I2S_D_OUT
AMP_I2S_BCLK
AMP_I2S_LRCK
AMP_INT_N
AMP_SYNC
R7007
R7008
R7009
R7010
R7011
R7016
TP7000
TP7001
TP7002
C7011
C7012
C7014
C7016
C7018
L7004
C7021
C7022
TA7001
C7033
C7034
C7035
C7036
C7074
SPK_TL_OUT_P
SPK_TL_OUT_N
SPK_TL_OUT_SENSE_P
SPK_TL_OUT_SENSE_N

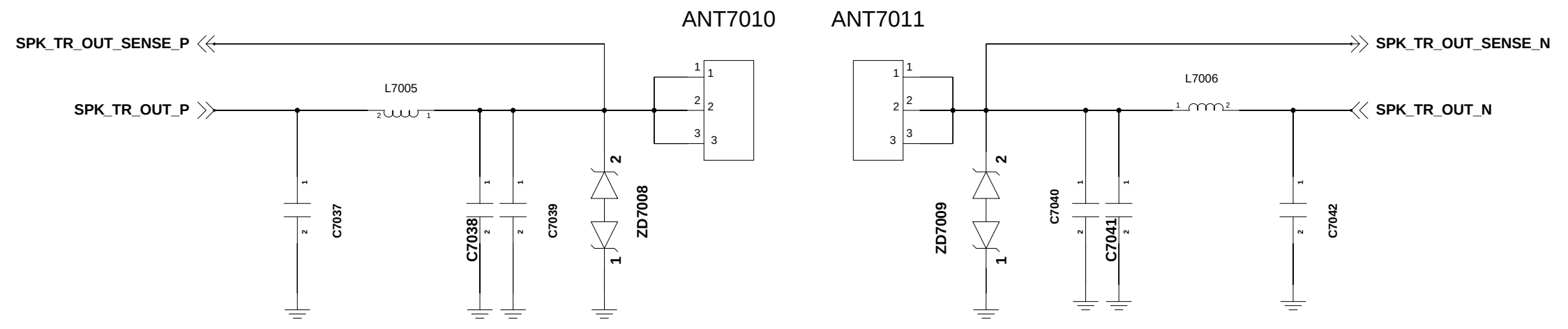
SPK TOP LEFT AMP


U7001
AMP_RST_N
AMP_SCL_1P8
AMP_SDA_1P8
AMP_I2S_D_IN
AMP_I2S_D_OUT
AMP_I2S_BCLK
AMP_I2S_LRCK
AMP_INT_N
AMP_SYNC
R7017
C7009
C7010
C7013
C7015
C7017
L7003
C7019
C7020
TA7000
C7029
C7030
C7031
C7032
C7075
SPK_TR_OUT_P
SPK_TR_OUT_N
SPK_TR_OUT_SENSE_P
SPK_TR_OUT_SENSE_N

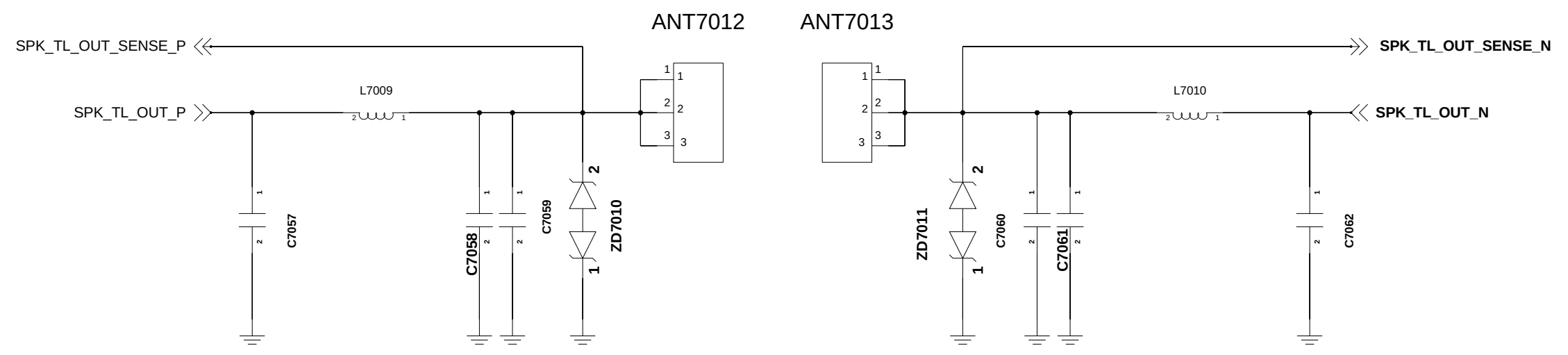
SPK TOP RIGHT AMP


ANT7008
ANT7009
SPK_BR_OUT_SENSE_P
SPK_BR_OUT_P
SPK_BR_OUT_SENSE_N
SPK_BR_OUT_N
L7001
L7002
C7023
C7024
C7025
ZD7006
ZD7007
C7026
C7027
C7028

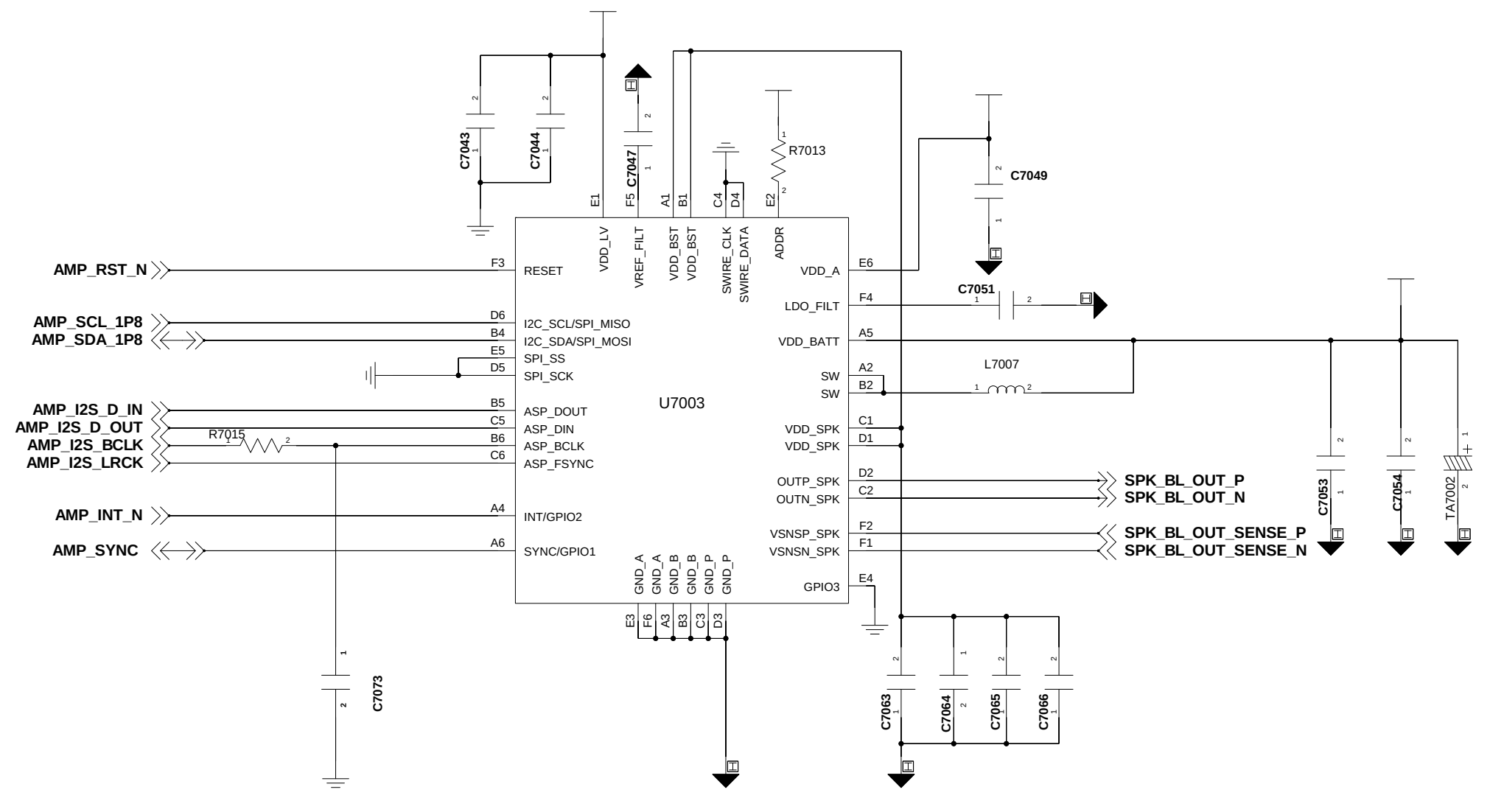
SPK BR CON


ANT7010
ANT7011
SPK_TR_OUT_SENSE_P
SPK_TR_OUT_P
SPK_TR_OUT_SENSE_N
SPK_TR_OUT_N
L7005
L7006
C7037
C7038
C7039
ZD7008
ZD7009
C7040
C7041
C7042

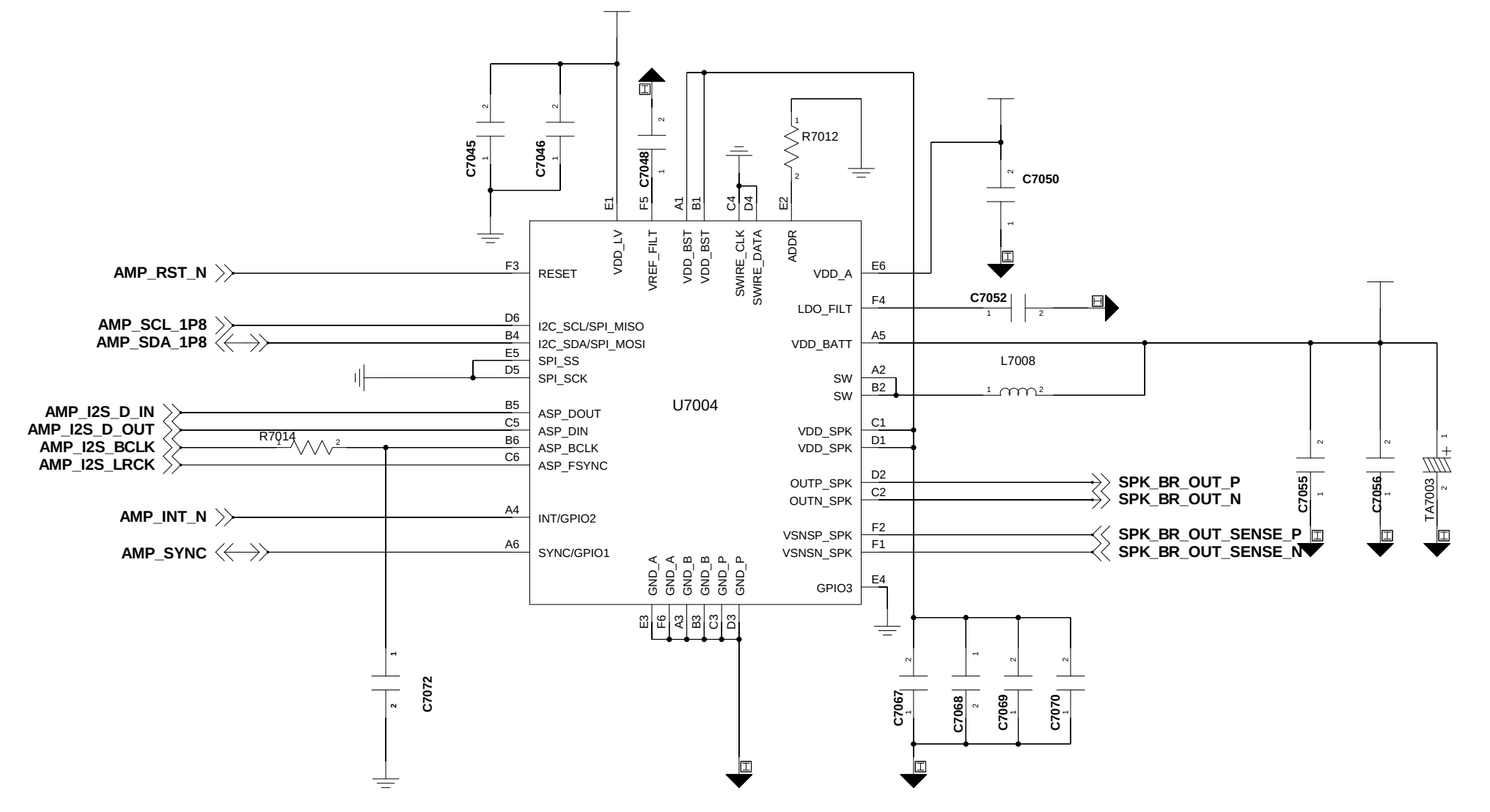
SPK TR CON


ANT7012
ANT7013
SPK_TL_OUT_SENSE_P
SPK_TL_OUT_P
SPK_TL_OUT_SENSE_N
SPK_TL_OUT_N
L7009
L7010
C7057
C7058
C7059
ZD7010
ZD7011
C7060
C7061
C7062

SPK TL CON


U7003
AMP_RST_N
AMP_SCL_1P8
AMP_SDA_1P8
AMP_I2S_D_IN
AMP_I2S_D_OUT
AMP_I2S_BCLK
AMP_I2S_LRCK
AMP_INT_N
AMP_SYNC
R7015
R7013
C7043
C7044
C7047
C7049
C7051
L7007
C7053
C7054
TA7002
C7063
C7064
C7065
C7066
C7073
SPK_BL_OUT_P
SPK_BL_OUT_N
SPK_BL_OUT_SENSE_P
SPK_BL_OUT_SENSE_N

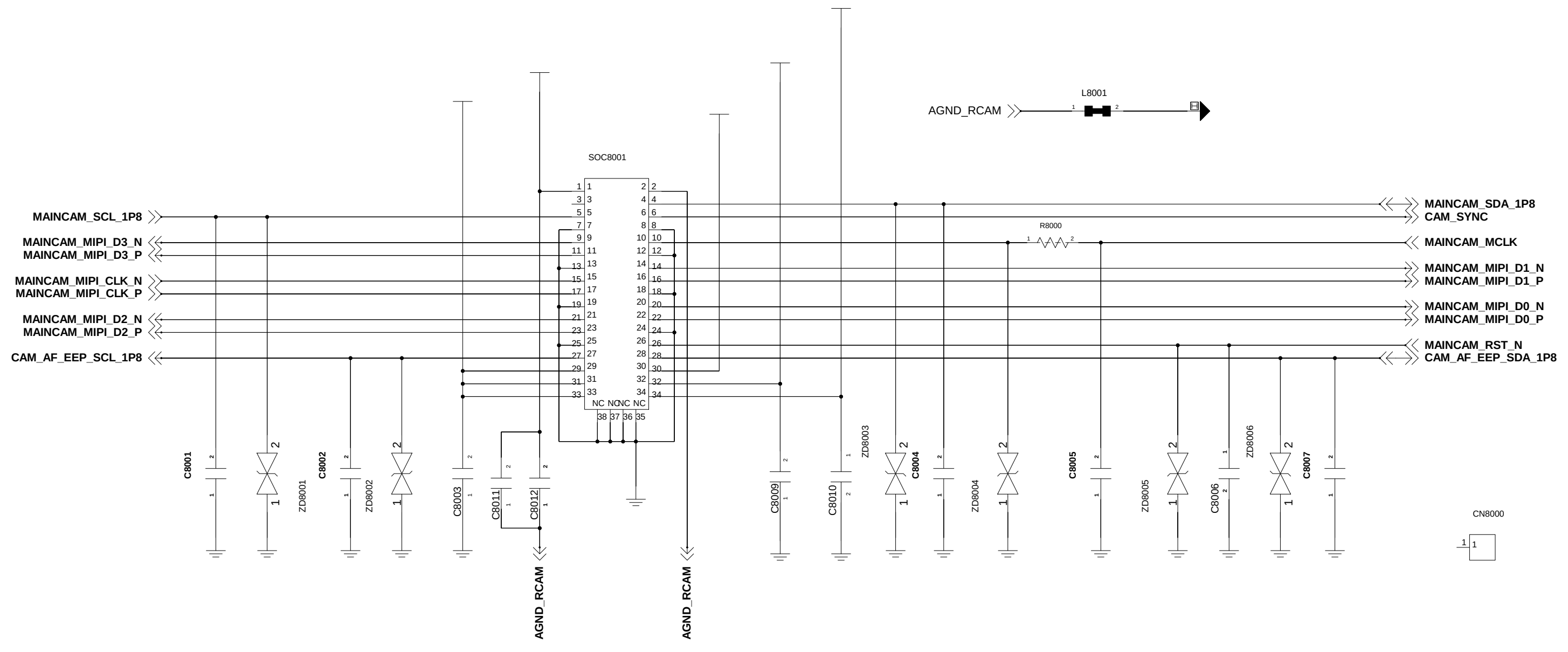
SPK BOTTOM LEFT AMP


U7004
AMP_RST_N
AMP_SCL_1P8
AMP_SDA_1P8
AMP_I2S_D_IN
AMP_I2S_D_OUT
AMP_I2S_BCLK
AMP_I2S_LRCK
AMP_INT_N
AMP_SYNC
R7014
R7012
C7045
C7046
C7048
C7050
C7052
L7008
C7055
C7056
TA7003
C7067
C7068
C7069
C7070
C7072
SPK_BR_OUT_P
SPK_BR_OUT_N
SPK_BR_OUT_SENSE_P
SPK_BR_OUT_SENSE_N

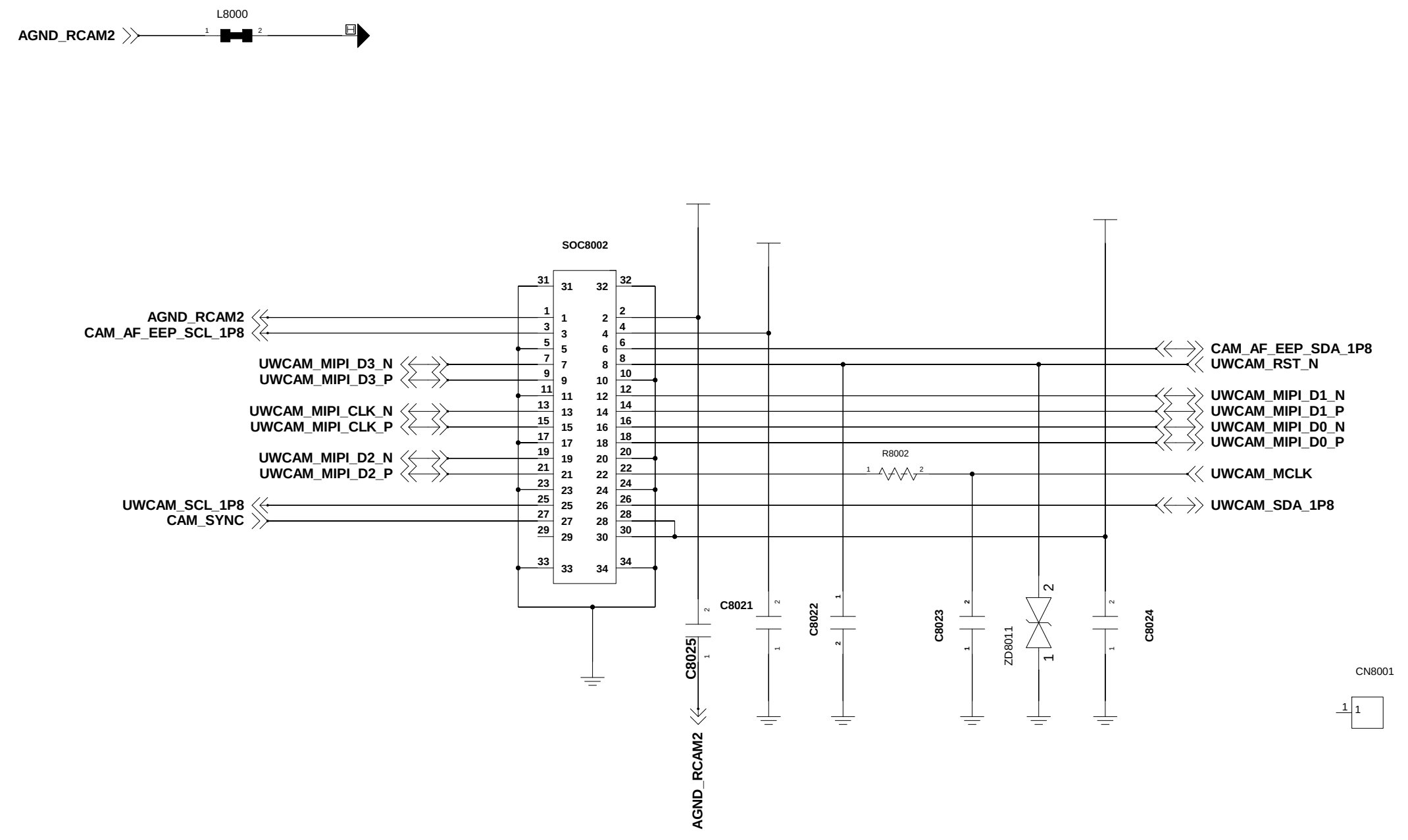
SPK BOTTOM RIGHT AMP

TITLE TabS7P
SIZE A1
DWG NO
REV
DRAWN BY User Name
SHEET 7 of 15
Date

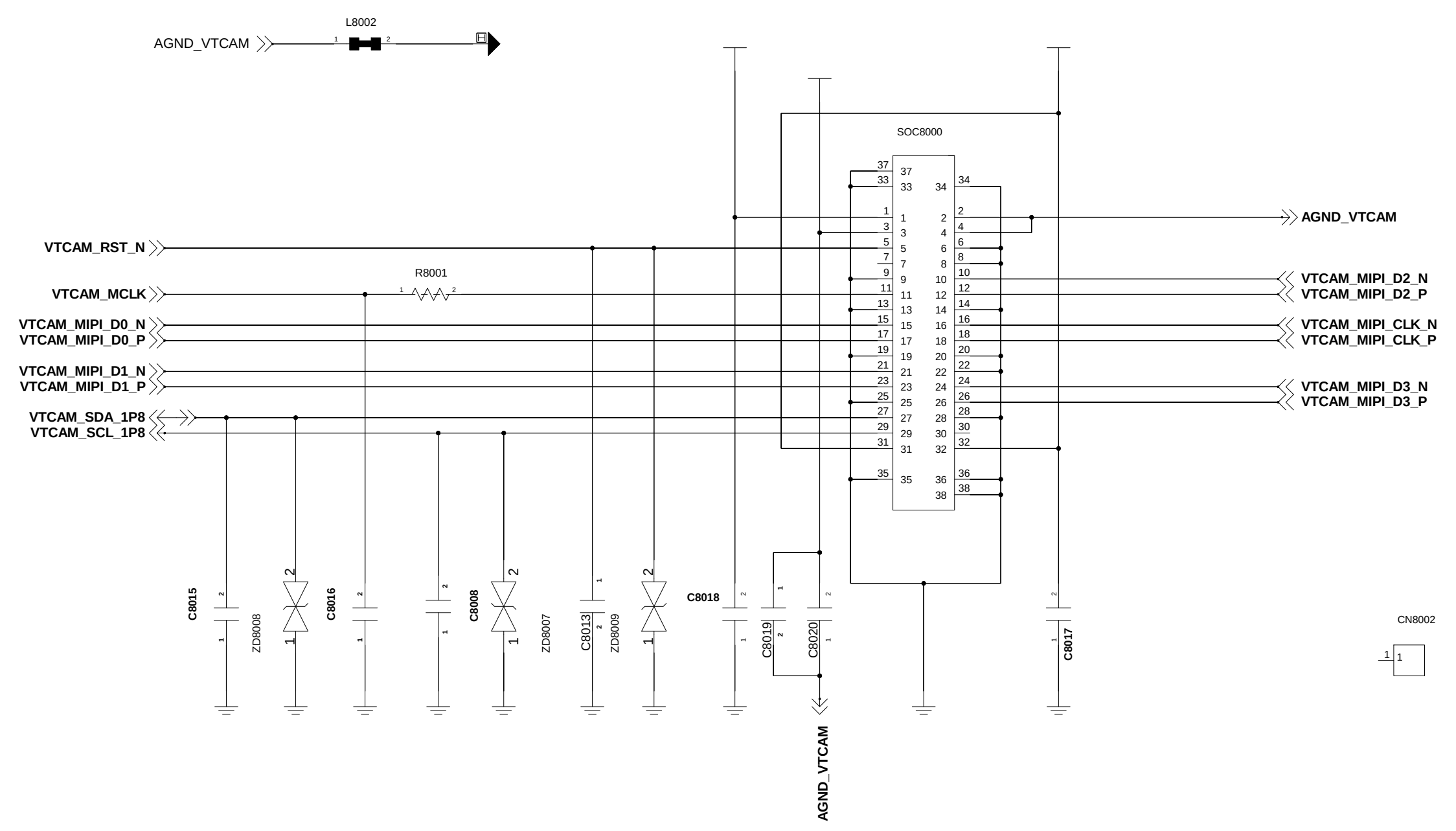
CAMERA


AGND_RCAM
L8001
SOC8001
MAINCAM_SCL_1P8
MAINCAM_MIPI_D3_N
MAINCAM_MIPI_D3_P
MAINCAM_MIPI_CLK_N
MAINCAM_MIPI_CLK_P
MAINCAM_MIPI_D2_N
MAINCAM_MIPI_D2_P
CAM_AF_EEP_SCL_1P8
MAINCAM_SDA_1P8
CAM_SYNC
R8000
MAINCAM_MCLK
MAINCAM_MIPI_D1_N
MAINCAM_MIPI_D1_P
MAINCAM_MIPI_D0_N
MAINCAM_MIPI_D0_P
MAINCAM_RST_N
CAM_AF_EEP_SDA_1P8
C8001
ZD8001
C8002
ZD8002
C8003
C8011
C8012
AGND_RCAM
AGND_RCAM
C8009
C8010
ZD8003
C8004
ZD8004
C8005
ZD8005
C8006
ZD8006
C8007
CN8000

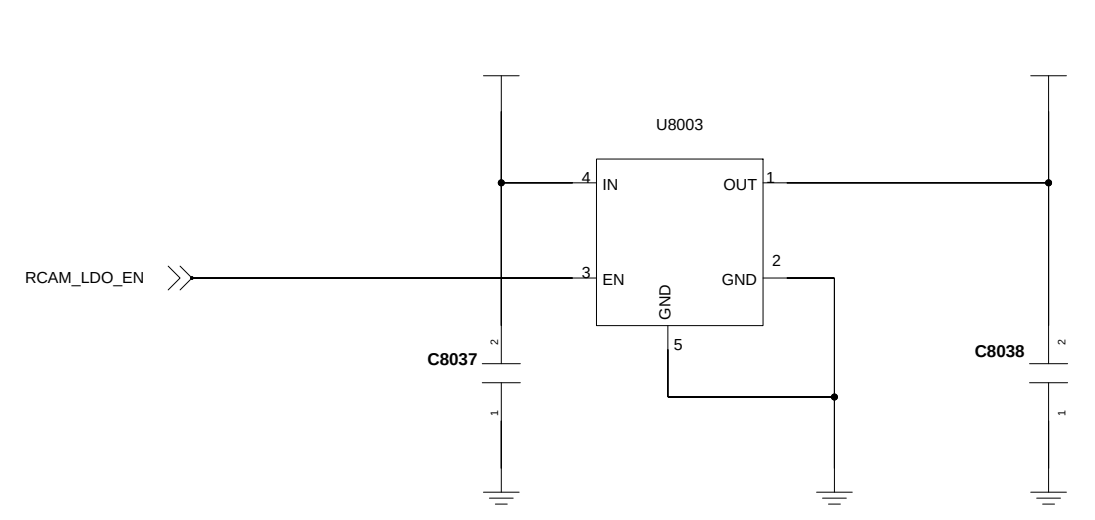
13M_CAM (RCAM)

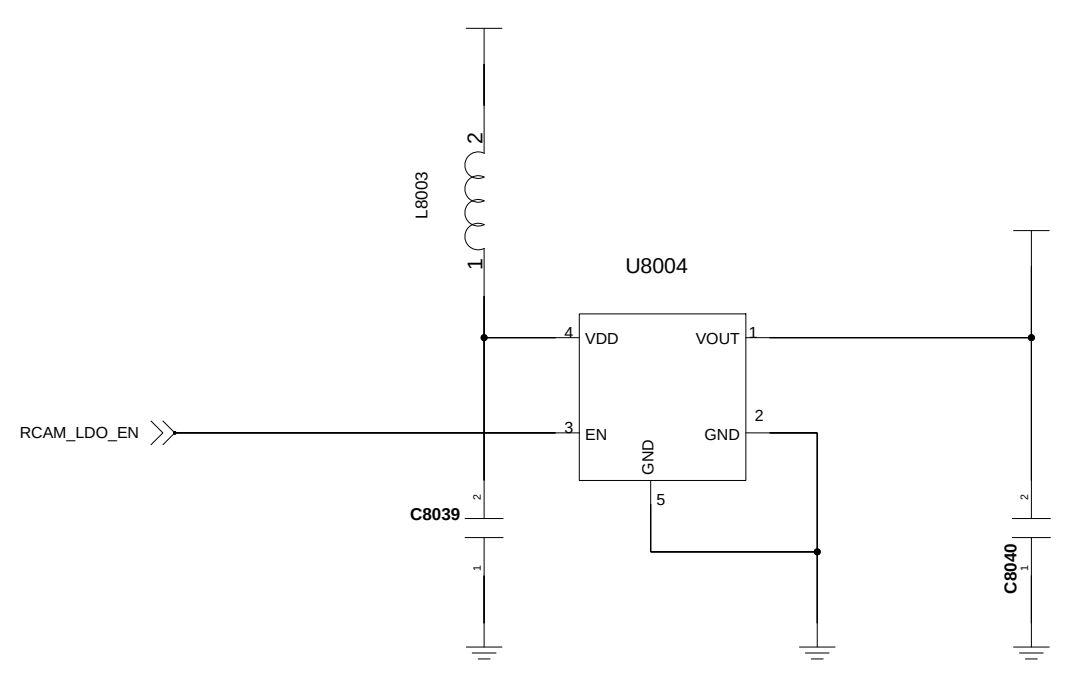

AGND_RCAM2
L8000
SOC8002
AGND_RCAM2
CAM_AF_EEP_SCL_1P8
UWCAM_MIPI_D3_N
UWCAM_MIPI_D3_P
UWCAM_MIPI_CLK_N
UWCAM_MIPI_CLK_P
UWCAM_MIPI_D2_N
UWCAM_MIPI_D2_P
UWCAM_SCL_1P8
CAM_SYNC
CAM_AF_EEP_SDA_1P8
UWCAM_RST_N
UWCAM_MIPI_D1_N
UWCAM_MIPI_D1_P
UWCAM_MIPI_D0_N
UWCAM_MIPI_D0_P
R8002
UWCAM_MCLK
UWCAM_SDA_1P8
C8025
C8021
C8022
C8023
ZD8011
C8024
AGND_RCAM2
CN8001

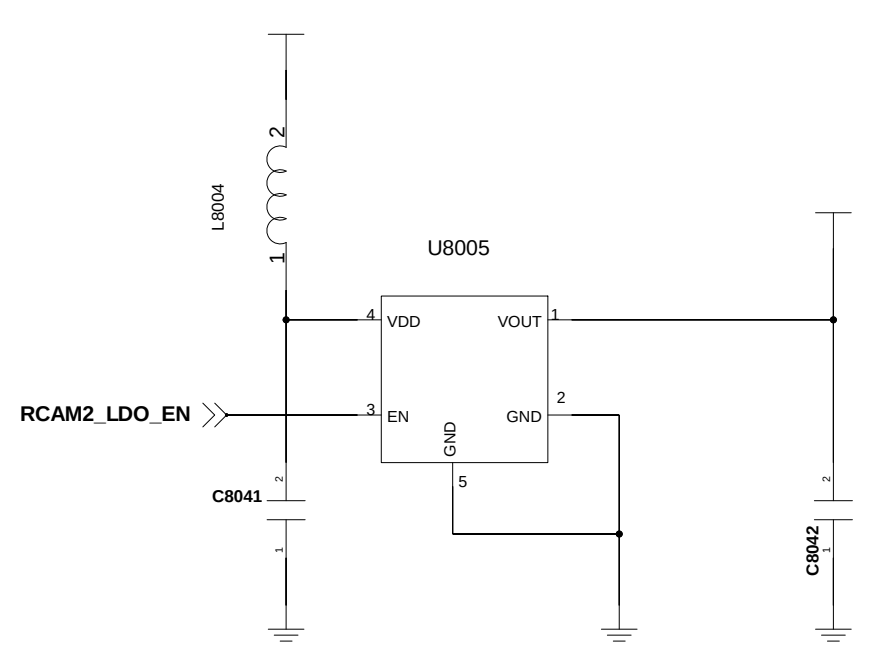
8M_UW CAM (RCAM2)

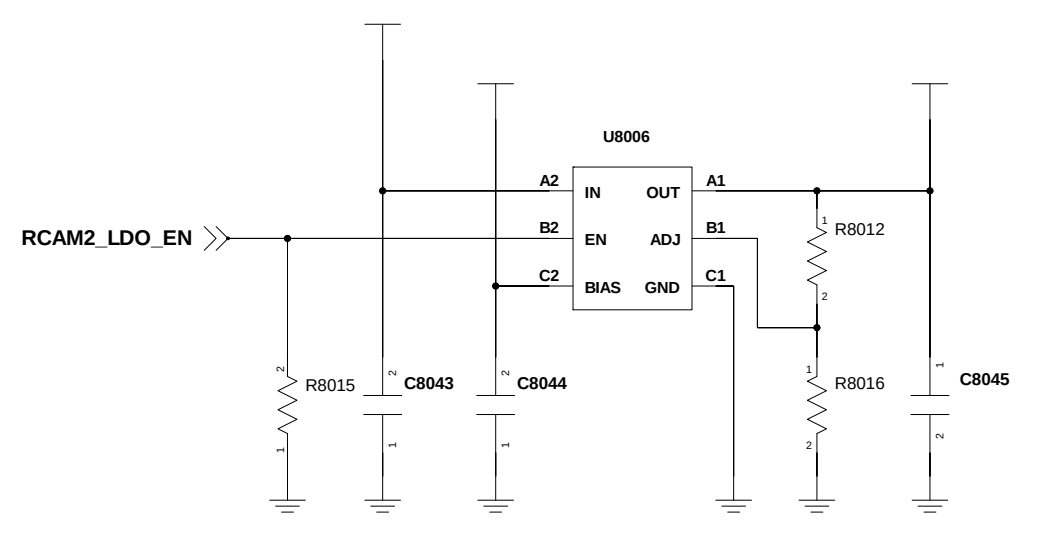

AGND_VTCAM
L8002
SOC8000
VTCAM_RST_N
R8001
VTCAM_MCLK
VTCAM_MIPI_D0_N
VTCAM_MIPI_D0_P
VTCAM_MIPI_D1_N
VTCAM_MIPI_D1_P
VTCAM_SDA_1P8
VTCAM_SCL_1P8
AGND_VTCAM
VTCAM_MIPI_D2_N
VTCAM_MIPI_D2_P
VTCAM_MIPI_CLK_N
VTCAM_MIPI_CLK_P
VTCAM_MIPI_D3_N
VTCAM_MIPI_D3_P
C8015
ZD8008
C8016
C8008
ZD8007
C8013
ZD8009
C8018
C8019
C8020
AGND_VTCAM
C8017
CN8002

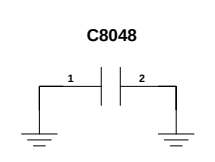
12M CAM (FCAM)


U8003
IN
OUT
EN
GND
GND
RCAM_LDO_EN
C8037
C8038

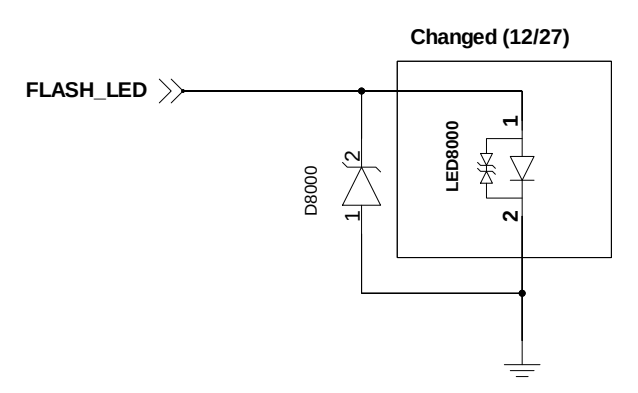

L8003
U8004
VDD
VOUT
EN
GND
GND
RCAM_LDO_EN
C8039
C8040


L8004
U8005
VDD
VOUT
EN
GND
GND
RCAM2_LDO_EN
C8041
C8042

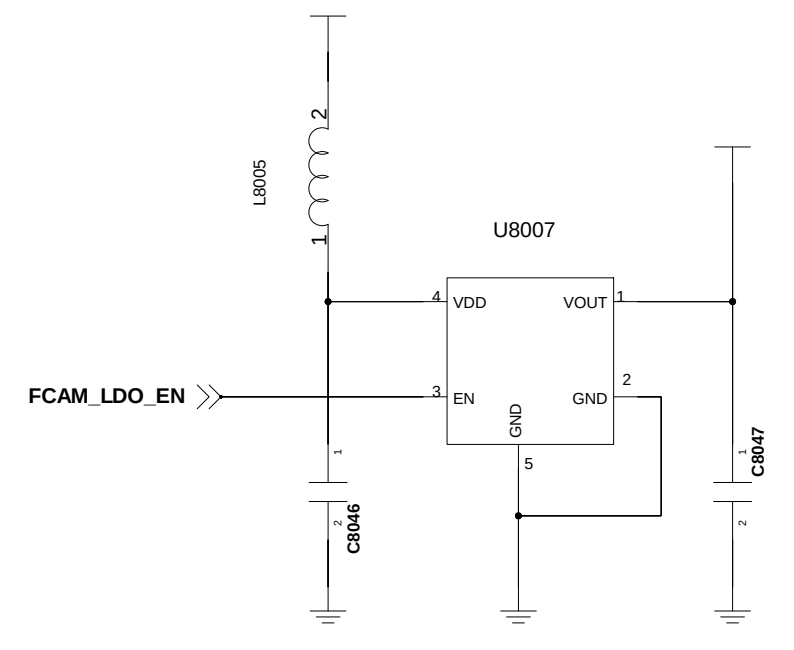

U8006
IN
OUT
EN
ADJ
BIAS
GND
RCAM2_LDO_EN
R8012
R8015
C8043
C8044
R8016
C8045



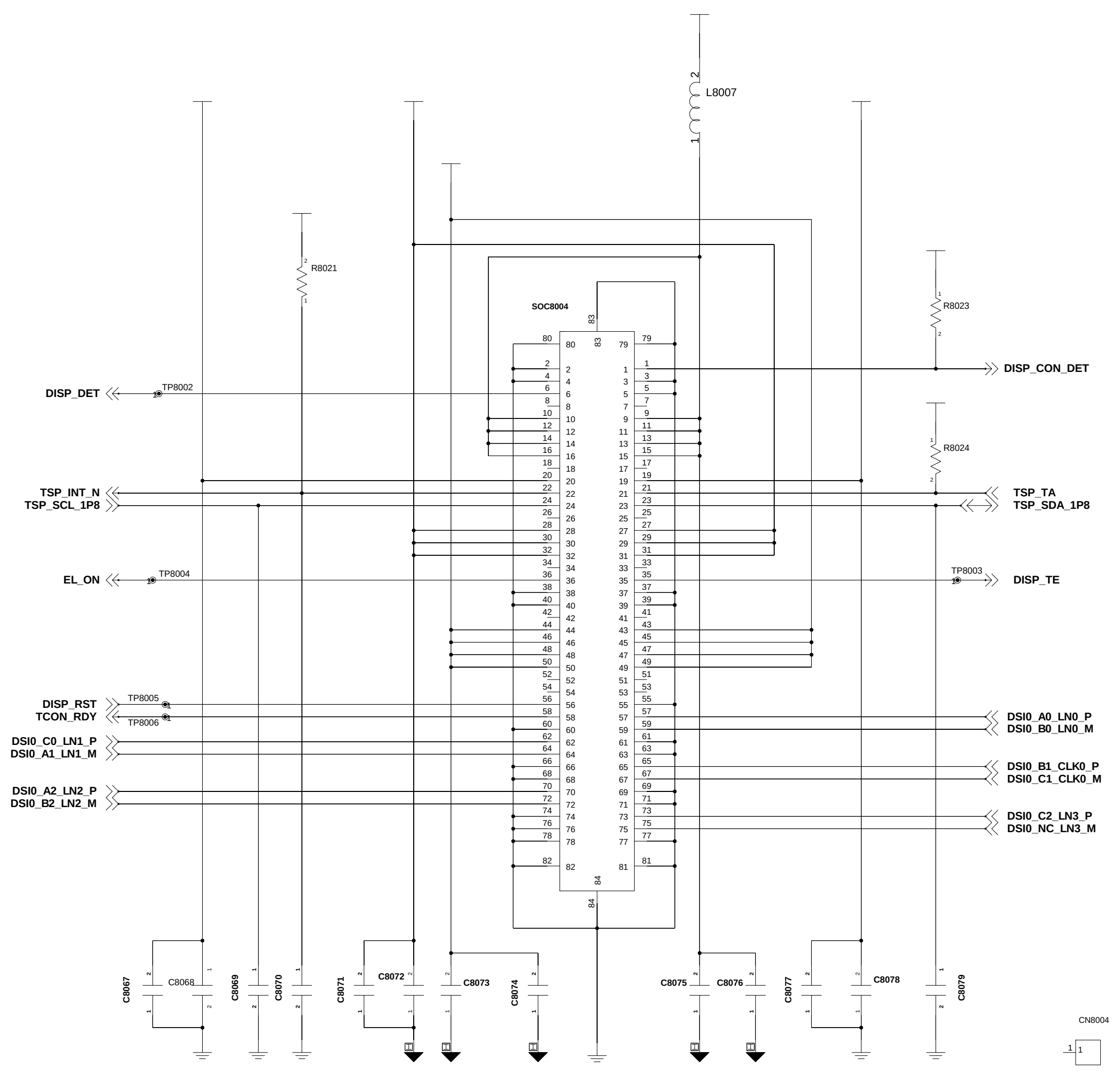
Dummy


Changed (12/27)
FLASH_LED
D8000
LED8000

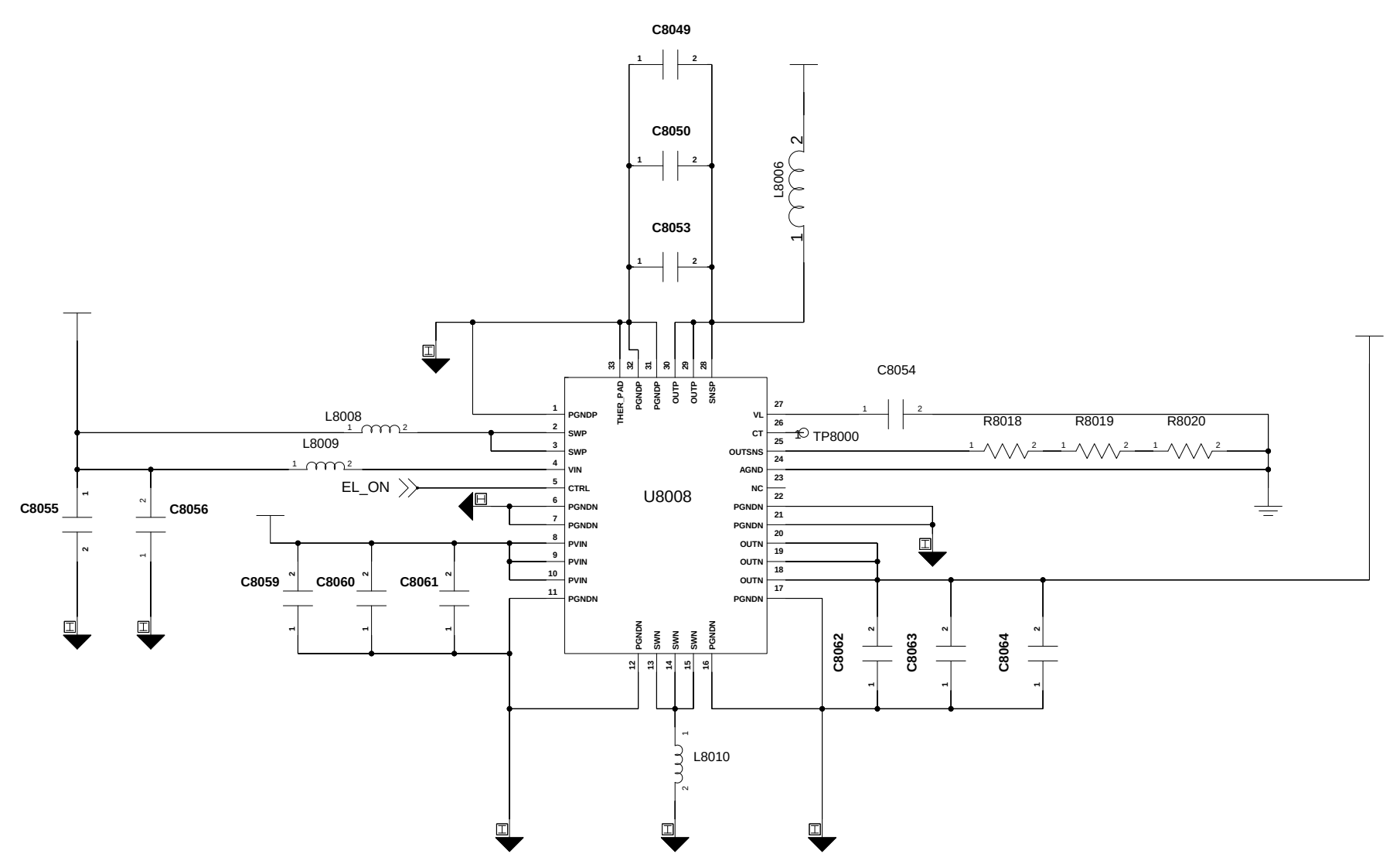
FLASH LED


L8005
U8007
VDD
VOUT
EN
GND
GND
FCAM_LDO_EN
C8046
C8047

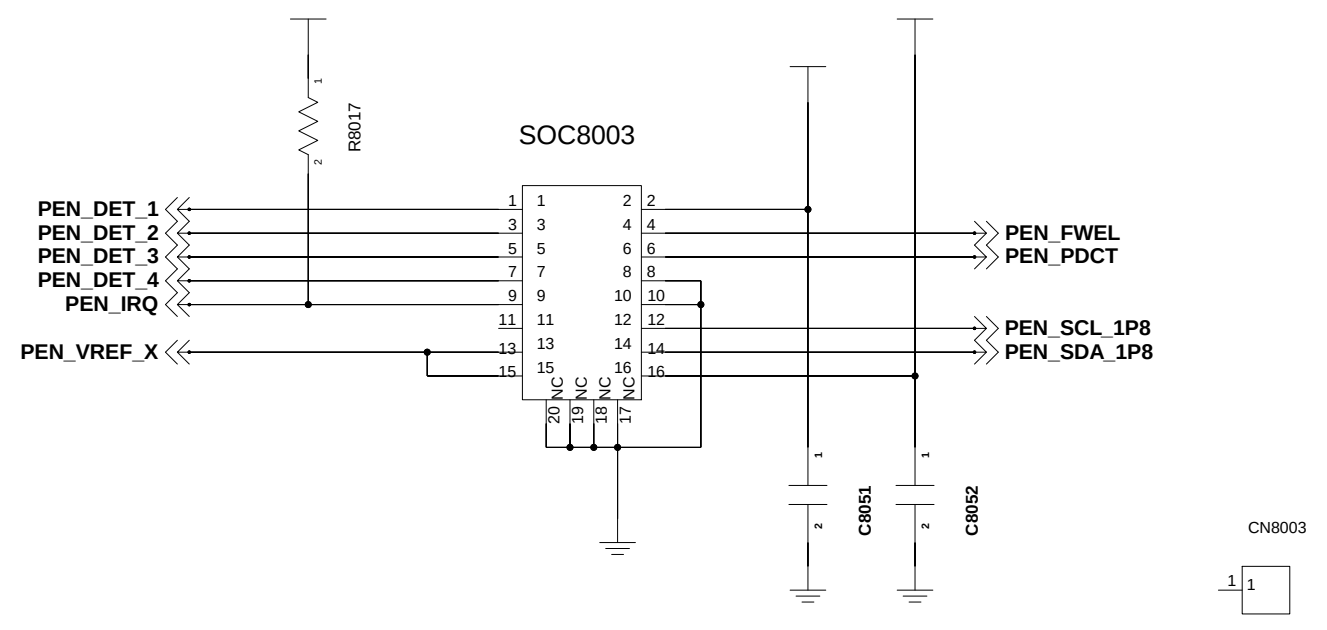
DISPLAY


L8007
R8021
SOC8004
DISP_DET
TP8002
TSP_INT_N
TSP_SCL_1P8
EL_ON
TP8004
DISP_RST
TCON_RDY
TP8005
TP8006
DSI0_C0_LN1_P
DSI0_A1_LN1_M
DSI0_A2_LN2_P
DSI0_B2_LN2_M
R8023
DISP_CON_DET
R8024
TSP_TA
TSP_SDA_1P8
TP8003
DISP_TE
DSI0_A0_LN0_P
DSI0_B0_LN0_M
DSI0_B1_CLK0_P
DSI0_C1_CLK0_M
DSI0_C2_LN3_P
DSI0_NC_LN3_M
C8067
C8068
C8069
C8070
C8071
C8072
C8073
C8074
C8075
C8076
C8077
C8078
C8079
CN8004

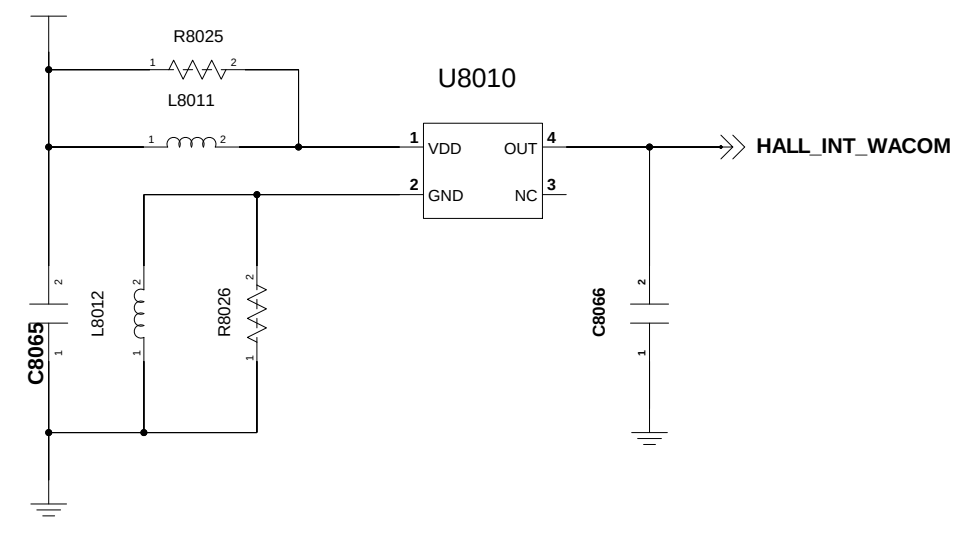
OCTA Con.


C8049
C8050
C8053
L8006
C8054
L8008
L8009
EL_ON
U8008
TP8000
R8018
R8019
R8020
C8055
C8056
C8059
C8060
C8061
C8062
C8063
C8064
L8010

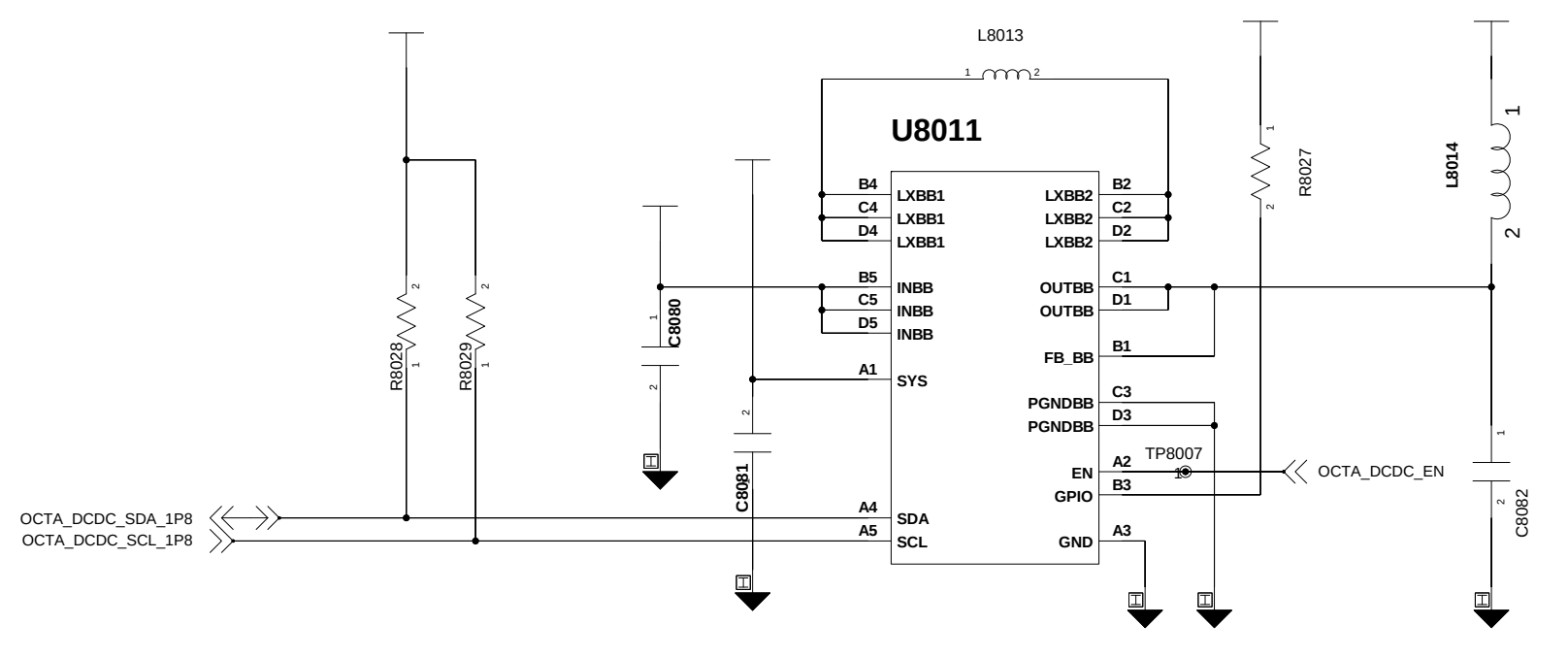
OCTA POWER


R8017
SOC8003
PEN_DET_1
PEN_DET_2
PEN_DET_3
PEN_DET_4
PEN_IRQ
PEN_VREF_X
PEN_FWEL
PEN_PDCT
PEN_SCL_1P8
PEN_SDA_1P8
C8051
C8052
CN8003

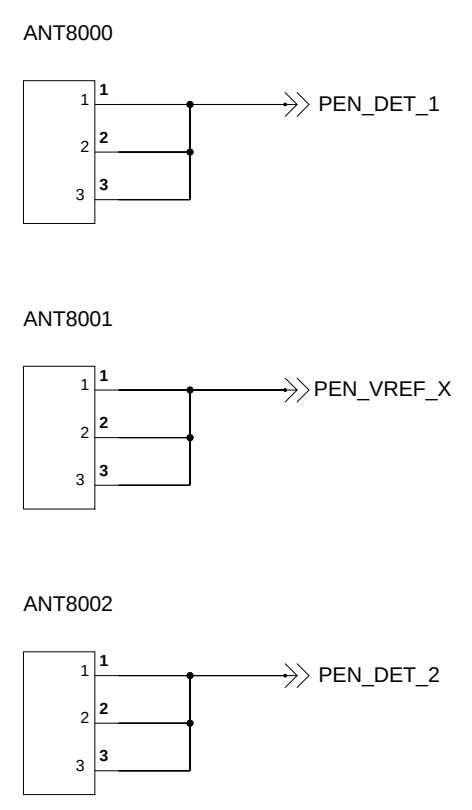
WACOM CONN
(MAIN <-> OCTA / WACOM)


R8025
L8011
U8010
VDD
OUT
GND
NC
HALL_INT_WACOM
C8065
L8012
R8026
C8066

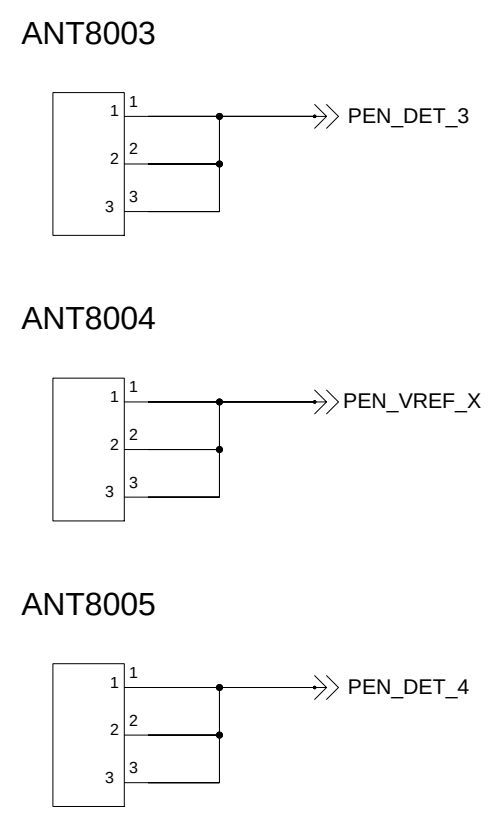
WACOM HALL_IC


L8013
U8011
LXBB1
LXBB2
INBB
OUTBB
FB_BB
SYS
PGNDBB
EN
GPIO
SDA
SCL
GND
R8027
L8014
R8028
R8029
C8080
C8081
C8082
TP8007
OCTA_DCDC_EN
OCTA_DCDC_SDA_1P8
OCTA_DCDC_SCL_1P8

TCON BUCK BOOSTER


ANT8000
PEN_DET_1
ANT8001
PEN_VREF_X
ANT8002
PEN_DET_2

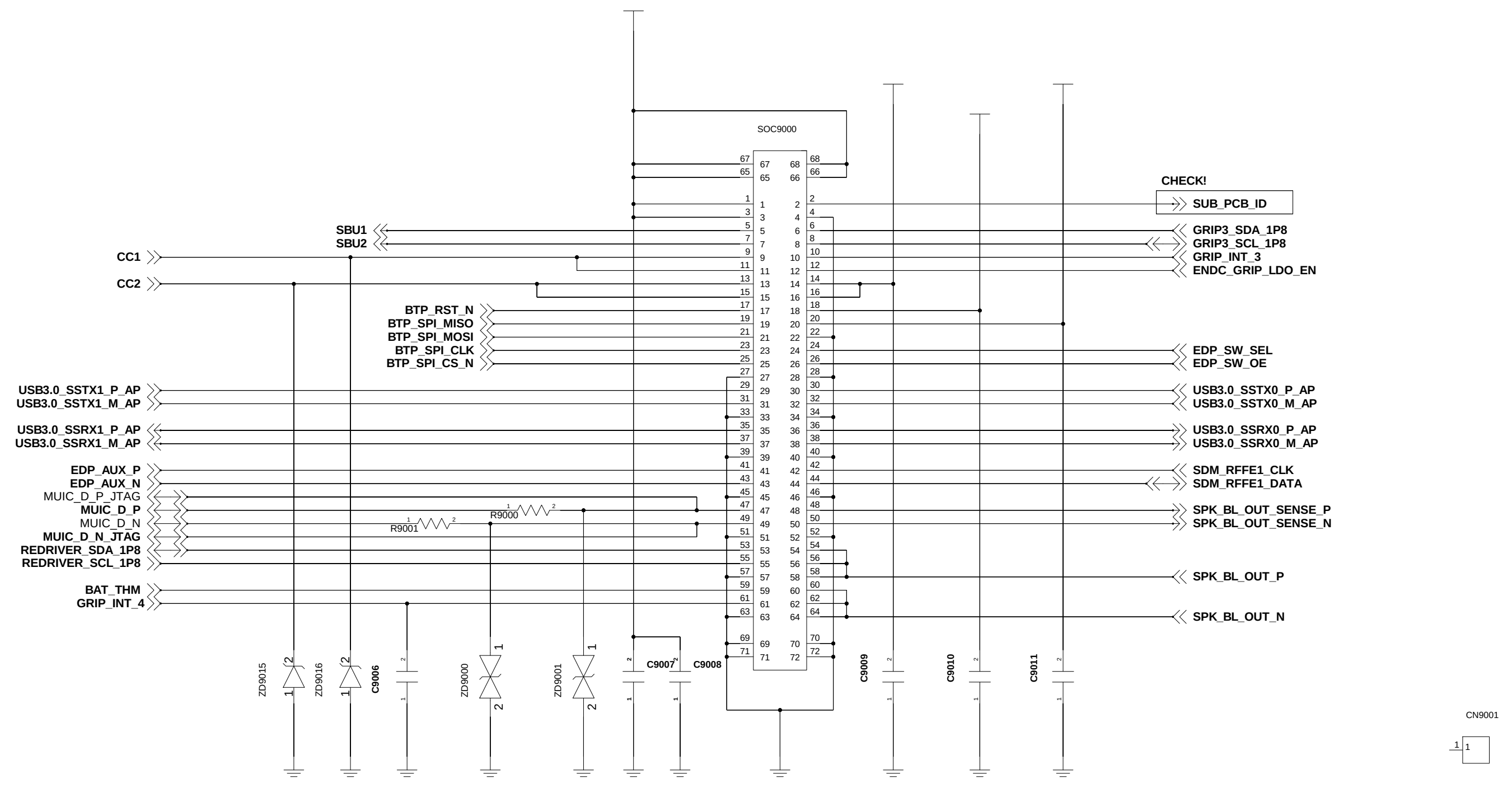
PEN CHG C/T


ANT8003
PEN_DET_3
ANT8004
PEN_VREF_X
ANT8005
PEN_DET_4

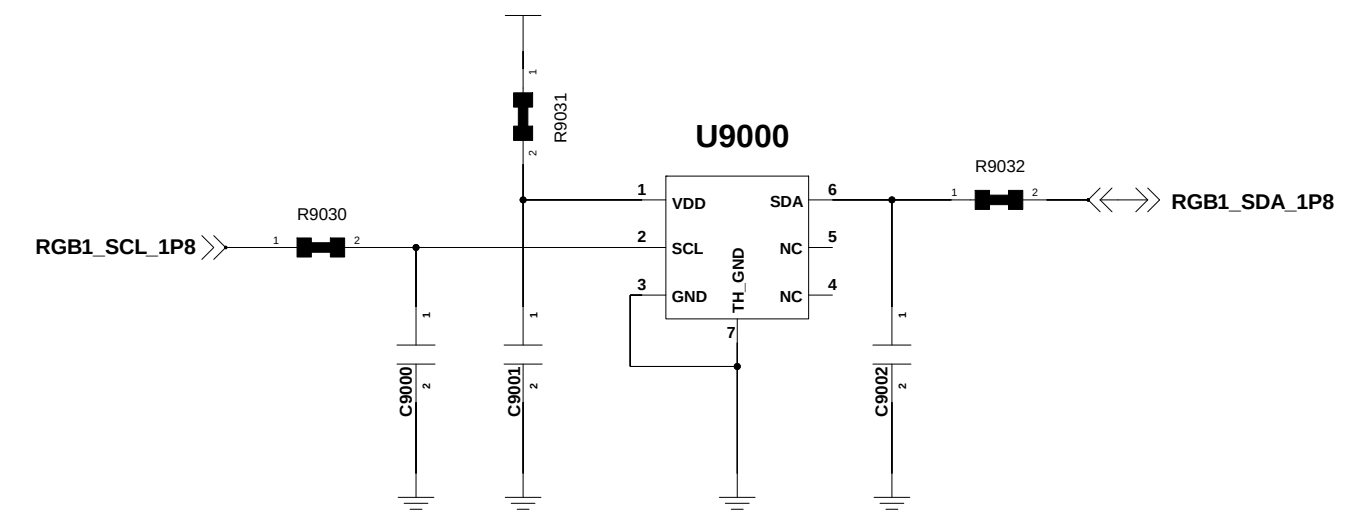
PEN CHG C/T_2

TITLE TabS7P
SIZE A1
DRAWN BY User Name
SHEET 8 of 15

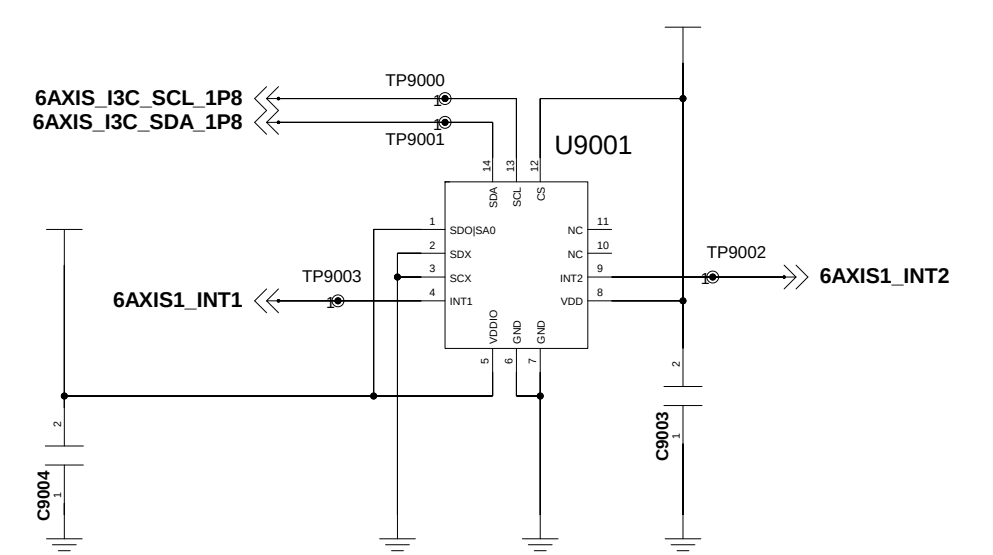
SENSOR / SD/SIM / KEY / I/F


SOC9000
CHECK!
SUB_PCB_ID
SBU1
SBU2
CC1
CC2
GRIP3_SDA_1P8
GRIP3_SCL_1P8
GRIP_INT_3
ENDC_GRIP_LDO_EN
BTP_RST_N
BTP_SPI_MISO
BTP_SPI_MOSI
BTP_SPI_CLK
BTP_SPI_CS_N
EDP_SW_SEL
EDP_SW_OE
USB3.0_SSTX1_P_AP
USB3.0_SSTX1_M_AP
USB3.0_SSRX1_P_AP
USB3.0_SSRX1_M_AP
USB3.0_SSTX0_P_AP
USB3.0_SSTX0_M_AP
USB3.0_SSRX0_P_AP
USB3.0_SSRX0_M_AP
EDP_AUX_P
EDP_AUX_N
MUIC_D_P_JTAG
MUIC_D_P
MUIC_D_N
MUIC_D_N_JTAG
REDRIVER_SDA_1P8
REDRIVER_SCL_1P8
BAT_THM
GRIP_INT_4
SDM_RFFE1_CLK
SDM_RFFE1_DATA
SPK_BL_OUT_SENSE_P
SPK_BL_OUT_SENSE_N
SPK_BL_OUT_P
SPK_BL_OUT_N
R9000
R9001
ZD9015
ZD9016
C9006
ZD9000
ZD9001
C9007
C9008
C9009
C9010
C9011
CN9001

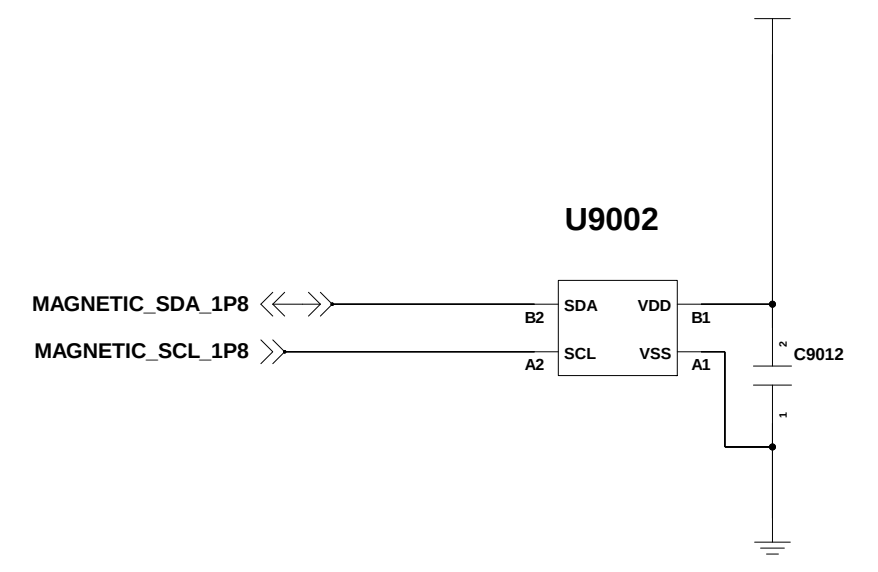
I/F CONN
(MAIN <-> USB)


U9000
R9031
R9030
R9032
RGB1_SCL_1P8
RGB1_SDA_1P8
VDD
SCL
GND
TH_GND
SDA
NC
NC
C9000
C9001
C9002

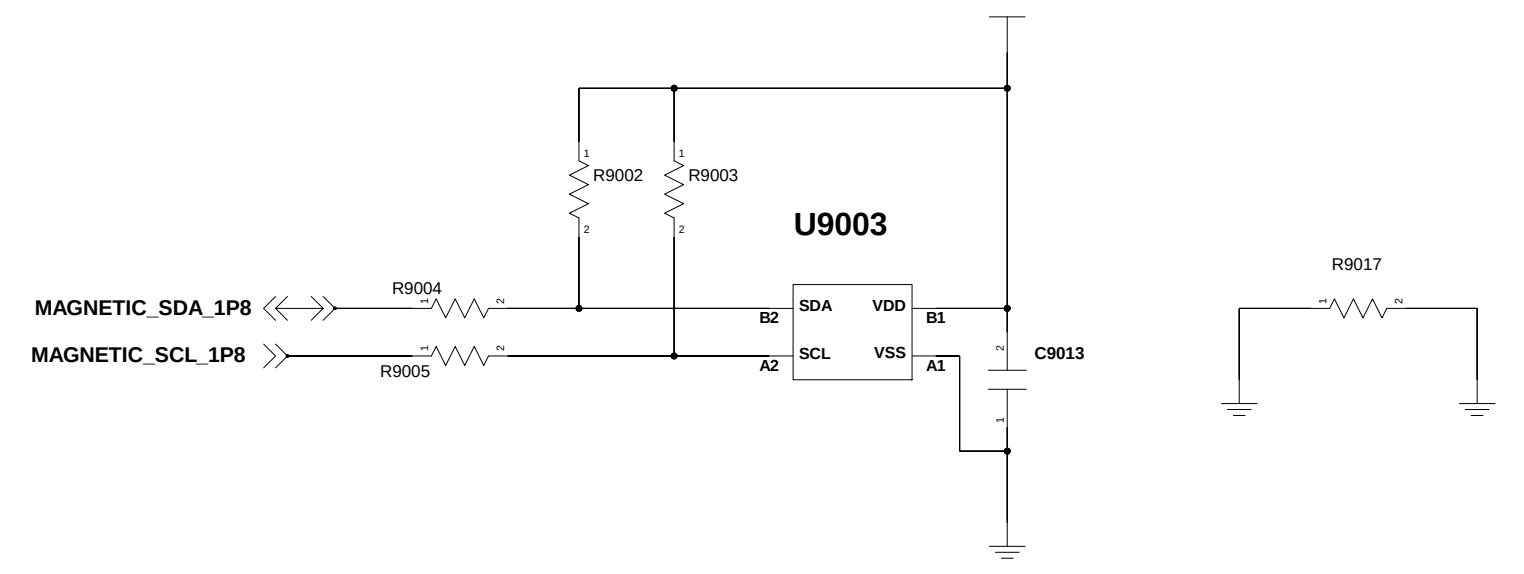
RGB Sensor 1


6AXIS_I3C_SCL_1P8
6AXIS_I3C_SDA_1P8
TP9000
TP9001
U9001
6AXIS1_INT1
TP9003
TP9002
6AXIS1_INT2
SDO/SA0
SDX
SCX
INT1
NC
NC
INT2
VDD
VDDIO
GND
GND
SDA
SCL
CS
C9003
C9004

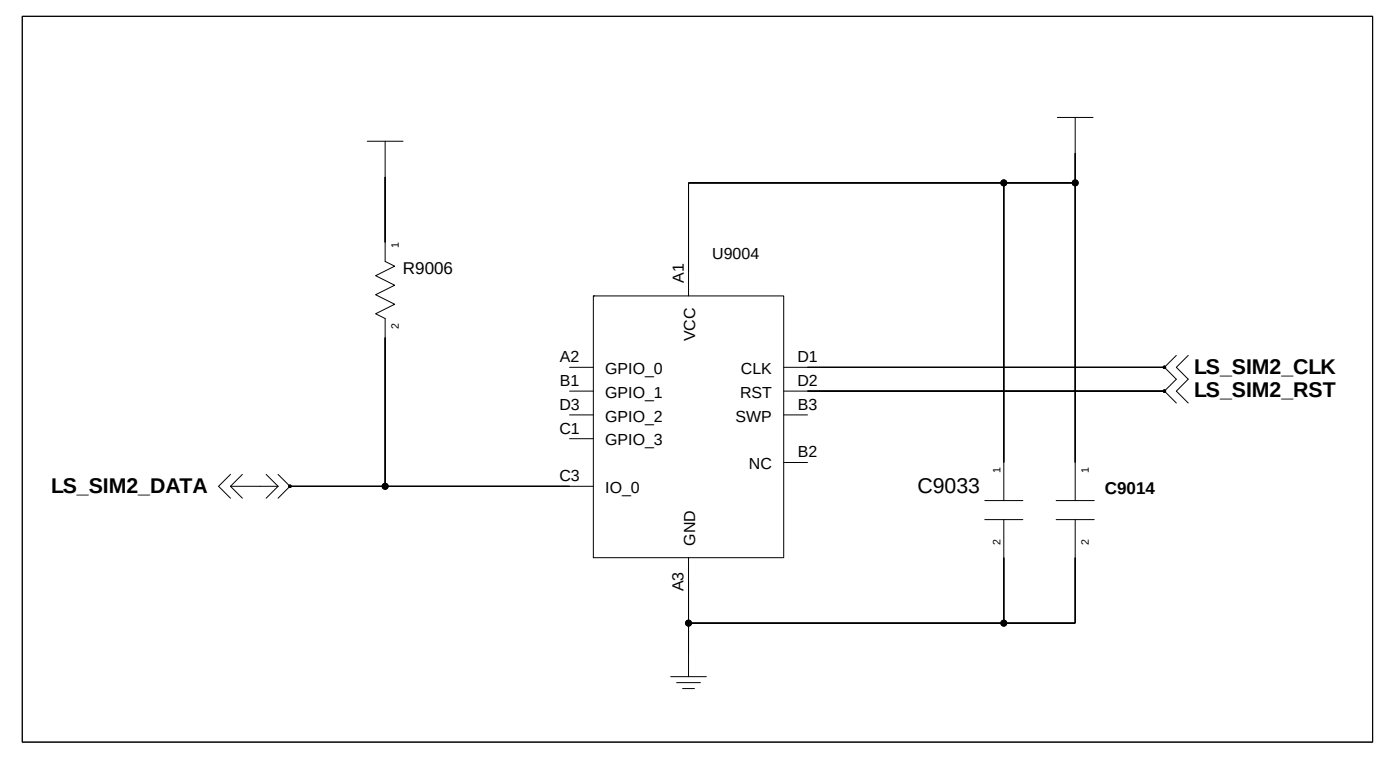
6 Axis (GYRO,ACCEL)


U9002
MAGNETIC_SDA_1P8
MAGNETIC_SCL_1P8
SDA
SCL
VDD
VSS
B2
A2
B1
A1
C9012

Magnetic SENSOR

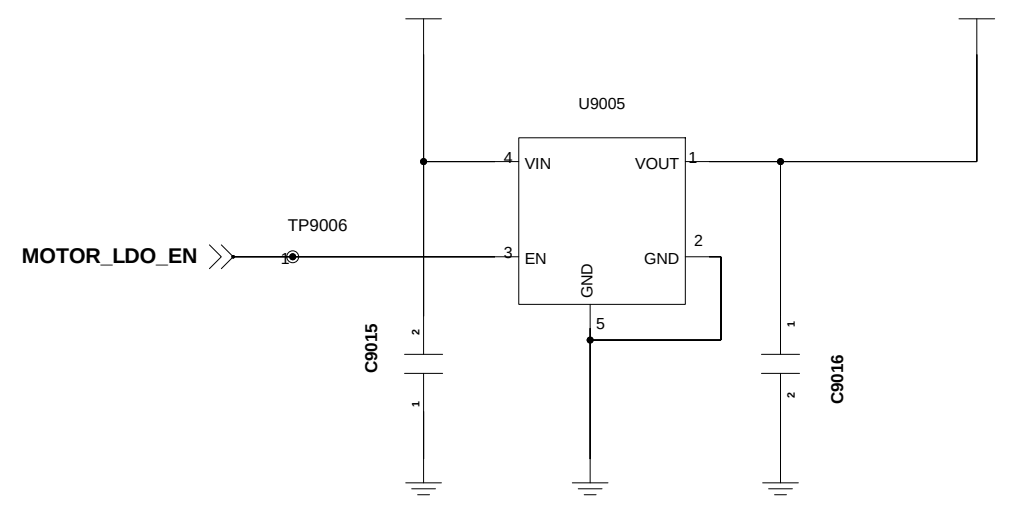

R9002
R9003
U9003
R9004
R9005
MAGNETIC_SDA_1P8
MAGNETIC_SCL_1P8
SDA
SCL
VDD
VSS
B2
A2
B1
A1
C9013
R9017

Magnetic SENSOR Option

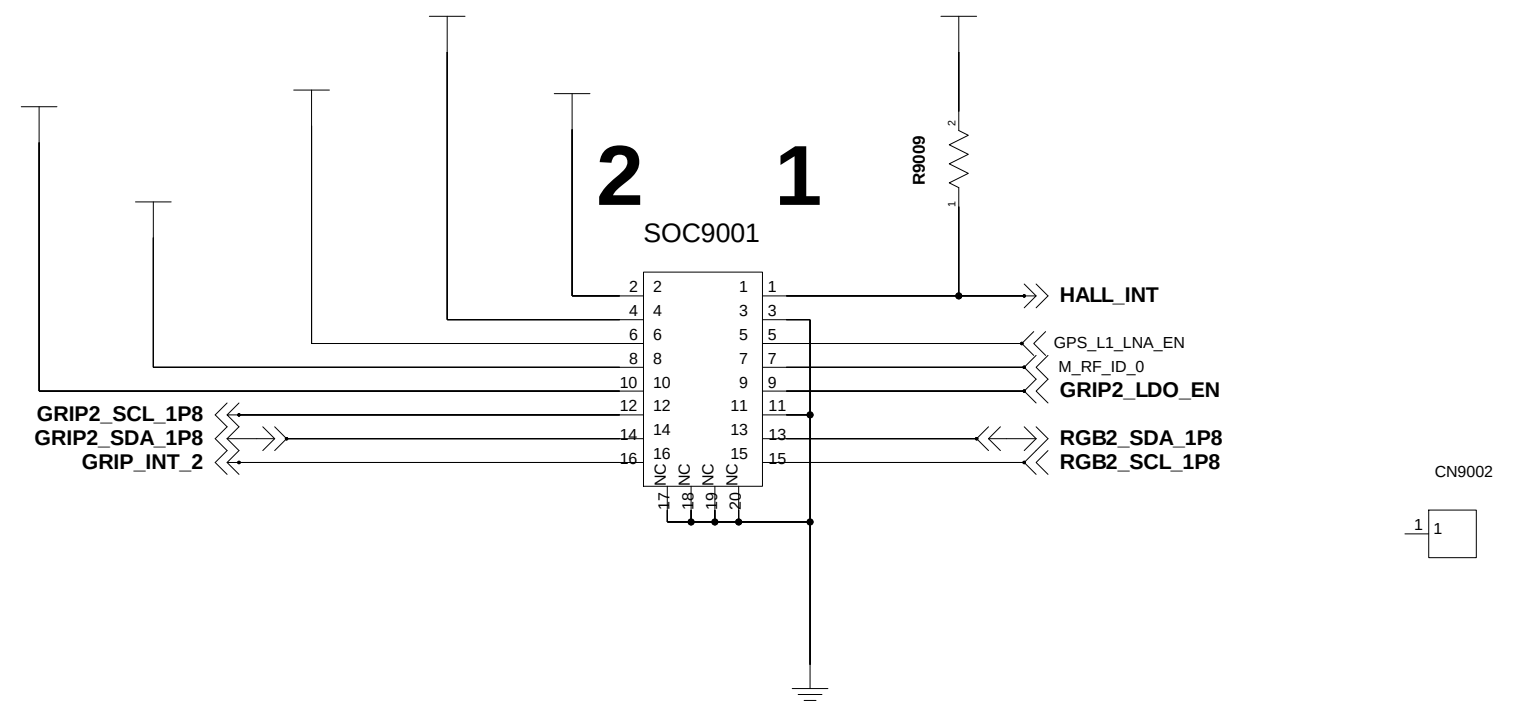

R9006
U9004
A1
VCC
A2
B1
D3
C1
C3
GPIO_0
GPIO_1
GPIO_2
GPIO_3
IO_0
CLK
RST
SWP
NC
D1
D2
B3
B2
GND
A3
LS_SIM2_DATA
LS_SIM2_CLK
LS_SIM2_RST
C9033
C9014

eSIM

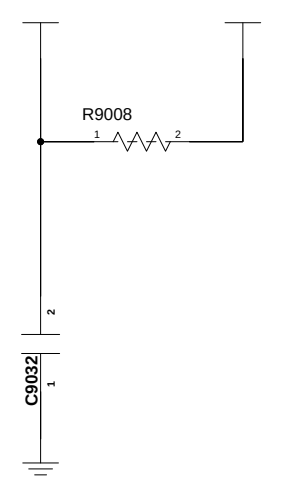
WIFI : delete


U9005
VIN
VOUT
EN
GND
GND
TP9006
MOTOR_LDO_EN
C9015
C9016

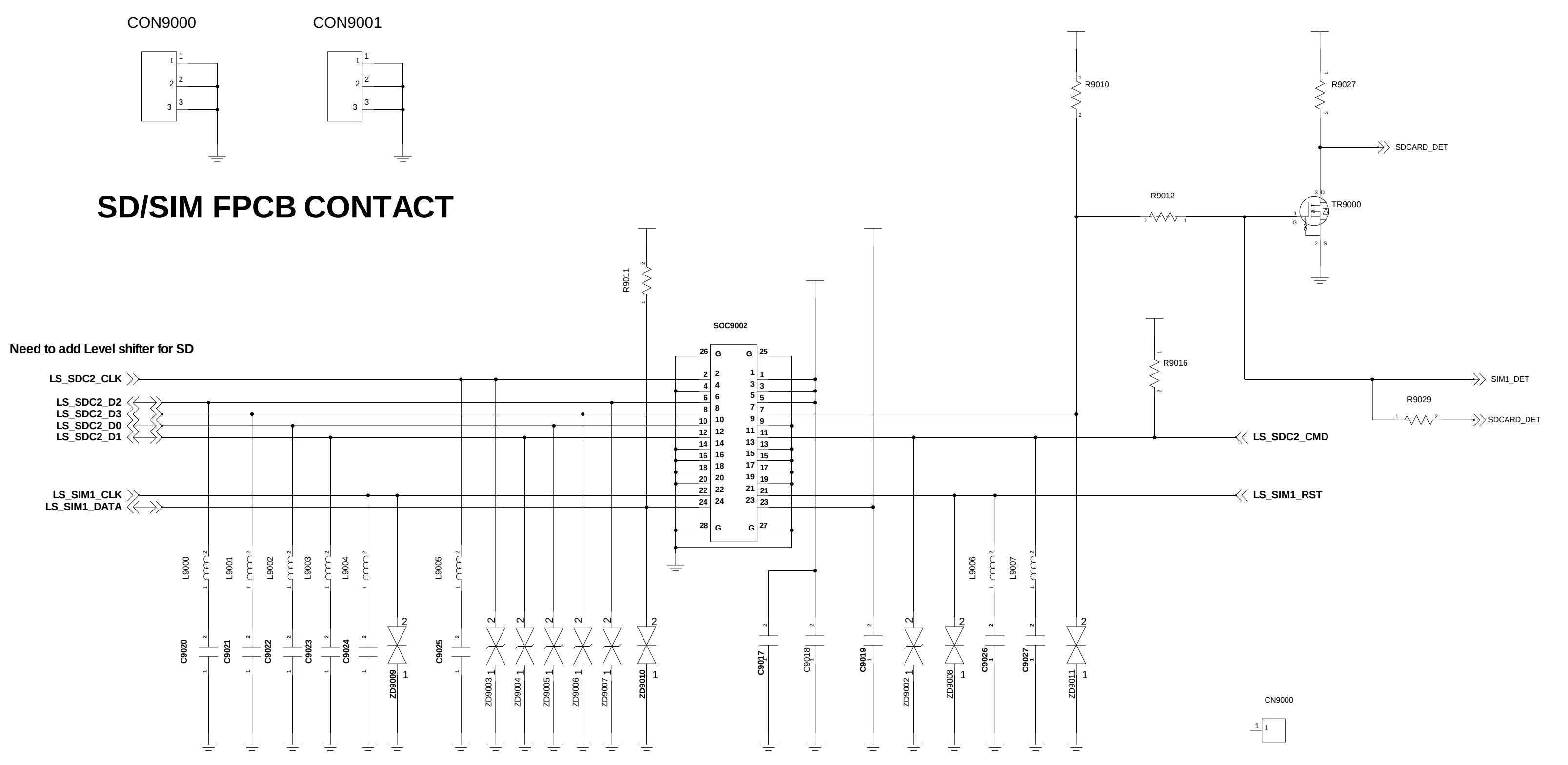
MOTOR LDO


2
1
SOC9001
R9009
HALL_INT
GPS_L1_LNA_EN
M_RF_ID_0
GRIP2_LDO_EN
RGB2_SDA_1P8
RGB2_SCL_1P8
GRIP2_SCL_1P8
GRIP2_SDA_1P8
GRIP_INT_2
NC
CN9002

SUB CONN
(MAIN <-> ANT SUB)

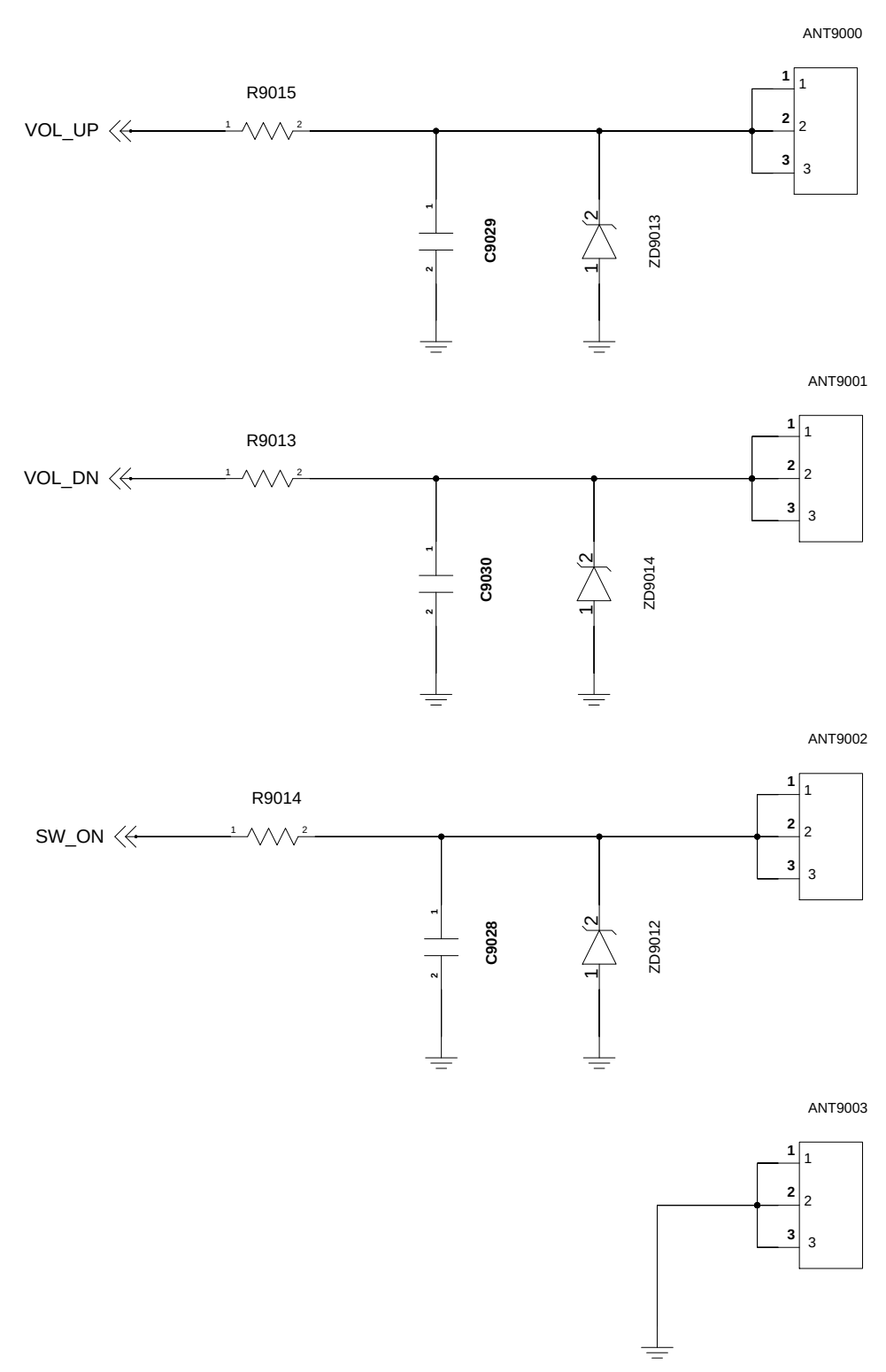

R9008
C9032

FOD


CON9000
CON9001
Need to add Level shifter for SD
LS_SDC2_CLK
LS_SDC2_D2
LS_SDC2_D3
LS_SDC2_D0
LS_SDC2_D1
LS_SIM1_CLK
LS_SIM1_DATA
SOC9002
R9011
R9010
R9012
R9016
R9027
R9029
TR9000
SDCARD_DET
SIM1_DET
SDCARD_DET
LS_SDC2_CMD
LS_SIM1_RST
L9000
L9001
L9002
L9003
L9004
L9005
L9006
L9007
C9020
C9021
C9022
C9023
C9024
C9025
C9017
C9018
C9019
C9026
C9027
ZD9009
ZD9003
ZD9004
ZD9005
ZD9006
ZD9007
ZD9010
ZD9002
ZD9008
ZD9011
CN9000

SD/SIM FPCB CONTACT

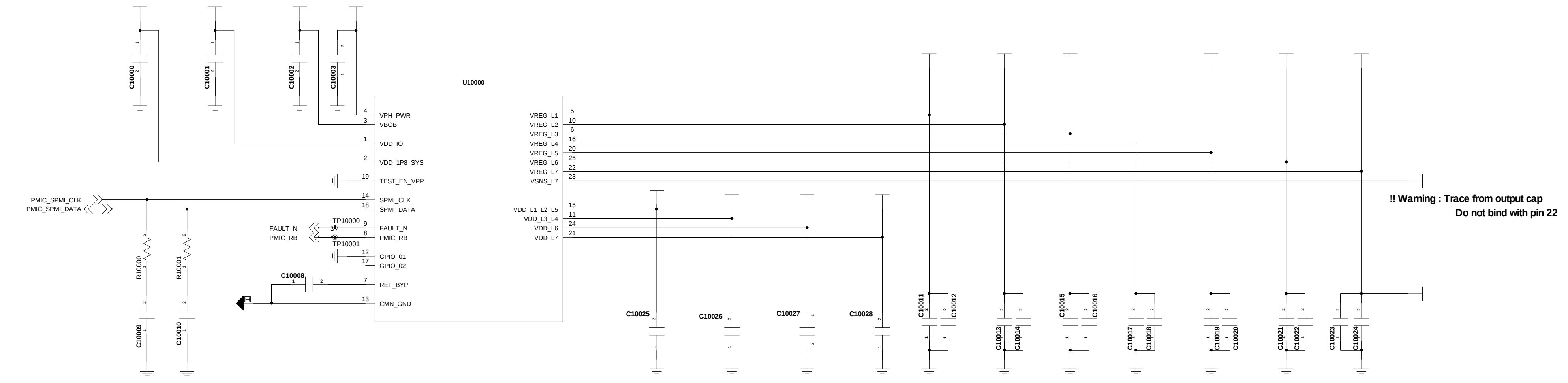
SIM/SD SOCKET Conn.


ANT9000
VOL_UP
R9015
C9029
ZD9013
ANT9001
VOL_DN
R9013
C9030
ZD9014
ANT9002
SW_ON
R9014
C9028
ZD9012
ANT9003

Key Contact

TITLE TabS7P
SIZE A1
DWG NO
REV
DRAWN BY User Name
SHEET 9 of 15
Date

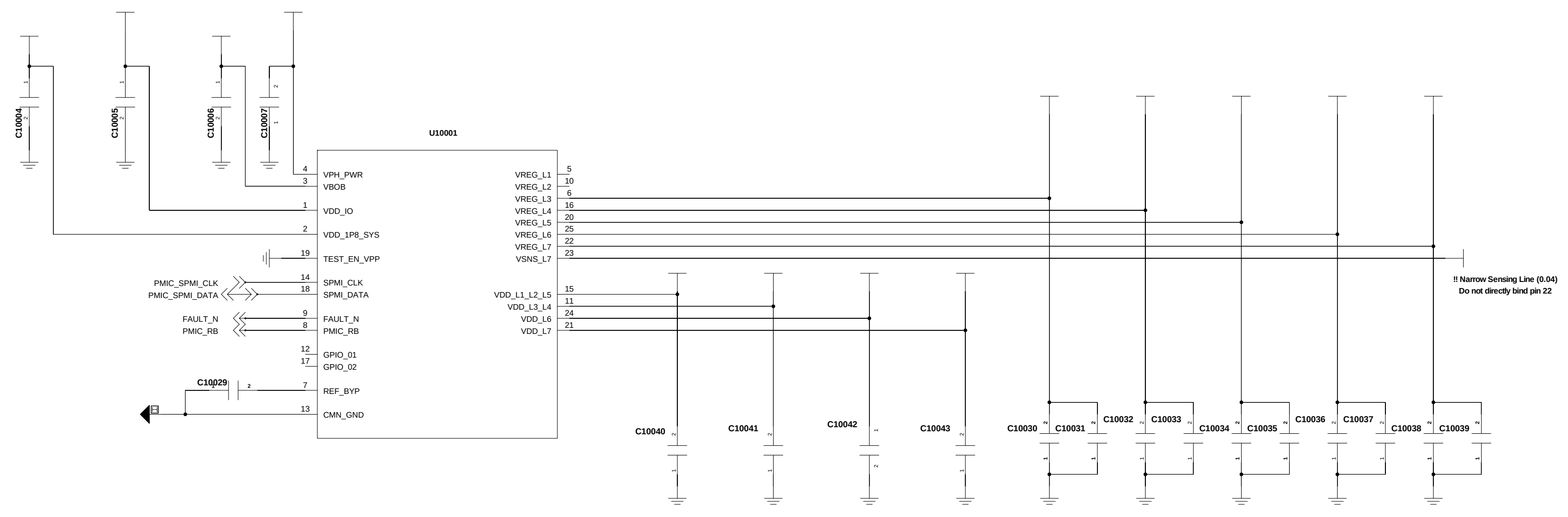
Power RF PMIC


C10000
C10001
C10002
C10003
U10000
VPH_PWR
VBOB
VDD_IO
VDD_1P8_SYS
TEST_EN_VPP
SPMI_CLK
SPMI_DATA
FAULT_N
PMIC_RB
GPIO_01
GPIO_02
REF_BYP
CMN_GND
VREG_L1
VREG_L2
VREG_L3
VREG_L4
VREG_L5
VREG_L6
VREG_L7
VSNS_L7
VDD_L1_L2_L5
VDD_L3_L4
VDD_L6
VDD_L7
PMIC_SPMI_CLK
PMIC_SPMI_DATA
FAULT_N
PMIC_RB
TP10000
TP10001
C10008
R10000
R10001
C10009
C10010
C10025
C10026
C10027
C10028
C10011
C10012
C10013
C10014
C10015
C10016
C10017
C10018
C10019
C10020
C10021
C10022
C10023
C10024
!! Warning : Trace from output cap
Do not bind with pin 22

PMR735D #0

SUB6 : Delete

WIFI : Delete

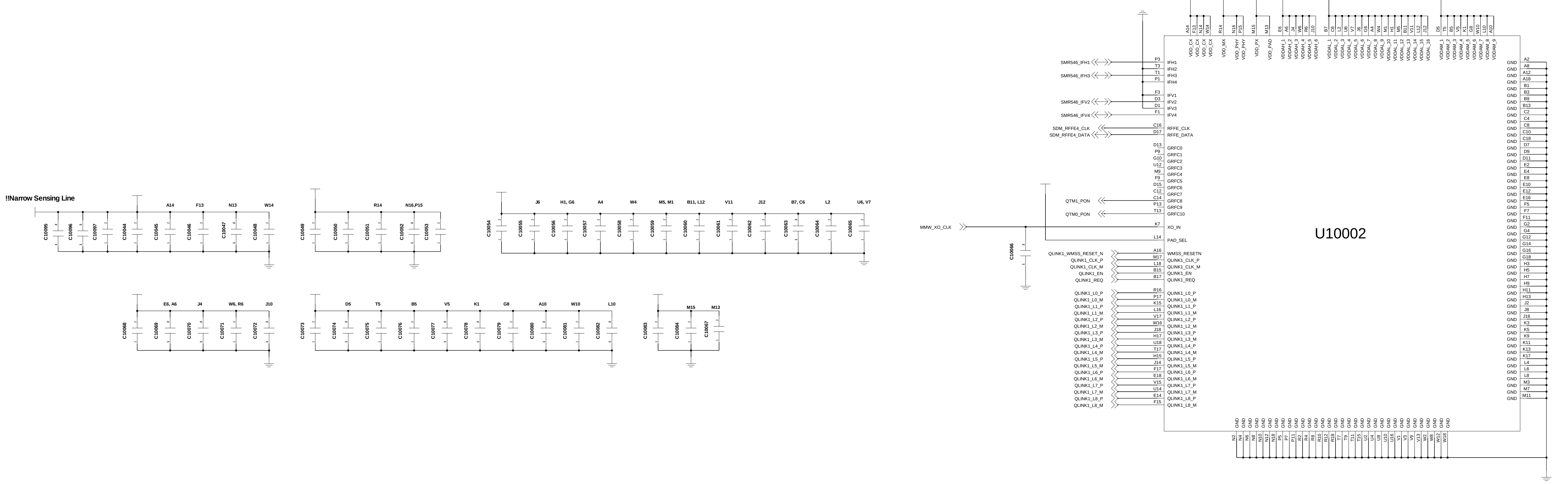

C10004
C10005
C10006
C10007
U10001
VPH_PWR
VBOB
VDD_IO
VDD_1P8_SYS
TEST_EN_VPP
SPMI_CLK
SPMI_DATA
FAULT_N
PMIC_RB
GPIO_01
GPIO_02
REF_BYP
CMN_GND
VREG_L1
VREG_L2
VREG_L3
VREG_L4
VREG_L5
VREG_L6
VREG_L7
VSNS_L7
VDD_L1_L2_L5
VDD_L3_L4
VDD_L6
VDD_L7
PMIC_SPMI_CLK
PMIC_SPMI_DATA
FAULT_N
PMIC_RB
C10029
C10040
C10041
C10042
C10043
C10030
C10031
C10032
C10033
C10034
C10035
C10036
C10037
C10038
C10039
!! Narrow Sensing Line (0.04)
Do not directly bind pin 22

PMR735D #1

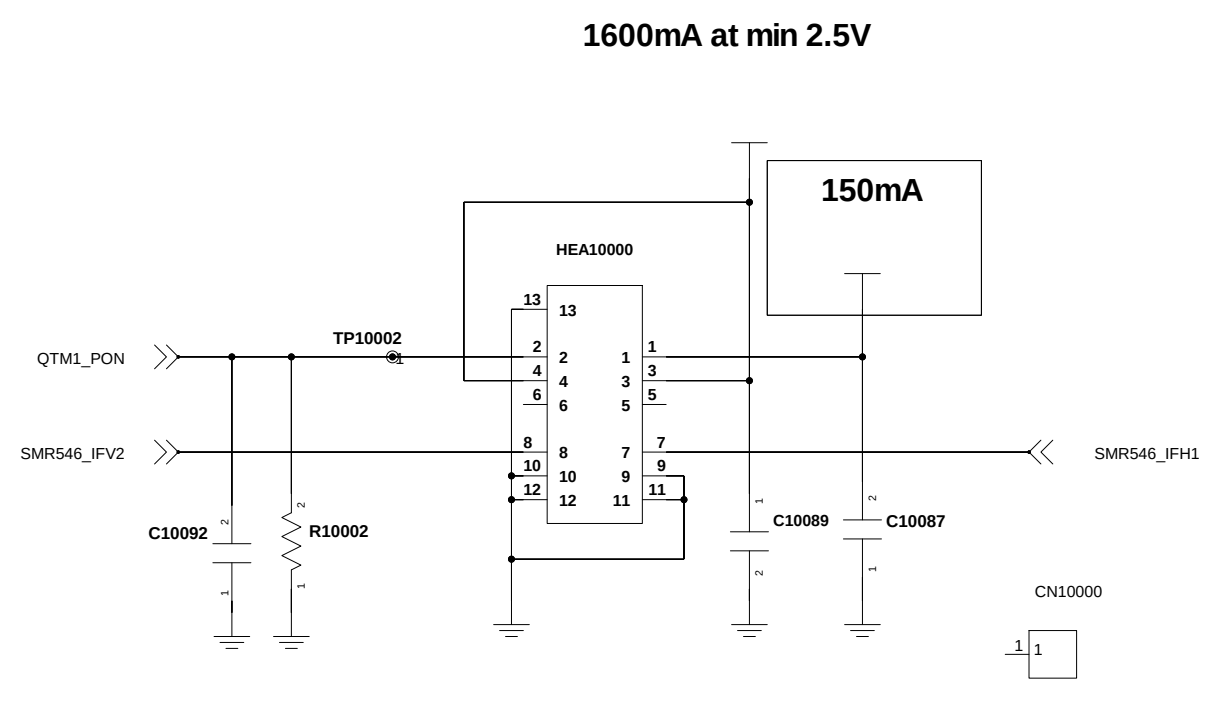
5G IFIC (SMR546)

SUB6 : Delete

WIFI : Delete


!!Narrow Sensing Line
C10095
C10096
C10097
C10044
C10045
C10046
C10047
C10048
A14
F13
N13
W14
C10049
C10050
C10051
C10052
C10053
R14
N16,P15
C10054
C10055
C10056
C10057
C10058
C10059
C10060
C10061
C10062
C10063
C10064
C10065
J6
H1, G6
A4
W4
M5, M1
B11, L12
V11
J12
B7, C6
L2
U6, V7
C10068
C10069
C10070
C10071
C10072
E6, A6
J4
W6, R6
J10
C10073
C10074
C10075
C10076
C10077
C10078
C10079
C10080
C10081
C10082
D5
T5
B5
V5
K1
G8
A10
W10
L10
C10083
C10084
C10067
M15
M13
MMW_XO_CLK
C10066
U10002
SMR546_IFH1
SMR546_IFH3
SMR546_IFV2
SMR546_IFV4
SDM_RFFE4_CLK
SDM_RFFE4_DATA
QTM1_PON
QTM0_PON
IFH1
IFH2
IFH3
IFH4
IFV1
IFV2
IFV3
IFV4
RFFE_CLK
RFFE_DATA
GRFC0
GRFC1
GRFC2
GRFC3
GRFC4
GRFC5
GRFC6
GRFC7
GRFC8
GRFC9
GRFC10
XO_IN
PAD_SEL
WMSS_RESETN
QLINK1_CLK_P
QLINK1_CLK_M
QLINK1_EN
QLINK1_REQ
QLINK1_WMSS_RESET_N
QLINK1_L0_P
QLINK1_L0_M
QLINK1_L1_P
QLINK1_L1_M
QLINK1_L2_P
QLINK1_L2_M
QLINK1_L3_P
QLINK1_L3_M
QLINK1_L4_P
QLINK1_L4_M
QLINK1_L5_P
QLINK1_L5_M
QLINK1_L6_P
QLINK1_L6_M
QLINK1_L7_P
QLINK1_L7_M
QLINK1_L8_P
QLINK1_L8_M
VDD_CX
VDD_MX
VDD_PHY
VDD_PX
VDD_PAD
VDDAH_1 … VDDAH_6
VDDAL_1 … VDDAL_16
VDDAM_1 … VDDAM_9
GND

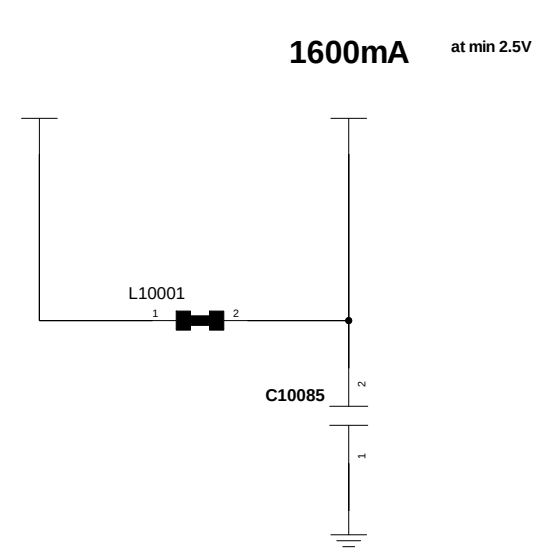
1600mA at min 2.5V


150mA
HEA10000
QTM1_PON
SMR546_IFV2
SMR546_IFH1
TP10002
C10092
R10002
C10089
C10087
CN10000

Need to check connector P/N and pin map

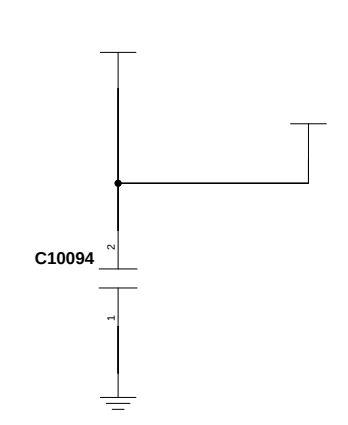
mmW_ANT_CONN (BACK)

1600mA at min 2.5V

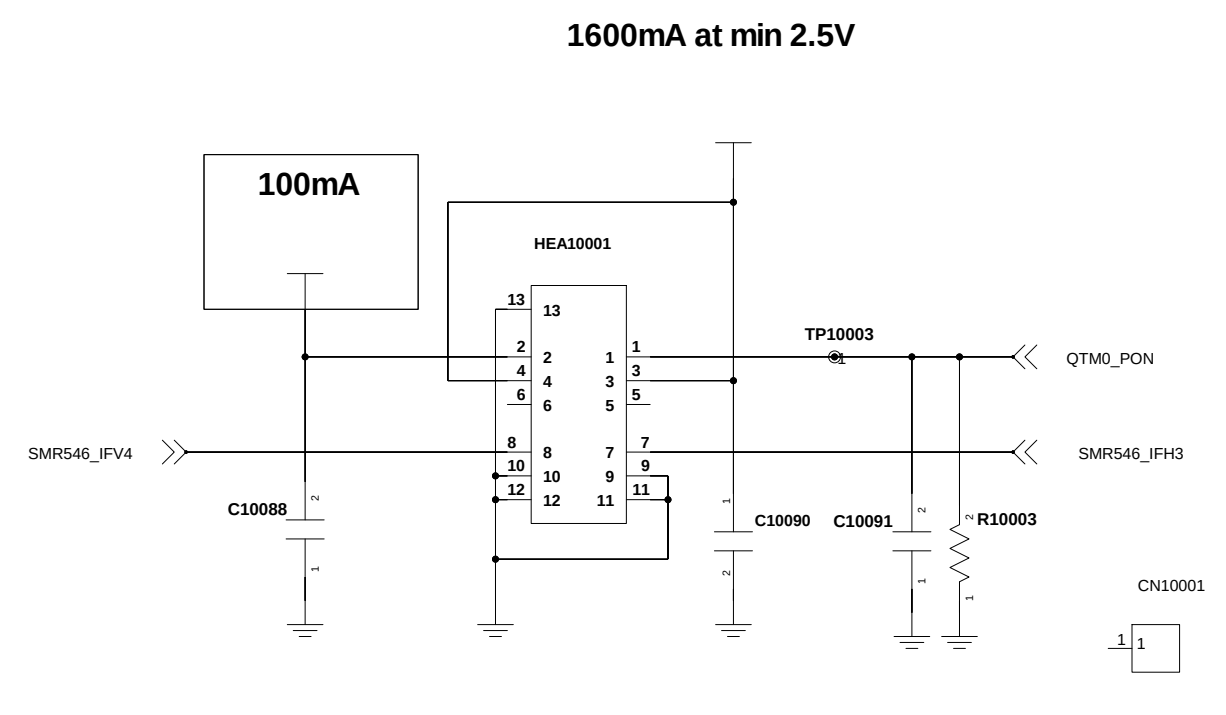

L10001
C10085

LOAD SWITCH

100mA



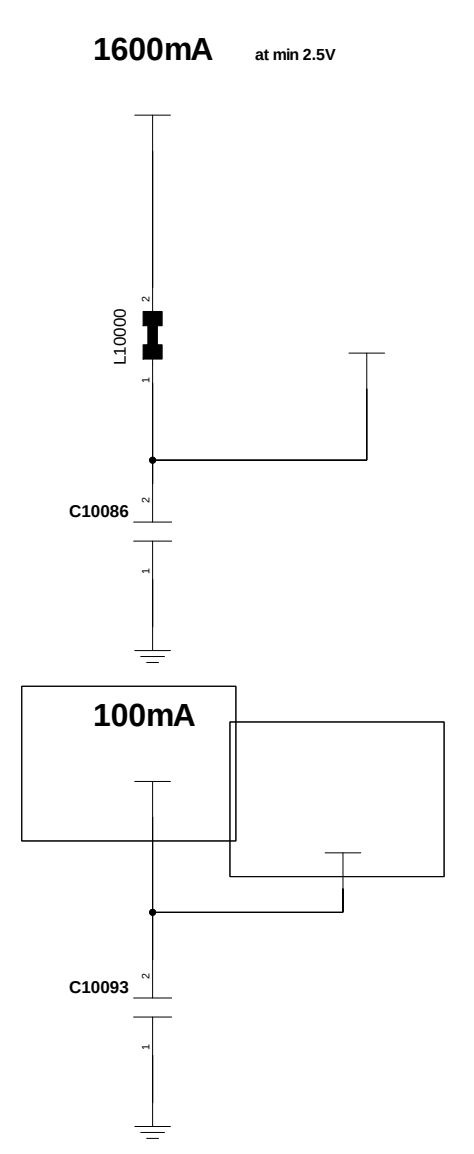
1600mA at min 2.5V


100mA
HEA10001
SMR546_IFV4
QTM0_PON
SMR546_IFH3
TP10003
C10088
C10090
C10091
R10003
CN10001

Need to check connector P/N and pin map

mmW_ANT_CONN (FRONT)

1600mA at min 2.5V


L10000
C10086
100mA
C10093

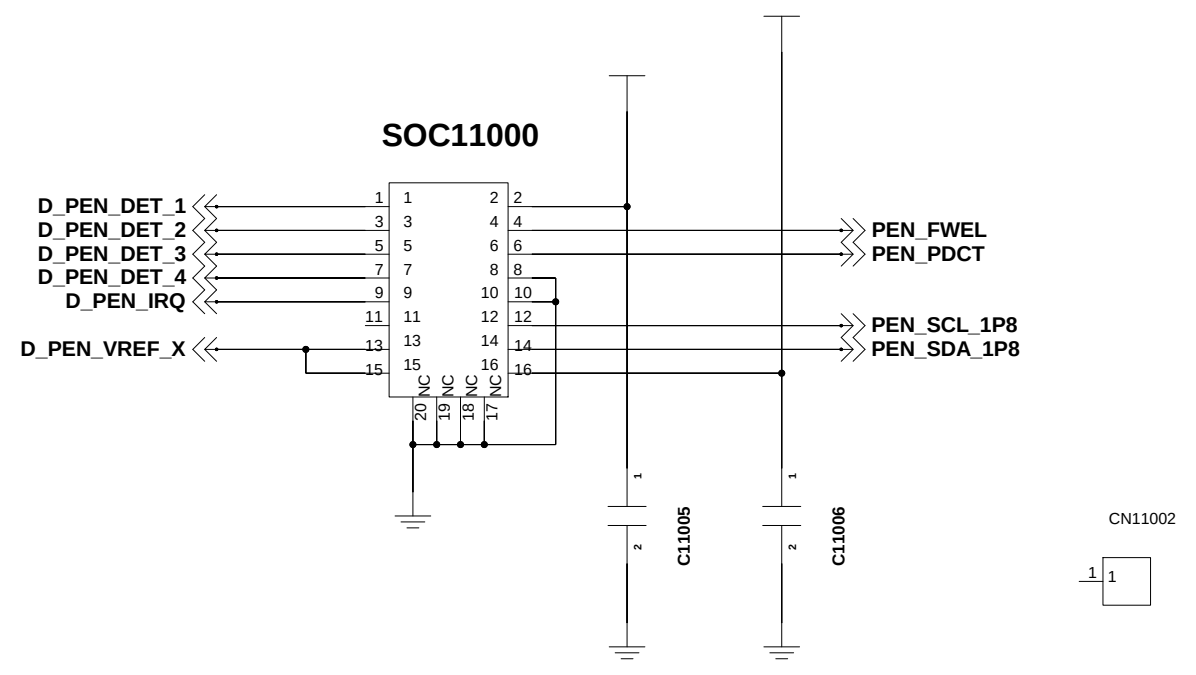
Radiation

TITLE	TabS7P	
SIZE A1	DWG NO PDM NUMBER	REV Rev
DRAWN BY User Name	SHEET 10 of 15	Date Drawing Date

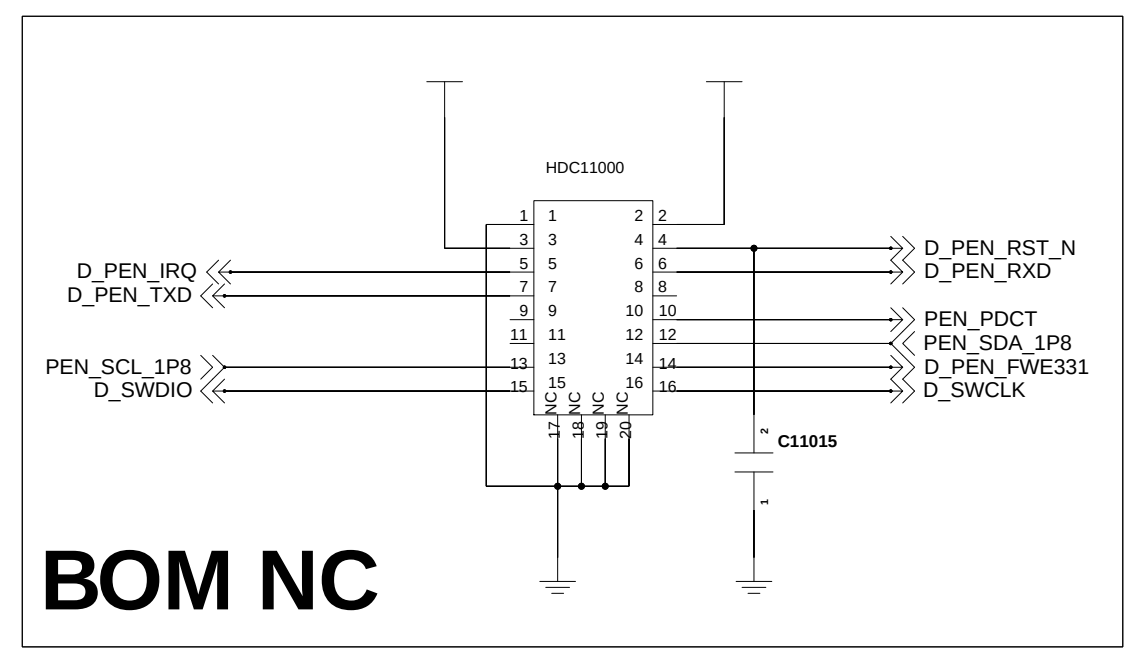
WACOM PCB (Ext.)

< SMD F/T >

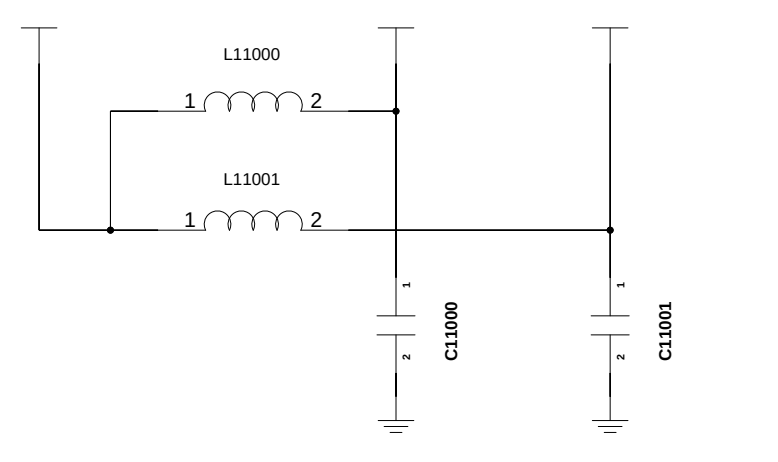
1. VREG_L13B_3P3
2. VREG_L15B_1P8
3. PEN_SCL_1P8
4. PEN_SDA_1P8
5. PEN_FWEL
6. PEN_PDCT
7. GND

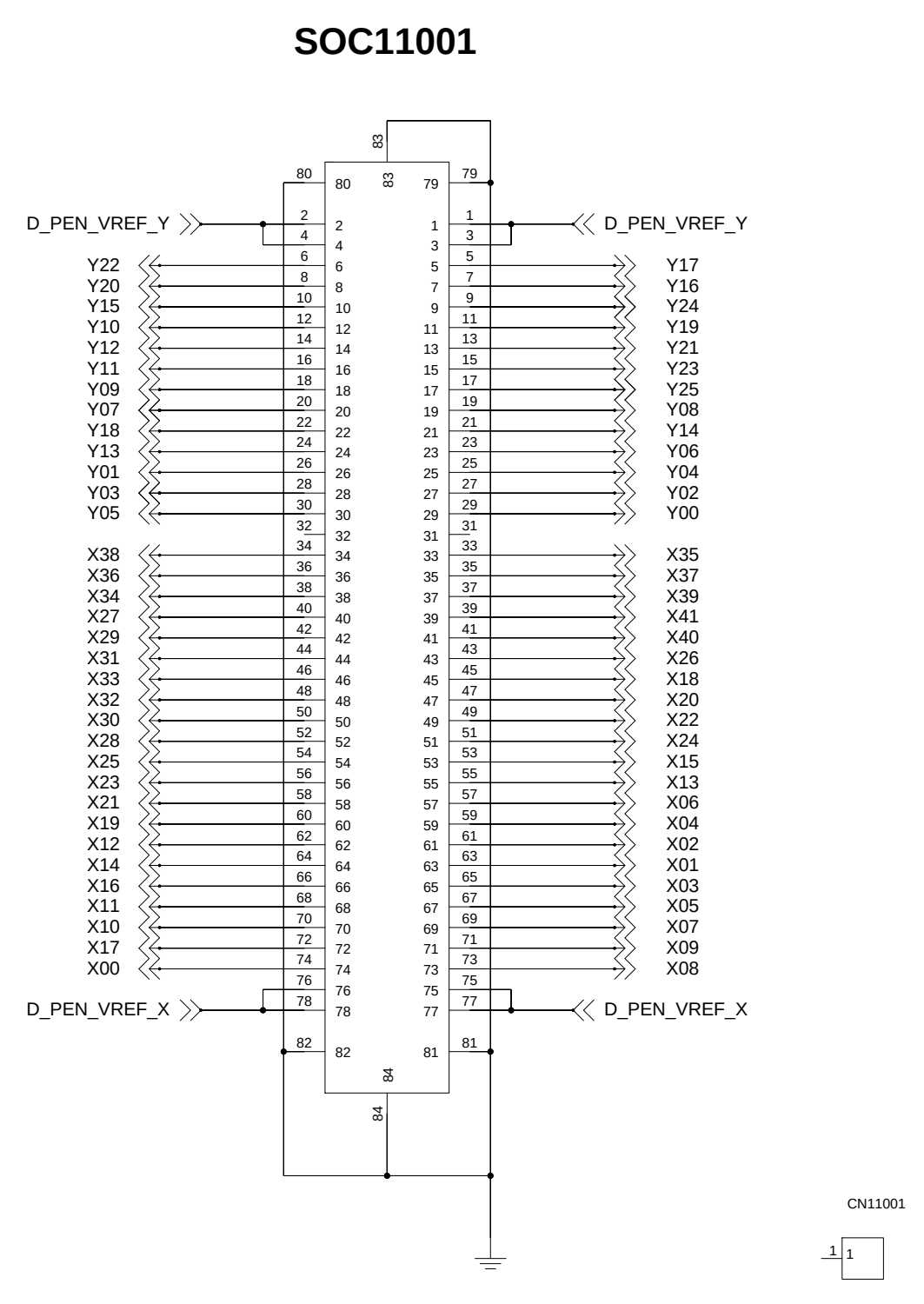

SOC11000
D_PEN_DET_1
D_PEN_DET_2
D_PEN_DET_3
D_PEN_DET_4
D_PEN_IRQ
D_PEN_VREF_X
PEN_FWEL
PEN_PDCT
PEN_SCL_1P8
PEN_SDA_1P8
C11005
C11006
CN11002

WACOM I/F CON.

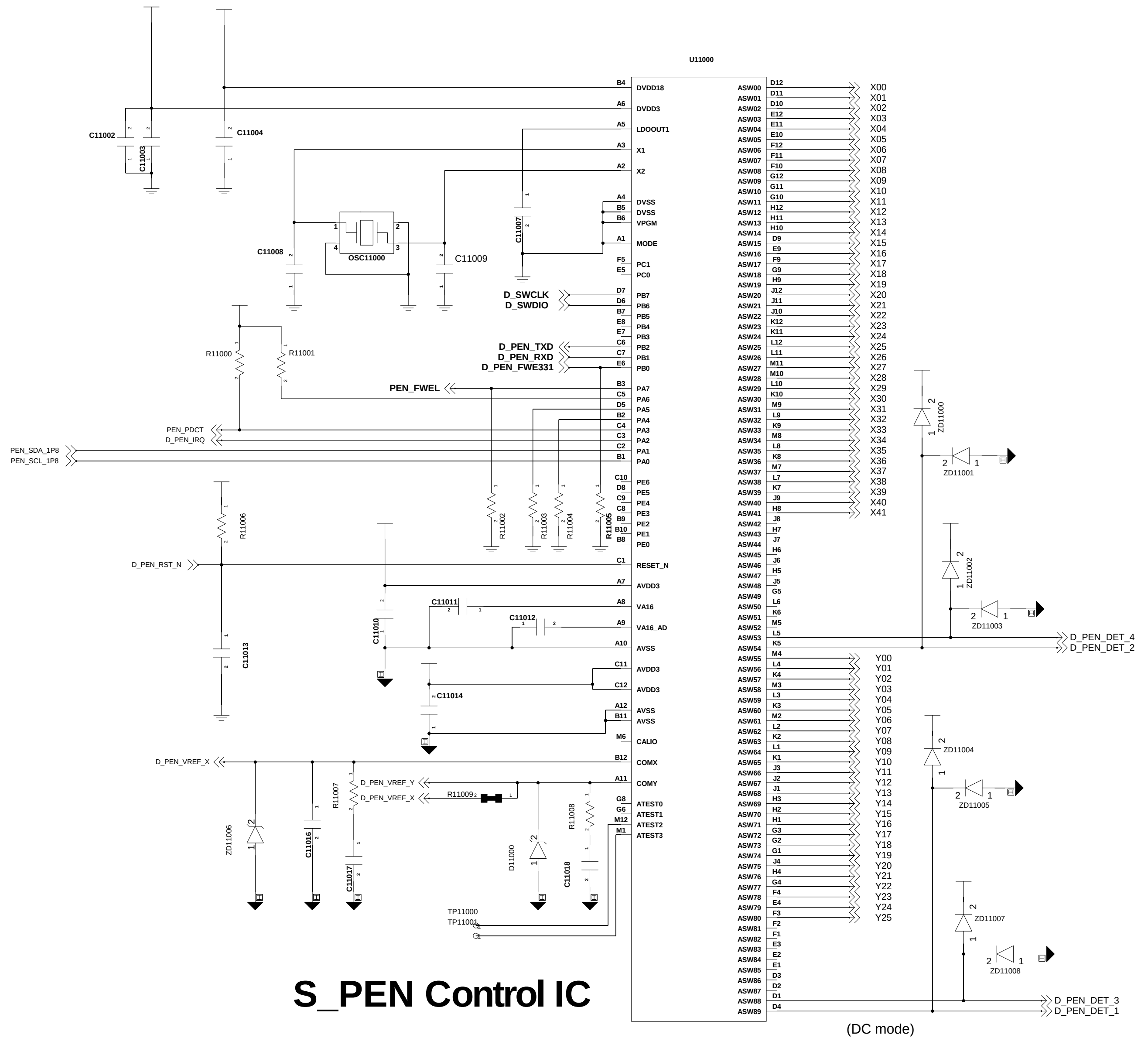

HDC11000
D_PEN_IRQ
D_PEN_TXD
PEN_SCL_1P8
D_SWDIO
D_PEN_RST_N
D_PEN_RXD
PEN_PDCT
PEN_SDA_1P8
D_PEN_FWE331
D_SWCLK
C11015
BOM NC

DEBUGER


L11000
L11001
C11000
C11001

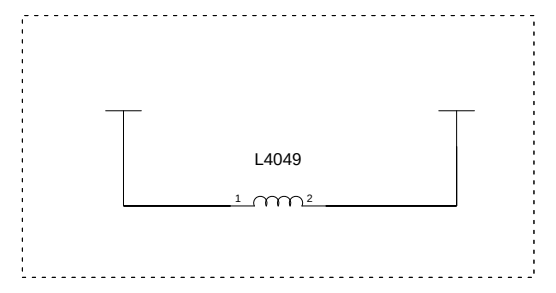

SOC11001
D_PEN_VREF_Y
Y22 Y20 Y15 Y10 Y12 Y11 Y09 Y07 Y18 Y13 Y01 Y03 Y05
X38 X36 X34 X27 X29 X31 X33 X32 X30 X28 X25 X23 X21 X19 X12 X14 X16 X11 X10 X17 X00
Y17 Y16 Y24 Y19 Y21 Y23 Y25 Y08 Y14 Y06 Y04 Y02 Y00
X35 X37 X39 X41 X40 X26 X18 X20 X22 X24 X15 X13 X06 X04 X02 X01 X03 X05 X07 X09 X08
D_PEN_VREF_X
CN11001

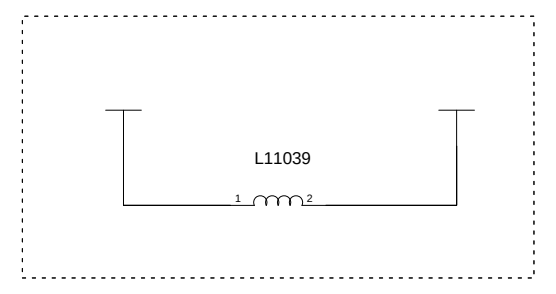
78pin S-PEN Connector

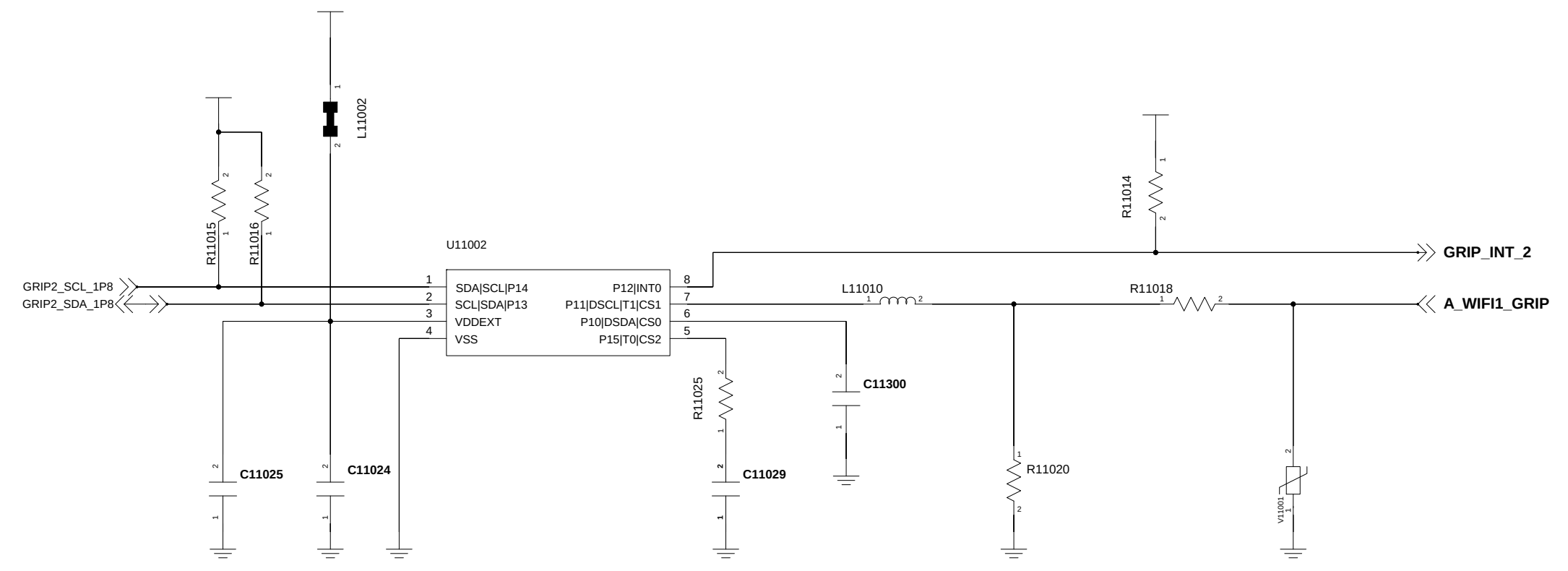

U11000
C11002
C11003
C11004
C11007
C11008
OSC11000
C11009
D_SWCLK
D_SWDIO
R11000
R11001
D_PEN_TXD
D_PEN_RXD
D_PEN_FWE331
PEN_FWEL
PEN_PDCT
D_PEN_IRQ
PEN_SDA_1P8
PEN_SCL_1P8
R11002
R11003
R11004
R11005
R11006
D_PEN_RST_N
C11010
C11011
C11012
C11013
C11014
D_PEN_VREF_X
D_PEN_VREF_Y
R11007
R11009
R11008
ZD11006
C11016
C11017
D11000
C11018
TP11000
TP11001
DVDD18 DVDD3 LDOOUT1 X1 X2 DVSS DVSS VPGM MODE PC1 PC0 PB7 PB6 PB5 PB4 PB3 PB2 PB1 PB0 PA7 PA6 PA5 PA4 PA3 PA2 PA1 PA0 PE6 PE5 PE4 PE3 PE2 PE1 PE0 RESET_N AVDD3 VA16 VA16_AD AVSS AVDD3 AVDD3 AVSS AVSS CALIO COMX COMY ATEST0 ATEST1 ATEST2 ATEST3
ASW00 … ASW89
X00 … X41
Y00 … Y25
ZD11000
ZD11001
ZD11002
ZD11003
ZD11004
ZD11005
ZD11007
ZD11008
D_PEN_DET_4
D_PEN_DET_2
D_PEN_DET_3
D_PEN_DET_1
(DC mode)

S_PEN Control IC

ANT SUB PCB (Ext.)



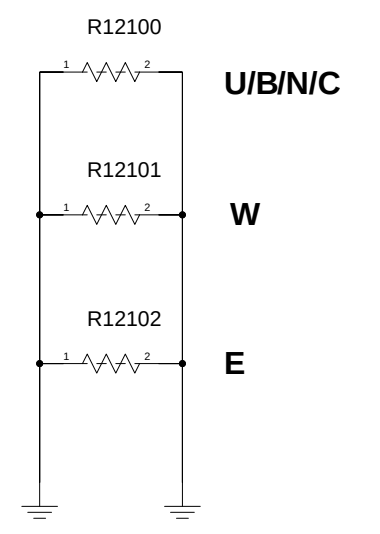



U11002
GRIP2_SCL_1P8
GRIP2_SDA_1P8
SDA|SCL|P14
SCL|SDA|P13
VDDEXT
VSS
P12|INT0
P11|DSCL|T1|CS1
P10|DSDA|CS0
P15|T0|CS2
R11015
R11016
L11002
C11025
C11024
R11025
C11029
C11300
L11010
R11020
R11014
R11018
V11001
GRIP_INT_2
A_WIFI1_GRIP

WIFI_GRIP_SENSOR

ANT PCB REGION

ANT PCB	SHORT
X818U (U), X816B/N/C (B/N/C)	R12100
X810 (W)	R12101
ETC (E)	R12102

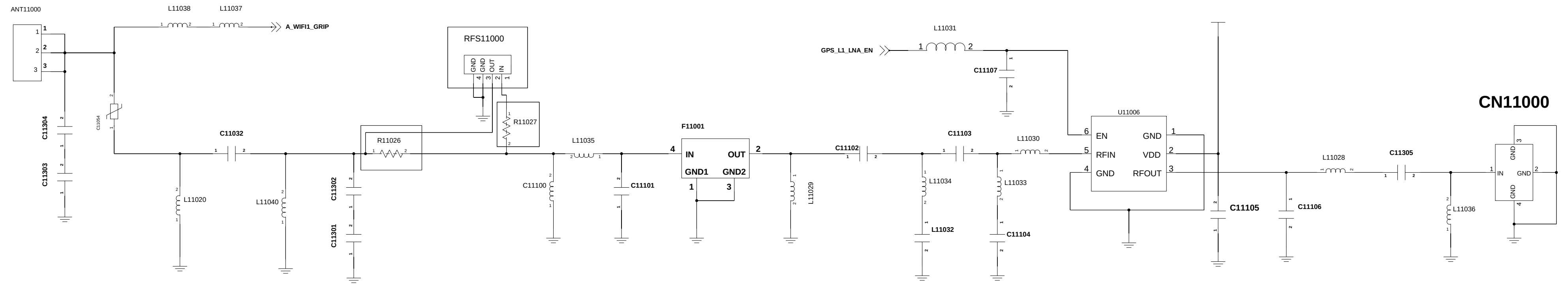

R12100
U/B/N/C
R12101
W
R12102
E

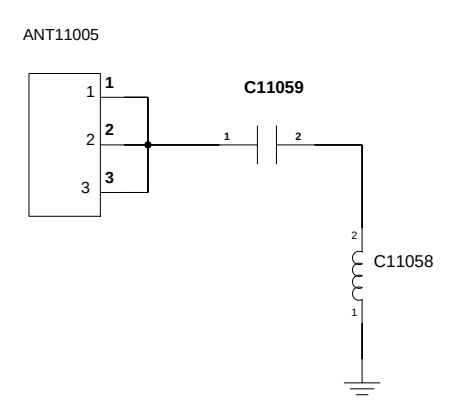
< SMD F/T >

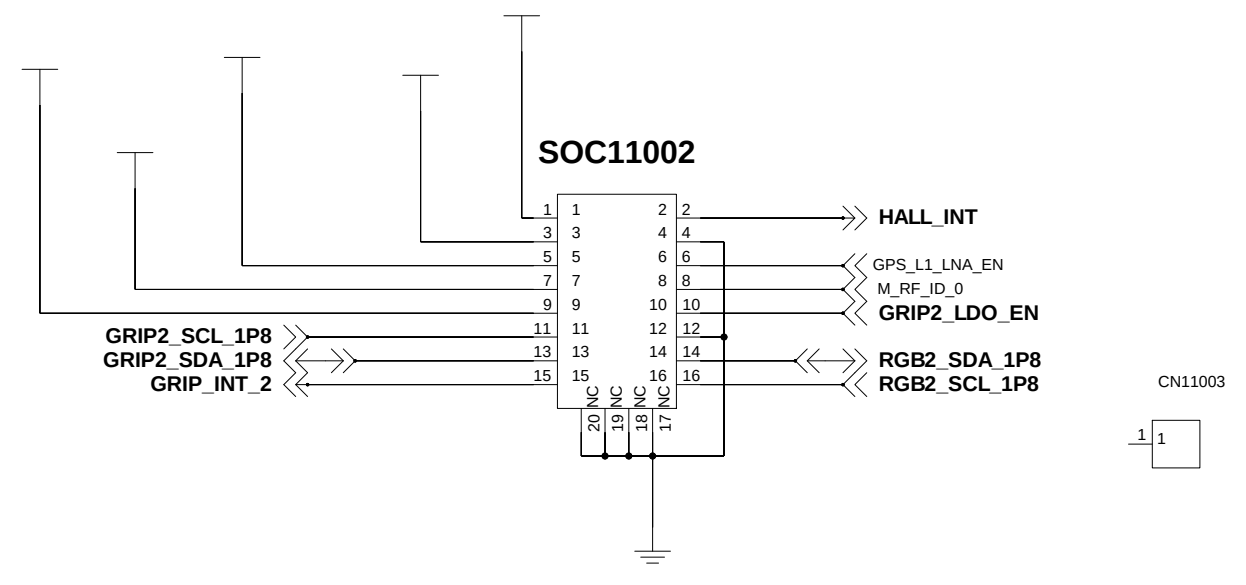
1. VDD_GRIP2_3P0_LDO
2. GRIP2_SCL_1P8
3. GRIP2_SDA_1P8
4. GRIP_INT_2
5. SDM_RFFE1_CLK
6. SDM_RFFE1_DATA
7. VREG_L15B_1P8
8. HALL_INT
9. VREG_L16B_3P0
10. RGB2_SCL_1P8
11. RGB2_SDA_1P8
12. GND

GPS ANT (GPS_L1)

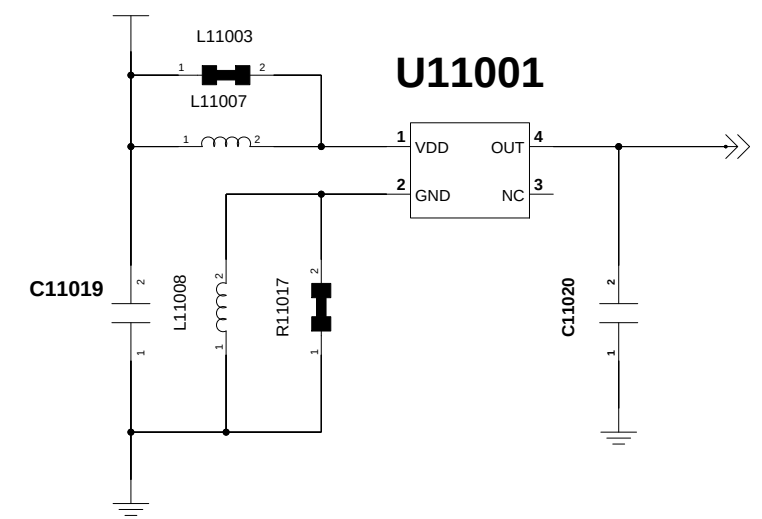
GPS_L1


ANT11000
L11038
L11037
A_WIFI1_GRIP
RFS11000
GND GND OUT IN
C11304
C11303
C11054
C11032
L11020
L11040
C11302
C11301
R11026
R11027
C11100
L11035
C11101
F11001
IN OUT GND1 GND2
L11029
GPS_L1_LNA_EN
L11031
C11107
C11102
C11103
L11030
L11034
L11032
L11033
C11104
U11006
EN RFIN GND GND VDD RFOUT
C11105
C11106
L11028
C11305
L11036
CN11000
IN GND

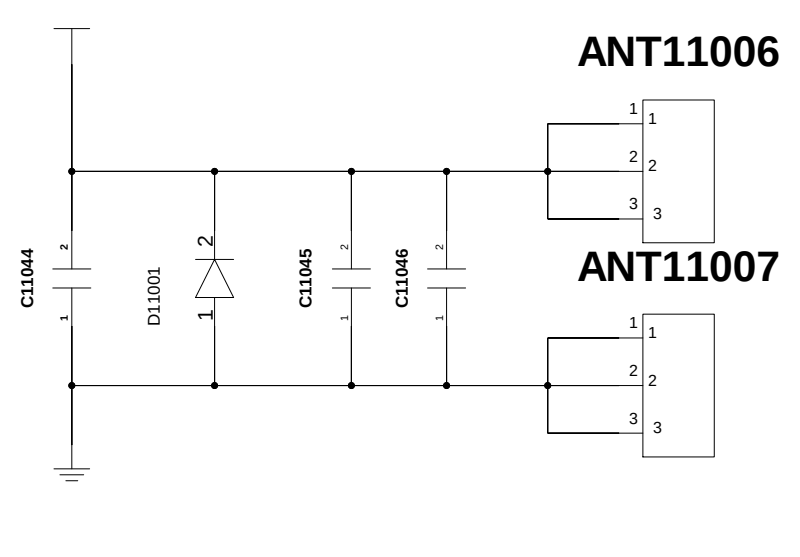

ANT11005
C11059
C11058


SOC11002
HALL_INT
GPS_L1_LNA_EN
M_RF_ID_0
GRIP2_LDO_EN
GRIP2_SCL_1P8
GRIP2_SDA_1P8
GRIP_INT_2
RGB2_SDA_1P8
RGB2_SCL_1P8
CN11003

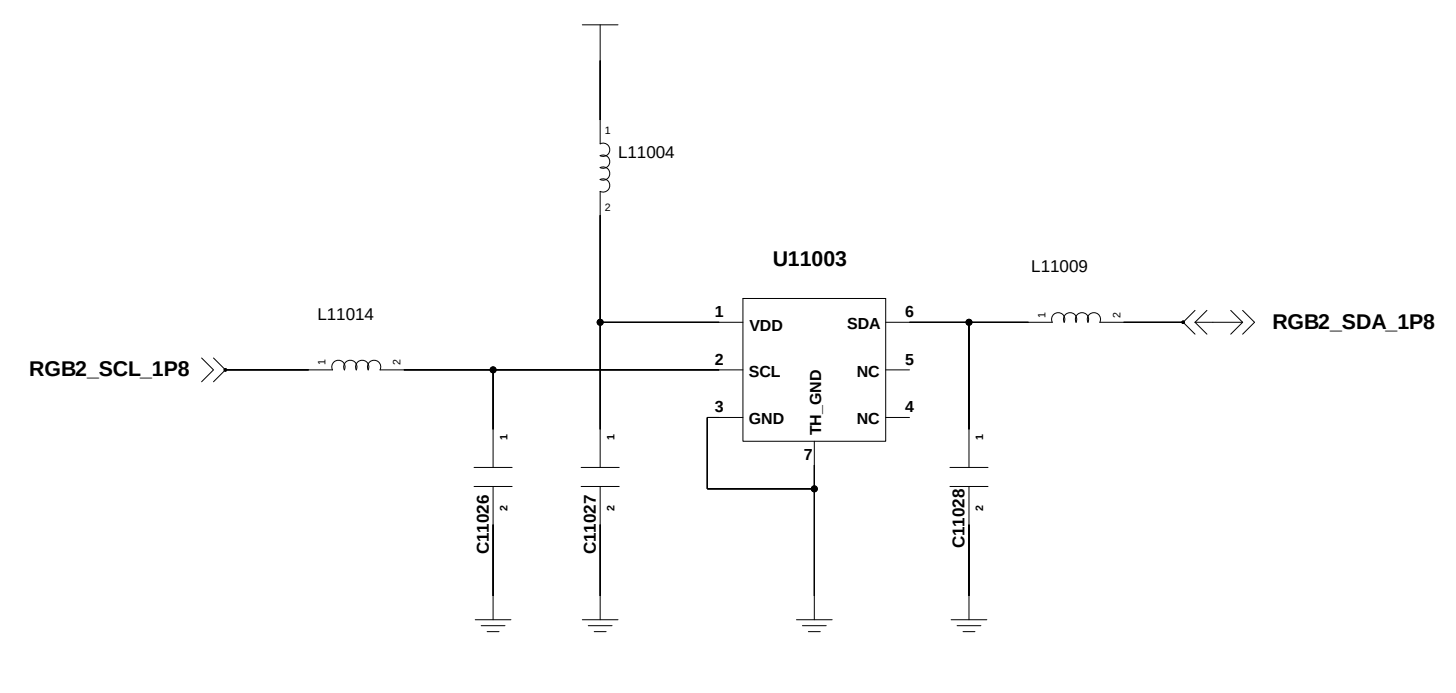
SUB CONN
(ANT SUB <-> MAIN)


U11001
L11003
L11007
VDD OUT GND NC
HALL_INT
C11019
L11008
R11017
C11020

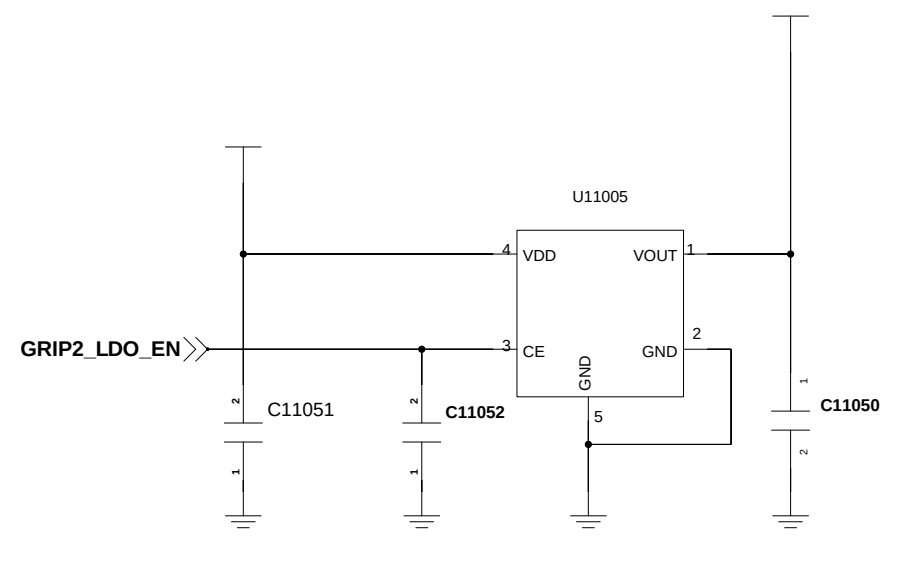
COVER_DET HALL_IC


ANT11006
ANT11007
C11044
D11001
C11045
C11046

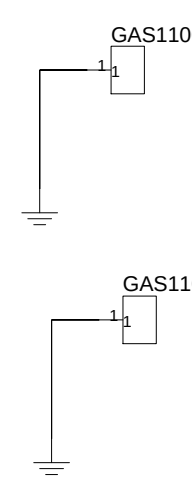
MOTOR


U11003
L11004
L11014
L11009
RGB2_SCL_1P8
RGB2_SDA_1P8
VDD SCL GND SDA NC NC TH_GND
C11026
C11027
C11028

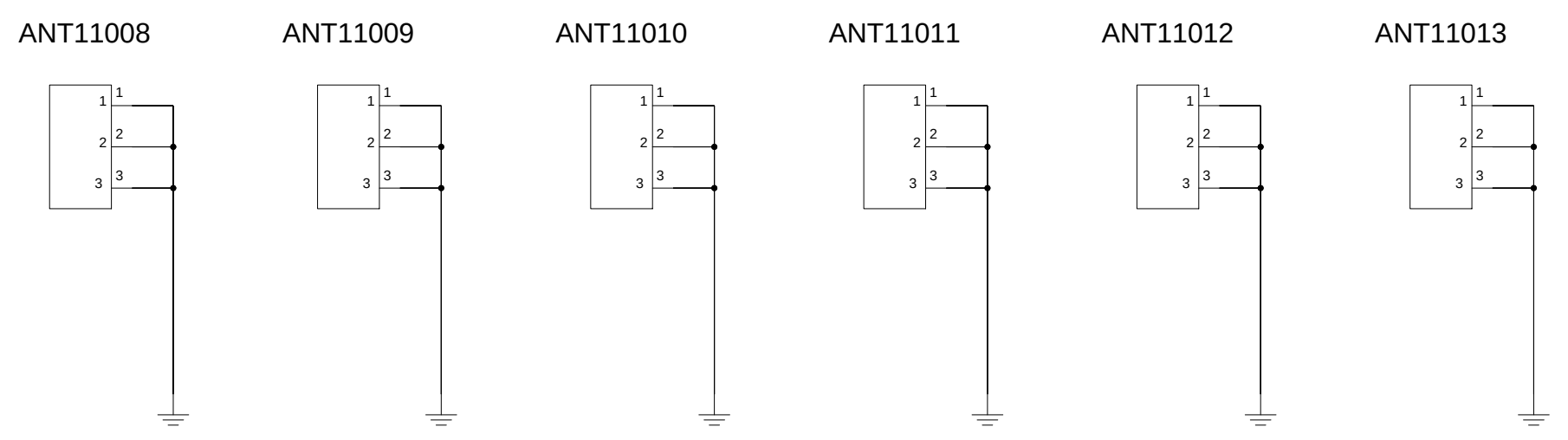
RGB Sensor 2


U11005
GRIP2_LDO_EN
VDD CE GND VOUT GND
C11051
C11052
C11050

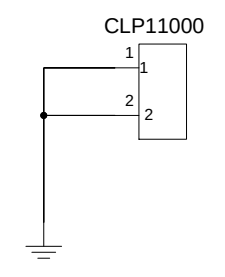
GRIP2 LDO


GAS11000
GAS11001

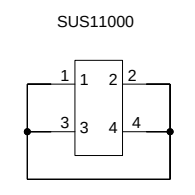
FRONT CONTACT (TOP)


ANT11008
ANT11009
ANT11010
ANT11011
ANT11012
ANT11013

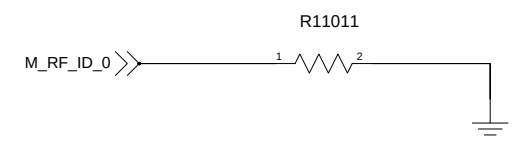
REAR CONTACT (BOT)



Cable CLIP

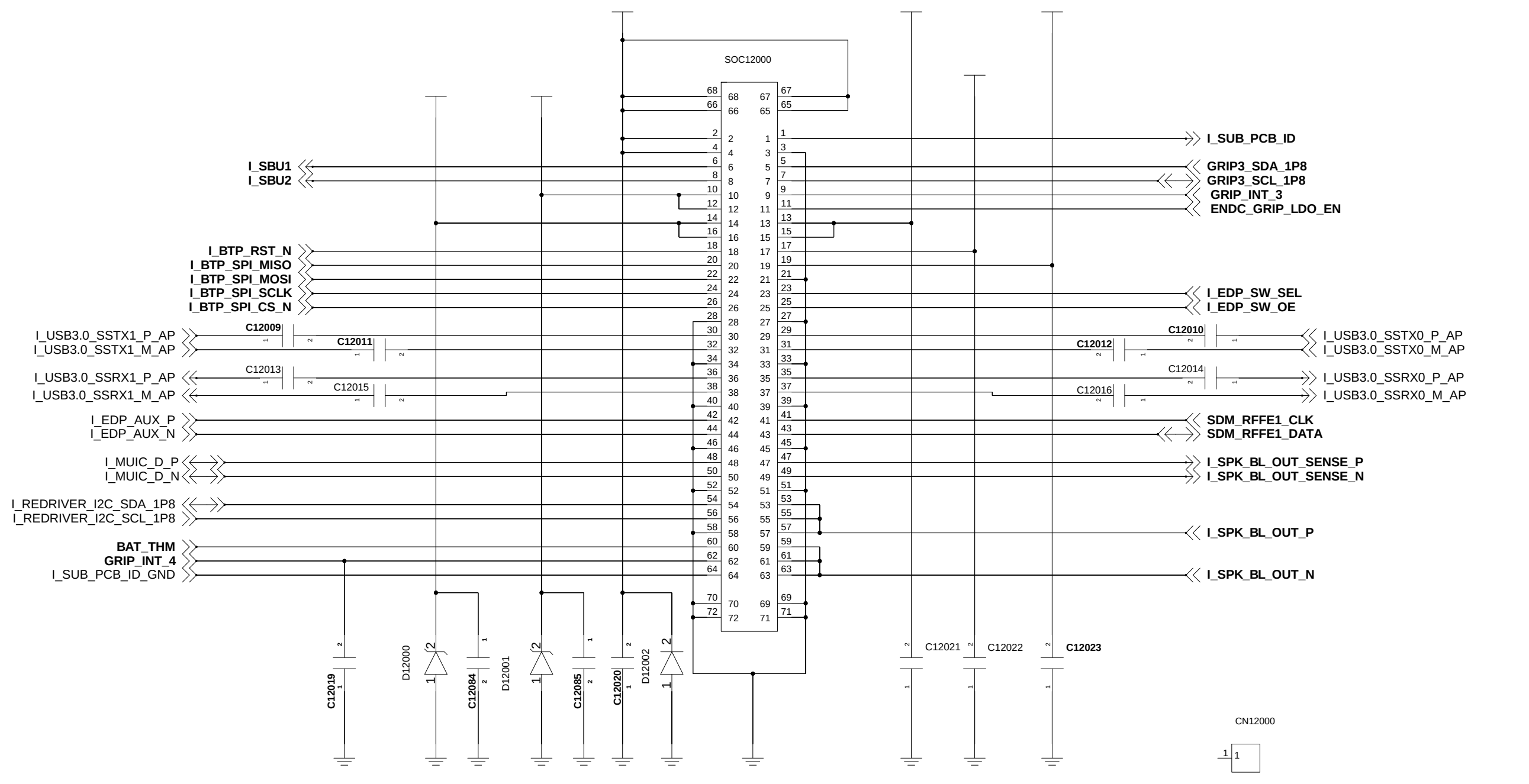


SCREW SUS (0.5mm)

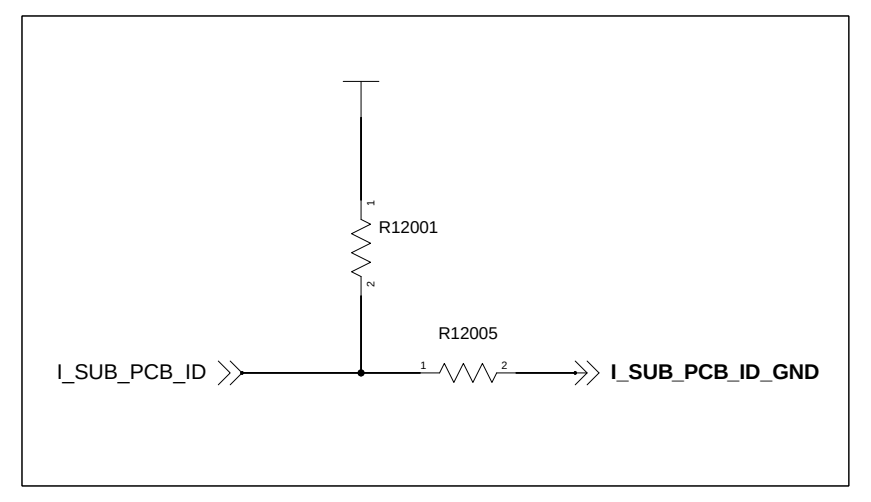

M_RF_ID_0
R11011

TITLE TabS7P
SIZE A1
DRAWN BY User Name
SHEET 11 of 15

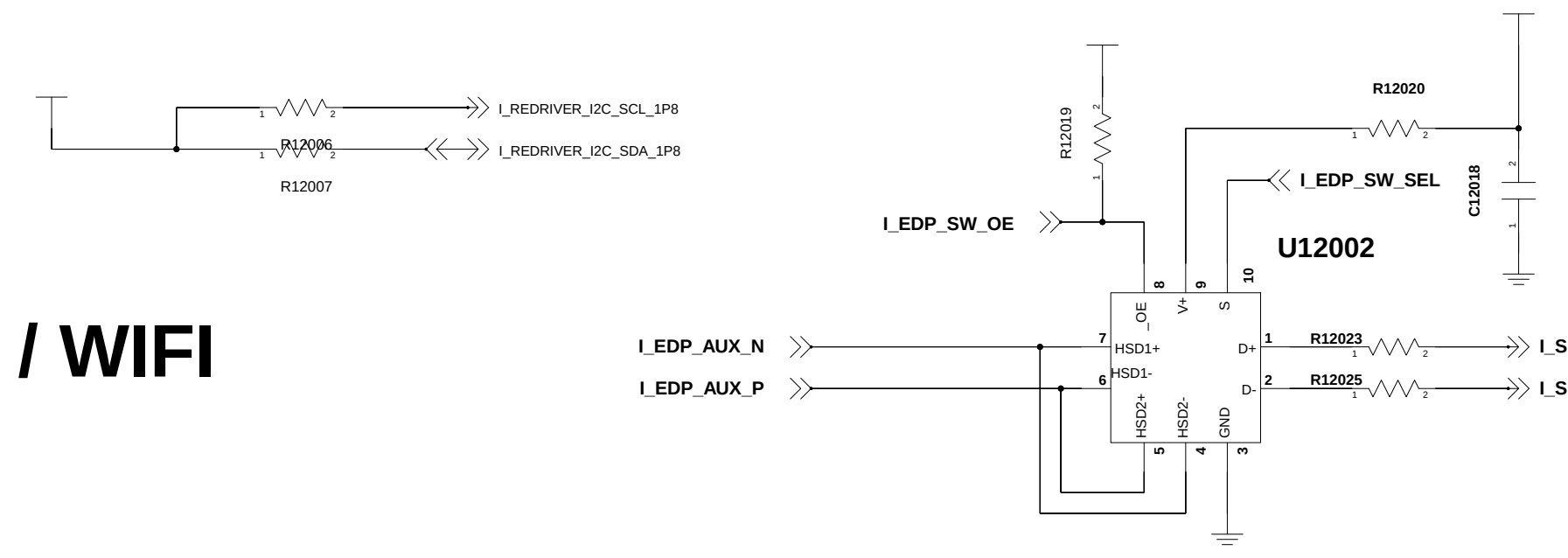
USB PCB (Ext.)


SOC12000
I_SBU1
I_SBU2
I_BTP_RST_N
I_BTP_SPI_MISO
I_BTP_SPI_MOSI
I_BTP_SPI_SCLK
I_BTP_SPI_CS_N
I_USB3.0_SSTX1_P_AP
I_USB3.0_SSTX1_M_AP
I_USB3.0_SSRX1_P_AP
I_USB3.0_SSRX1_M_AP
I_EDP_AUX_P
I_EDP_AUX_N
I_MUIC_D_P
I_MUIC_D_N
I_REDRIVER_I2C_SDA_1P8
I_REDRIVER_I2C_SCL_1P8
BAT_THM
GRIP_INT_4
I_SUB_PCB_ID_GND
I_SUB_PCB_ID
GRIP3_SDA_1P8
GRIP3_SCL_1P8
GRIP_INT_3
ENDC_GRIP_LDO_EN
I_EDP_SW_SEL
I_EDP_SW_OE
I_USB3.0_SSTX0_P_AP
I_USB3.0_SSTX0_M_AP
I_USB3.0_SSRX0_P_AP
I_USB3.0_SSRX0_M_AP
SDM_RFFE1_CLK
SDM_RFFE1_DATA
I_SPK_BL_OUT_SENSE_P
I_SPK_BL_OUT_SENSE_N
I_SPK_BL_OUT_P
I_SPK_BL_OUT_N
C12009
C12011
C12013
C12015
C12010
C12012
C12014
C12016
C12019
D12000
C12084
D12001
C12085
C12020
D12002
C12021
C12022
C12023
CN12000

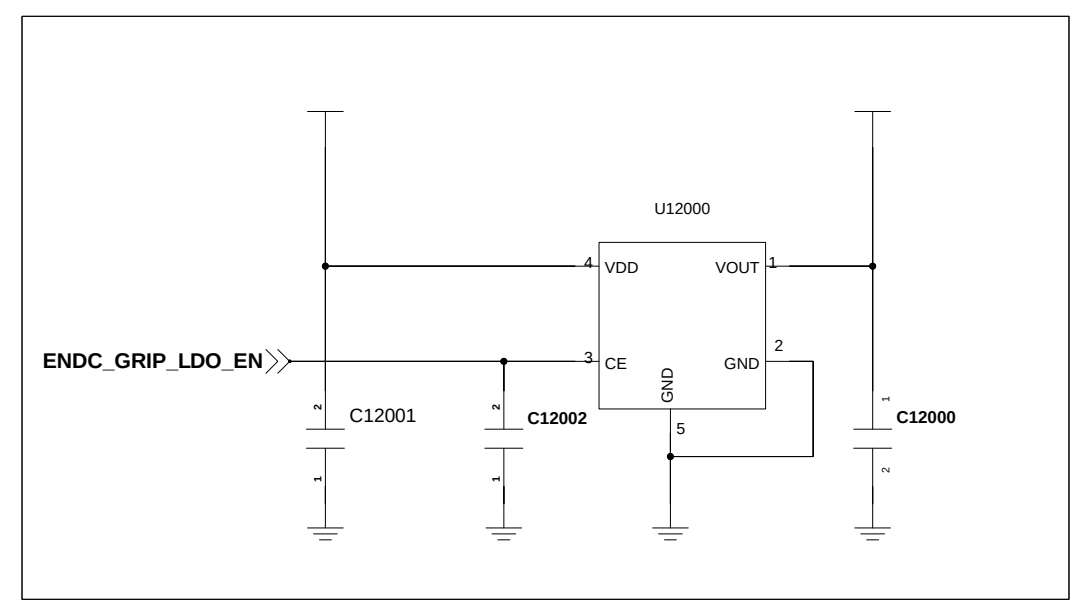
I/F CONN (USB)


R12001
R12005
I_SUB_PCB_ID
I_SUB_PCB_ID_GND

mmW,Sub6 / WIFI

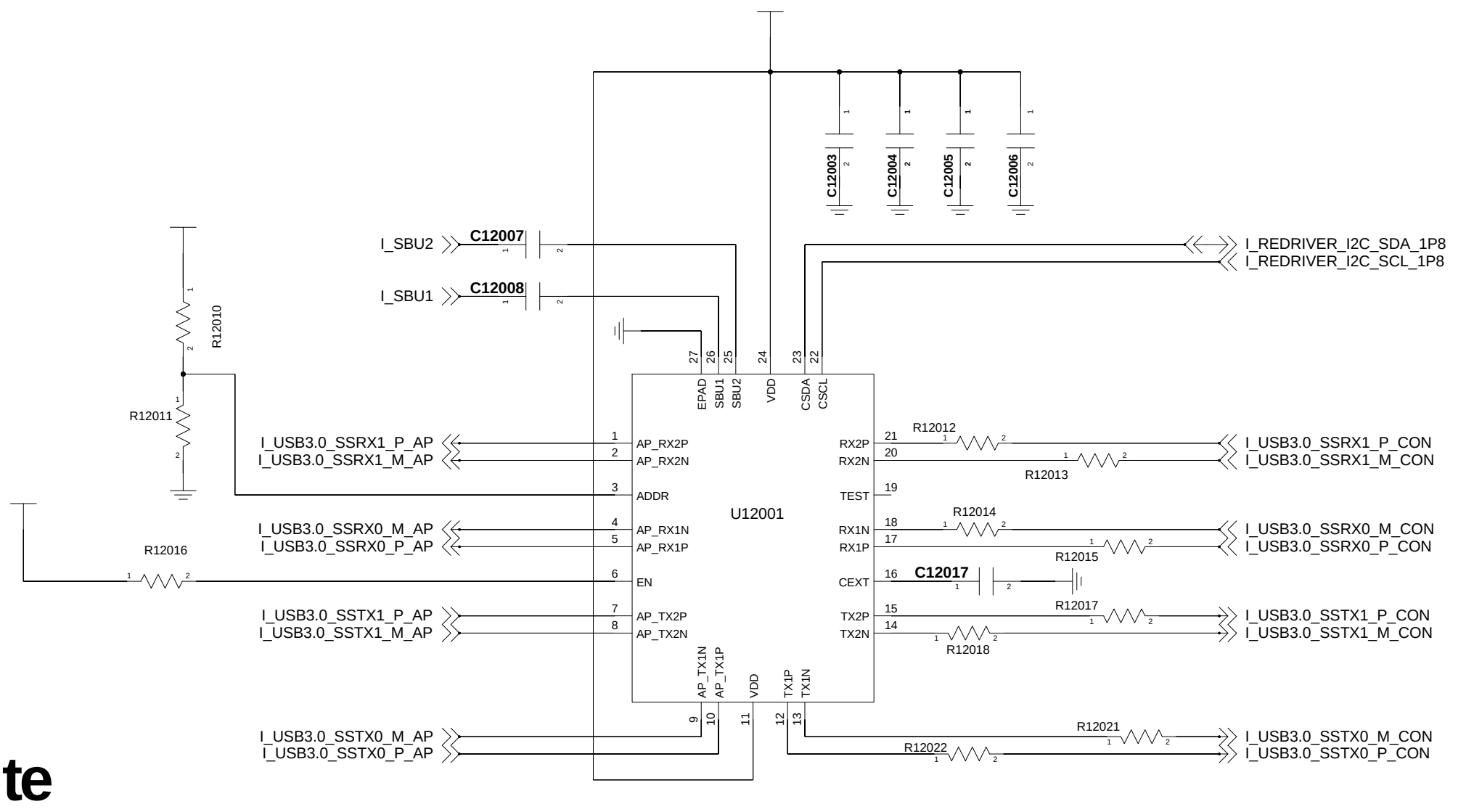

R12006
R12007
I_REDRIVER_I2C_SCL_1P8
I_REDRIVER_I2C_SDA_1P8
R12019
R12020
C12018
I_EDP_SW_SEL
I_EDP_SW_OE
U12002
I_EDP_AUX_N
I_EDP_AUX_P
HSD1+
HSD1-
HSD2+
HSD2-
OE
V+
S
D+
D-
GND
R12023
R12025
I_SBU1
I_SBU2

DP Switch


U12000
VDD
VOUT
CE
GND
ENDC_GRIP_LDO_EN
C12001
C12002
C12000

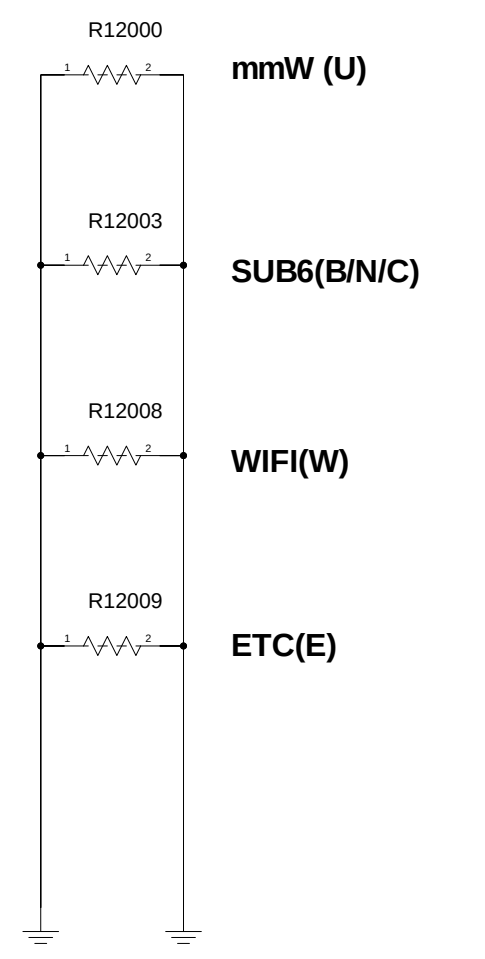
WIFI delete

GRIP3 LDO


C12003
C12004
C12005
C12006
I_SBU2
C12007
I_SBU1
C12008
I_REDRIVER_I2C_SDA_1P8
I_REDRIVER_I2C_SCL_1P8
R12010
R12011
R12016
U12001
EPAD
SBU1
SBU2
VDD
CSDA
CSCL
AP_RX2P
AP_RX2N
ADDR
AP_RX1N
AP_RX1P
EN
AP_TX2P
AP_TX2N
AP_TX1N
AP_TX1P
TX1P
TX1N
RX2P
RX2N
TEST
RX1N
RX1P
CEXT
TX2P
TX2N
C12017
R12012
R12013
R12014
R12015
R12017
R12018
R12021
R12022
I_USB3.0_SSRX1_P_AP
I_USB3.0_SSRX1_M_AP
I_USB3.0_SSRX0_M_AP
I_USB3.0_SSRX0_P_AP
I_USB3.0_SSTX1_P_AP
I_USB3.0_SSTX1_M_AP
I_USB3.0_SSTX0_M_AP
I_USB3.0_SSTX0_P_AP
I_USB3.0_SSRX1_P_CON
I_USB3.0_SSRX1_M_CON
I_USB3.0_SSRX0_M_CON
I_USB3.0_SSRX0_P_CON
I_USB3.0_SSTX1_P_CON
I_USB3.0_SSTX1_M_CON
I_USB3.0_SSTX0_M_CON
I_USB3.0_SSTX0_P_CON

REDRIVER & DP

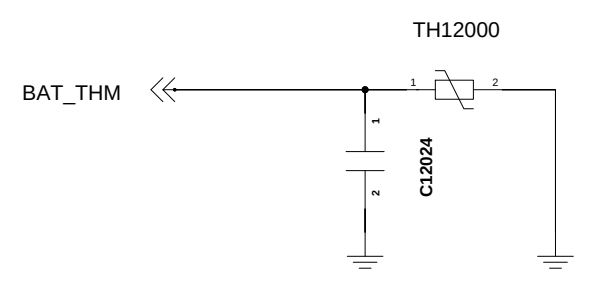
USB PCB REGION


R12000
mmW (U)
R12003
SUB6(B/N/C)
R12008
WIFI(W)
R12009
ETC(E)

USB PCB	SHORT
mmW(U)	R12000
SUB6(B/N/C)	R12003
WIFI(W)	R12008
ETC(E)	R12009

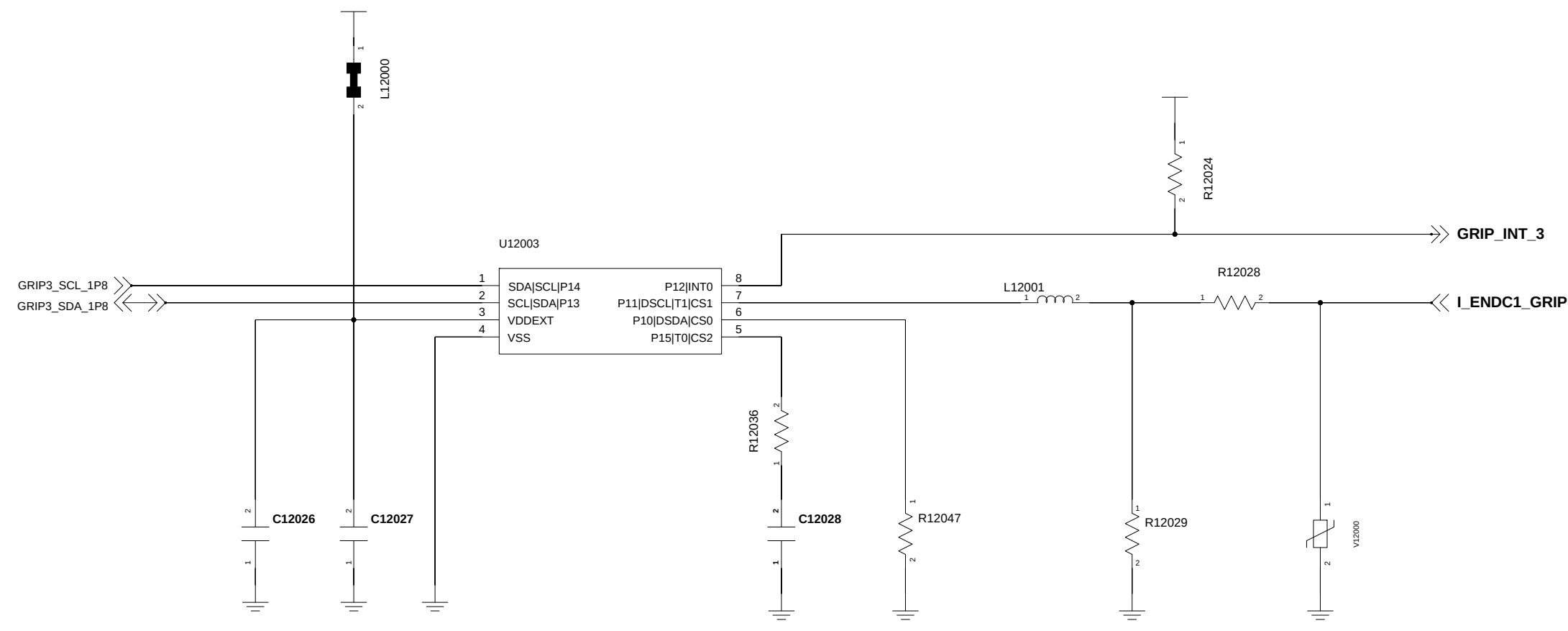
< USB PCB - SMD F/T >

1. VREG_L15B_1P8
2. BAT_THM
3. SDM_RFFE1_CLK
4. SDM_RFFE1_DATA
5. GND
6. VREG_BOB1_3P3
7. GRIP3_SCL_1P8
8. GRIP3_SDA_1P8
9. ENDC_GRIP_LDO_EN

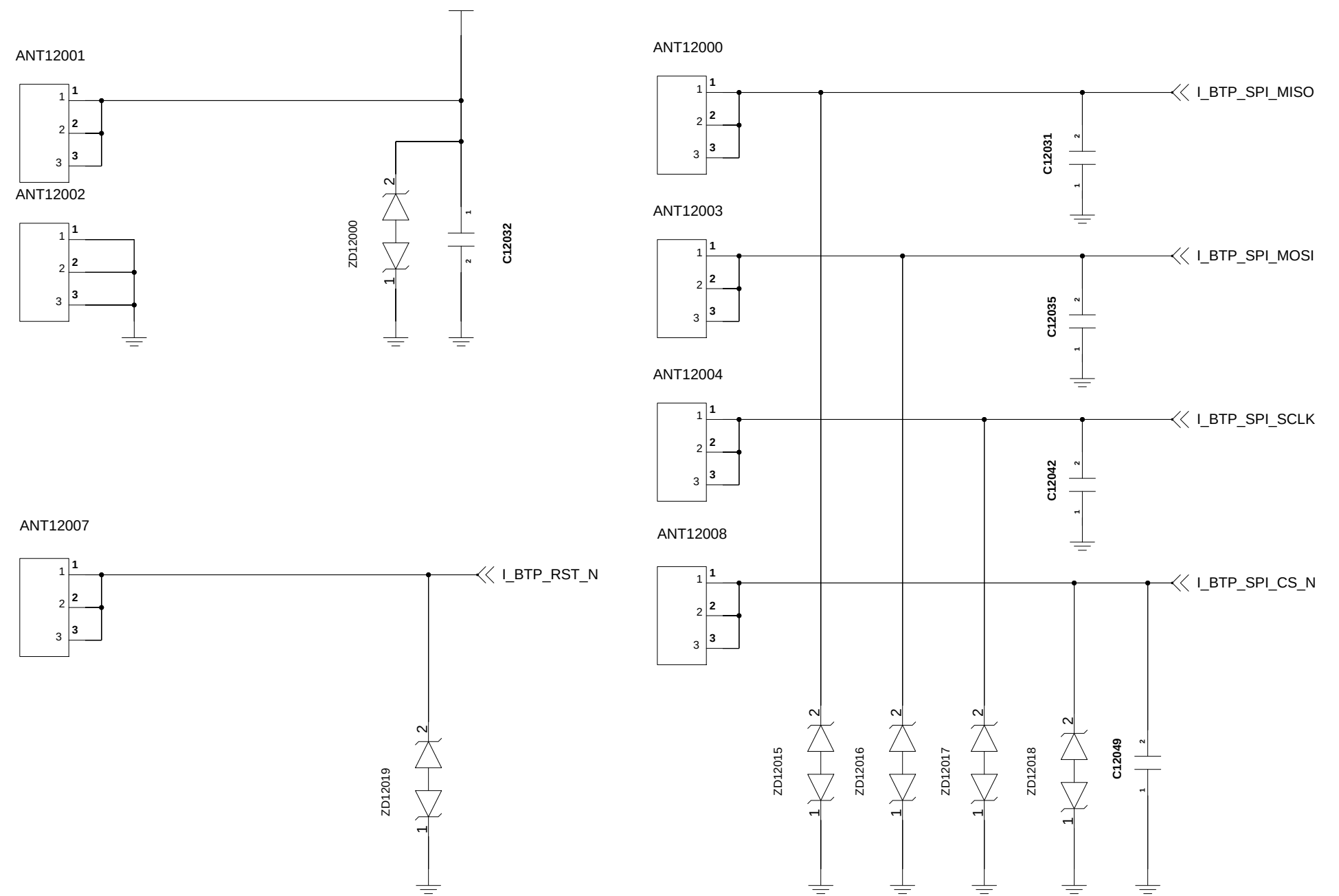

BAT_THM
TH12000
C12024

BAT THM

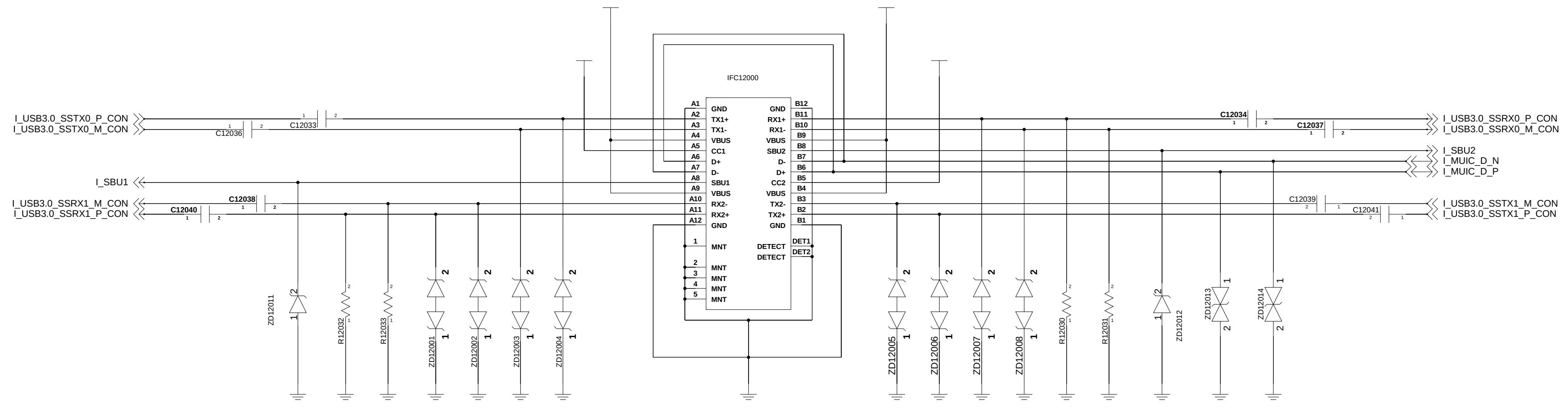
WIFI : Delete


L12000
R12024
U12003
GRIP3_SCL_1P8
GRIP3_SDA_1P8
SDA|SCL|P14
SCL|SDA|P13
VDDEXT
VSS
P12|INT0
P11|DSCL|T1|CS1
P10|DSDA|CS0
P15|T0|CS2
GRIP_INT_3
L12001
R12028
I_ENDC1_GRIP
R12036
C12026
C12027
C12028
R12047
R12029
V12000

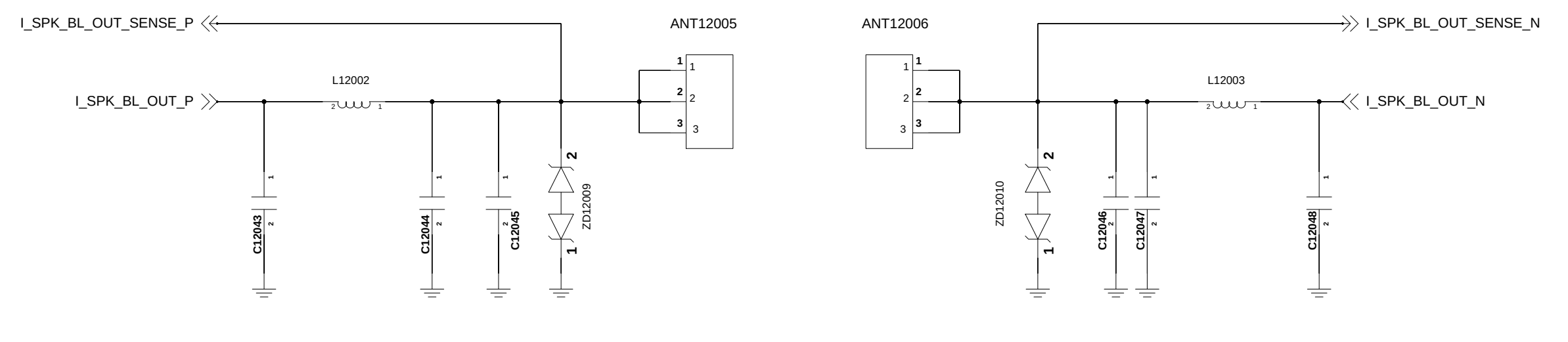
ENDC_GRIP_SENSOR


ANT12001
ANT12002
ZD12000
C12032
ANT12007
I_BTP_RST_N
ZD12019
ANT12000
ANT12003
ANT12004
ANT12008
I_BTP_SPI_MISO
I_BTP_SPI_MOSI
I_BTP_SPI_SCLK
I_BTP_SPI_CS_N
C12031
C12035
C12042
C12049
ZD12015
ZD12016
ZD12017
ZD12018

FOD CON.

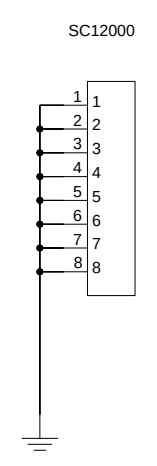

IFC12000
I_USB3.0_SSTX0_P_CON
I_USB3.0_SSTX0_M_CON
C12036
C12033
I_SBU1
I_USB3.0_SSRX1_M_CON
I_USB3.0_SSRX1_P_CON
C12038
C12040
GND
TX1+
TX1-
VBUS
CC1
D+
D-
SBU1
RX2-
RX2+
MNT
RX1+
RX1-
SBU2
CC2
TX2-
TX2+
DETECT
DET1
DET2
C12034
C12037
C12039
C12041
I_USB3.0_SSRX0_P_CON
I_USB3.0_SSRX0_M_CON
I_SBU2
I_MUIC_D_N
I_MUIC_D_P
I_USB3.0_SSTX1_M_CON
I_USB3.0_SSTX1_P_CON
ZD12011
R12032
R12033
ZD12001
ZD12002
ZD12003
ZD12004
ZD12005
ZD12006
ZD12007
ZD12008
R12030
R12031
ZD12012
ZD12013
ZD12014

Type-C Conn.


I_SPK_BL_OUT_SENSE_P
I_SPK_BL_OUT_P
L12002
ANT12005
ANT12006
I_SPK_BL_OUT_SENSE_N
I_SPK_BL_OUT_N
L12003
C12043
C12044
C12045
ZD12009
ZD12010
C12046
C12047
C12048

SPK BL CON.

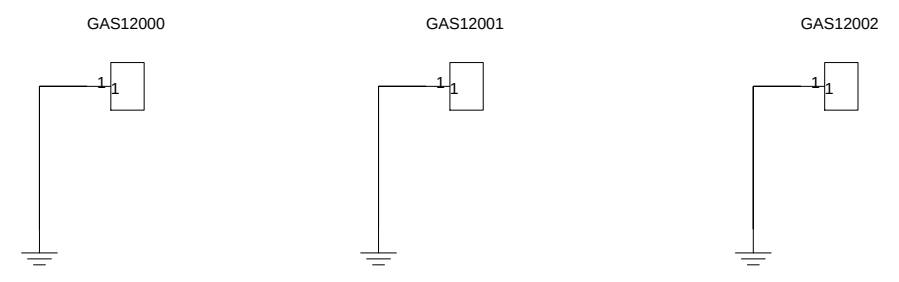
SHIELD CAN



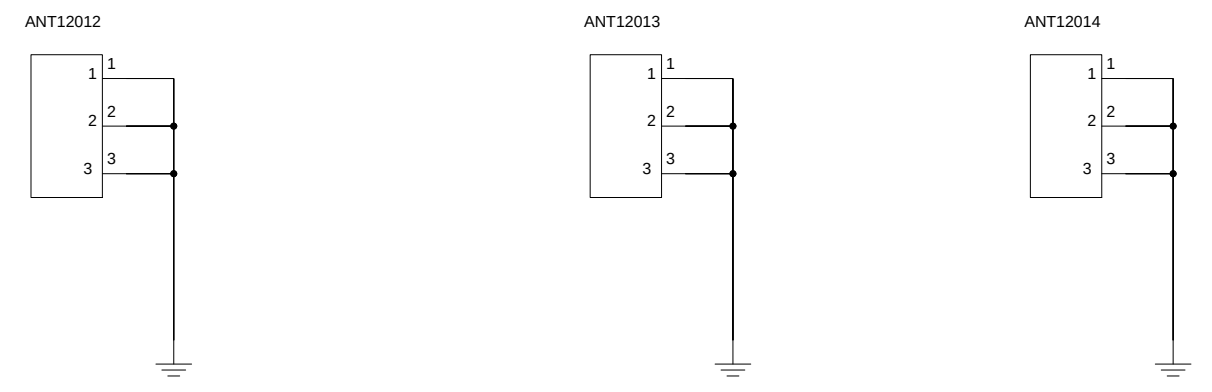
Re-Driver

< BOTTOM S/C >

BOTHHOLE/CONTACT

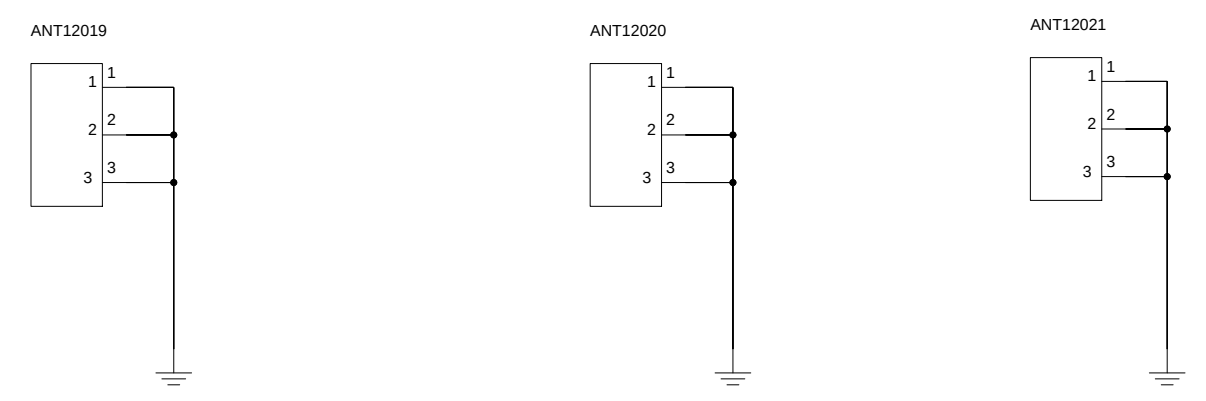

GAS12000
GAS12001
GAS12002

Front CONTACT (JSM 1.3T)


ANT12012
ANT12013
ANT12014

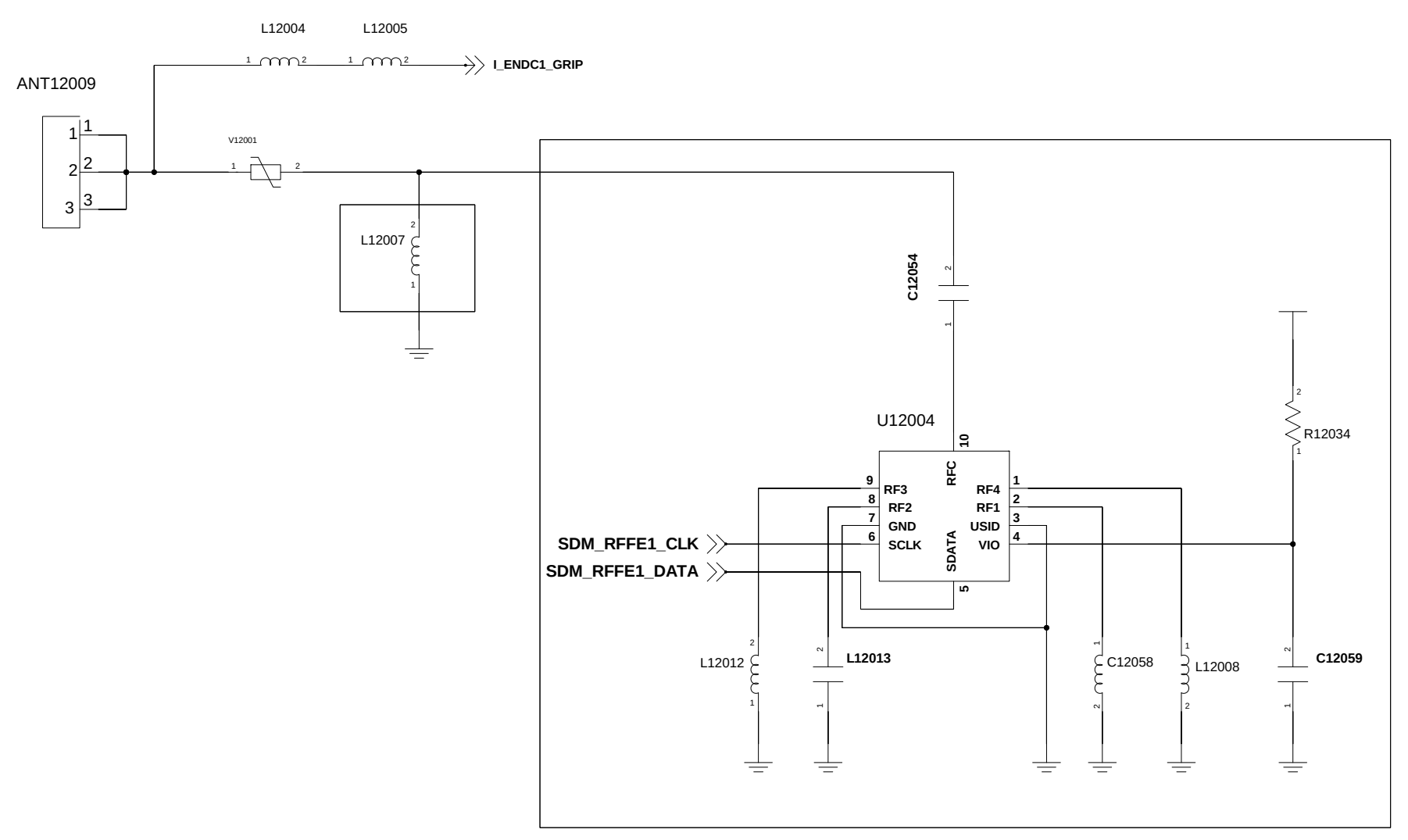
ANT12015
ANT12022

REAR CONTACT (1.5mm)


ANT12019
ANT12020
ANT12021

REAR CONTACT (1.3mm)

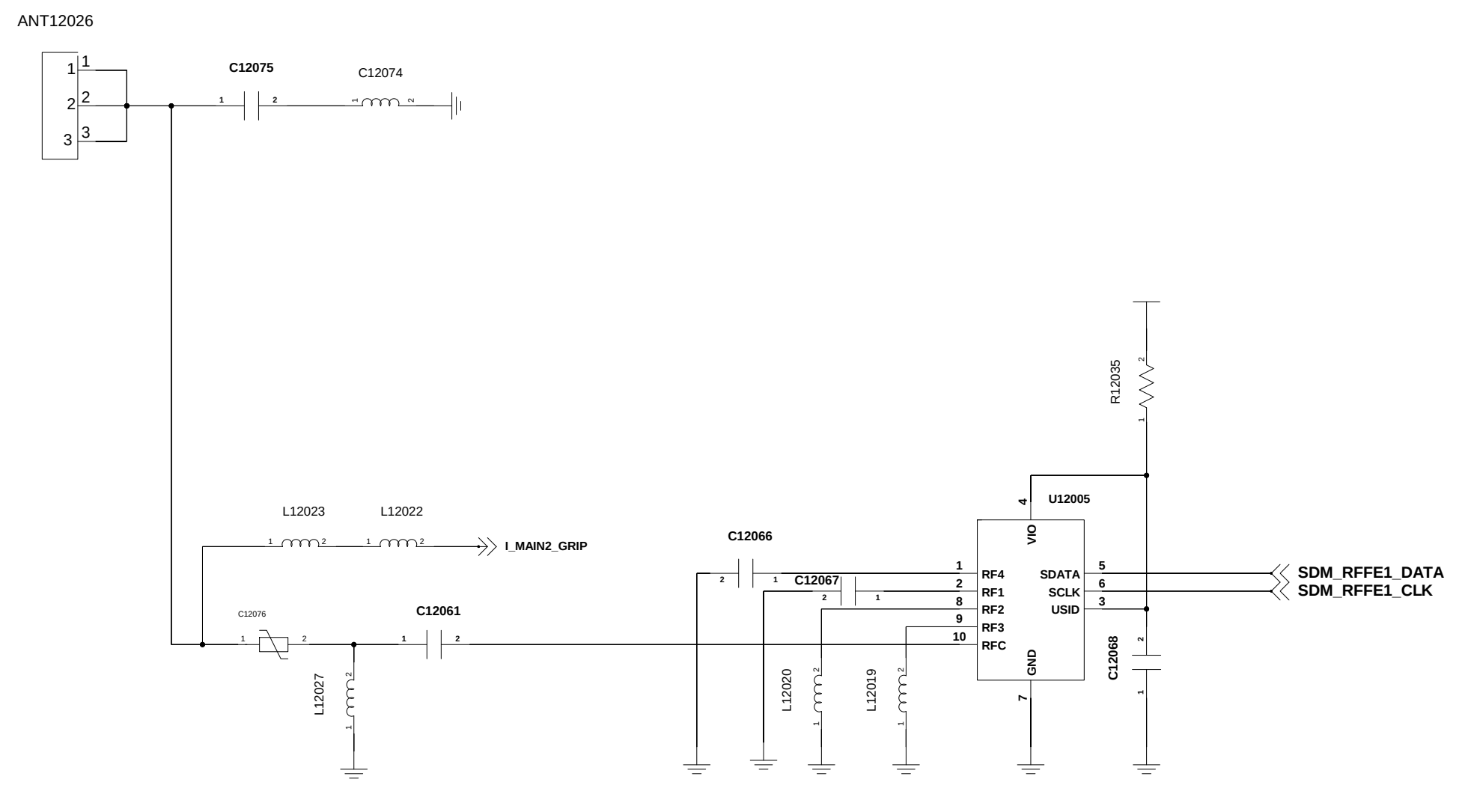
WIFI : Delete


ANT12009
L12004
L12005
I_ENDC1_GRIP
V12001
L12007
C12054
U12004
RF3
RF2
GND
SCLK
RFC
SDATA
RF4
RF1
USID
VIO
SDM_RFFE1_CLK
SDM_RFFE1_DATA
R12034
L12012
L12013
C12058
L12008
C12059

Sub6 delete

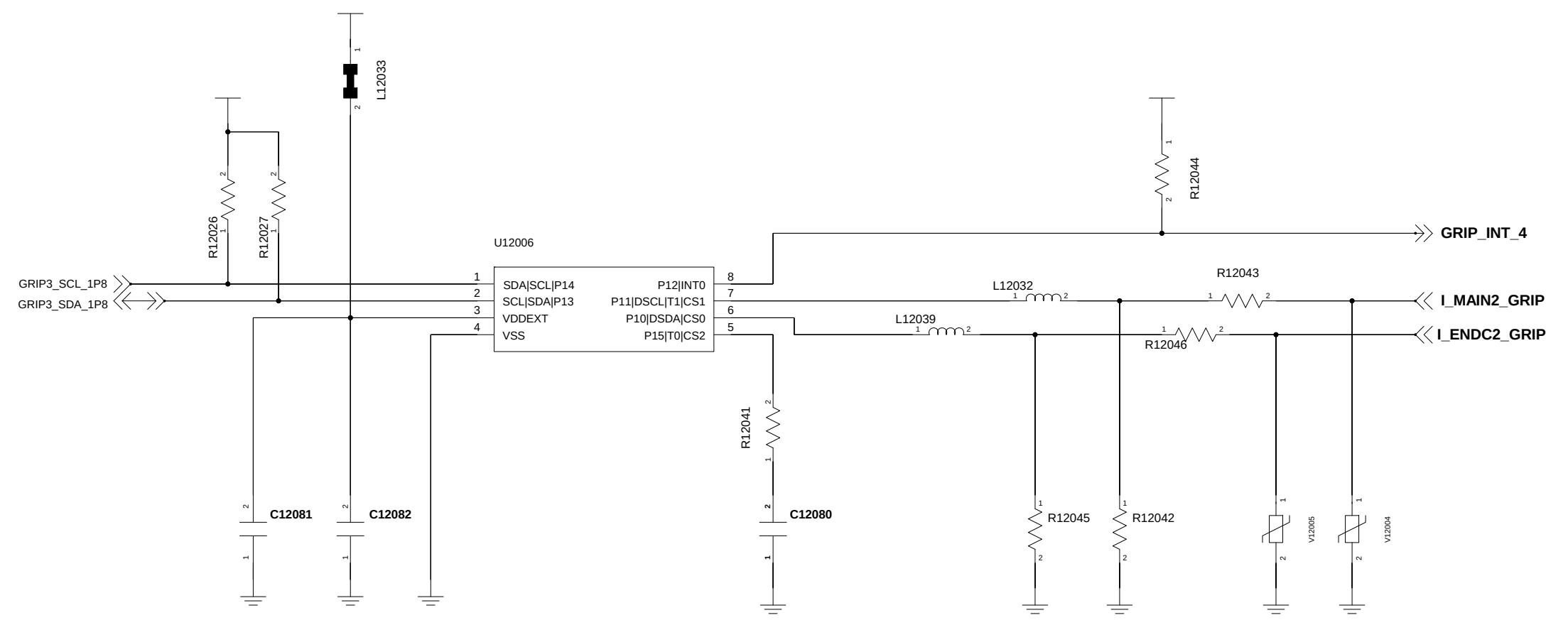
SUB1 ANT

(DRX_L/M/HB, ENDC_L/M/HB, N41 SRS)

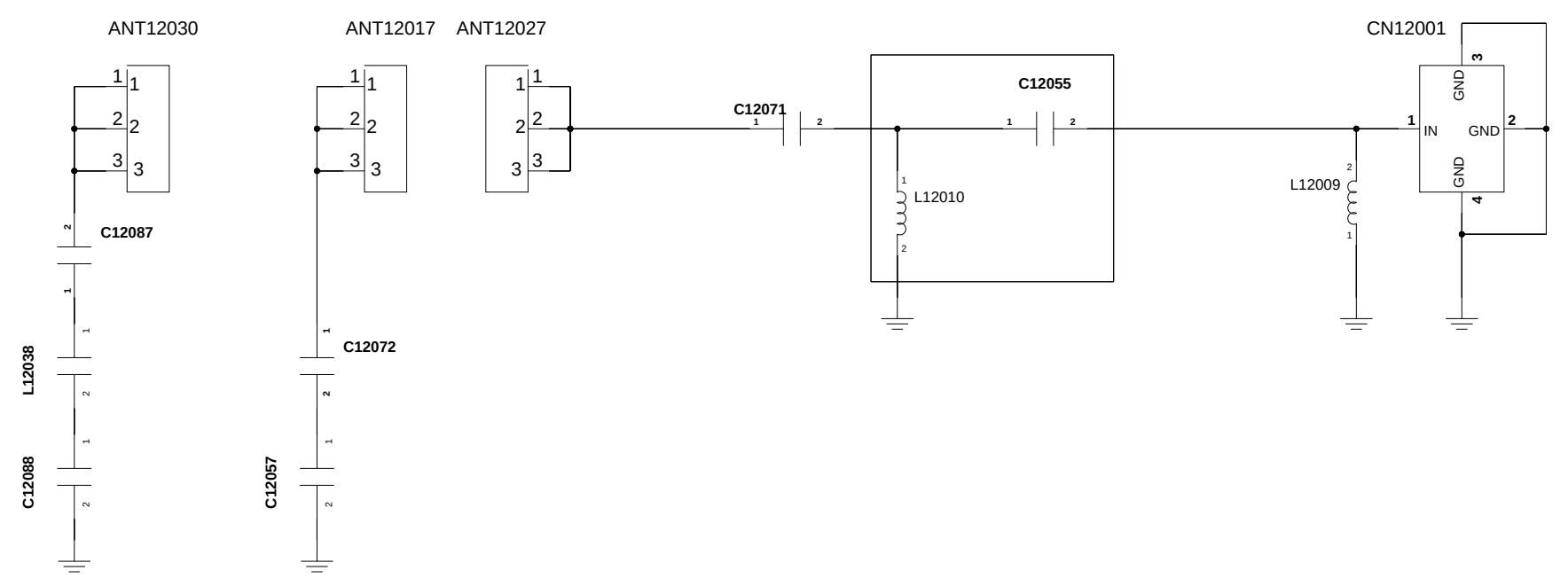

ANT12026
C12075
C12074
L12023
L12022
I_MAIN2_GRIP
C12076
C12061
L12027
C12066
C12067
L12020
L12019
U12005
RF4
RF1
RF2
RF3
RFC
VIO
GND
SDATA
SCLK
USID
R12035
C12068
SDM_RFFE1_DATA
SDM_RFFE1_CLK

MAIN2 ANT XGND SW / GRIP

(N77, N78, B48 TRX)

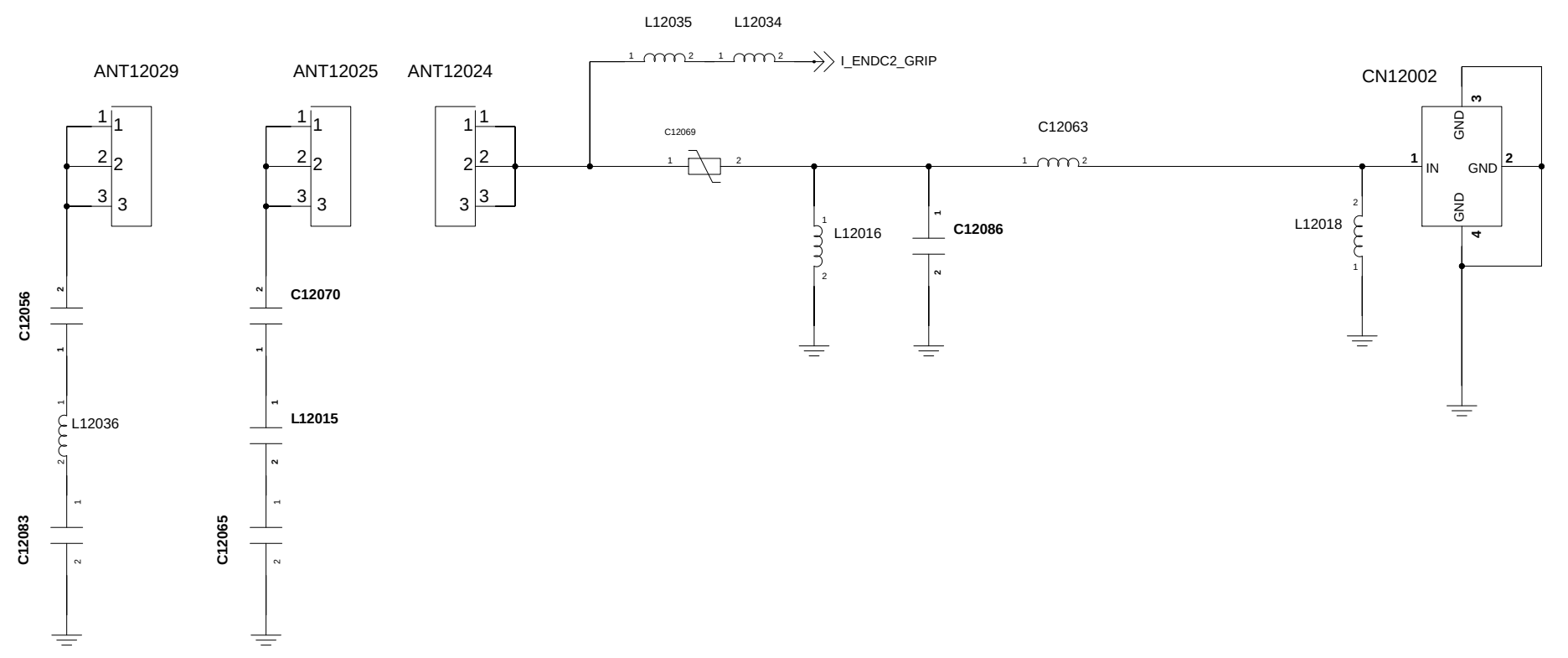

L12033
R12026
R12027
U12006
GRIP3_SCL_1P8
GRIP3_SDA_1P8
SDA|SCL|P14
SCL|SDA|P13
VDDEXT
VSS
P12|INT0
P11|DSCL|T1|CS1
P10|DSDA|CS0
P15|T0|CS2
R12044
GRIP_INT_4
L12032
R12043
I_MAIN2_GRIP
L12039
R12046
I_ENDC2_GRIP
R12041
C12081
C12082
C12080
R12045
R12042
V12005
V12004

MAIN2_GRIP_SENSOR


ANT12030
ANT12017
ANT12027
C12087
L12038
C12088
C12072
C12057
C12071
C12055
L12010
CN12001
L12009
IN
GND

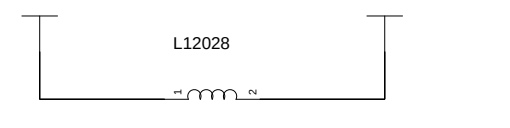
SUB2 ANT

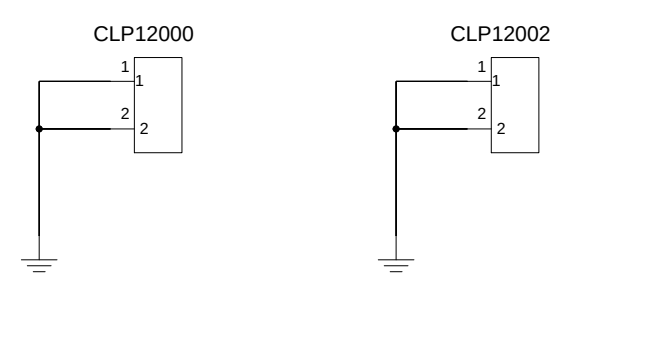
(DRX_MIMO_M/HB, DRX_n77 / n78, N41 SRS)


ANT12029
ANT12025
ANT12024
L12035
L12034
I_ENDC2_GRIP
C12069
C12063
L12016
C12086
CN12002
L12018
IN
GND
C12056
L12036
C12083
C12070
L12015
C12065

SUB4 ANT

(PRX_MIMO_MB / HB, DRX_MIMO_n77 / n78)



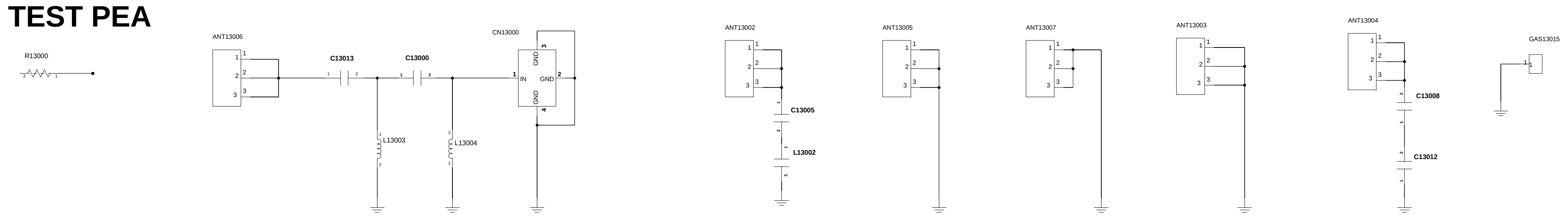

CLP12000
CLP12002

Cable CLIP

TITLE TabS7P
SIZE A1
DWG NO
REV
DRAWN BY User Name
SHEET 12 of 15
Date

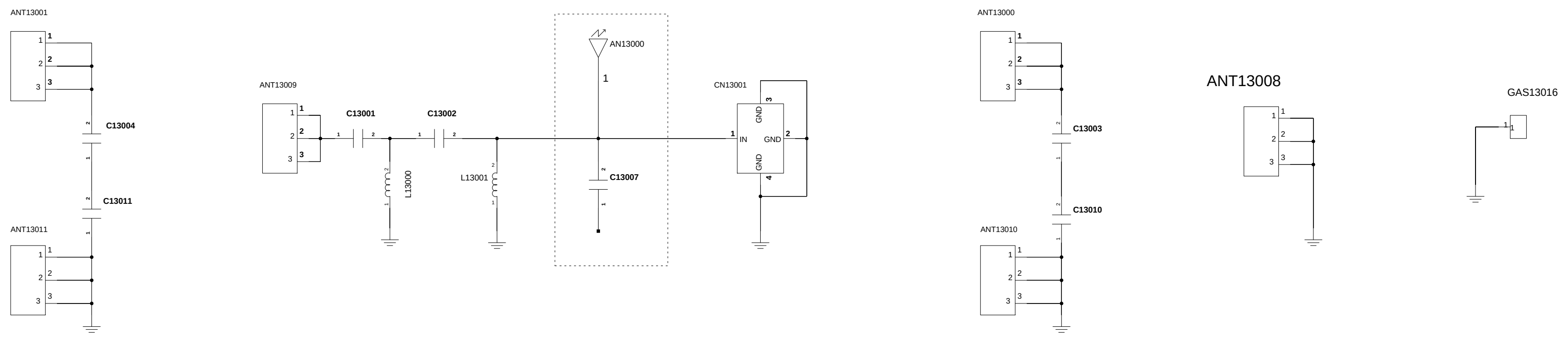
WIFI TL PCB (Ext.)

5G TEST PEA

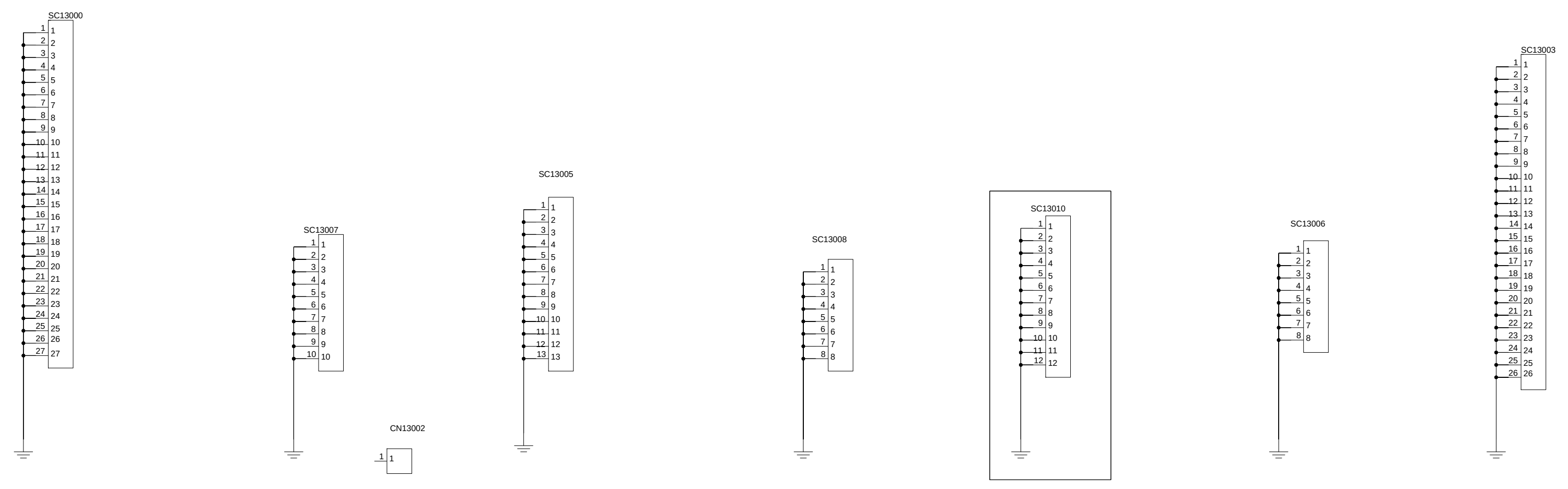

R13000
ANT13006
C13013
C13000
CN13000
IN
GND
GND
GND
L13003
L13004
ANT13002
C13005
L13002
ANT13005
ANT13007
ANT13003
ANT13004
C13008
C13012
GAS13015

WIFI TR PCB (Ext.)

PEA ANT


ANT13001
C13004
C13011
ANT13011
ANT13009
C13001
C13002
L13000
L13001
AN13000
C13007
CN13001
IN
GND
GND
GND
ANT13000
C13003
C13010
ANT13010
ANT13008
GAS13016

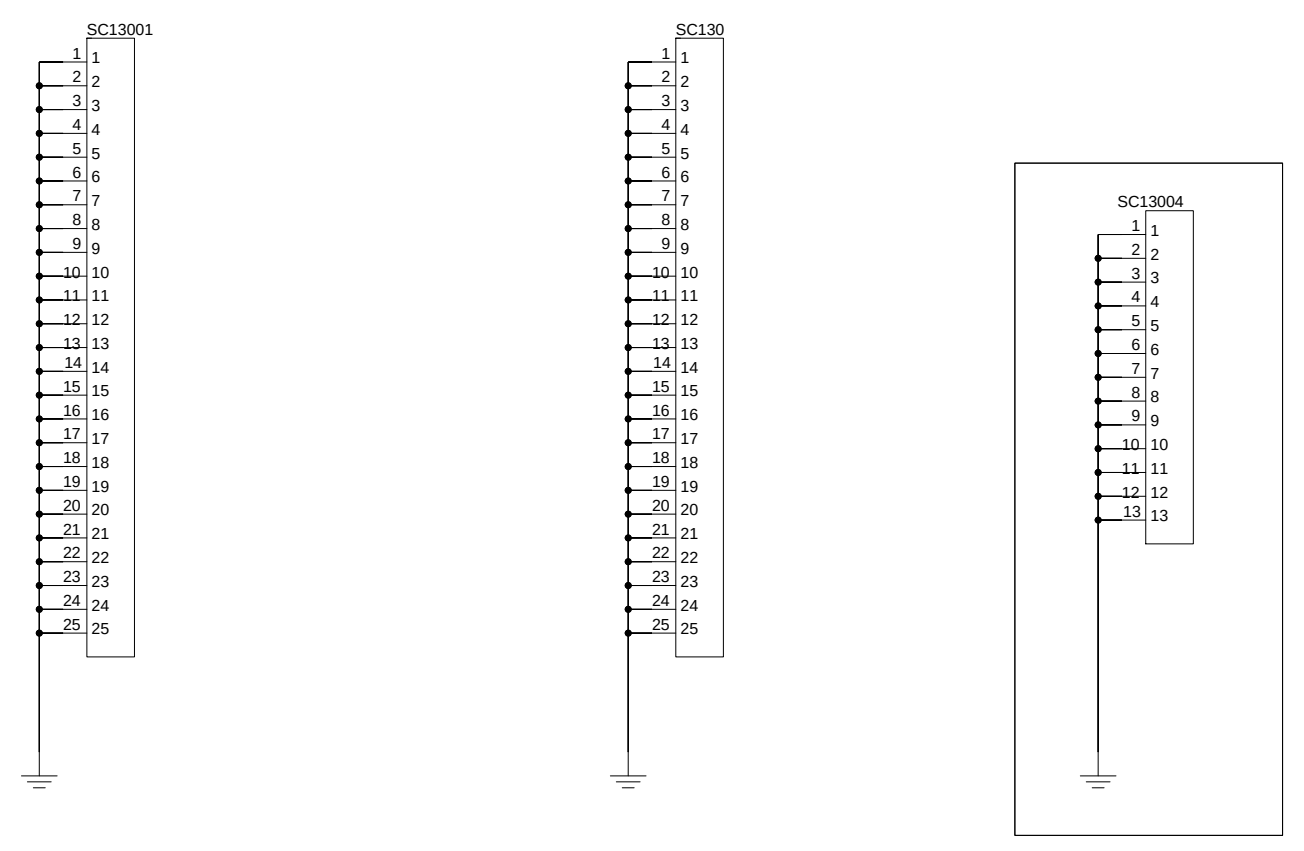
CLIP / SMR / SUS / SHIELDCAN for Main PCB


SC13000
SC13007
CN13002
SC13005
SC13008
SC13010
SC13006
SC13003

RF1 **OSC** **SPK** **UFS** **5G** **OCTA** **OVP**

mmW:ADD
SUB6, WIFI:OPEN

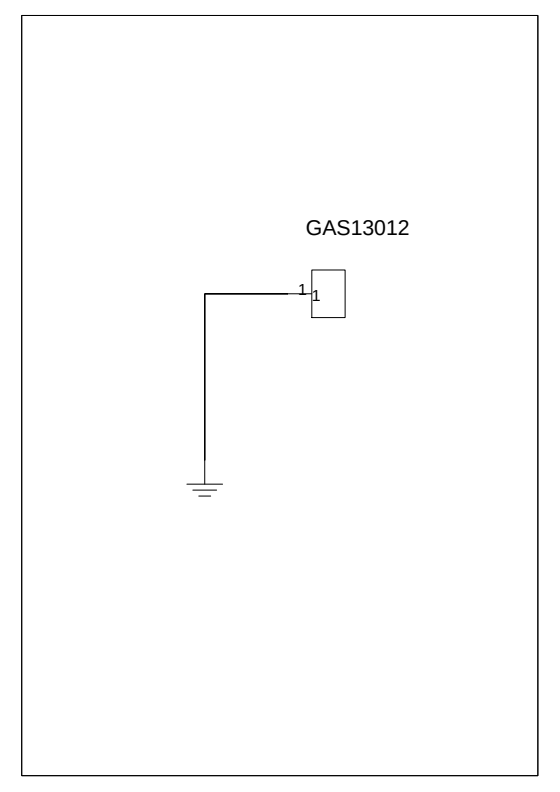
< TOP S/C >


SC13001
SC130
SC13004

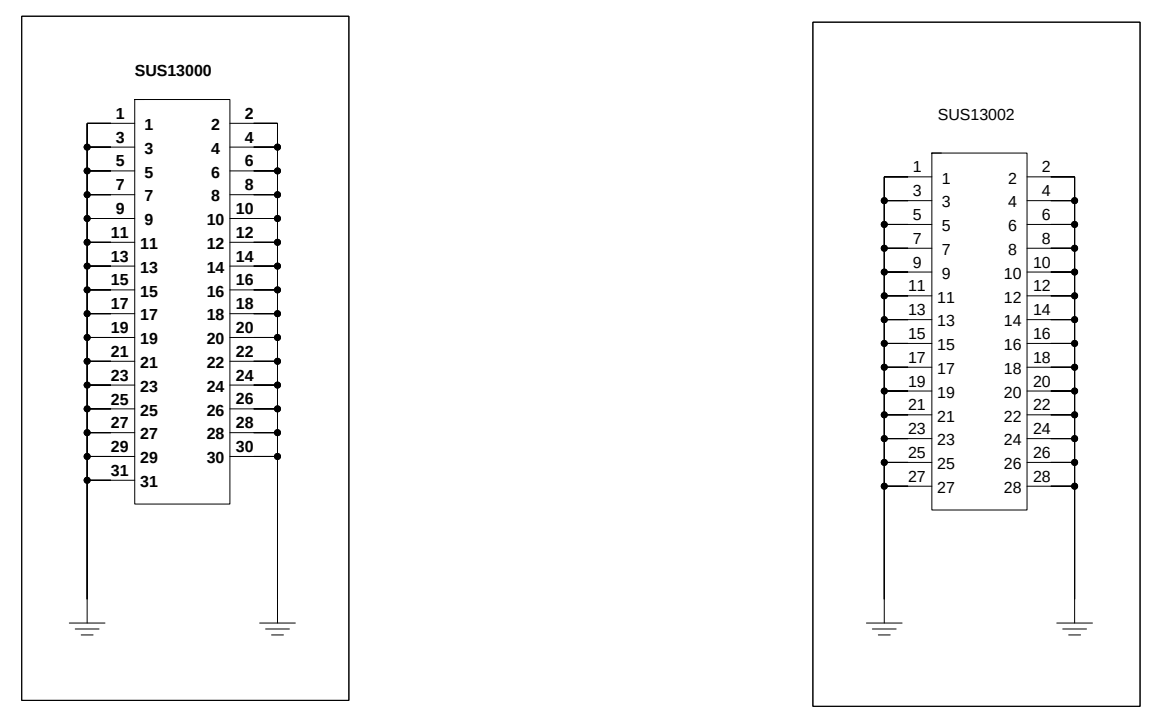
WIFI **AP** **RF2**

mmW, SUB6:ADD
WIFI:OPEN

< BOTTOM S/C >



need for CAM

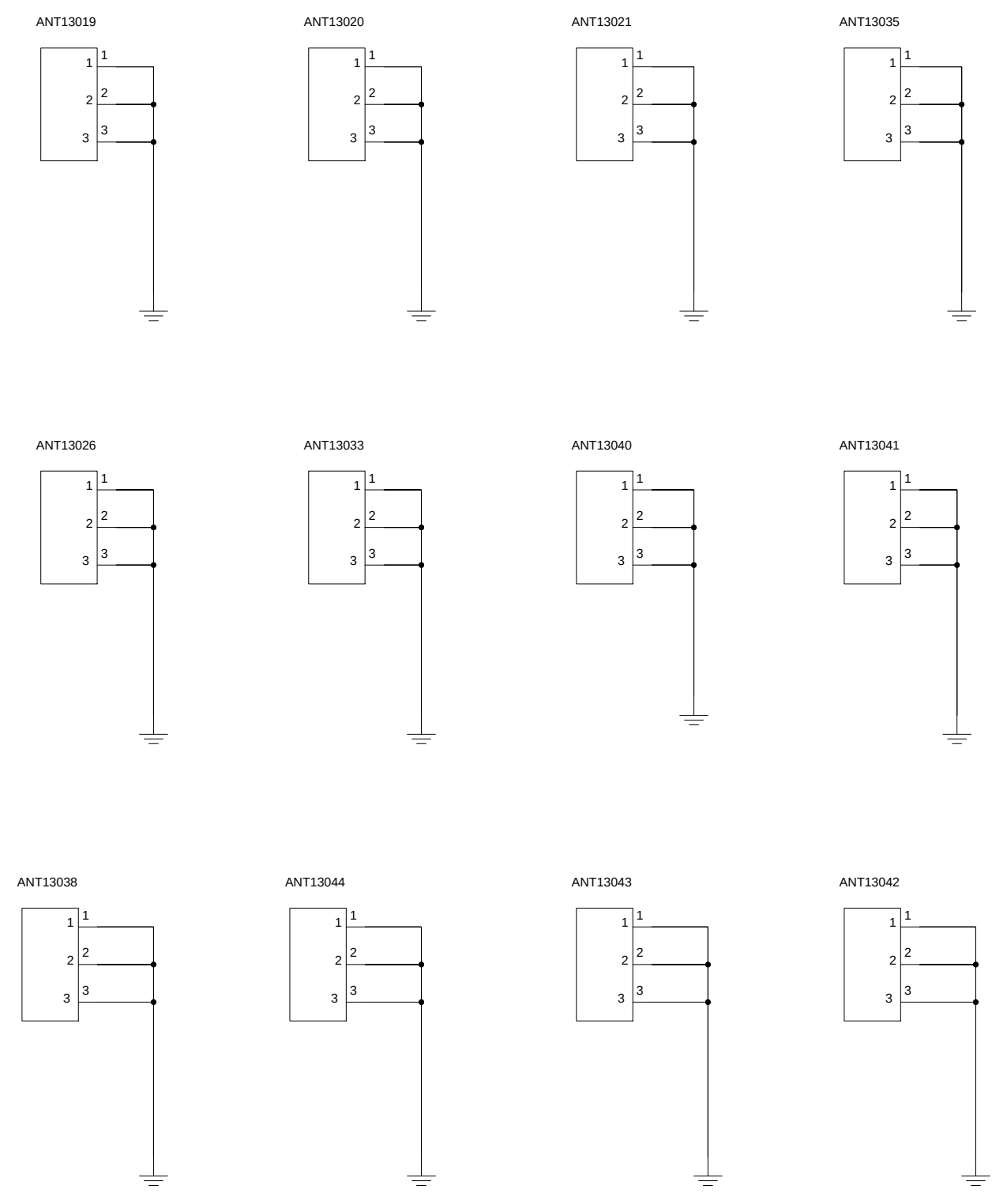

SUS13000
SUS13002

mmWAVE SUS
mmW:ADD
SUB6, WIFI:OPEN

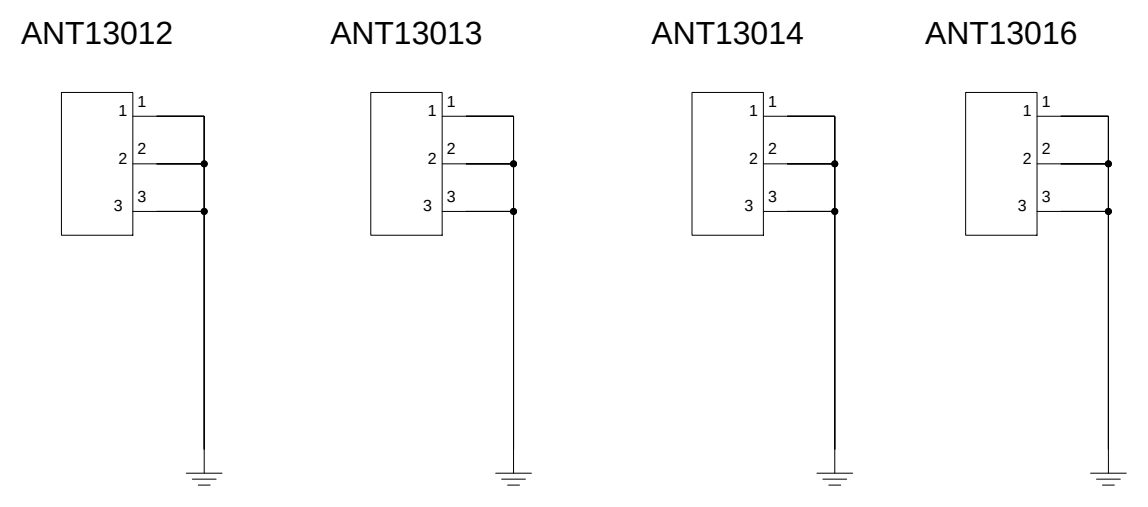
PEN-DECO SUS
mmW:OPEN
SUB6, WIFI:ADD

< TOP SUS >

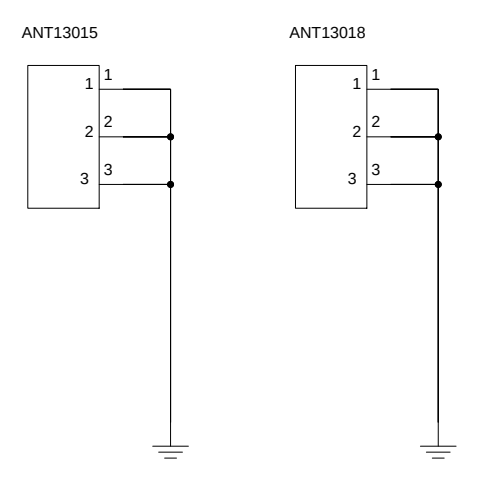
< BOT SUS >


ANT13019
ANT13020
ANT13021
ANT13035
ANT13026
ANT13033
ANT13040
ANT13041
ANT13038
ANT13044
ANT13043
ANT13042

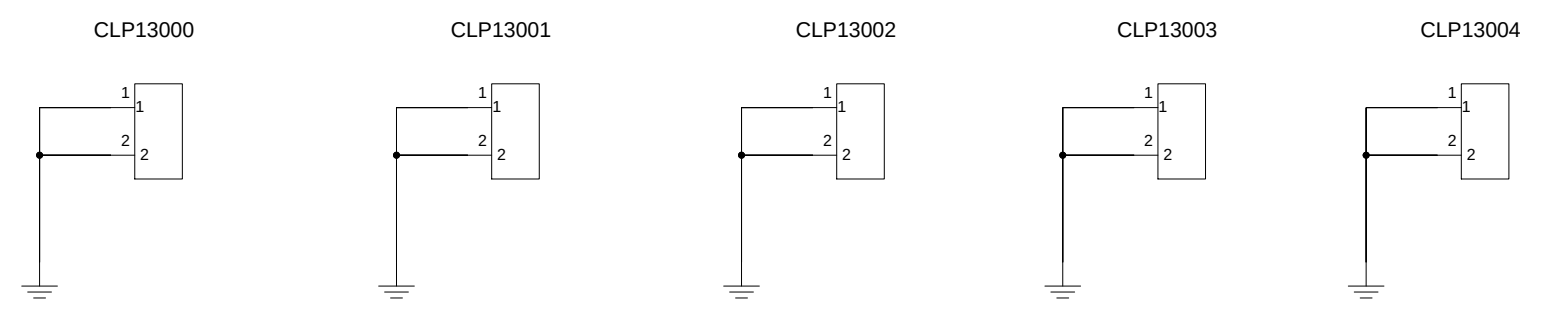
REAR CONTACT (1.5T)


ANT13012
ANT13013
ANT13014
ANT13016

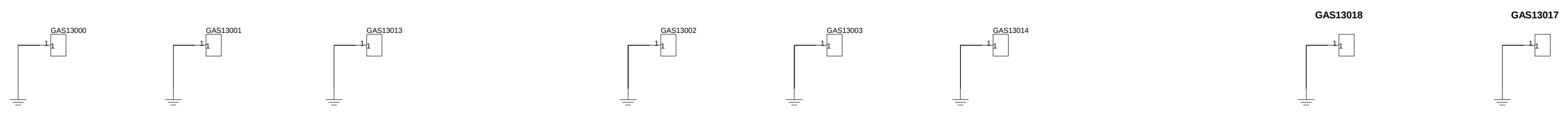
REAR CONTACT (1.1T)


ANT13015
ANT13018

REAR CONTACT (1.5T)

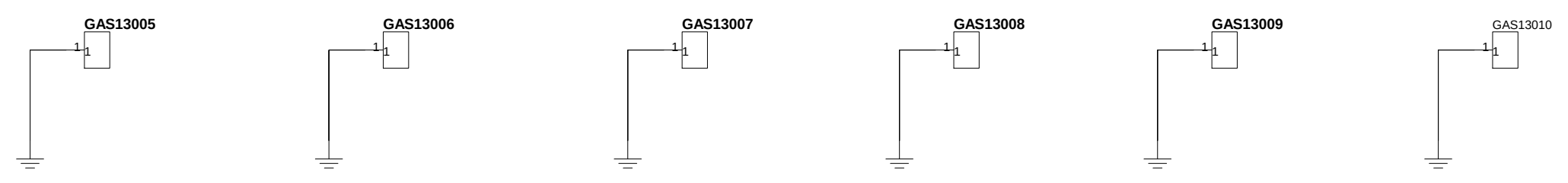

CLP13000
CLP13001
CLP13002
CLP13003
CLP13004

Cable CLIP (1.1T)


GAS13000
GAS13001
GAS13013
GAS13002
GAS13003
GAS13014
GAS13018
GAS13017

REAR CONTACT (SMR 1.4T) **REAR CONTACT (SMR 1.1T-OPEN)** **REAR CONTACT (SMR 0.8T)**

need for Core PCB


GAS13005
GAS13006
GAS13007
GAS13008
GAS13009
GAS13010

Front CONTACT (2.2T)

TITLE TabS7P
SIZE A1 DWG NO REV
DRAWN BY User Name
SHEET 13 of 15 Date

Sencondary PCB PAD (Main PCB)

PCB14000
(1/2)
PCB14000
(2/2)
PCIE0_TX1_M_WLAN
PCIE0_TX0_M_WLAN
PCIE0_REFCLK_M_WLAN
PCIE0_RX1_M_WLAN
PCIE0_RX0_M_WLAN
CC_SCL_1P8
CC_SDA_1P8
6AXIS1_INT2
6AXIS1_INT1
GRIP2_SCL_1P8
BT_UART_RXD
BT_UART_TXD
PCIE0_TX1_P_WLAN
PCIE0_TX0_P_WLAN
PCIE0_REFCLK_P_WLAN
PCIE0_RX1_P_WLAN
PCIE0_RX0_P_WLAN
REDRIVER_SDA_1P8
UWCAM_SDA_1P8
CAM_AF_EEP_SDA_1P8
GRIP2_SDA_1P8
WCN_SW_CTRL
BT_EN
WCN_SW_CTRL_WL_CX
UW_VTCAM_SDA_1P8
PMIC_SPMI_DATA
PMIC_SPMI_CLK
UWCAM_MIPI_D3_P
UWCAM_MIPI_D3_N
SDC2_CLK
REDRIVER_SCL_1P8
UWCAM_SCL_1P8
MAINCAM_SDA_1P8
MAINCAM_RST_N
BT_UART_RTS
BT_UART_CTS
CC_DIR
M_RF_ID_1
BAT_ID
SLEEP_CLK
WLAN_BT_SLIMBUS_CLK
WLAN_BT_SLIMBUS_DATA
UWCAM_MIPI_D2_P
UWCAM_MIPI_D2_N
MAINCAM_SCL_1P8
GRIP_INT_3
CAM_AF_EEP_SCL_1P8
MCU_SCL_1P8
UW_VTCAM_SCL_1P8
HALL_INT
CABLE_DET_GPS
M_RF_ID_2
M_RF_ID_0
3RD_MIC_CLK
3RD_MIC_DATA
UWCAM_MIPI_D1_P
UWCAM_MIPI_D1_N
UW_VTCAM_MCLK
UW_VTCAM_MIPI_D1_N
UW_VTCAM_MIPI_CLK_N
UW_VTCAM_MIPI_D0_N
UW_VTCAM_MIPI_D3_P
UW_VTCAM_MIPI_D3_N
VTCAM_MIPI_CLK_N
VTCAM_MIPI_D0_N
VTCAM_MIPI_D1_N
VTCAM_MIPI_D2_N
VTCAM_MIPI_D3_N
SDC2_D1
SDC2_CMD
SDC2_D2
CABLE_DET_MAIN2
PEN_IRQ
DSI0_NC_LN3_M
DSI0_C2_LN3_P
DSI0_A2_LN2_P
FAULT_N
DSI0_A1_LN1_M
DSI0_B1_CLK0_P
DSI0_B2_LN2_M
DSI0_C0_LN1_P
DSI0_C1_CLK0_M
DSI0_A0_LN0_P
DSI0_B0_LN0_M
MAINCAM_MIPI_D0_P
MAINCAM_MIPI_D0_N
MAINCAM_MIPI_CLK_N
PMIC_RB
MAINCAM_MIPI_D1_N
MAINCAM_MIPI_D1_P
MAINCAM_MIPI_CLK_P
MAINCAM_MIPI_D3_N
MAINCAM_MIPI_D2_N
MAINCAM_MIPI_D3_P
MAINCAM_MIPI_D2_P
MAINCAM_MCLK
UWCAM_MIPI_CLK_P
UWCAM_MIPI_CLK_N
MAIN_MIC_DATA
MAIN_MIC_CLK
PON_RESET_N
IF_PMIC_SDA_1P8
IF_PMIC_SCL_1P8
PCIE0_RST_N
PCIE0_WAKE_N
CABLE_DET_WIFI0
MCU_SWCLK
MCU_SDA_1P8
WLAN_EN
SECURE_NVM_SCL_1P8
VTCAM_RST_N
SDC2_D3
SDC2_D0
VTCAM_MCLK
VTCAM_MIPI_D3_P
VTCAM_MIPI_D2_P
VTCAM_MIPI_D1_P
VTCAM_MIPI_D0_P
VTCAM_MIPI_CLK_P
UW_VTCAM_MIPI_D2_P
UW_VTCAM_MIPI_D2_N
UW_VTCAM_MIPI_D0_P
UW_VTCAM_MIPI_CLK_P
UW_VTCAM_MIPI_D1_P
UWCAM_MCLK
UWCAM_MIPI_D0_N
UWCAM_MIPI_D0_P
SUB_MIC_CLK
SUB_MIC_DATA
PMK_CLK6_CXO
PCIE0_CLK_REQ_N
KBD_PWR_EN
MCU_nRST
DC_IRQ_N
SECURE_NVM_SDA_1P8
UW_VTCAM_RST_N
UWCAM_RST_N
BTP_RST_N
BTP_SPI_CS_N
BTP_SPI_CLK
BTP_SPI_MOSI
BTP_SPI_MISO
GRIP_INT_1
EDP_AUX_N
USB3.0_SSTX1_M_AP
USB3.0_SSTX0_M_AP
USB3.0_SSRX0_P_AP
USB3.0_SSRX1_P_AP
eUSB_AP_D_P
UW_VTCAM_LDO_EN
EDP_AUX_P
USB3.0_SSTX1_P_AP
USB3.0_SSTX0_P_AP
USB3.0_SSRX0_M_AP
USB3.0_SSRX1_M_AP
eUSB_AP_D_N
QLINK2_EN
QLINK2_REQ
QLINK2_WMSS_RESET_N
FCAM_LDO_EN
JTAG_TDI
QLINK2_L1_P
QLINK2_L3_P
JTAG_TMS
JTAG_TDO
QLINK2_L1_M
QLINK2_L3_M
JTAG_TCK
RCAM2_LDO_EN
SPK_AMP_LDO_EN
JTAG_RST_OUT_N
QLINK2_L0_M
QLINK2_L2_P
KBD_DCDC_SCL_1P8
RCAM_LDO_EN
JTAG_TRST_N
MOTOR_LDO_EN
QLINK2_L0_P
QLINK2_L2_M
KBD_DCDC_SDA_1P8
GRIP3_SCL_1P8
OCTA_DCDC_SCL_1P8
AMP_INT_N
MCU_CONN_INT
MCU_INT
DIRECT_SDA_1P8
GRIP3_SDA_1P8
OCTA_DCDC_SDA_1P8
QLINK2_L4_P
QLINK2_CLK_M
DIRECT_SCL_1P8
AMP_SDA_1P8
AMP_SCL_1P8
QLINK2_L4_M
QLINK2_CLK_P
QLINK1_EN
QLINK1_REQ
QLINK1_WMSS_RESET_N
QLINK0_EN
QLINK0_REQ
QLINK0_WMSS_RESET_N
QLINK1_L0_M
QLINK1_L0_P
QLINK1_L2_M
QLINK1_L2_P
QLINK0_L3_P
QLINK0_L0_P
QLINK0_L3_M
QLINK0_L0_M
QLINK1_L4_P
QLINK1_L4_M
QLINK1_L7_P
QLINK1_L7_M
QLINK0_L1_P
QLINK0_CLK_M
QLINK1_L1_P
QLINK1_L1_M
QLINK1_CLK_P
QLINK1_CLK_M
QLINK0_L1_M
QLINK0_CLK_P
QLINK1_L5_M
QLINK1_L5_P
QLINK1_L3_M
QLINK1_L3_P
QLINK0_L2_P
QLINK0_L4_P
QLINK0_L2_M
QLINK0_L4_M
QLINK1_L8_P
QLINK1_L8_M
QLINK1_L6_P
QLINK1_L6_M
SIM1_EN
BLANKET_THM
SDM_RFFE0_CLK
SDM_RFFE0_DATA
AMP_I2S_LRCK
AMP_I2S_D_IN
AMP_I2S_D_OUT
AMP_I2S_BCLK
DISP_RST
DISP_DET
SIM2_EN
FLASH_LED
FLASH_LED
SDM_RFFE1_CLK
SDM_RFFE1_DATA
OCTA_DCDC_EN
VTCAM_SDA_1P8
VTCAM_SCL_1P8
FLASH_LED
WCN_HOST_SOL_OUT
GRIP_INT_2
HALL_INT_WACOM
SDM_RFFE4_DATA
SDM_RFFE4_CLK
BAT_THM
CHG_THM
SUB_CON_DET
WCN_WLAN_SOL_IN
SUB_PCB_ID
SIM1_DET
SIM2_DET
MCU_ACC_SDA_1P8
MCU_ACC_SCL_1P8
RGB1_SCL_1P8
GRIP1_LDO_EN
VOL_UP
CHG_DET
SDCARD_DET
RGB1_SDA_1P8
RGB2_SCL_1P8
MAGNETIC_SCL_1P8
MAGNETIC_SDA_1P8
UFS0_TX0_P
UFS0_TX1_P
UFS0_RX0_P
UFS0_RX1_P
UFS0_RESET_N
UFS0_TX0_M
UFS0_TX1_M
UFS0_RX0_M
UFS0_RX1_M
CABLE_DET_WIFI1
SIM1_DATA_1P2
SIM1_CLK_1P2
BTP_LDO_EN
SM_PS_HOLD
WLAN_COEX_SDXUART_RX
TSP_SCL_1P8
GPS_L1_LNA_EN
LAA_AS_EN
LAA_EN_0
TSP_SDA_1P8
WLAN_COEX_SDXUART_TX
PEN_LDO_EN
SIM2_DATA_1P2
SIM2_CLK_1P2
ENDC_GRIP_LDO_EN
TSP_INT_N
MCU_ACC_INT
eUSB_SCL_1P8
PEN_SDA_1P8
HWID_LRPST_THM
GRIP2_LDO_EN
AP_UART_TXD
AP_UART_RXD
eUSB_SDA_1P8
OTG_DET
VBUS_DISCHARGING_EN
PEN_SCL_1P8
CAM_IO_LDO_EN
TYPE_C_INT
AMP_RST_N
AP_USB_BOOT
PEN_PDCT
SIM1_RST_1P2
SIM2_RST_1P2
TCON_RDY
PEN_FWEL
DISP_TE
CABLE_DET_SUB1
EDP_SW_OE
CABLE_DET_SUB2
CABLE_DET_SUB4
IF_PMIC_INTB
CABLE_DET_SUB3
GRIP1_SDA_1P8
DISP_CON_DET
HWID_LRPST_THM_SEL
OVP_PWR_FLAGB
EDP_SW_SEL
GRIP1_SCL_1P8
eUSB_RPTR_RST_N
RGB2_SDA_1P8
6AXIS_I3C_SCL_1P8
6AXIS_I3C_SDA_1P8
GRIP_INT_4

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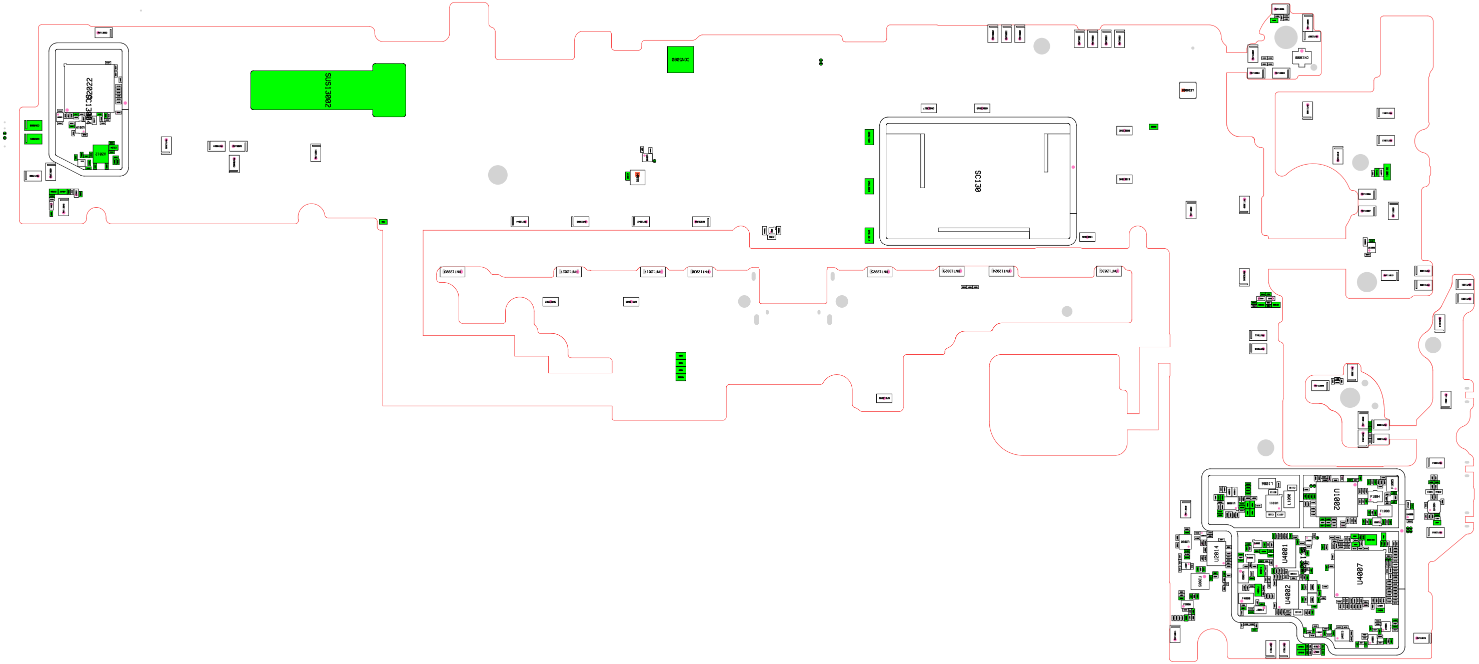
SIZE A1
DWG NO PDM NUMBER
REV Rev
DRAWN BY User Name
SHEET 14 of 15
Date Drawing Date



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U2012
U2019
CON5000
SC130
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ANT12027
ANT12017
ANT12038
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ANT12029
ANT12024
ANT12026
CN13008
U1002
L1086
L1090
U1011
F1004
F1008
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F2006
U4001
U4002
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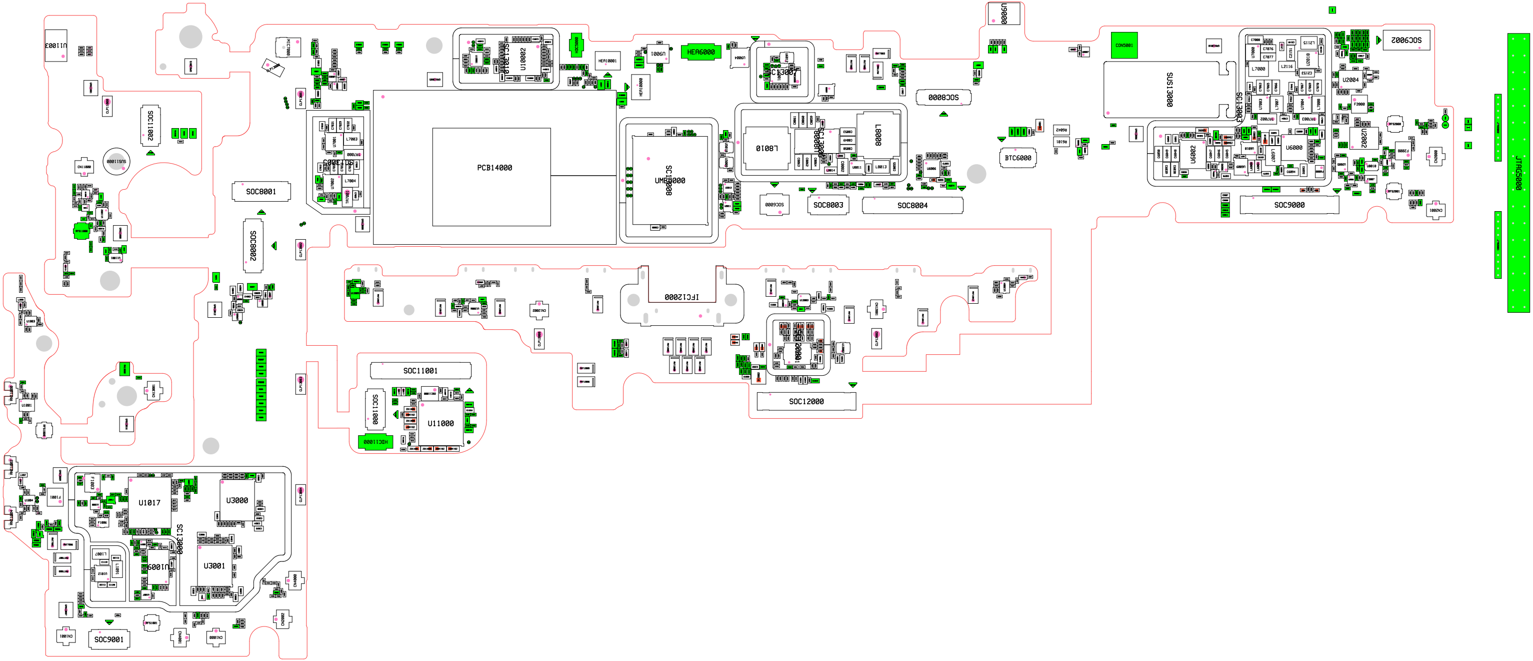
U4804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MIC7000
CLP13000
SC13010
U10002
HDC5000
HEA10001
U9001
HEA6000
U9004
SC13007
U9000
CON5001
SUS13000
SC13003
SOC9002
U2004
U2018
L7000
L2115
L2116
C7876
C7877
C7000
C2151
C2153
U7003
U7004
U7007
L7008
TA7002
TA7003
U2002
F2000
F2008
F2007
RFS2000
RFS2001
CN2000
CN2001
JTAG5000
SOC11002
SUS11000
CN11000
CLP11000
SOC8001
SOC8002
CLP13001
CLP13002
SC13005
U7001
U7002
L7003
L7004
TA7000
TA7001
PCB14000
HEA10000
UME5000
SC13008
L8010
SC13006
L8008
C8853
C8849
C8850
C8082
L8013
C8055
SOC8000
SOC6000
SOC8003
SOC8004
BTC6000
R6042
R6101
U6000
U6002
U6006
U6010
L6007
C6094
SOC9000
IFC12000
CN12002
CLP12000
CLP12002
CN12001
SC12000
SOC12000
SOC11001
SOC11000
U11000
HDC11000
CLP13003
CLP13004
U1003
U1001
ANT1000
ANT1002
ANT1004
U1004
F1001
CN13001
RFS1000
F1003
U1017
U3000
SC13000
F1006
L1087
U1812
L1091
U1009
U3001
CN4000
CN2002
RFS1001
SOC9001
CN1001
CN4001
CN1000

TAB_S9_CORE PCB_REV0.5 (S9, S9+, S9 Ultra)

- CORE PCB TYPE -

2023.05.26

sheet01 : AP_SM8550

sheet02 : PMIC

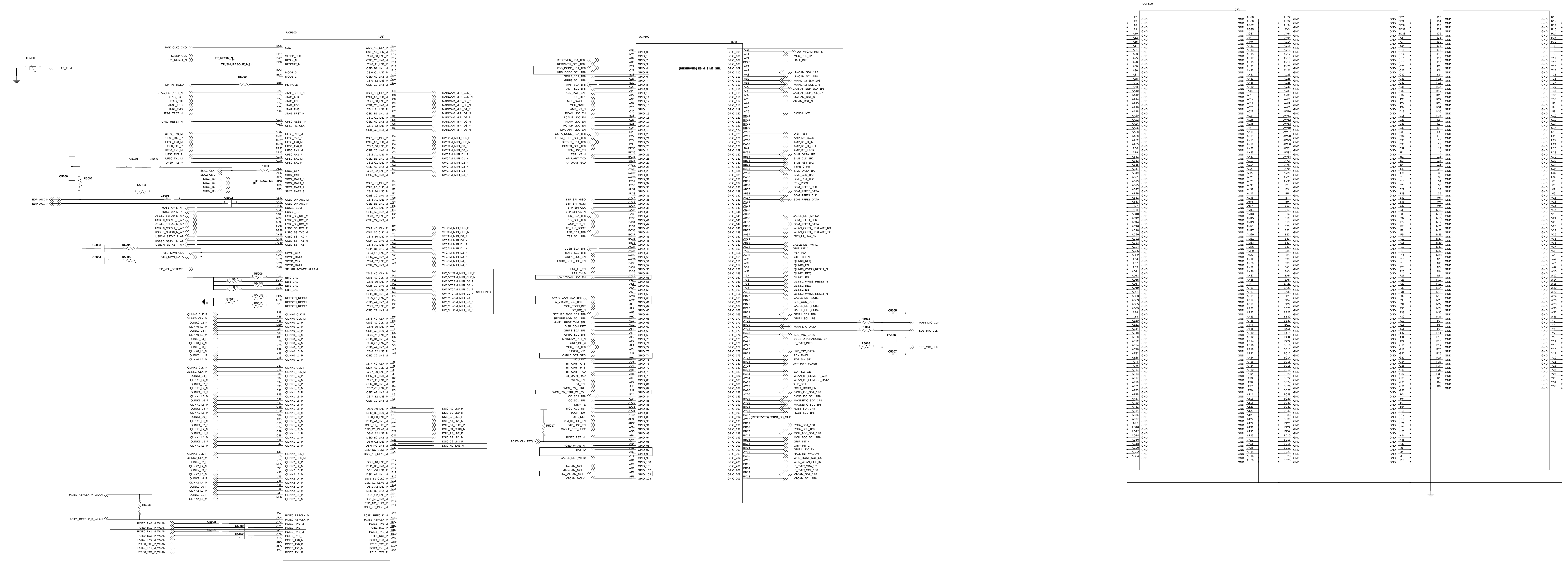
sheet03 : CORE PCB

TITLE TabS9_CORE
SIZE A1
DWG NO PDM NUMBER
REV Rev
DRAWN BY User Name
SHEET 1 of 4
Date Drawing Date

SM8550 (INTERFACE/GPIO) (Core PCB)

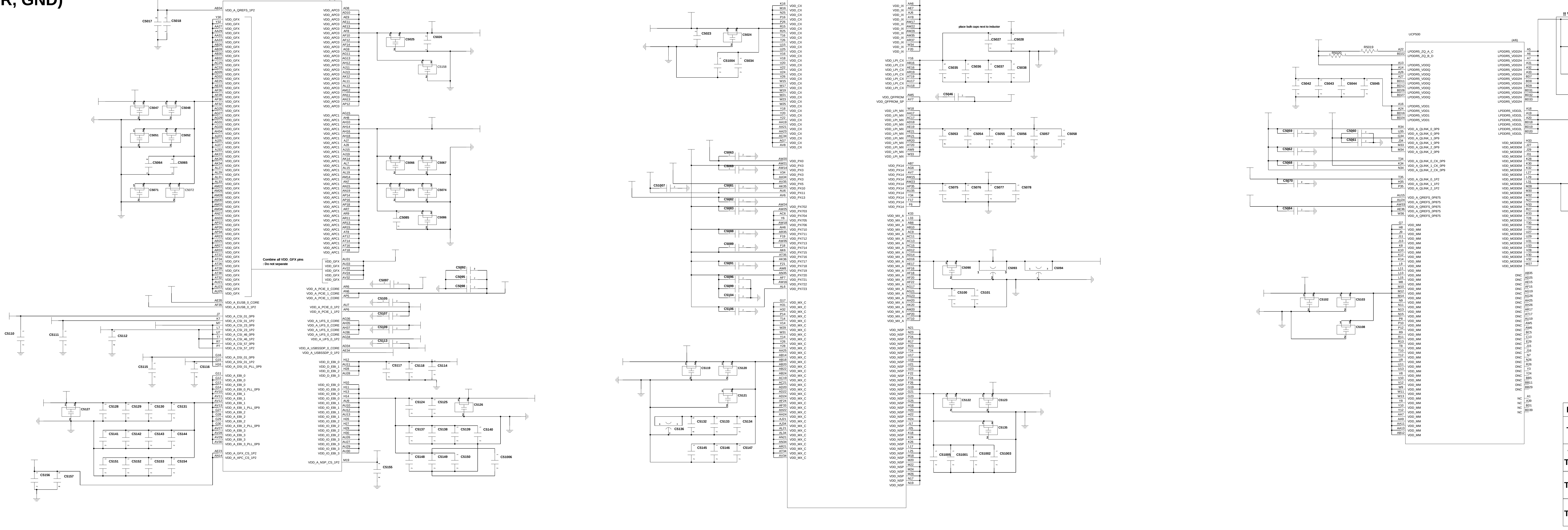
mmWAVE, SUB6 : SM8550

System


UCP500
(1/6)
PMK_CLK6_CXO
SLEEP_CLK
PON_RESET_N
TP_RESIN_N
TP_SM_RESOUT_N
TH5000
AP_THM
R5000
SM_PS_HOLD
JTAG_RST_OUT_N
JTAG_TCK
JTAG_TDI
JTAG_TDO
JTAG_TMS
JTAG_TRST_N
UFS0_RESET_N
UFS0_RX0_M
UFS0_RX0_P
UFS0_TX0_M
UFS0_TX0_P
UFS0_RX1_M
UFS0_RX1_P
UFS0_TX1_M
UFS0_TX1_P
C5160
L5000
R5001
SDC2_CLK
SDC2_CMD
SDC2_D0
SDC2_D1
SDC2_D2
SDC2_D3
TP_SDC2_D1
C5000
R5002
R5003
C5001
C5002
EDP_AUX_N
EDP_AUX_P
eUSB_AP_D_N
eUSB_AP_D_P
USB3.0_SSRX0_M_AP
USB3.0_SSRX0_P_AP
USB3.0_SSRX1_M_AP
USB3.0_SSRX1_P_AP
USB3.0_SSTX0_M_AP
USB3.0_SSTX0_P_AP
USB3.0_SSTX1_M_AP
USB3.0_SSTX1_P_AP
C5003
R5004
C5004
R5005
PMIC_SPMI_CLK
PMIC_SPMI_DATA
SP_VPH_DETECT
R5006
R5007
R5008
R5009
R5010
R5011
R5012
QLINK0_CLK_P
QLINK0_CLK_M
QLINK0_L2_P
QLINK0_L2_M
QLINK0_L3_P
QLINK0_L3_M
QLINK0_L4_P
QLINK0_L4_M
QLINK0_L0_P
QLINK0_L0_M
QLINK0_L1_P
QLINK0_L1_M
QLINK1_CLK_P
QLINK1_CLK_M
QLINK1_L4_P
QLINK1_L4_M
QLINK1_L7_P
QLINK1_L7_M
QLINK1_L5_P
QLINK1_L5_M
QLINK1_L6_P
QLINK1_L6_M
QLINK1_L8_P
QLINK1_L8_M
QLINK1_L0_P
QLINK1_L0_M
QLINK1_L2_P
QLINK1_L2_M
QLINK1_L1_P
QLINK1_L1_M
QLINK1_L3_P
QLINK1_L3_M
QLINK2_CLK_P
QLINK2_CLK_M
QLINK2_L2_P
QLINK2_L2_M
QLINK2_L3_P
QLINK2_L3_M
QLINK2_L4_P
QLINK2_L4_M
QLINK2_L0_P
QLINK2_L0_M
QLINK2_L1_P
QLINK2_L1_M
PCIE0_REFCLK_M_WLAN
R5018
PCIE0_REFCLK_P_WLAN
PCIE0_RX0_M_WLAN
PCIE0_RX0_P_WLAN
PCIE0_RX1_M_WLAN
PCIE0_RX1_P_WLAN
PCIE0_TX0_M_WLAN
PCIE0_TX0_P_WLAN
PCIE0_TX1_M_WLAN
PCIE0_TX1_P_WLAN
C5008
C5009
C5161
C5162
MAINCAM_MIPI_CLK_P
MAINCAM_MIPI_CLK_N
MAINCAM_MIPI_D0_P
MAINCAM_MIPI_D0_N
MAINCAM_MIPI_D1_P
MAINCAM_MIPI_D1_N
MAINCAM_MIPI_D2_P
MAINCAM_MIPI_D2_N
MAINCAM_MIPI_D3_P
MAINCAM_MIPI_D3_N
UWCAM_MIPI_CLK_P
UWCAM_MIPI_CLK_N
UWCAM_MIPI_D0_P
UWCAM_MIPI_D0_N
UWCAM_MIPI_D1_P
UWCAM_MIPI_D1_N
UWCAM_MIPI_D2_P
UWCAM_MIPI_D2_N
UWCAM_MIPI_D3_P
UWCAM_MIPI_D3_N
VTCAM_MIPI_CLK_P
VTCAM_MIPI_CLK_N
VTCAM_MIPI_D0_P
VTCAM_MIPI_D0_N
VTCAM_MIPI_D1_P
VTCAM_MIPI_D1_N
VTCAM_MIPI_D2_P
VTCAM_MIPI_D2_N
VTCAM_MIPI_D3_P
VTCAM_MIPI_D3_N
UW_VTCAM_MIPI_CLK_P
UW_VTCAM_MIPI_CLK_N
UW_VTCAM_MIPI_D0_P
UW_VTCAM_MIPI_D0_N
UW_VTCAM_MIPI_D1_P
UW_VTCAM_MIPI_D1_N
UW_VTCAM_MIPI_D2_P
UW_VTCAM_MIPI_D2_N
UW_VTCAM_MIPI_D3_P
UW_VTCAM_MIPI_D3_N
S9U_ONLY
DSI0_A0_LN0_P
DSI0_B0_LN0_M
DSI0_C0_LN1_P
DSI0_A1_LN1_M
DSI0_B1_CLK0_P
DSI0_C1_CLK0_M
DSI0_A2_LN2_P
DSI0_B2_LN2_M
DSI0_C2_LN3_P
DSI0_NC_LN3_M
PCIE0_CLK_REQ_N
UCP500
(5/6)
(RESERVED) ESIM_SIM2_SEL
REDRIVER_SDA_1P8
REDRIVER_SCL_1P8
KBD_DCDC_SDA_1P8
KBD_DCDC_SCL_1P8
GRIP3_SDA_1P8
GRIP3_SCL_1P8
AMP_SDA_1P8
AMP_SCL_1P8
KBD_PWR_EN
CC_DIR
MCU_SWCLK
MCU_nRST
AMP_INT_N
RCAM_LDO_EN
RCAM2_LDO_EN
FCAM_LDO_EN
MOTOR_LDO_EN
SPK_AMP_LDO_EN
OCTA_DCDC_SDA_1P8
OCTA_DCDC_SCL_1P8
DIRECT_SDA_1P8
DIRECT_SCL_1P8
PEN_LDO_EN
TSP_INT_N
AP_UART_TXD
AP_UART_RXD
BTP_SPI_MISO
BTP_SPI_MOSI
BTP_SPI_CLK
BTP_SPI_CS_N
PEN_SDA_1P8
PEN_SCL_1P8
AMP_RST_N
AP_USB_BOOT
TSP_SDA_1P8
TSP_SCL_1P8
eUSB_SDA_1P8
eUSB_SCL_1P8
GRIP2_LDO_EN
ENDC_GRIP_LDO_EN
LAA_AS_EN
LAA_EN_0
UW_VTCAM_LDO_EN
UW_VTCAM_SDA_1P8
UW_VTCAM_SCL_1P8
MCU_CONN_INT
DC_IRQ_N
SECURE_NVM_SDA_1P8
SECURE_NVM_SCL_1P8
HWID_LRPST_THM_SEL
DISP_CON_DET
GRIP2_SDA_1P8
GRIP2_SCL_1P8
MAINCAM_RST_N
GRIP_INT_3
MCU_SDA_1P8
6AXIS1_INT1
CABLE_DET_GPS
MCU_INT
BT_UART_CTS
BT_UART_RTS
BT_UART_TXD
BT_UART_RXD
WLAN_EN
BT_EN
WCN_SW_CTRL
WCN_SW_CTRL_WL_CX
CC_SDA_1P8
CC_SCL_1P8
DISP_TE
MCU_ACC_INT
TCON_RDY
OTG_DET
CAM_IO_LDO_EN
BTP_LDO_EN
CABLE_DET_SUB2
R5017
PCIE0_RST_N
PCIE0_WAKE_N
BAT_ID
CABLE_DET_WIFI0
UWCAM_MCLK
MAINCAM_MCLK
UW_VTCAM_MCLK
VTCAM_MCLK
UW_VTCAM_RST_N
MCU_SCL_1P8
HALL_INT
UWCAM_SDA_1P8
UWCAM_SCL_1P8
MAINCAM_SDA_1P8
MAINCAM_SCL_1P8
CAM_AF_EEP_SDA_1P8
CAM_AF_EEP_SCL_1P8
UWCAM_RST_N
VTCAM_RST_N
6AXIS1_INT2
DISP_RST
AMP_I2S_BCLK
AMP_I2S_D_IN
AMP_I2S_D_OUT
AMP_I2S_LRCK
SIM1_DATA_1P2
SIM1_CLK_1P2
SIM1_RST_1P2
TYPE_C_INT
SIM2_DATA_1P2
SIM2_CLK_1P2
SIM2_RST_1P2
PEN_PDCT
SDM_RFFE0_CLK
SDM_RFFE0_DATA
SDM_RFFE1_CLK
SDM_RFFE1_DATA
CABLE_DET_MAIN2
SDM_RFFE4_CLK
SDM_RFFE4_DATA
WLAN_COEX_SDXUART_RX
WLAN_COEX_SDXUART_TX
GPS_L1_LNA_EN
CABLE_DET_WIFI1
GRIP_INT_1
PEN_IRQ
BTP_RST_N
QLINK0_REQ
QLINK0_EN
QLINK0_WMSS_RESET_N
QLINK1_REQ
QLINK1_EN
QLINK1_WMSS_RESET_N
QLINK2_REQ
QLINK2_EN
QLINK2_WMSS_RESET_N
CABLE_DET_SUB1
SUB_CON_DET
CABLE_DET_SUB3
CABLE_DET_SUB4
GRIP1_SDA_1P8
GRIP1_SCL_1P8
R5013
C5005
MAIN_MIC_CLK
MAIN_MIC_DATA
R5014
SUB_MIC_CLK
C5006
SUB_MIC_DATA
VBUS_DISCHARGING_EN
IF_PMIC_INTB
R5016
3RD_MIC_CLK
C5007
3RD_MIC_DATA
PEN_FWEL
EDP_SW_SEL
OVP_PWR_FLAGB
EDP_SW_OE
WLAN_BT_SLIMBUS_CLK
WLAN_BT_SLIMBUS_DATA
DISP_DET
OCTA_DCDC_EN
6AXIS_I3C_SDA_1P8
6AXIS_I3C_SCL_1P8
MAGNETIC_SDA_1P8
MAGNETIC_SCL_1P8
RGB1_SDA_1P8
RGB1_SCL_1P8
(RESERVED) COPR_SS_SUB
RGB2_SDA_1P8
RGB2_SCL_1P8
MCU_ACC_SDA_1P8
MCU_ACC_SCL_1P8
GRIP_INT_4
GRIP_INT_2
GRIP1_LDO_EN
HALL_INT_WACOM
WCN_HOST_SOL_OUT
WCN_WLAN_SOL_IN
IF_PMIC_SDA_1P8
IF_PMIC_SCL_1P8
VTCAM_SDA_1P8
VTCAM_SCL_1P8
UCP500
(6/6)
GND

SM8550

(POWER, GND)


UCP500
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VDD_A_QREFS_1P2
C5017
C5018
VDD_GFX
VDD_APC0
VDD_APC1
C5025
C5026
C5158
C5047
C5048
C5051
C5052
C5064
C5065
C5071
C5072
C5066
C5067
C5073
C5074
C5085
C5086
Combine all VDD_GFX pins
: Do not separate
C5092
C5095
C5097
C5098
VDD_A_PCIE_0_CORE
VDD_A_PCIE_1_CORE
VDD_A_PCIE_0_1P2
VDD_A_PCIE_1_1P2
C5105
C5107
VDD_A_UFS_0_CORE
VDD_A_UFS_0_1P2
C5109
C5113
VDD_A_USBSSDP_0_CORE
VDD_A_USBSSDP_0_1P2
VDD_A_EUSB_0_CORE
VDD_A_EUSB_0_1P2
VDD_A_CSI_01_0P9
VDD_A_CSI_01_1P2
VDD_A_CSI_23_0P9
VDD_A_CSI_23_1P2
VDD_A_CSI_46_0P9
VDD_A_CSI_46_1P2
VDD_A_CSI_57_0P9
VDD_A_CSI_57_1P2
C5110
C5111
C5112
VDD_A_DSI_01_0P9
VDD_A_DSI_01_1P2
VDD_A_DSI_01_PLL_0P9
C5115
C5116
VDD_A_EBI_0
VDD_A_EBI_0_PLL_0P9
VDD_A_EBI_1
VDD_A_EBI_1_PLL_0P9
VDD_A_EBI_2
VDD_A_EBI_2_PLL_0P9
VDD_A_EBI_3
VDD_A_EBI_3_PLL_0P9
VDD_A_GFX_CS_1P2
VDD_A_APC_CS_1P2
C5127
C5128
C5129
C5130
C5131
C5141
C5142
C5143
C5144
C5151
C5152
C5153
C5154
C5156
C5157
VDD_D_EBI_0
VDD_D_EBI_1
VDD_D_EBI_2
VDD_D_EBI_3
C5117
C5118
C5114
VDD_IO_EBI_0
VDD_IO_EBI_1
VDD_IO_EBI_2
VDD_IO_EBI_3
C5124
C5125
C5126
C5137
C5138
C5139
C5140
C5148
C5149
C5150
C51006
VDD_A_NSP_CS_1P2
C5155
UCP500
(3/6)
VDD_CX
C5023
C5024
C51004
C5034
VDD_PX0
VDD_PX3
VDD_PX5
VDD_PX10
VDD_PX11
VDD_PX13
VDD_PX702
VDD_PX703
VDD_PX704
VDD_PX705
VDD_PX706
VDD_PX710
VDD_PX711
VDD_PX712
VDD_PX713
VDD_PX714
VDD_PX715
VDD_PX716
VDD_PX717
VDD_PX718
VDD_PX719
VDD_PX720
VDD_PX721
VDD_PX722
VDD_PX723
C5063
C5069
C51007
C5081
C5082
C5083
C5088
C5089
C5091
C5096
C5099
C5104
C5106
VDD_MX_C
C5119
C5120
C5121
C5132
C5133
C5134
C5136
C5145
C5146
C5147
VDD_IX
C5012
C5013
C5014
C5015
C5016
place bulk caps next to inductor
C5027
C5028
VDD_LPI_CX
C5035
C5036
C5037
C5038
VDD_QFPROM
VDD_QFPROM_SP
C5046
VDD_LPI_MX
C5053
C5054
C5055
C5056
C5057
C5058
VDD_PX14
C5075
C5076
C5077
C5078
VDD_MX_A
C5090
C5093
C5094
C5100
C5101
VDD_NSP
C5122
C5123
C5135
C51005
C51001
C51002
C51003
UCP500
(4/6)
R5019
R5020
LPDDR5_ZQ_A_C
LPDDR5_ZQ_B_D
LPDDR5_VDDQ
C5042
C5043
C5044
C5045
LPDDR5_VDD1
VDD_A_QLINK_0_0P9
VDD_A_QLINK_1_0P9
VDD_A_QLINK_2_0P9
VDD_A_QLINK_0_CK_0P9
VDD_A_QLINK_1_CK_0P9
VDD_A_QLINK_2_CK_0P9
VDD_A_QLINK_0_1P2
VDD_A_QLINK_1_1P2
VDD_A_QLINK_2_1P2
C5059
C5060
C5061
C5062
C5068
C5070
C5084
VDD_A_QREFS_0P875
VDD_MM
C5102
C5103
C5108
LPDDR5_VDD2H
!! VDD2H decap CASE 2
C5019
C5020
C5021
C5022
C5029
C5030
C5031
C5032
C5033
C5039
C5040
C5041
LPDDR5_VDD2L
C5049
C5050
R5022
VDD_MODEM
C5079
C5080
C5087
DNC
NC

DRAM_POP (CORE)

Model	Memory	Marking	TOP	BOT
TabS9	8G	8G	TP_8G_T	TP_8G_B
TabS9 TabS9P	12G	12G	TP_12G_T	TP_12G_B
TabS9U	12G	12U	TP_12U_T	TP_12U_B
TabS9U	16G	16U	TP_16U_T	TP_16U_B

8G	
12G	
16G	

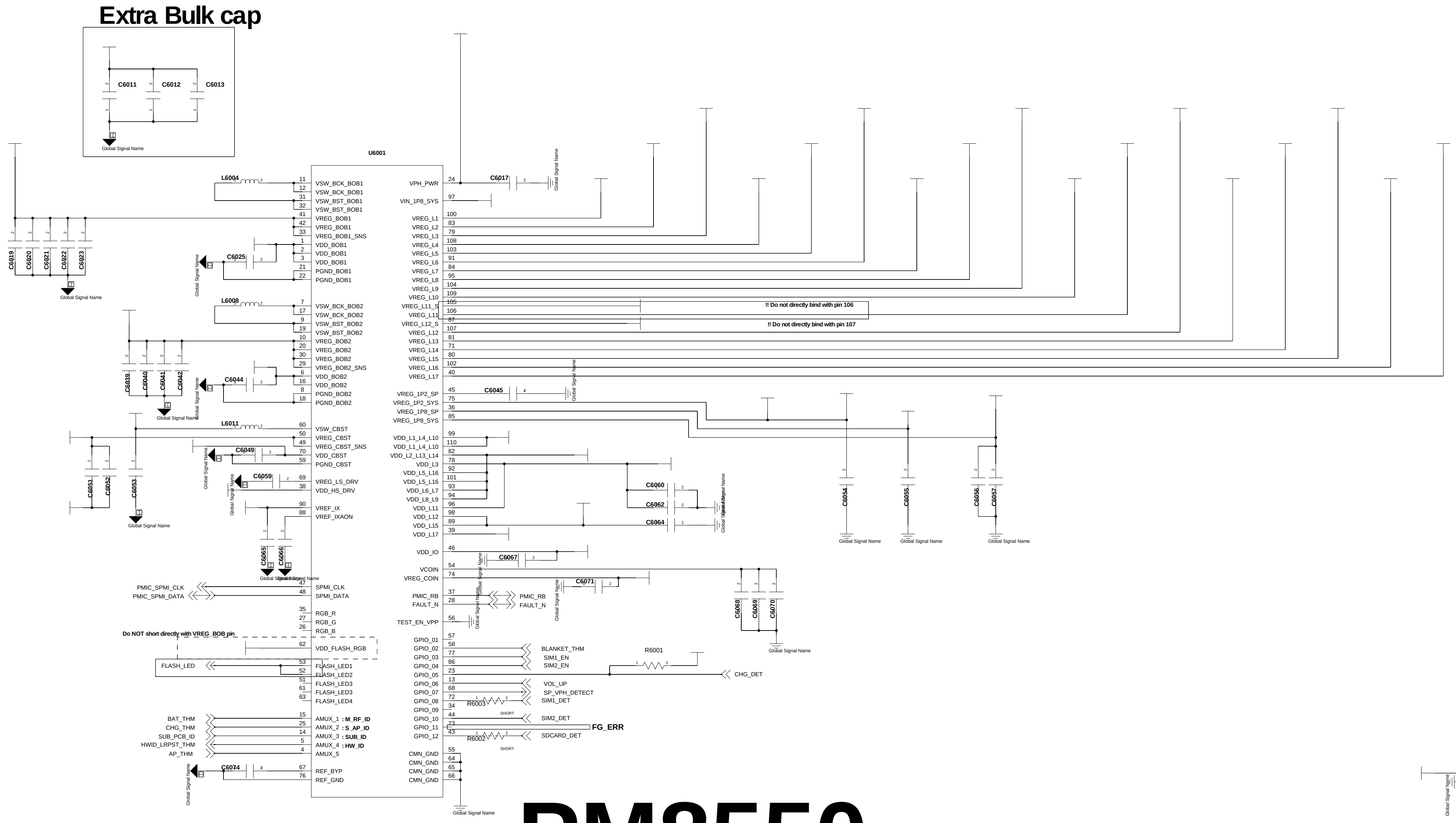
TITLE
TabS9_CORE
SIZE A1
DWG NO PDM NUMBER
REV Rev
DRAWN BY User Name
SHEET 2 of 4
Date Drawing Date

AP PMIC set (Core PCB)

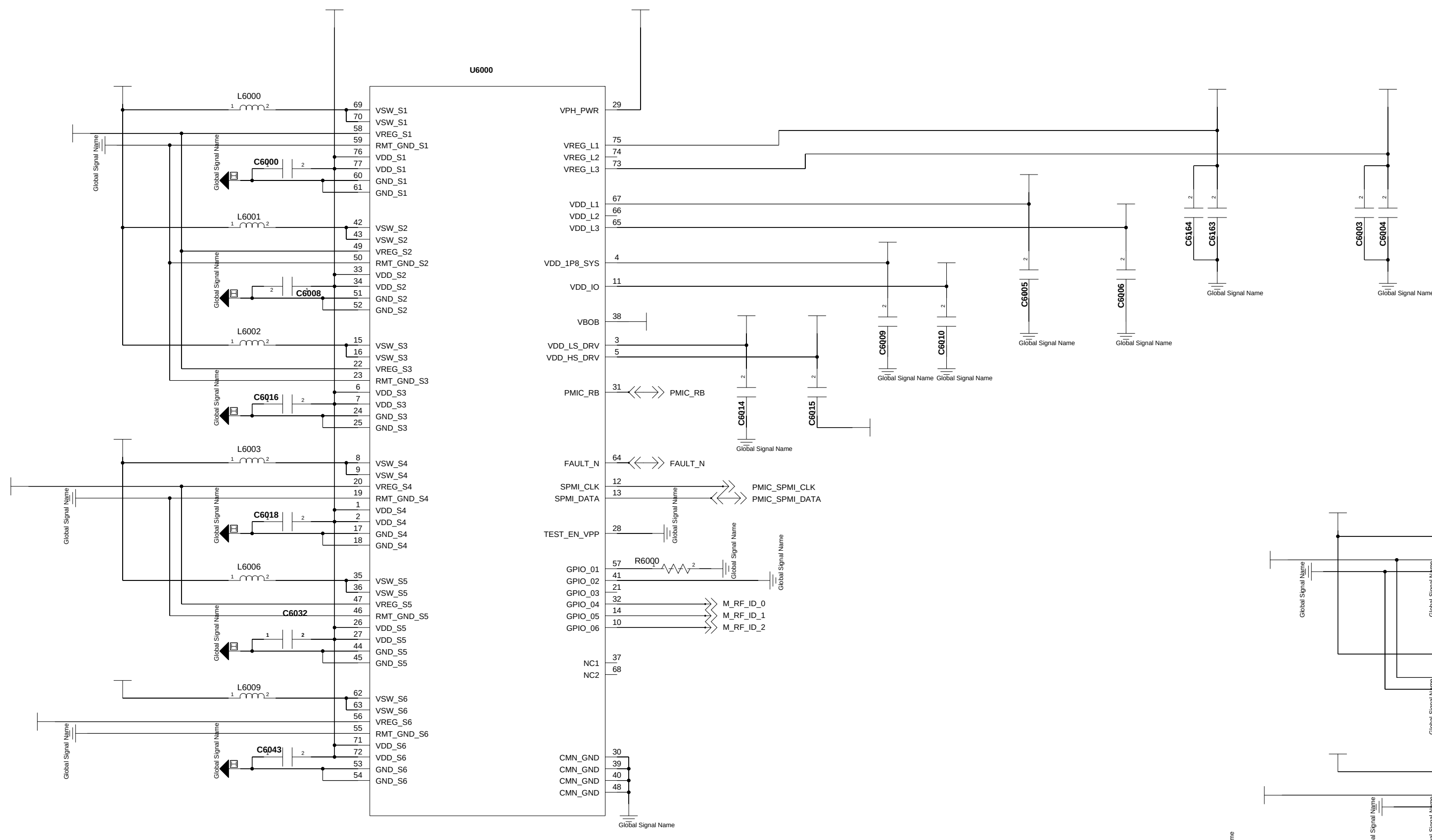
PMIC (PM8550/8550VS/8550VE)

!! CAUTION !!

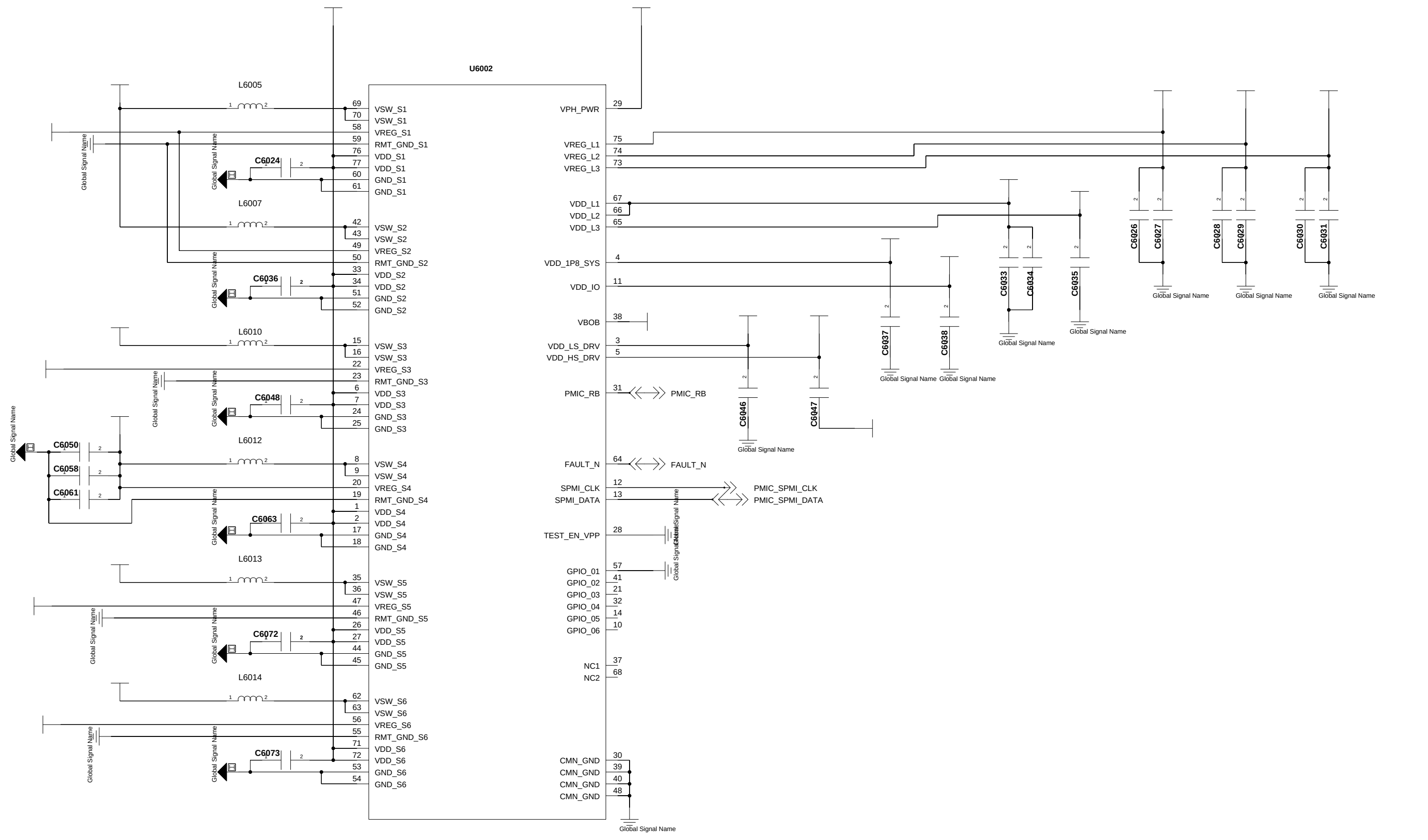
Place Cin 10uF Bulk Caps at corner of PM8550VS and PM8550VE as close as possible
- PM8550VS : VDD_S1, S3, S4, S6
- PM8550VE : VDD_S1, S4, S5, S8


Extra Bulk cap
C6011
C6012
C6013
U6001
L6004
C6019
C6020
C6021
C6022
C6023
C6025
L6008
C6039
C6040
C6041
C6042
C6044
L6011
C6051
C6052
C6053
C6049
C6059
C6065
C6066
VSW_BCK_BOB1
VSW_BST_BOB1
VREG_BOB1
VREG_BOB1_SNS
VDD_BOB1
PGND_BOB1
VSW_BCK_BOB2
VSW_BST_BOB2
VREG_BOB2
VREG_BOB2_SNS
VDD_BOB2
PGND_BOB2
VSW_CBST
VREG_CBST
VREG_CBST_SNS
VDD_CBST
PGND_CBST
VREG_LS_DRV
VDD_HS_DRV
VREF_IX
VREF_IXAON
SPMI_CLK
SPMI_DATA
PMIC_SPMI_CLK
PMIC_SPMI_DATA
RGB_R
RGB_G
RGB_B
Do NOT short directly with VREG_BOB pin
VDD_FLASH_RGB
FLASH_LED
FLASH_LED1
FLASH_LED2
FLASH_LED3
FLASH_LED4
BAT_THM
CHG_THM
SUB_PCB_ID
HWID_LRPST_THM
AP_THM
AMUX_1 : M_RF_ID
AMUX_2 : S_AP_ID
AMUX_3 : SUB_ID
AMUX_4 : HW_ID
AMUX_5
C6074
REF_BYP
REF_GND
VPH_PWR
C6017
VIN_1P8_SYS
VREG_L1
VREG_L2
VREG_L3
VREG_L4
VREG_L5
VREG_L6
VREG_L7
VREG_L8
VREG_L9
VREG_L10
VREG_L11_S
VREG_L11
VREG_L12_S
VREG_L12
VREG_L13
VREG_L14
VREG_L15
VREG_L16
VREG_L17
!! Do not directly bind with pin 106
!! Do not directly bind with pin 107
VREG_1P2_SP
VREG_1P2_SYS
VREG_1P8_SP
VREG_1P8_SYS
C6045
VDD_L1_L4_L10
VDD_L2_L13_L14
VDD_L3
VDD_L5_L16
VDD_L6_L7
VDD_L8_L9
VDD_L11
VDD_L12
VDD_L15
VDD_L17
C6060
C6062
C6064
C6054
C6055
C6056
C6057
VDD_IO
C6067
VCOIN
VREG_COIN
C6071
C6068
C6069
C6070
PMIC_RB
FAULT_N
TEST_EN_VPP
GPIO_01
GPIO_02
GPIO_03
GPIO_04
GPIO_05
GPIO_06
GPIO_07
GPIO_08
GPIO_09
GPIO_10
GPIO_11
GPIO_12
BLANKET_THM
SIM1_EN
SIM2_EN
R6001
CHG_DET
VOL_UP
SP_VPH_DETECT
R6003
SIM1_DET
SHORT
SIM2_DET
FG_ERR
R6002
SDCARD_DET
CMN_GND
Global Signal Name

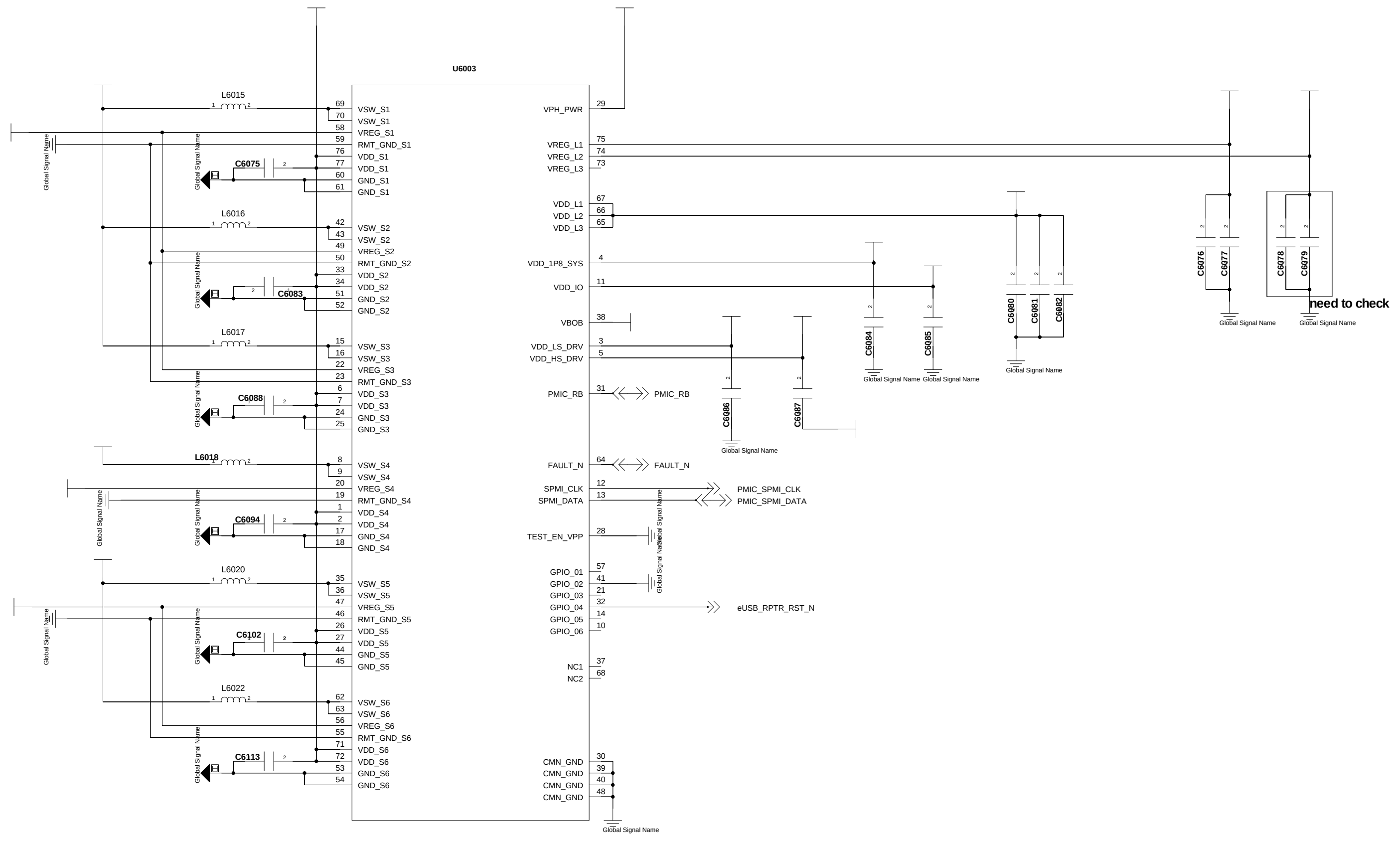
PM8550


U6000
L6000
C6000
L6001
C6008
L6002
C6016
L6003
C6018
L6006
C6032
L6009
C6043
VSW_S1
VREG_S1
RMT_GND_S1
VDD_S1
GND_S1
VSW_S2
VREG_S2
RMT_GND_S2
VDD_S2
GND_S2
VSW_S3
VREG_S3
RMT_GND_S3
VDD_S3
GND_S3
VSW_S4
VREG_S4
RMT_GND_S4
VDD_S4
GND_S4
VSW_S5
VREG_S5
RMT_GND_S5
VDD_S5
GND_S5
VSW_S6
VREG_S6
RMT_GND_S6
VDD_S6
GND_S6
VPH_PWR
VREG_L1
VREG_L2
VREG_L3
VDD_L1
VDD_L2
VDD_L3
VDD_1P8_SYS
VDD_IO
VBOB
VDD_LS_DRV
VDD_HS_DRV
PMIC_RB
FAULT_N
SPMI_CLK
SPMI_DATA
PMIC_SPMI_CLK
PMIC_SPMI_DATA
TEST_EN_VPP
R6000
GPIO_01
GPIO_02
GPIO_03
GPIO_04
GPIO_05
GPIO_06
M_RF_ID_0
M_RF_ID_1
M_RF_ID_2
NC1
NC2
CMN_GND
C6009
C6010
C6014
C6015
C6005
C6006
C6164
C6163
C6003
C6004
Global Signal Name

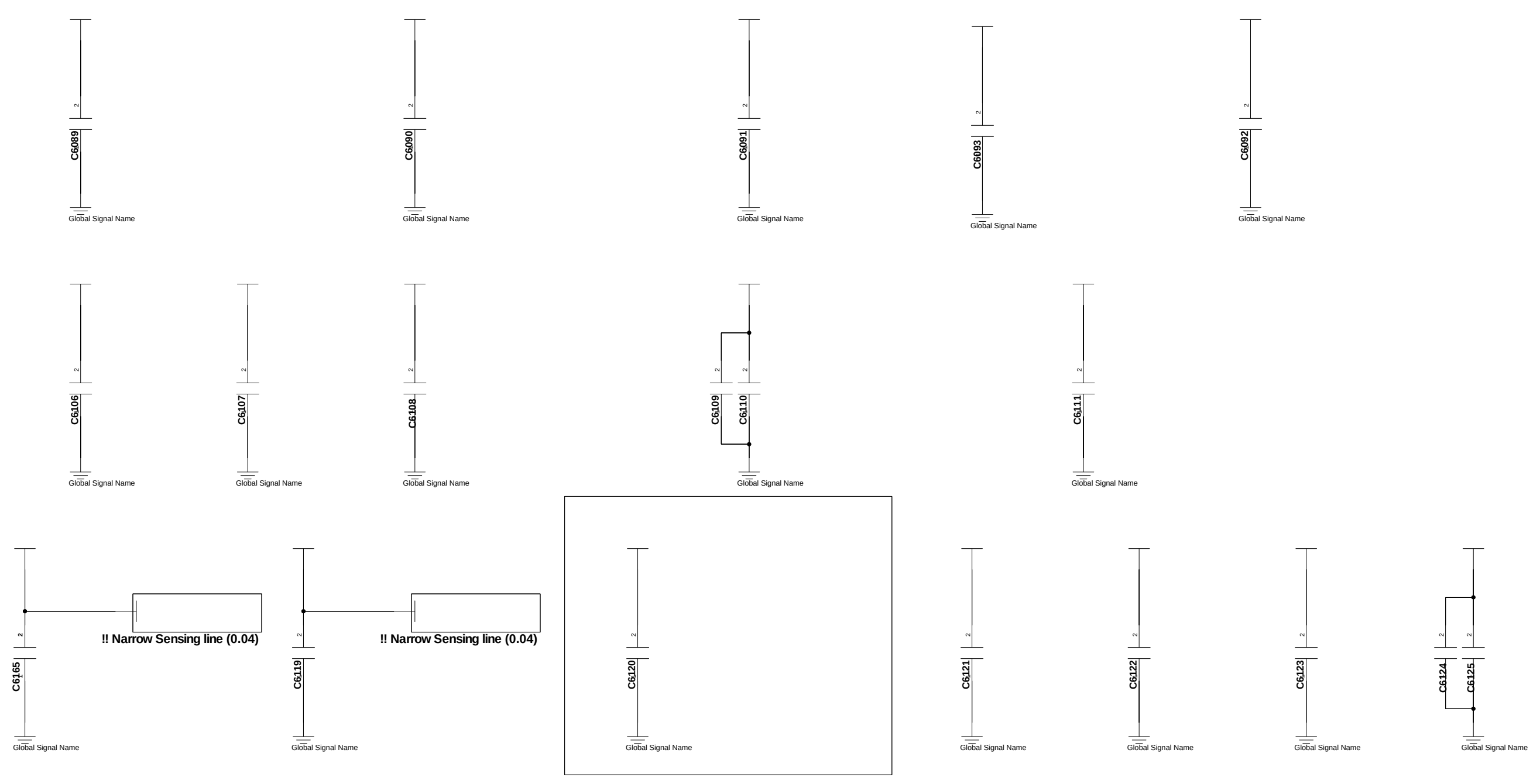
PM8550VS_C


U6002
L6005
C6024
L6007
C6036
L6010
C6048
L6012
C6050
C6058
C6061
C6063
L6013
C6072
L6014
C6073
VSW_S1
VREG_S1
RMT_GND_S1
VDD_S1
GND_S1
VSW_S2
VREG_S2
RMT_GND_S2
VDD_S2
GND_S2
VSW_S3
VREG_S3
RMT_GND_S3
VDD_S3
GND_S3
VSW_S4
VREG_S4
RMT_GND_S4
VDD_S4
GND_S4
VSW_S5
VREG_S5
RMT_GND_S5
VDD_S5
GND_S5
VSW_S6
VREG_S6
RMT_GND_S6
VDD_S6
GND_S6
VPH_PWR
VREG_L1
VREG_L2
VREG_L3
VDD_L1
VDD_L2
VDD_L3
VDD_1P8_SYS
VDD_IO
VBOB
VDD_LS_DRV
VDD_HS_DRV
PMIC_RB
FAULT_N
SPMI_CLK
SPMI_DATA
PMIC_SPMI_CLK
PMIC_SPMI_DATA
TEST_EN_VPP
GPIO_01
GPIO_02
GPIO_03
GPIO_04
GPIO_05
GPIO_06
NC1
NC2
CMN_GND
C6026
C6027
C6028
C6029
C6030
C6031
C6033
C6034
C6035
C6037
C6038
C6046
C6047
Global Signal Name

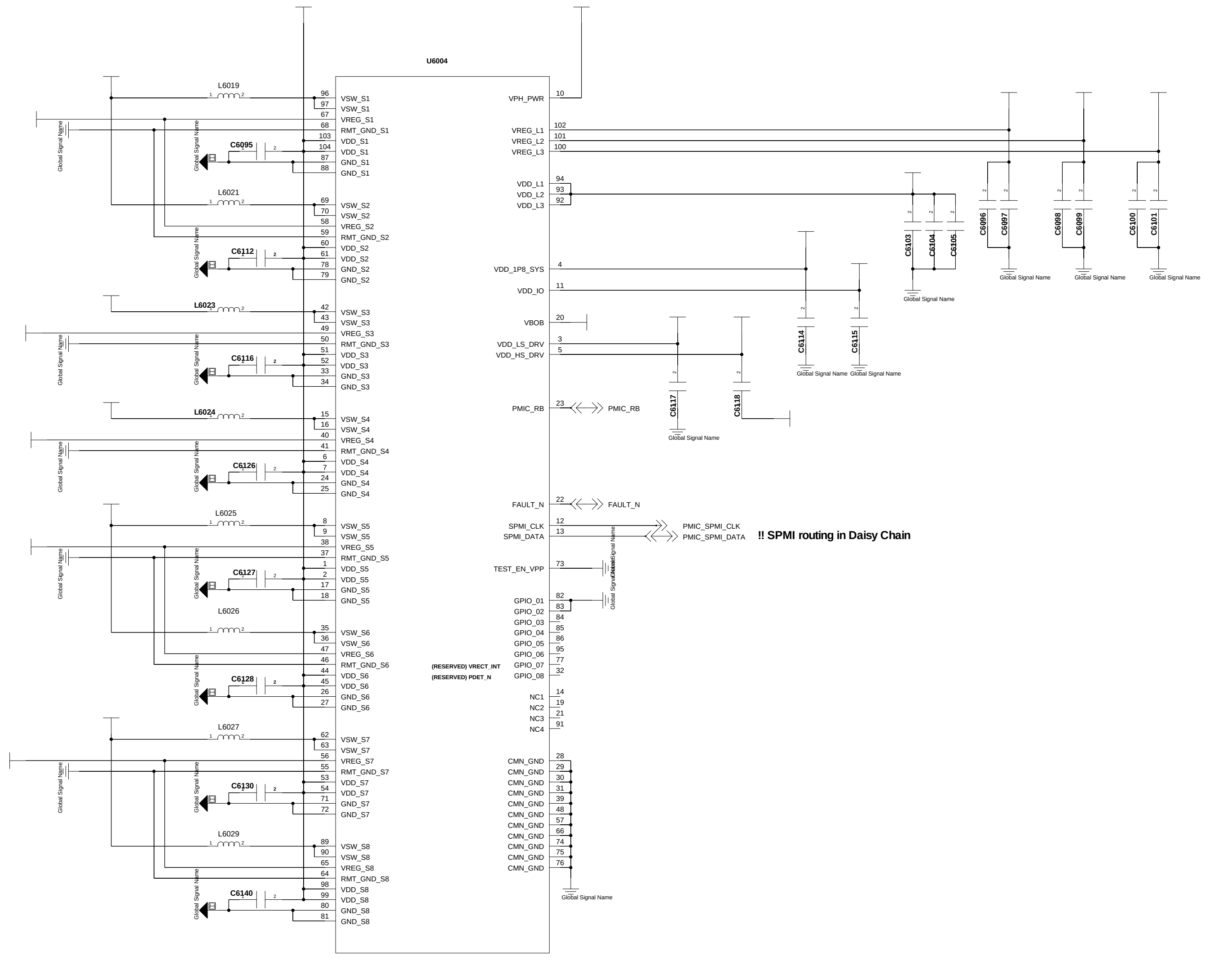
PM8550VS_E


U6003
L6015
C6075
L6016
C6083
L6017
C6088
L6018
C6094
L6020
C6102
L6022
C6113
VSW_S1
VREG_S1
RMT_GND_S1
VDD_S1
GND_S1
VSW_S2
VREG_S2
RMT_GND_S2
VDD_S2
GND_S2
VSW_S3
VREG_S3
RMT_GND_S3
VDD_S3
GND_S3
VSW_S4
VREG_S4
RMT_GND_S4
VDD_S4
GND_S4
VSW_S5
VREG_S5
RMT_GND_S5
VDD_S5
GND_S5
VSW_S6
VREG_S6
RMT_GND_S6
VDD_S6
GND_S6
VPH_PWR
VREG_L1
VREG_L2
VREG_L3
VDD_L1
VDD_L2
VDD_L3
VDD_1P8_SYS
VDD_IO
VBOB
VDD_LS_DRV
VDD_HS_DRV
PMIC_RB
FAULT_N
SPMI_CLK
SPMI_DATA
PMIC_SPMI_CLK
PMIC_SPMI_DATA
TEST_EN_VPP
GPIO_01
GPIO_02
GPIO_03
GPIO_04
GPIO_05
GPIO_06
eUSB_RPTR_RST_N
NC1
NC2
CMN_GND
C6084
C6085
C6086
C6087
C6080
C6081
C6082
C6076
C6077
C6078
C6079
need to check
Global Signal Name

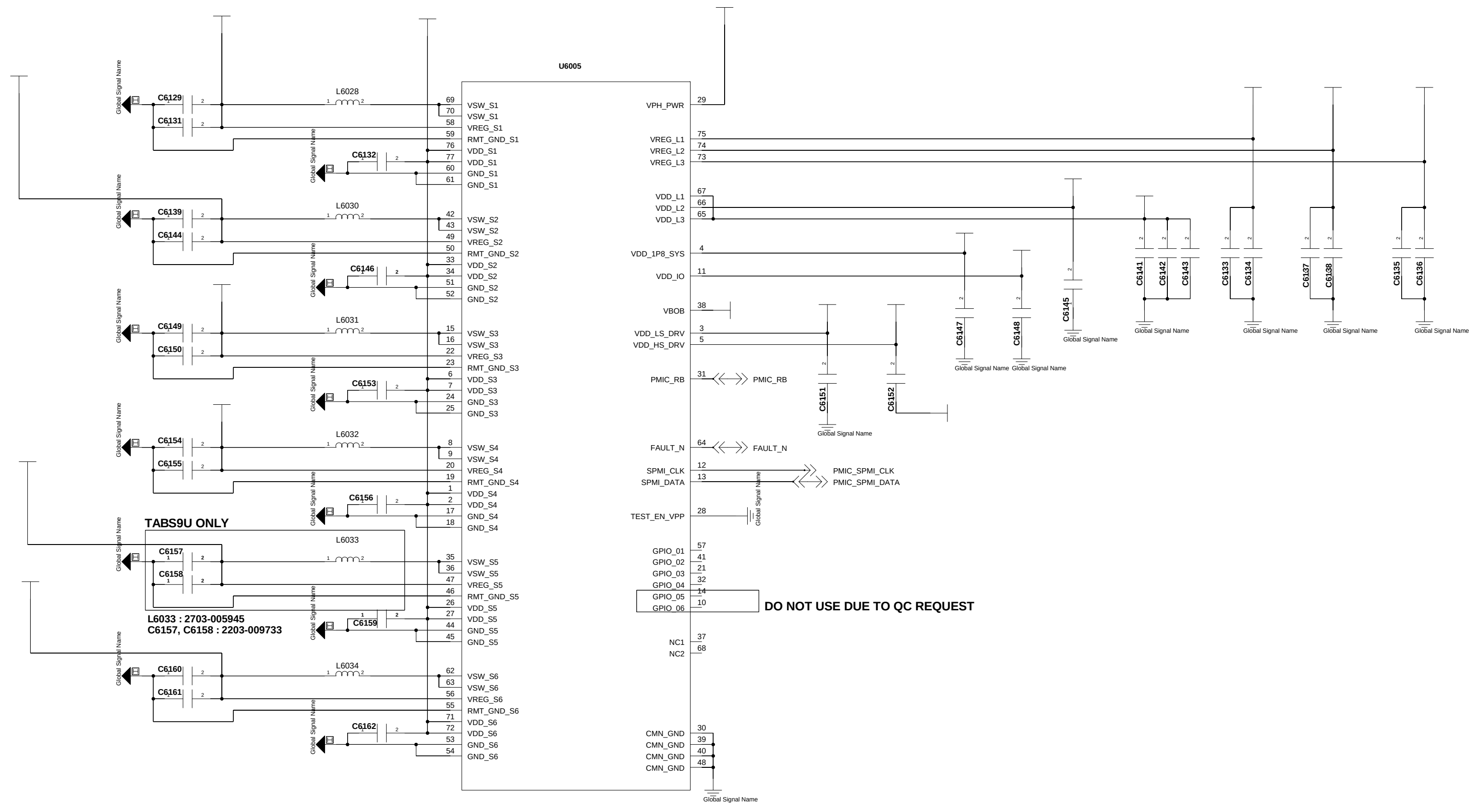
PM8550VS_D


C6089
C6090
C6091
C6093
C6092
C6106
C6107
C6108
C6109
C6110
C6111
C6165
!! Narrow Sensing line (0.04)
C6119
!! Narrow Sensing line (0.04)
C6120
C6121
C6122
C6123
C6124
C6125
Global Signal Name

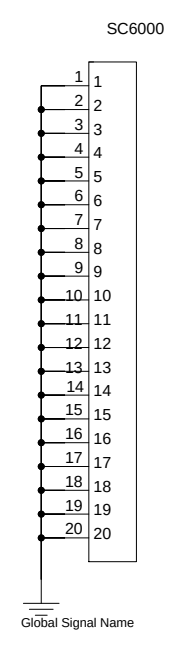
PM8550 LDO OUT CAP


U6004
L6019
C6095
L6021
C6112
L6023
C6116
L6024
C6126
L6025
C6127
L6026
C6128
L6027
C6130
L6029
C6140
VSW_S1
VREG_S1
RMT_GND_S1
VDD_S1
GND_S1
VSW_S2
VREG_S2
RMT_GND_S2
VDD_S2
GND_S2
VSW_S3
VREG_S3
RMT_GND_S3
VDD_S3
GND_S3
VSW_S4
VREG_S4
RMT_GND_S4
VDD_S4
GND_S4
VSW_S5
VREG_S5
RMT_GND_S5
VDD_S5
GND_S5
VSW_S6
VREG_S6
RMT_GND_S6
VDD_S6
GND_S6
VSW_S7
VREG_S7
RMT_GND_S7
VDD_S7
GND_S7
VSW_S8
VREG_S8
RMT_GND_S8
VDD_S8
GND_S8
VPH_PWR
VREG_L1
VREG_L2
VREG_L3
VDD_L1
VDD_L2
VDD_L3
VDD_1P8_SYS
VDD_IO
VBOB
VDD_LS_DRV
VDD_HS_DRV
PMIC_RB
FAULT_N
SPMI_CLK
SPMI_DATA
PMIC_SPMI_CLK
PMIC_SPMI_DATA
!! SPMI routing in Daisy Chain
TEST_EN_VPP
GPIO_01
GPIO_02
GPIO_03
GPIO_04
GPIO_05
GPIO_06
GPIO_07
GPIO_08
(RESERVED) VRECT_INT
(RESERVED) PDET_N
NC1
NC2
NC3
NC4
CMN_GND
C6114
C6115
C6117
C6118
C6103
C6104
C6105
C6096
C6097
C6098
C6099
C6100
C6101
C6135
C6136
Global Signal Name

PM8550VE_F


U6005
C6129
C6131
L6028
C6132
C6139
C6144
L6030
C6146
C6149
C6150
L6031
C6153
C6154
C6155
L6032
C6156
TABS9U ONLY
C6157
C6158
L6033
L6033 : 2703-005945
C6157, C6158 : 2203-009733
C6159
C6160
C6161
L6034
C6162
VSW_S1
VREG_S1
RMT_GND_S1
VDD_S1
GND_S1
VSW_S2
VREG_S2
RMT_GND_S2
VDD_S2
GND_S2
VSW_S3
VREG_S3
RMT_GND_S3
VDD_S3
GND_S3
VSW_S4
VREG_S4
RMT_GND_S4
VDD_S4
GND_S4
VSW_S5
VREG_S5
RMT_GND_S5
VDD_S5
GND_S5
VSW_S6
VREG_S6
RMT_GND_S6
VDD_S6
GND_S6
VPH_PWR
VREG_L1
VREG_L2
VREG_L3
VDD_L1
VDD_L2
VDD_L3
VDD_1P8_SYS
VDD_IO
VBOB
VDD_LS_DRV
VDD_HS_DRV
PMIC_RB
FAULT_N
SPMI_CLK
SPMI_DATA
PMIC_SPMI_CLK
PMIC_SPMI_DATA
TEST_EN_VPP
GPIO_01
GPIO_02
GPIO_03
GPIO_04
GPIO_05
GPIO_06
DO NOT USE DUE TO QC REQUEST
NC1
NC2
CMN_GND
C6147
C6148
C6145
C6151
C6152
C6141
C6142
C6143
C6133
C6134
C6137
C6138
C6135
C6136
Global Signal Name

PM8550VS_G


SC6000
1
2
3
4
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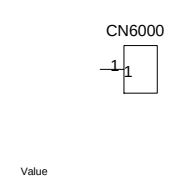
16
17
18
19
20
Global Signal Name


CN6000
1
Value

SHIELD CAN

2D BARCORD

TITLE TabS9_CORE
SIZE A1 | DWG NO PDM NUMBER | REV Rev
DRAWN BY User Name | SHEET 3 of 4 | Date Drawing Date

COREPCB PAD(Core PCB)

PCB100
(1/2)
PCIE0_TX1_M_WLAN
PCIE0_TX0_M_WLAN
PCIE0_REFCLK_M_WLAN
PCIE0_RX1_M_WLAN
PCIE0_RX0_M_WLAN
CC_SCL_1P8
CC_SDA_1P8
6AXIS1_INT2
6AXIS1_INT1
GRIP2_SCL_1P8
BT_UART_RXD
BT_UART_TXD
Global Signal Name
PCIE0_TX1_P_WLAN
PCIE0_TX0_P_WLAN
PCIE0_REFCLK_P_WLAN
PCIE0_RX1_P_WLAN
PCIE0_RX0_P_WLAN
REDRIVER_SDA_1P8
UWCAM_SDA_1P8
CAM_AF_EEP_SDA_1P8
GRIP2_SDA_1P8
WCN_SW_CTRL
BT_EN
WCN_SW_CTRL_WL_CX
UW_VTCAM_SDA_1P8
PMIC_SPMI_DATA
PMIC_SPMI_CLK
UWCAM_MIPI_D3_P
UWCAM_MIPI_D3_N
SDC2_CLK
REDRIVER_SCL_1P8
UWCAM_SCL_1P8
MAINCAM_SDA_1P8
MAINCAM_RST_N
BT_UART_RTS
BT_UART_CTS
CC_DIR
M_RF_ID_1
BAT_ID
SLEEP_CLK
WLAN_BT_SLIMBUS_CLK
WLAN_BT_SLIMBUS_DATA
UWCAM_MIPI_D2_P
UWCAM_MIPI_D2_N
MAINCAM_SCL_1P8
GRIP_INT_3
CAM_AF_EEP_SCL_1P8
MCU_SCL_1P8
UW_VTCAM_SCL_1P8
HALL_INT
CABLE_DET_GPS
M_RF_ID_2
M_RF_ID_0
3RD_MIC_CLK
3RD_MIC_DATA
UWCAM_MIPI_D1_P
UWCAM_MIPI_D1_N
UW_VTCAM_MCLK
UW_VTCAM_MIPI_D1_N
UW_VTCAM_MIPI_CLK_N
UW_VTCAM_MIPI_D0_N
UW_VTCAM_MIPI_D3_P
UW_VTCAM_MIPI_D3_N
VTCAM_MIPI_CLK_N
VTCAM_MIPI_D0_N
VTCAM_MIPI_D1_N
VTCAM_MIPI_D2_N
VTCAM_MIPI_D3_N
SDC2_D1
SDC2_CMD
SDC2_D2
Global Signal Name
CABLE_DET_MAIN2
PEN_IRQ
ground
DSI0_NC_LN3_M
DSI0_C2_LN3_P
DSI0_A2_LN2_P
FAULT_N
DSI0_A1_LN1_M
DSI0_B1_CLK0_P
DSI0_B2_LN2_M
DSI0_C0_LN1_P
DSI0_C1_CLK0_M
DSI0_A0_LN0_P
DSI0_B0_LN0_M
MAINCAM_MIPI_D0_P
MAINCAM_MIPI_D0_N
MAINCAM_MIPI_CLK_N
PMIC_RB
MAINCAM_MIPI_D1_N
MAINCAM_MIPI_D1_P
MAINCAM_MIPI_CLK_P
MAINCAM_MIPI_D3_N
MAINCAM_MIPI_D2_N
MAINCAM_MIPI_D3_P
MAINCAM_MIPI_D2_P
MAINCAM_MCLK
UWCAM_MIPI_CLK_P
UWCAM_MIPI_CLK_N
MAIN_MIC_DATA
MAIN_MIC_CLK
PON_RESET_N
IF_PMIC_SDA_1P8
IF_PMIC_SCL_1P8
PCIE0_RST_N
PCIE0_WAKE_N
CABLE_DET_WIFI0
MCU_SWCLK
MCU_SDA_1P8
WLAN_EN
SECURE_NVM_SCL_1P8
VTCAM_RST_N
SDC2_D3
SDC2_D0
VTCAM_MCLK
VTCAM_MIPI_D3_P
VTCAM_MIPI_D2_P
VTCAM_MIPI_D1_P
VTCAM_MIPI_D0_P
VTCAM_MIPI_CLK_P
UW_VTCAM_MIPI_D2_P
UW_VTCAM_MIPI_D2_N
UW_VTCAM_MIPI_D0_P
UW_VTCAM_MIPI_CLK_P
UW_VTCAM_MIPI_D1_P
UWCAM_MCLK
UWCAM_MIPI_D0_N
UWCAM_MIPI_D0_P
SUB_MIC_CLK
SUB_MIC_DATA
PMK_CLK6_CXO
PCIE0_CLK_REQ_N
KBD_PWR_EN
MCU_NRST
DC_IRQ_N
SECURE_NVM_SDA_1P8
UW_VTCAM_RST_N
UWCAM_RST_N
PCB100
(2/2)
QLINK2_EN
QLINK2_REQ
QLINK2_WMSS_RESET_N
FCAM_LDO_EN
JTAG_TDI
QLINK2_L1_P
QLINK2_L3_P
JTAG_TMS
JTAG_TDO
QLINK2_L1_M
QLINK2_L3_M
JTAG_TCK
RCAM2_LDO_EN
SPK_AMP_LDO_EN
JTAG_RST_OUT_N
QLINK2_L0_M
QLINK2_L2_P
KBD_DCDC_SCL_1P8
RCAM_LDO_EN
JTAG_TRST_N
MOTOR_LDO_EN
QLINK2_L0_P
QLINK2_L2_M
KBD_DCDC_SDA_1P8
GRIP3_SCL_1P8
OCTA_DCDC_SCL_1P8
AMP_INT_N
MCU_CONN_INT
MCU_INT
DIRECT_SDA_1P8
GRIP3_SDA_1P8
OCTA_DCDC_SDA_1P8
QLINK2_L4_P
QLINK2_CLK_M
DIRECT_SCL_1P8
AMP_SDA_1P8
AMP_SCL_1P8
QLINK2_L4_M
QLINK2_CLK_P
QLINK1_EN
QLINK1_REQ
QLINK1_WMSS_RESET_N
QLINK0_EN
QLINK0_REQ
QLINK0_WMSS_RESET_N
QLINK1_L0_M
QLINK1_L0_P
QLINK1_L2_M
QLINK1_L2_P
QLINK0_L3_P
QLINK0_L0_P
QLINK0_L3_M
QLINK0_L0_M
QLINK1_L4_P
QLINK1_L4_M
QLINK1_L7_P
QLINK1_L7_M
QLINK0_L1_P
QLINK0_CLK_M
QLINK1_L1_P
QLINK1_L1_M
QLINK1_CLK_P
QLINK1_CLK_M
QLINK0_L1_M
QLINK0_CLK_P
Global Signal Name
QLINK1_L5_M
QLINK1_L5_P
QLINK1_L3_M
QLINK1_L3_P
QLINK0_L2_P
QLINK0_L4_P
QLINK0_L2_M
QLINK0_L4_M
QLINK1_L8_P
QLINK1_L8_M
QLINK1_L6_P
QLINK1_L6_M
SIM1_EN
BLANKET_THM
SDM_RFFE0_CLK
SDM_RFFE0_DATA
AMP_I2S_LRCK
AMP_I2S_D_IN
AMP_I2S_D_OUT
AMP_I2S_BCLK
DISP_RST
DISP_DET
SIM2_EN
FLASH_LED
FLASH_LED
SDM_RFFE1_CLK
SDM_RFFE1_DATA
OCTA_DCDC_EN
VTCAM_SDA_1P8
VTCAM_SCL_1P8
FLASH_LED
WCN_HOST_SOL_OUT
GRIP_INT_2
HALL_INT_WACOM
SDM_RFFE4_DATA
SDM_RFFE4_CLK
BAT_THM
CHG_THM
SUB_CON_DET
WCN_WLAN_SOL_IN
SUB_PCB_ID
SIM1_DET
SIM2_DET
MCU_ACC_SDA_1P8
MCU_ACC_SCL_1P8
RGB1_SCL_1P8
GRIP1_LDO_EN
VOL_UP
CHG_DET
SDCARD_DET
RGB1_SDA_1P8
RGB2_SCL_1P8
MAGNETIC_SCL_1P8
MAGNETIC_SDA_1P8
Global Signal Name
BTP_RST_N
BTP_SPI_CS_N
BTP_SPI_CLK
BTP_SPI_MOSI
BTP_SPI_MISO
GRIP_INT_1
EDP_AUX_N
USB3.0_SSTX1_M_AP
USB3.0_SSTX0_M_AP
USB3.0_SSRX0_P_AP
USB3.0_SSRX1_P_AP
eUSB_AP_D_P
UW_VTCAM_LDO_EN
EDP_AUX_P
USB3.0_SSTX1_P_AP
USB3.0_SSTX0_P_AP
USB3.0_SSRX0_M_AP
USB3.0_SSRX1_M_AP
eUSB_AP_D_N
Global Signal Name
UFS0_TX0_P
UFS0_TX1_P
UFS0_RX0_P
UFS0_RX1_P
UFS0_RESET_N
UFS0_TX0_M
UFS0_TX1_M
UFS0_RX0_M
UFS0_RX1_M
CABLE_DET_WIFI1
SIM1_DATA_1P2
SIM1_CLK_1P2
BTP_LDO_EN
SM_PS_HOLD
WLAN_COEX_SDXUART_RX
TSP_SCL_1P8
GPS_L1_LNA_EN
LAA_AS_EN
LAA_EN_0
TSP_SDA_1P8
WLAN_COEX_SDXUART_TX
PEN_LDO_EN
SIM2_DATA_1P2
SIM2_CLK_1P2
ENDC_GRIP_LDO_EN
TSP_INT_N
MCU_ACC_INT
eUSB_SCL_1P8
PEN_SDA_1P8
HWID_LRPST_THM
GRIP2_LDO_EN
AP_UART_TXD
AP_UART_RXD
eUSB_SDA_1P8
OTG_DET
VBUS_DISCHARGING_EN
PEN_SCL_1P8
CAM_IO_LDO_EN
TYPE_C_INT
AMP_RST_N
AP_USB_BOOT
PEN_PDCT
SIM1_RST_1P2
SIM2_RST_1P2
TCON_RDY
PEN_FWEL
DISP_TE
CABLE_DET_SUB1
EDP_SW_OE
CABLE_DET_SUB2
CABLE_DET_SUB4
IF_PMIC_INTB
CABLE_DET_SUB3
GRIP1_SDA_1P8
DISP_CON_DET
HWID_LRPST_THM_SEL
OVP_PWR_FLAGB
EDP_SW_SEL
GRIP1_SCL_1P8
eUSB_RPTR_RST_N
RGB2_SDA_1P8
6AXIS_I3C_SCL_1P8
6AXIS_I3C_SDA_1P8
GRIP_INT_4

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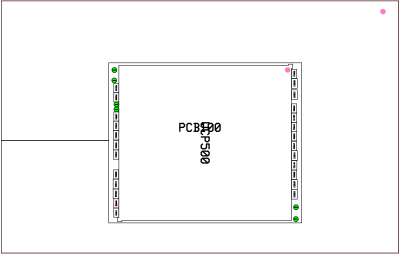
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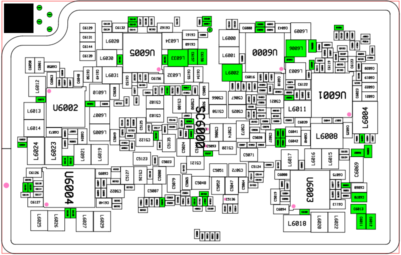
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